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Decommissioning and Waste Management Considerations for Fusion Facilities

Deuterium–Tritium Toroidal Magnetic Confinement Facilities

DECOMMISSIONING AND WASTE
MANAGEMENT CONSIDERATIONS
FOR FUSION FACILITIES

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DECOMMISSIONING AND WASTE MANAGEMENT CONSIDERATIONS FOR FUSION FACILITIES

DEUTERIUM–TRITIUM TOROIDAL MAGNETIC
CONFINEMENT FACILITIES

INTERNATIONAL ATOMIC ENERGY AGENCY
VIENNA, 2026

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FOREWORD

The decommissioning of deuterium–tritium magnetic confinement fusion facilities has several unique aspects arising from their complex design and mode of operation, with neutron activated components being of particular importance. This reflects the high emission rate of energetic neutrons, which are emitted with energies of about 14 MeV, characteristic of deuterium–tritium plasmas. Another important factor is the use and production of tritium during operation. The high mobility of this isotope and its tendency to diffuse into facility components can result in significant tritium inventories and volumes of tritium contaminated materials that need to be managed.

Many experimental fusion facilities around the world have already been operated and subsequently closed, although not necessarily decommissioned. These facilities typically had a low radiological source term because they mostly operated without tritium, generally using hydrogen or deuterium plasmas instead. However, several experimental fusion facilities currently under development or in operation plan to use deuterium–tritium plasma, primarily because of its lower ignition temperature (i.e. the temperature at which the plasma becomes self-sustaining) compared with alternative fuels.

These facilities aim to support the development of next generation facilities, including demonstration plants, pilot plants and eventually fusion power plants, all of which are referred to as fusion facilities in this publication. To ensure effective decommissioning, issues related to the high neutron flux environment and tritium usage, particularly in fusion power plants, will need increased attention. Key considerations include workforce radiation exposure during dismantling, tritium handling, remote operations and the management of radioactive waste.

There is currently only limited international experience with the dismantling of fusion facilities that have operated with deuterium–tritium plasmas, although valuable insights have been gained from the dismantling of experimental facilities such as the Tokamak Fusion Test Reactor (TFTR) at the Princeton Plasma Physics Laboratory in the United States of America. Additional experience is expected to be gained in the coming years from the dismantling of the Joint European Torus (JET) in the United Kingdom.

The overall dimensions of the international ITER facility, currently under construction in France, as well as those of other fusion facilities being planned, are significantly larger than those of TFTR and JET. The systems, structures and components that make up these large facilities are expected to generate substantial quantities of radioactive waste upon dismantling, owing to both material activation and tritium contaminated components and materials, which will necessitate effective safety measures.

This publication sets out technical considerations, presents lessons learned and provides Member States with other practical information relevant to the future decommissioning and waste management of toroidal magnetic confinement fusion facilities, especially those involving deuterium–tritium operation. This work is intended to support broader international efforts to plan effectively for the full life cycle of fusion facilities, from construction and operation through to safe and efficient decommissioning.

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1. INTRODUCTION

1.1. BACKGROUND

Experimental toroidal magnetic confinement fusion facilities, such as tokamaks, have played a key role in advancing fusion research and informing the design of future power-producing plants. As some of these facilities are being decommissioned, initial experience is being gained in dismantling activated components and managing radioactive waste, including tritiated materials. This experience provides a foundation for developing technical methodologies and waste management strategies that will, for future facilities, be required at an increasingly larger scale and complexity due to the higher neutron fluences and more extensive tritium inventories. These include facilities such as the International Thermonuclear Experimental Reactor (ITER), aiming for net scientific energy gain ($Q>1$), but also net power output demonstration (DEMO) facilities, and commercial-scale fusion plants beyond this. The net power output goal is typically expected to be realized either through generation of electricity (by converting the heat generated from fusion reactions) or through direct application of the generated heat, for example to industrial processes such as materials and chemical manufacturing or hydrogen production processes [1].

Deuterium-tritium (D-T) fuelled fusion facilities produce energetic neutrons (14.1 MeV) as a by-product of fusion reactions taking place within the plasma, leading to significant activation of the vessel components. The post-operation radiological environment resulting from neutron activation of materials during operation will be particularly important for materials comprising the first wall, but also tritium breeding blankets, radiological and thermal shields, the vacuum vessel structure and magnet components – see Section 3. Significant tritium inventories are expected to be present in the first wall and vessel structures. The dismantling of neutron activated components and the management of tritiated materials therefore constitutes the primary focus of decommissioning fusion facilities.

Current experience in the decommissioning of fusion facilities is limited to activities at a small number of magnetic confinement fusion-based experimental facilities globally. Indeed, only the Joint European Torus (JET) facility in the United Kingdom and Tokamak Fusion Test Reactor (TFTR) in the United States of America have operated with D-T fuel¹ at megawatt-scale fusion power levels. Although these operations were limited to short pulses lasting only a few seconds, they generated 14.1 MeV neutrons in sufficient quantities to cause appreciable material activation, necessitating approaches to decommissioning that required significant radiological consideration. Planned DEMOs and commercial fusion power plants are expected to operate at much higher fusion power levels, and over extended periods with higher availability compared to TFTR, JET and even ITER. This will result in significantly greater decommissioning challenges due to larger radiological inventories accumulated over time. In addition to managing substantial quantities of tritiated materials, decommissioning may also involve handling hazardous waste, such as beryllium and lead.

Fusion facilities are not expected to produce long-lived actinides, except in limited cases such as where beryllium with trace uranium impurities is used as a plasma-facing material or as a neutron multiplying medium in breeding blankets. Accordingly, waste management plans do not generally need to include disposal provisions which contain radioactivity over long term frames. More significant quantities of long-lived radionuclides may be expected to occur in hybrid fusion-fission concepts, though these fall outside the scope of this publication. While D-T fusion machines facilities are expected to produce substantial volumes of radioactive waste generated through neutron activation and tritiation, the radionuclides produced via activation are expected to be short lived, which means their activity levels will decay during a reasonably period of time. Hence storage consideration for such waste prior to clearance or authorized discharge are expected. The benefit of storing the radioactive waste, is that the decayed waste could be retrieved for clearance, processing and/or disposal at a later time, or, in the case

¹ Approximately 100 grams of tritium were consumed at TFTR, with than 1000 grams consumed at JET [6].

of effluent, for authorized discharge [Appendix I and Ref. 25]. These combined strategies are expected to reduce the need for high level waste which will need a geological repository.

1.2. OBJECTIVES

The objective of this publication is to report on experience gained and current good practice in the planning and implementation of decommissioning of toroidal magnetic confinement fusion facilities, including the management of resulting materials and radioactive waste. The publication is intended to support Member States involved in the operation or development of fusion plants, including DEMO facilities, D-T-based testing facilities and future fusion power plants. While the focus is on these facility types, the guidance on current good practice is also expected to be relevant to other fusion facilities.

1.3. SCOPE

The publication addresses experience gained from implementation of past and ongoing decommissioning activities and associated radioactive waste management, together with requirements related to general good practice in decommissioning planning and associated regulatory frameworks. Accordingly, the scope includes early-stage planning, relevant technologies, and strategies applicable to decommissioning and waste handling. It also covers research and development needs, along with the implications for authorization processes and regulatory oversight in the implementation of decommissioning of waste treatment systems.

1.4. STRUCTURE OF THE DOCUMENT

This publication comprises seven main sections, structured as follows:

Section 2 summarizes the basic principles of fusion energy and the main current fusion facilities.

Section 3 outlines the overarching factors influencing the decommissioning of fusion facilities taking account of experiences already gained from decommissioning of toroidal magnetic fusion facilities and describes the origin and properties of materials from these facilities.

Section 4 discusses the overall regulatory framework, including aspects related to national policy and associated strategic objectives and infrastructure.

Section 5 addresses the planning and implementation of decommissioning. It provides an overview of technologies and tools relevant to the decommissioning of industrial fusion facilities. It also highlights specific research and development needed to ensure availability of the required technologies.

Section 6 focuses on the management of radioactive and hazardous materials and waste, taking into account their characteristics and functions within fusion facilities.

Section 7 presents a summary of the main conclusions.

Appendix I provides overview of the IAEA relevant safety standards and guides. Annexes I-V summarize practical experience by Member States applicable to the topic of this publication.

2. FUSION BASICS AND OVERVIEW OF WORLDWIDE FUSION FACILITIES

Magnetic confinement fusion facilities (e.g. tokamaks) involve thermonuclear fusion reactions taking place in a hot plasma, typically at temperatures exceeding 100M K. The plasma is a state of matter characterized by positive fusion fuel ions and free-moving electrons with unique properties that are distinct from solids, liquids, or gases.

2.1. CATEGORIES OF FUSION FACILITY CONCEPTS

The fusion reactions most widely studied are shown in Table 1 [2], with most of the fusion plant concepts under development being expected to use D-T fuel² [3]. This is because D-T fusion is regarded as a relatively easier route to achieving fusion breakeven as its reactivity is significantly higher at lower plasma temperatures (typically in the range 10-20 keV) than other fuels. In addition, its Q-value of 17.6 MeV per fusion reaction is relatively high. D-T fuelling is currently the most researched option by both public and private sector actors. Fig. 1 shows a comparison of the most widely studied fusion reactions.

TABLE 1. LIST OF THE MOST WIDELY STUDIED FUSION REACTIONS (*reproduced from [2]*)

Reactants	Products
D-T	${}^4\text{He}$ (3.5 MeV) + n (14.1 MeV)
D-D	T (1.01 MeV) + p (3.02 MeV) (50%)
	${}^3\text{He}$ (0.82 MeV) + n (2.45 MeV) (50%)
D- ${}^3\text{He}$	${}^4\text{He}$ (3.6 MeV) + p (14.7 MeV)
T-T	${}^4\text{He}$ + 2n + 11.3 MeV
${}^3\text{He}$ - ${}^3\text{He}$	${}^4\text{He}$ + 2p
${}^3\text{He}$ -T	${}^4\text{He}$ + p + n + 12.1 MeV (51%)
	${}^4\text{He}$ (4.8 MeV) + D (9.5 MeV) (43%)
	${}^4\text{He}$ (0.5 MeV) + n (1.9 MeV) + p (11.9 MeV) (6%)
D- ${}^6\text{Li}$	2 ${}^4\text{He}$ + 22.4 MeV
p- ${}^6\text{Li}$	${}^4\text{He}$ (1.7 MeV) + ${}^3\text{He}$ (2.3 MeV)
${}^3\text{He}$ - ${}^6\text{Li}$	2 ${}^4\text{He}$ + p + 16.9 MeV
p- ${}^{11}\text{B}$	3 ${}^4\text{He}$ + 8.7 MeV



FIG. 1. Fuel sources considered by the private sector as of 2025 [3] (Courtesy of Fusion Industry Association).

Various fusion facility concepts are currently under research and development, in the public and private sectors and as public-private partnerships (see Fig. 2). These can be broadly grouped into the following categories:

- Magnetic confinement fusion (e.g. tokamaks, stellarators/heliotrons, which confine the plasma using strong magnetic fields), representing about 63% of the operating, under construction or planned fusion facilities by the private sector [1];
- Inertial confinement fusion (which use intense energy beams to compress and heat fuel pellets), representing around 8% of the fusion facilities (including planned ones) by the private sector [1];

² A number of private sector actors including in the United States of America and in Australia are researching the potential use of proton-boron-11 (p- ${}^{11}\text{B}$) fuel.

- Alternative confinement concepts around 28% (e.g. dense plasma focus; field reversed configuration; magnetic mirror machine; magnetized target fusion).

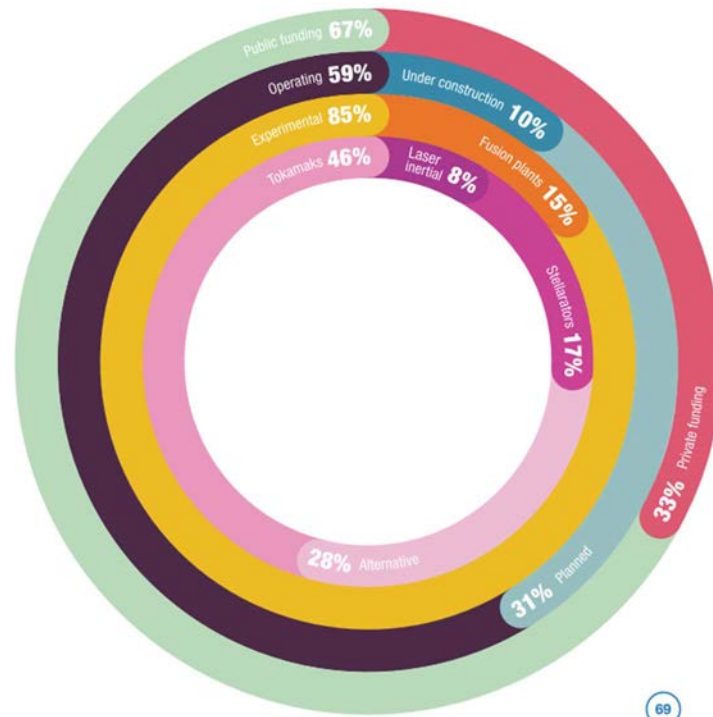


FIG. 2. Range of fusion facilities operating, under construction and planned, as of 2025 (Reproduced from [1]).

2.2. TYPICAL FUSION SYSTEMS AND FUNCTIONS OF A MAGNETIC CONFINEMENT FUSION FACILITY

Magnetic confinement fusion facilities comprise structures, systems and components in a complex configuration, typically involving increased complexity with increasing facility power.

2.2.1. Vacuum vessel

The vacuum vessel is a metallic chamber that houses the fusion plasma and serves as the first confinement barrier. It is maintained under high vacuum conditions to create the necessary environment for sustaining fusion reactions.

2.2.2. In-vessel components

The in-vessel components comprise the systems and components located inside the vacuum vessel. For a typical facility using magnetic confinement (e.g. tokamak, stellarator), these include:

- The first wall – the divertor, shielding structures, diagnostic and plasma control hardware, heating system antennae and launchers, embedded coolant piping, together with the breeding blanket modules;
- Cooling systems and associated piping.

Their composition and designs vary depending on the purpose of the fusion system, whether it is an experimental facility, a DEMO or a commercial Facility. In DEMO and commercial-scale fusion facilities, a critical in-vessel component is the tritium breeding blanket, positioned directly behind the first wall. The tritium breeding blanket is the largest in-vessel component; it is intended to capture fusion-generated neutrons to generate high-grade heat and facilitate nuclear reactions with lithium or

lithium compounds to produce tritium. It also serves as a shielding function, protecting sensitive components such as superconducting magnets and the outer wall of the vacuum vessel from radiation damage and nuclear heating.

The second largest in-vessel component is the divertor, located in the vacuum vessel. The role of the divertor is to handle the high power plasma heat exhaust and to remove impurities, including helium ash, a by-product of the fusion reactions [4].

2.2.3. Magnets

Strong magnetic fields are essential in magnetic confinement fusion facilities to confine and shape the plasma, but they also induce significant mechanical stresses on the supporting structures. In tokamaks, these magnetic fields are generated by a combination of magnetic systems, including the central solenoid, toroidal field coils, and poloidal field coils, some of which operate using copper-based magnets and some using superconducting magnet technology.

2.2.4. Fuel systems

Fusion facilities using D-T fuel will consume tritium at a rate of 55.8 kg tritium per GW fusion power per year [5]. Depending on the design the following features apply [6]:

- Tritium is expected to be produced by the breeding blankets by absorption of the fusion neutrons and consumption of lithium;
- Vacuum pumping systems remove the exhaust including the helium ash and the unburnt D-T fraction, keeping the burning plasma under vacuum, as well as generating the vacuum condition to start a new plasma discharge between plasma pulses;
- fuel injection systems provide the fuel to the plasma³ in sufficient amounts to keep the target core density constant at the level needed to sustain a burning plasma and deliver plasma enhancement gases to radiate power in order to control the heat flux to plasma facing components.

The above systems are located in the tokamak building, with the tritium processing systems being in a separate building – the tritium plant. These systems clean up the exhaust, recovering and extracting the unburnt tritium⁴, removal of impurities including the helium, and providing the required 50/50 D-T mix to be delivered to the fuel injection systems – see Fig. 3.

Tritium does not occur naturally in significant quantities. As a result, it needs to be produced (or ‘bred’) through fusion reactions between fusion-generated neutrons and lithium-based breeder materials, which in many designs are positioned behind the first wall surrounding the plasma. Tritium can permeate through dissolution and diffusion processes that accelerate at elevated temperatures, resulting in both surface and bulk contamination.

³ E.g. a Pellet Enhanced Gas (PEG) fuelling system.

⁴ Typical fusion reaction yields are currently of the order of 1%, which underlines the importance of the fuel cycle.

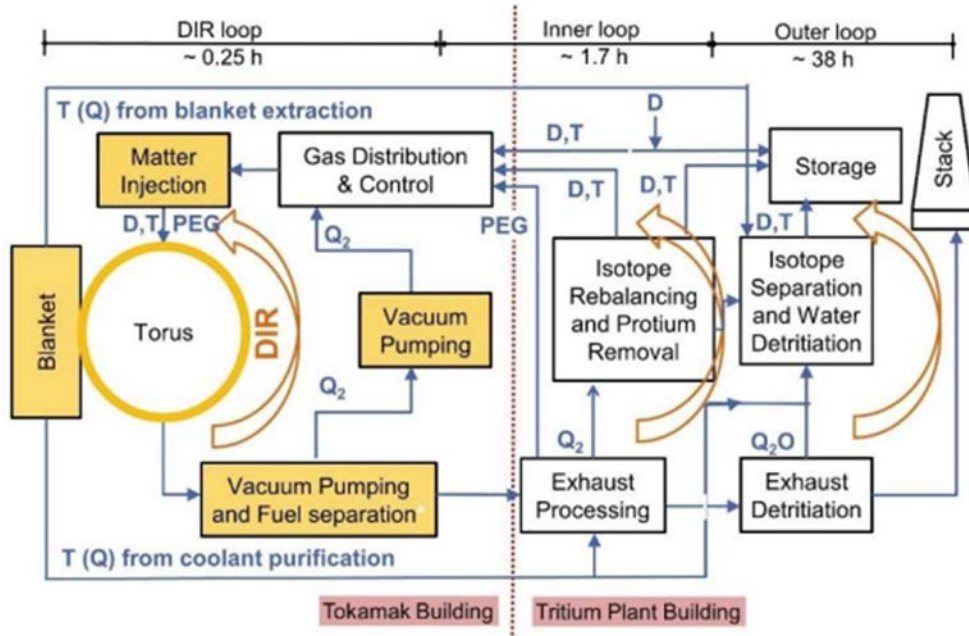


FIG. 3. D-T Fuel cycle (Reproduced from [6]) (Q denotes any of the hydrogen isotopes H/D/T).

2.2.5. Energy extraction and power generation circuit

Depending on the power level of the plant, active cooling may be required for the in-vessel components and vacuum vessel. In future fusion power plants, thermal energy extraction is expected to be integrated with an energy conversion system to generate electricity or provide high-grade process heat.

2.2.6. Heating systems / diagnostics

To initiate and control plasma operating conditions, several heating systems are typically employed. In some plants, these include high energy neutral particle beams that penetrate the magnetic confinement field (neutral beam injection systems), as well as ion and electron cyclotron resonance heating systems. Diagnostics are conventionally an array of instrumentation systems which monitor key plasma and system characteristics such that operations are sustained and safe.

Both heating and diagnostics systems can be large and complex components. In the case of ITER, some of these systems exceed 10 metres in length and span several meters in other dimensions.

2.2.7. Maintenance and nuclear services

Due to the presence of highly irradiated and tritiated components, dedicated remote handling tools may be required for maintenance operations. The tools need to be capable of operating not only inside the vacuum vessel but also in adjacent areas and within hot cells, which serve to contain irradiation and prevent the spread of tritium-contamination. Examples of key systems are presented in Figs. 3 to 7, with the internal structure of the JET tokamak shown in Fig. 4, and the main components of the ITER tokamak shown in Fig. 5.

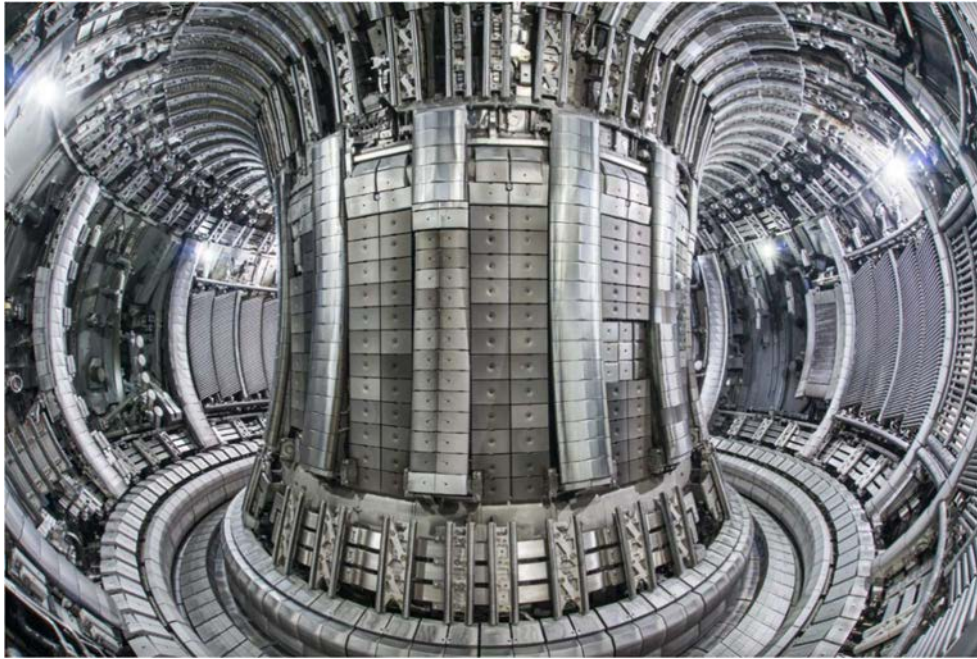


FIG. 4. Internal structure of the JET tokamak (Courtesy of UKAEA).

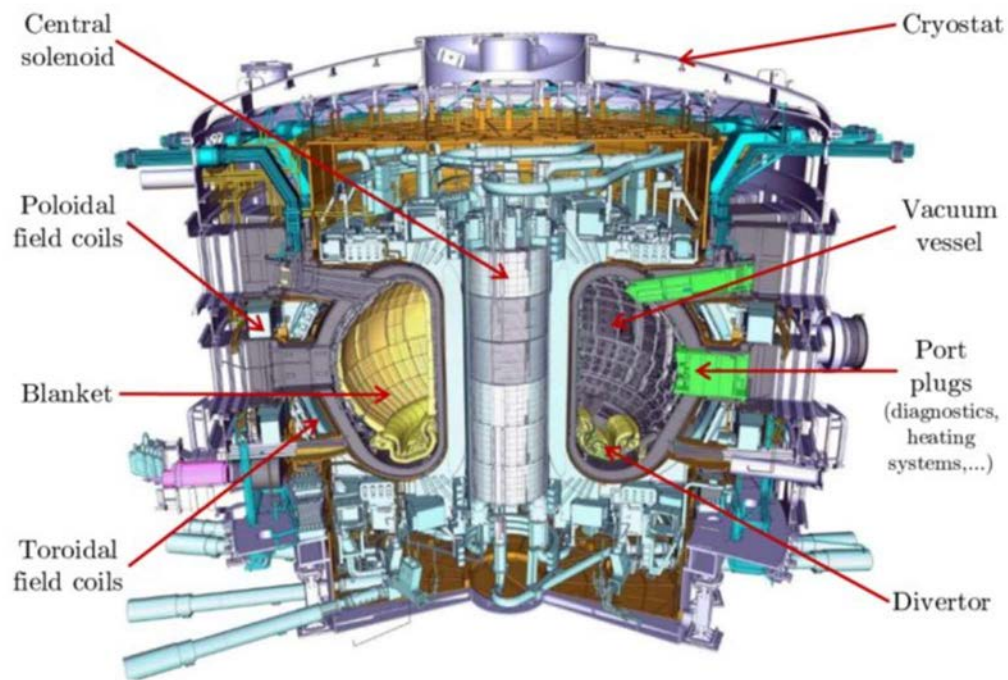


FIG. 5. Main components of the ITER tokamak (Courtesy of ITER Organisation).

Surrounding the tokamak, various areas are designated for supporting systems such as diagnostics, cooling water, cryogenics, steady state and pulsed power systems, vacuum pumping and control systems. An example of this layout is provided in Fig. 6.

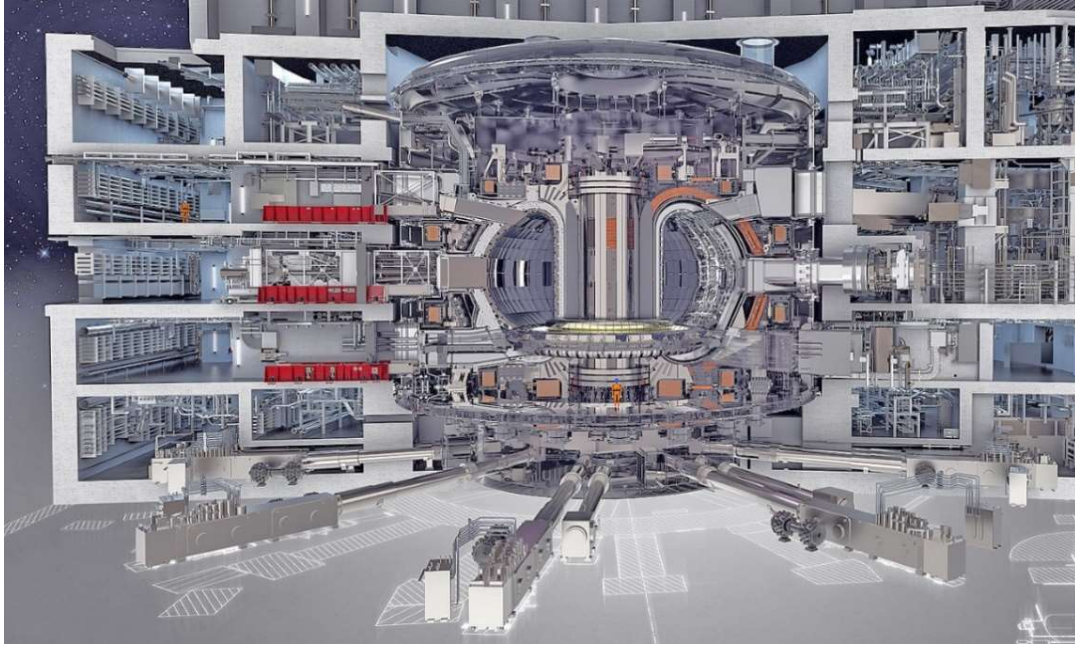


FIG. 6. ITER tokamak building with its associated supporting systems (Courtesy of ITER Organisation).

The infrastructure of a fusion facility extends well beyond the tokamak and its core components. As illustrated in Fig. 7, which shows the JET facility, a fusion facility encompasses a wide range of associated systems and support areas.

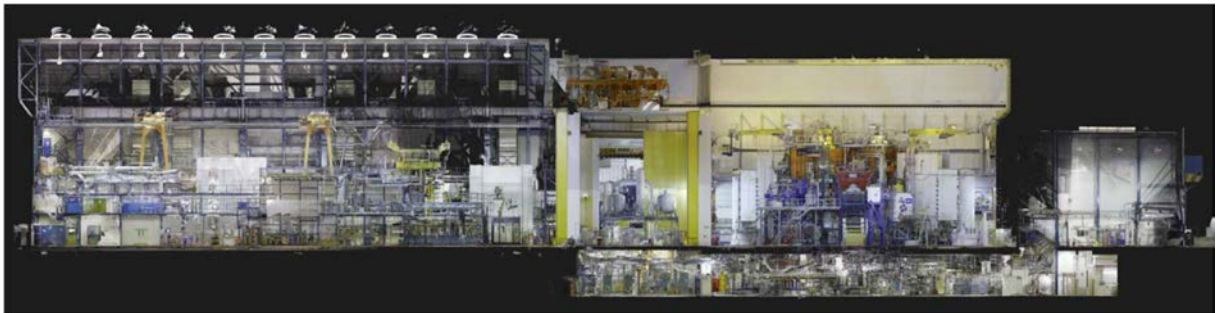


FIG. 7. 3D scan cut-away image of the JET torus hall, assembly hall, basement and hot cell (Courtesy of UKAEA).

3. KEY CONSIDERATIONS ON DECOMMISSIONING AND WASTE MANAGEMENT

The design of systems using D-T plasma introduces fusion specific issues not typically encountered in fission reactors, including:

- Overall architecture of the fusion systems;
- Size;
- Presence of significant amounts of tritium⁵;
- Requirement to remotely dismantle large and heavy structures.

Decommissioning and waste management considerations for such plants are dominated by the effects of high levels of neutron activation and of tritium contamination of structures, systems, and components during the operational phase of the facility (see also Section 5). Such considerations apply despite the

⁵ Fusion power plants may consume in the order of 100 kg of tritium or more per year [6].

expected differences in material choices, size, and operational characteristics. In light of this, fusion facilities need to be designed, operated and decommissioned with due consideration both of specific radiological and chemical properties of tritium and of the combined effects of tritium-contamination due to its permeation and neutron activation. The latter may have important implications for radiation protection and radioactive material management throughout the facility lifecycle.

3.1. DECOMMISSIONING EXPERIENCES OF REPRESENTATIVE TOROIDAL MAGNETIC FUSION FACILITIES

Although significant decommissioning experience has been gained for research-scale facilities which have predominantly used D-D fuelling (see Annex I), these activities have provided limited analogy to the decommissioning to future large scale D-T systems due to the absence of significant tritium contamination and 14.1 MeV neutron activation of materials. Relevant information on the decommissioning of a D-D tokamak in Japan is presented in Annex I-1, including the sequence of dismantling activities and classification of the removed components.

In terms of D-T systems, a key example of decommissioning is provided by the TFTR system operated by the Princeton Physics Laboratory in the United States of America. This tokamak and associated infrastructure were decommissioned after completion of D-T operations. Key considerations such as the breaching of tritium systems, experience of 14.1 MeV neutron activation, and size reduction activities are highly relevant to future fusion decommissioning projects. A summary of the TFTR experience is provided in Annex I-2. Decommissioning of the JET tokamak systems commenced in 2024, providing an important current example of decommissioning considerations of a D-T fuelled system. The system operated with tritium inventories during operations (~grams), exhibiting corresponding tritium permeation into materials and corresponding contamination. JET was the first facility to deploy detritiation techniques for waste materials in the context of dismantling operations (See Annexes I-2 and V-1).

Projections for industrial fusion facilities are difficult to present comprehensively due to the large diversity of projects. However, the quantity of low-level waste arising during decommissioning of fusion facilities is likely to be substantial and dedicated approaches such as clearance and recycling may need further development (see Section 6). In the same manner, even if the tritium content may vary among the different concepts or designs of fusion facilities, there will inevitably be parts of components (circuits of the tritium processing loop for example) that will have a high tritium specific activity, potentially needing detritiation techniques⁶ to simplify their management. This may include the establishment of a suitable disposal route and or/closing out the fuel cycle (see Section 6.2).

3.2. RADIOACTIVE AND HAZARDOUS MATERIALS AND WASTE IN FUSION FACILITIES

3.2.1. Origin

Fusion radioactive and hazardous waste may arise from numerous radioactive and hazardous materials (such as beryllium or lead-lithium) in a wide range of concentrations and in a variety of physical and chemical forms. Radioactive wastes generated in fusion facilities are significantly different from those from fission nuclear power plants (NPPs), not only in terms of quantities and radioisotope composition but also due to the absence of transuranic elements, the presence of significant quantities of tritium, lower decay heat removal requirement and lower radioactivity content overall of a fusion power plant when compared to NPPs. There is also the possibility of generation of transuranic elements due to impurities, e.g. of beryllium.

⁶ Detritiation techniques used in future machines may include ‘baking’, glow discharge cleaning, ion/electron cyclotron and discharge cleaning (see Section 5.3).

The high energy neutrons from D-T fusion pose considerable challenges for the structural materials within and including the vacuum vessel. Radioactive waste generated in fusion facilities comprises primarily tritiated materials and activation products, i.e. radioisotopes generated from neutron activation of structural materials such as steels (e.g. EUROFER97, F82H, and AISI 316L). Some alloying and impurity elements may produce long-lived isotopes upon irradiation, potentially complicating disposal. Elements subject to neutronic activation include carbon, chromium, manganese, copper, aluminium, together with impurities (especially niobium, molybdenum, rhenium, iridium and silver⁷), which might preclude their disposal in near surface disposal facilities – see Annex III. Reuse, recycling and clearance efforts are therefore of crucial importance.

3.2.2. Classification of radioactive waste

An overall waste management strategy — covering the entire facility lifecycle — needs to be developed at the design stage of a fusion project. A typical prerequisite for this is the establishment of a system of waste classification, usually associated with the regulations for management of radioactive waste. Different approaches to radioactive waste classification are used internationally, usually involving some combination of activity, half-life and specific requirements associated with the waste.

An international classification system has been developed by IAEA based on activity levels and half-life, considering long-term safety [7]. This approach classifies waste as: high level (heat generating, requiring geological disposal), intermediate level (requiring geological disposal), low level (suitable for disposal in near surface engineered repositories), very low level (suitable for disposal in surface non-engineered repositories), very short lived waste (requiring storage for decay) and exempt waste (which may be cleared from regulatory control or are exempt from such control due to their very low hazard potential). Although this approach is not applied universally, national approaches generally follow broadly compatible approaches, e.g. France applies nuclide specific limits as part of the classification system.

An important characteristic of fusion energy is that large quantities of waste may be produced (e.g. to the scale of structures, buildings, tokamaks, material handling infrastructure), but these are mainly classified as low level or very low-level waste. Some intermediate level waste may also arise, though this will generally decay to low level waste after a period of one hundred years or so (see e.g. Annex II and Annex V). This phenomenon has several important implications: a large proportion of the waste is potentially suitable for clearance from regulatory control⁸, and the relatively small proportion of intermediate level waste which may arise is not long lived and therefore does not require to be disposed in geological formations to secure long-term protection of people or the environment. Nonetheless, the intermediate level waste does require management for a significant period, such as its placement in a secure storage environment.

An important objective of materials selection is limiting neutron activation, such that even highly activated components ultimately result in only low-level waste (LLW). While precise waste classifications are country specific, LLW is generally distinguished by limited amounts of long-lived radionuclides, such that it requires robust isolation for only a few hundred years and is therefore suitable for near-surface (as opposed to geological) disposal.

3.2.3. Main management challenges

When high energy neutrons interact with the structures and component material, they induce activation, making the materials radioactive. An important objective of selection of materials used in fusion

⁷ Future facilities may be expected to minimize or even exclude these elements.

⁸ In this publication recycling is generally taken to mean that the material can be reused within the nuclear industry, while clearance means that the material can be released from regulatory control for conventional uses, sometimes subject to restrictions (specific clearance). A variety of national approaches to regulation of clearance currently exist – see Annex II.

reactors is to minimize the potential for creation of long lived radionuclides through this process, such that even highly activated components ultimately result in only low-level waste (LLW).

Waste from fusion facilities that initially exceeds LLW criteria may be able to meet following specific treatment. The experience at JET is instructive, even if the overall neutron budget is significantly lower than in ITER and even lower compared to a nuclear fusion power plant, it is foreseen that there will be no ILW because of neutron activation after 10 years of service lifetime. Tritium permeation, due to its high mobility, could still lead to ILW if no treatment were undertaken; detecting and measuring it presents an important challenge.

Detritiation technologies could be used to recover tritium for reuse as fuel, thereby enabling the diversion of significant quantities of radioactive material for disposal and reducing overall waste management costs. Such an approach is currently challenging as it could require the separation of some materials before treatment and the development of high treatment capacities (in some cases with an industrialization of the existing processes such as thermal treatment), together with establishment of systems for the efficient management of secondary waste (Annex V-1).

3.3. TIMELINE FOR DISMANTLING AND DISPOSAL – TRADE OFF AND POTENTIAL PROVISION OF INTERIM STORAGE FACILITIES

A key consideration for decommissioning is the duration of the period from permanent shutdown of a facility to the commencement of dismantling of its essential components. An ambition to complete decommissioning in a timely manner may conflict with a desire to delay dismantling until the high radiation levels present after shutdown have attenuated, thus reducing risks to the workforce and the risk of mobilization and release of radioactive materials.

Where dismantling of a fusion system is undertaken prior to the availability of facilities for disposal of the resulting waste, interim storage arrangements will need to be considered. Such facilities are highly relevant for fusion applications due to the comparatively short half-life of fusion relevant radionuclides and the benefits of interim storage for decay. Such an approach is current practice for particle accelerator complexes with highly activated materials [8].

Interim decay storage of materials at the fusion facility site may provide an option for some operation and decommissioning programmes, though it will be important to gain support for this approach from local communities. Decay storage brings the following potential benefits:

- Limits transport of radioactive material;
- Likely availability of a competent local workforce pool;
- Promotes reuse of capabilities (such as abatement, shielding and handling infrastructure) after fusion operations have ceased.

Alternatively, materials from fusion facilities may be consolidated for decay storage along with other radioactive materials in national storage facilities, if compatible with tritium inventories. While this may alleviate local community opposition and might enable the early reuse of a fusion site, there are significant challenges with regard to material transport and waste ownership / duty of care. National strategies for consolidated storage are currently not in place for fusion applications, such that retrieval and reuse / recycling of any materials from these facilities is likely to be beyond the designed scope of existing waste management facilities.

Decommissioning of large-scale fusion facilities immediately after permanent shutdown of the facility is likely to be very challenging due to the need to dismantle components that are highly radioactive, some with dose rates of hundreds or thousands of Sv/h or more (see Annexes I-3 and V-3). This would require radiation-resistant remote handling equipment operable at these dose rates, which may challenge the limits of many remote handling components. Such systems have not been demonstrated at the scale

and complexity required for fusion, hence adaptation to requirements for decommissioning of fusion facilities is necessary due to the large size of components and complexity of operation.

Conversely, after longer decay times (possibly ~100 years), the dose rates from most in-vessel reduced-activation steel (e.g. F82H or EUROFER97) components fall sharply to near or below limits typical of current commercial recycling facilities, e.g. 0.2 mSv/h contact dose rate. This results from the relatively short half-life (<30 years) of some of the key activation products⁹ in its structure systems and components, e.g. Co-57, Co-58, Co-60, Fe-55, Fe-59 and Mn-54. In such facilities, some long-lived radionuclides such as C-14, Nb-94 and some Mo isotopes, will be present. Decay behaviour and the resultant reduction in dose rates may be defined based on simulations of each fusion device and later confirmed by measurements performed on the plant following its permanent shutdown.

Subject to the above, a prolonged period of preparation before physical dismantling of the tokamak (of the order of a few decades) may be appropriate for fusion facilities. This need not be a dormant period but rather can be taken as an opportunity to commence site reconfiguration, create facilities supporting decommissioning and waste management, and perform radiological characterization. This will enable the reduction of the dose rate to operators and radiation hardness requirements for remote handling tools that remain necessary for dismantling operations of vacuum vessel components and other internal structures. There are likely to be significant parts of the site infrastructure (such as power supplies, gas and cryogenic systems) which can be decommissioned during this timeframe.

A prolonged period of preparation can also be used as an opportunity to optimize waste packaging and transport. As an example, following a decay period of a few decades, the dose resulting from Co-60 will have diminished, allowing the amount of waste that can be placed into a waste disposal container to be increased substantially, while respecting the 2 mSv/h contact dose rate limit from transport regulations.

Optimizing the overall decommissioning strategy requires a balance to be drawn from the benefits available from decay of short lived radioactivity, which supports delayed dismantling of certain parts of the facility, and the cost benefit available from immediate dismantling, e.g. leading to reduced power consumption and building maintenance costs. The management of decommissioning projects is also influenced by the annual financing capacity as well as the overall cost of the project. Striking a good balance is essential, considering the costs, risks induced by radioactivity, societal acceptability, options for reuse of the site, transport and storage capacities and availability of a disposal facility. The optimal approach to finding the right balance will be country specific. Preserving knowledge archives for future generations is necessary to maintain knowledge of the final status of plant systems. For fusion facilities like ITER, the lifetime of the vacuum vessel and in-vessel components will coincide with the operational lifetime of the facility. It is conceivable that future facilities will aim to replace some components during the facility lifetime, requiring emplacement of the disused components in storage facilities.

4. REGULATORY FRAMEWORK FOR DECOMMISSIONING AND WASTE MANAGEMENT AND MAIN DECOMMISSIONING PHASES

Implementing the decommissioning and management of the associated radioactive waste of a fusion facility will be in line with each Member State national policy and legal considerations, together with the existing relevant regulatory and institutional framework governing decommissioning and the management of radioactive and hazardous materials. In addition, the requirement for financing decommissioning of the fusion facility is an important consideration during the design phase, together

⁹ Some facility designs may minimize or exclude these radionuclides.

with national norms in terms of public involvement in decisions relating to its site release following its permanent shutdown and decommissioning.

Radioactive waste is generated not only from decommissioning, but also arises during the operation phase, e.g. gaseous or liquid radioactive waste (mostly tritium contaminated) and components that are removed or replaced, typically during maintenance.

While comprehensive international safety and regulatory guidance exist for facilities and activities [6] (all sources of ionizing radiation, e.g. irradiation facilities, nuclear power plants and activities such as radioactive waste management, decommissioning, respectively) in Member States, comparable guidance for fusion facilities is currently lacking. Nonetheless, the international safety standards developed by the IAEA for facilities safety requirements offer valuable insights that can assist countries in developing regulatory frameworks for decommissioning and waste management of fusion facilities – see Appendix I.

4.1. REGULATORY FRAMEWORK FOR DECOMMISSIONING AND WASTE MANAGEMENT OF FUSION FACILITIES

Current regulatory frameworks related to the decommissioning of fusion facilities and management of resulting waste in selected countries are summarized in Annex I and II – China (Annex II-1), France (Annex II-2), Japan (Annex II-3), Russian Federation (Annex II-4 and Annex II-5), United Kingdom [9, 10] and United States of America (Annex I-6).

Although there is currently no globally harmonized definition of a fusion power plant, many jurisdictions recognize the need to establish clear regulatory frameworks for fusion facilities intended to produce electricity or heat for commercial use. Existing regulatory processes for research scale fusion devices often serve as the foundation and precedent, and in some cases, may also apply to future fusion plants, either directly or with tailored modifications.

Many countries are actively considering approaches to regulating fusion. The following pages highlight recent regulatory developments that have been publicly announced or documented.

It may be assumed that these arrangements will evolve significantly with the evolution of fusion technology worldwide, i.e. with the construction and operation of fusion power plant facilities. The design choices of fusion technologies (operational characteristics, scale, source term) and the materials employed are likely, in turn, to impact the management of radioactive waste and decommissioning, in line with applicable national regulatory frameworks.

4.1.1. Regulatory approaches for oversight of decommissioning

In some Member States the approach taken is to designate a fusion facility as a nuclear facility and to apply the same overall regulatory framework as applied to facilities using nuclear fission, typically in line with [11]. In others, a bespoke regulatory approach for fusion facilities is being discussed or developed in which fusion facilities are classified as radiation, rather than nuclear facilities. Some jurisdictions combine elements of both approaches, depending on the power level of the facility, applying a graded approach to regulation, in the level of regulatory effort, detail, and regulatory oversight to decommissioning projects. Under most regimes the primary regulatory authority, both for facility operation and for decommissioning, is typically the nuclear or radiation safety regulator.

Countries which have adopted, or are considering to adopt, an approach to regulation of large fusion facilities which is similar to that applied for nuclear installations include China, France and the Russian Federation:

- In China the National Nuclear Safety Administration (NNSA) regulates nuclear and radiation safety, including regulatory oversight of fusion facilities. The existing regulatory framework

applies primarily for fission reactors. The regulatory authorities are actively working to develop a fusion specific safety regulatory framework. Additional details can be found in Annex II-1, indicating that the preliminary assessment suggests that the planned China Fusion Engineering Test Reactor (CFETR) will be regulated under the Law on Nuclear Safety, in accordance with underlying regulations, still to be developed, specific to fusion.

- The ITER facility is regulated as a licensed nuclear facility, under a similar regime as applied for fission, with the primary regulator being the Authority for Nuclear Safety and Radiation Protection (ASNR) – see Annex II-2. A preliminary description of the decommissioning strategy was produced in the early phases of the project (Preliminary Safety Report) and progressively developed further. Accordingly, a detailed decommissioning plan is submitted for consideration by the regulator as a condition of being granted a decree to decommission the facility. The authorities need to be notified in advance of the permanent shutdown of the facility and, in accordance with [12], it is required that this declaration be accompanied by an update of the decommissioning plan [13].
- The Russian Federation currently has no specific regulations in force relating to the operation or decommissioning of fusion facilities. Legislation has been drafted specifying that fusion facilities will be regulated under the main primary legislation on atomic energy use (Federal Law No. 170 on Atomic Energy Use), which covers nuclear installations, radiation sources and radioactive waste storage facilities. These facilities are regulated by Rostekhnadzor, the Federal Service for Environmental, Industrial and Nuclear Supervision. The draft law is currently undergoing readings in the Russian Parliament. According to the draft legislation fusion facilities could be classified either as nuclear facilities (facilities containing nuclear reactors or fissile materials) or as radiation sources (facilities containing radioactive materials or generating ionizing radiation), depending on the specific level of associated hazard, thus facilitating a graded approach to their regulation. Subject to approval of the draft regulation, the decommissioning of fusion facilities will in future be regulated under current regulations for nuclear facilities and radiation sources. Additional details are provided in Annex II-4.

Some countries, such as Japan the United States of America, which have a single regulatory authority for nuclear and radiation safety, have designated currently existing facilities as radiation facilities. A similar approach is envisaged in the United Kingdom, though not administered by the nuclear safety regulator. The intent in each case is to regulate fusion facilities as radiation facilities differentiated from nuclear fission facilities:

- The JT-60 device in Japan is regulated by the Nuclear Regulatory Authority (NRA) under the regulatory framework applied to radiation facilities using only non-fission materials throughout its lifecycle (i.e. the Radioisotopes Regulation Act), including decommissioning. A decommissioning plan for a radiation facility needs to be notified to NRA before decommissioning may proceed. The regulatory framework for larger fusion facilities, including DEMO type facilities, has not yet been determined; meanwhile, facilities under development will continue to be regulated under the Radioisotopes Regulation Act – see Annex II-3 and [14]¹⁰.
- In the UK fusion is currently designated as a ‘radioactive substances activity’, akin to large scale industrial irradiators or cyclotrons in the medical sector. The Energy Act (2023) excludes fusion facilities from compliance with the requirement of the Nuclear Installations Act such as the requirement to obtain a nuclear site licence. Accordingly, the nuclear safety regulator (Office for Nuclear Regulation) is not responsible for oversight of decommissioning, which responsibility is undertaken by the Environment Agency and Health and Safety Executive, which regulates operating radiation facilities [9, 10]. The JET facility serves as a pilot for the development of requirements for fusion decommissioning in the UK, e.g. a decommissioning strategy and plan is required [15].

¹⁰ Japan Fusion Energy Council, Press release: Fusion Energy White Paper Released as part of its activities to contribute to the spread of fusion energy, J-Fusion (2025) – <https://jfusion.jp/news/2025/0630501/>

- In the United States of America, facilities utilizing nuclear materials or byproduct material (radioactive material, tailings, and certain naturally occurring radioactive materials) for civilian purposes are regulated under the Atomic Energy Act, primarily by the Nuclear Regulatory Commission (NRC). Although there is currently no reference to fusion energy systems in the U.S. Code of Federal Regulations (CFRs), NRC staff are currently developing a new rule addressing both operations and decommissioning. This limited-scope rulemaking will amend the NRC's byproduct material regulations (10 CFR Part 30). It aims to establish regulatory framework requirements for fusion systems to address near-term fusion systems by providing appropriate fusion-related definitions, develop content of application requirements of a licensing application, and determine other appropriate requirements. The decommissioning of TFTR was performed with oversight from the Department of Energy's (DOEs) Office of Science.

4.1.2. Regulations for classification and management of specific radioactive waste from fusion facilities and material recycling

The management of large quantities of materials with low levels of radioactive contamination will generally involve a combination of strategies aimed at ensuring that the quantity of material requiring disposal in radioactive waste repositories is minimized. In line with this ambition, management options will typically need to include reuse and/or recycling of material and clearance of material (without significant contamination or activation) from regulatory control. Where these options are not feasible, the materials need to be managed as radioactive waste, i.e. processing, storage and disposal.

4.1.2.1. Clearance

Fusion is expected to generate a large quantity of VLLW/LLW and ILW radioactive materials (plus a small amount of HLW in some cases) – see Annex III. Some materials, such as those outside the vacuum vessel, are shielded from the high levels of neutron radiation seen within in-vessel components or affected only by surface contamination or limited tritium permeation. Nonetheless, these volumes require careful management, including during decommissioning activities¹¹. For lower severity radioactive materials, decontamination to the point of clearance may be pursued (refer to Annex V-2 for a discussion of material management in the context of the EU DEMO project). Clearance may be applied where radioactive material can be sufficiently decontaminated (or its radioactivity reduced by decay over a specific cooling period) to allow free release from regulatory control such that it no longer classified as radioactive material.

National approaches to clearance and threshold activity limits vary significantly. International clearance limits have been proposed by IAEA on the basis of the radiation dose potentially received from cleared materials being of the order of 10 μ Sv or less for reasonably foreseeable circumstances and 1 mSv for 'low probability' scenarios [16]; activity concentration levels for generic clearance of key radioisotopes are provided in [17]. A subset of these¹², reproduced in Table 2, have also been adopted by the French Ministry of Public Health [18].

¹¹ Fusion facility design may enable some component replacement during operation. In such cases management of radioactive materials, including clearance and recycling, may also occur during operation.

¹² Under French regulations, "the use of substances from an installation in which a nuclear activity is carried out or has been carried out, when they are contaminated, activated or likely to be contaminated by radionuclides [13], may be subject to an exemption [13] since these substances are first the subject of a recovery operation carried out in an installation mentioned in [13] the environmental code. If the products resulting from the recovery operation contain a radionuclide, the mass activity concentration of the radionuclide may not exceed the corresponding limit value defined in Table 5. When the exemption is granted, the products resulting from the recovery operation are no longer radioactive substances as defined in [13] the environmental code and no longer justify radioprotection controls following analysis under conditions consistent with those set by the derogation. At present this provision is applicable only to metallic radwaste." [18]

TABLE 2. CLEARANCE / EXEMPTION LEVELS FOR SELECTED RADIOISOTOPES

Isotope	IAEA Clearance Levels (Bq/g) [17]	France Exemption Levels (Bq/g) [18]	UK ‘Out-of-Scope’ levels for radioactive materials ¹³ [19]) (Bq/g)
Be-10			***
C-14	1	1	10
Co-60	0.1	0.1	0.1
Fe-55	1000	1000	1000
H-3	100	100	100
Mn-53	100	100	100
Mo-93	10	10	10
Nb-94	0.1	0.1	0.1
Ni-63	100	100	100
Ni-59	100	100	100
Tc-99	1	1	1
Nb-93m	10	10	10

A lack of harmonization between different countries could limit the interest of industrial companies in developing an international market. Clearance is not a universal practice, although regulations already exist in many countries and standards have been established in the European Union, Russian Federation, generally in line with those proposed by the IAEA. European countries (Sweden, Spain, Belgium and Italy) have cleared materials in industrial quantities, while the USA applies clearance on a case-by-case basis [20]. In the United States, past attempts by the NRC to establish general clearance limits have been met with significant public opposition [21].

Depending on jurisdictional limits, a significant volume of waste that is neutron activated but sufficiently shielded from high neutron fluence exposure and tritium contamination may meet clearance criteria without additional processing, following a period of decay. In the optimistic scenario presented in Annex V-2, large components, such as the bioshield and the cryostat, could be cleared after ~10 and ~30 years, respectively, with exposure to a dose limited to 10 µSv/year. Clearance of highly radioactive materials within the vacuum vessel and in-vessel components will not be feasible even after 100 years of decay time, though low activated materials may be cleared after ~10–30 years, depending on the component composition [22].

Some materials may become clearable following specific treatment, e.g. as undertaken at the Studsvik facility in Sweden described in [23], which achieves clearance of a significant fraction of components such as PWR steam generators, boilers, turbine components, and BWR heat exchangers (contaminated but non-activated primary circuit components) by melting and resultant separation of radionuclides. The expected nuclide distribution is predicted to be similar to that for contaminated metals, with tritium leaving the metal during the metal heat-up and being released; the process may need to be tailored to capture it in the ventilation system if the quantity of retained tritium in the metals is above releasable

¹³ Environmental Permitting (England and Wales) Regulations 2016 as amended — environmental legislation within the UK is devolved in Wales, Scotland and Northern Ireland. The example provided is from legislation within England and Wales, however the figures for clearance (i.e. ‘out of scope’ levels) are the same across the UK. Under UK legislation, not all radionuclides are given clearance limits however radionuclides of this category (such as Be-10) are subject to a limit of 0.01 Bq/g or a limit that can be demonstrated to give rise to a dose to a member of the public of up to 10 µSv per year calculated by reference to IAEA Safety Standards [17]

limits or of sufficient value to be recovered¹⁴. Certain radionuclides, if used for structural and in-vessel component compositions (e.g. Al-26 and Nb-94), may be captured and removed in slag by this process, while others (Ni-59, Ni-63, Co-60) cannot, and would require a storage and disposal.

4.1.2.2. Recycling

For more highly activated materials from fusion facilities such as in-vessel components, reduction of activity to clearance levels is unlikely to be feasible. For such materials, disposal as waste might still be avoided if they can be recycled. Here, recycling means that the material, while too radioactive to be cleared to the markets, might nevertheless be re-usable within the nuclear industry. Such a strategy requires that the design facilitates the separation (e.g. mechanically) of different materials in decommissioning to the extent possible and further requires that techniques be developed to separate of chemical species during processing.

Facilitating recycling and reuse requires that several diverse challenges are addressed, including the segregation of mixed activated materials, the application of chemical processes to remove detrimental isotopes, the management of secondary waste, and the expected properties of recycled materials, the manufacturing of components using radioactive materials as raw materials (see further details in Annex V-7) and addressing the build-up of radioactivity in the recycled materials over time.

The economics of recycling are likely to be influenced significantly by the desired time horizon. During the first decades after shutdown of a fusion reactor, some of the waste to be recycled could be comparable to spent fuel from a fission reactor in terms of short term dose rate, in which case the required industrial facilities may need be similar to the ones used for fuel reprocessing (hot cells, remote handling, etc.). The economic balance needs to be assessed, with an overall analysis considering the resource savings and reduction of the cost of storage or disposal but including the cost of treatment and disposal of the secondary waste streams that will be produced. Prospects for recycling and experience feedback are discussed further in [20].

The feasibility of recycling is also likely to depend significantly on the material in question. For the most highly activated components such as the blanket, divertor, and surrounding shielding, structural materials such as RAFM steel will comprise a significant waste volume. A technical analysis [22] considers the melting process described above and in [23] applied to these highly activated components; the radioactivity generated by RAFM steel activation (especially Fe-55 and Co-60, which are the primary dose contributors many decades after shutdown) tends to remain in the steel melt rather than separating into the slag which is removed by the process. Furthermore, because the slag is formed by the addition of limestone and oxygen, its volume exceeds that of the original recycled components. Even if this could be reduced, a relatively modest (2.5x) reduction would concentrate activation products to the point that LLW criteria would be exceeded, and deep geological disposal of this secondary waste stream would be required. The raw material (iron/steel) recovered by the process is unlikely to justify this cost and complexity.

Other materials may have better prospects for recycling, in particular beryllium, lead and tungsten, which are used as neutron multipliers in solid ceramic and other breeder blanket concepts. The cost of these materials is high, and future breeding blanket concepts may require a large supply relative to present production. Recycling may be a necessity to support deployment of many fusion reactors based on these concepts. From a waste and decommissioning perspective, the choice of using beryllium needs to be carefully considered because introduces not only toxic risks but also some long-term radiological risks due to the potential presence of uranium impurities¹⁵. The ore used to manufacture beryllium contains varying degrees of uranium, up to 120 wppm, some fertile ores would be closer to 100 wppm

¹⁴ Detrition techniques used in future machines may include ‘baking’, glow discharge cleaning, ion/electron cyclotron and discharge cleaning, with tritium being removed through the vacuum pumping and detritiation systems (see Section 5.3). Subsequent removal of tritium is done during decommissioning of the in-vessel components.

¹⁵ Beryllium has been used in JET for first wall tiles requiring additional considerations for handling.

[24]. Beryllium, when exposed to a significant neutron flux, will then inevitably lead to the production of long-lived radioactive transuranic isotopes that may additionally be chemically toxic, e.g. uranium, plutonium, americium and curium. To use beryllium in fusion facilities, efforts are likely needed to reduce uranium impurity levels¹⁶. Disposing of this beryllium and recycling it in accordance with national waste regulations is likely to be challenging.

4.1.2.3. Management of radioactive waste

Current regulations for management of radioactive waste aim to protect human beings from any hazards — short term and long term — associated with such waste. In general, different countries apply a range of different requirements, associated with their overarching legal frameworks, national customs, available infrastructure for waste management and facilities for disposal of waste. The requirements address pre-treatment, treatment, conditioning, transport, storage and disposal of radioactive waste, including the development of waste acceptance criteria for waste packages being placed in storage and/or disposal [25].

In the United States, criteria used for classifying LLW are based on a scenario in which institutional control over the site is assumed for 100 years, and the waste remains ‘stable’ and recognizable as such for 500 years. Class C LLW (the highest tier of LLW) is defined by activity concentration limits. Radionuclides listed in relevant US Regulations are shown in Table 3 [26]. The methodology was expanded by researchers to incorporate fusion-relevant activation products [27, 28].

TABLE 3. SPECIFIC ACTIVITY LIMITS FOR CLASS C LLW IN THE UNITED STATES (*reproduced from [26]*)

Isotope	Specific Activity Limit GBq/m ³
C-14	296
C-14 [†]	2960
Cm-242	740 000
Cs-137	170 200
I-129	2.96
Nb-94 [†]	7.4
Ni-59 [†]	8140
Ni-63	25 900
Ni-63 [†]	259 000
Sr-90	259 000
Tc-99	111
α -emitting transuranics, $t_{1/2} > 5$ years	3700
Pu-241	129 500

[†] In activated metal.

In the UK, LLW limits are radionuclide independent and instead consider bulk activity concentration of material: any material below 12 kBq/g (β/γ) and 4 kBq/g (α) as described by the LLW repository Waste Acceptance Criteria [19]. There are also strict chemical and physical constraints on waste disposal via LLW routes. These limits are historic and consider dose uptake to persons during emplacement and storage scenarios. There are also limits on ‘discrete items’ which are highly engineered / durable and may survive longer than the repository does. Hence there are separate dose

¹⁶ New processing techniques under development in Japan may reduce uranium impurities by several orders of magnitude, such that production of long lived radionuclides may be minimized – see <https://conferences.iaea.org/event/406/contributions/37085/attachments/20804/35186/Masaru%20-%20Miressio.pdf>

considerations for these items accounting for dose uptake if persons rediscover materials after the repository has ceased to exist.

Further information on national approaches to radioactive waste classification is given in:

- Canada (see Annex V-5);
- China (see Annex II-1);
- France [see Annex II-2, 29];
- Japan (see Annex IV-4);
- Russian Federation (see Annex II-5);
- Spain [30].

4.2. STRATEGIC OBJECTIVES

Depending on the size and power of a fusion facility, its eventual decommissioning will inevitably represent a significant industrial project, with associated safety, environmental, financial and socio-economic impacts. Implementation of the decommissioning project, together with the intended end state of the facility and the expected future use of the site on which it is located, will therefore be of interest to a wide range of stakeholders. Given this background, project planning will need to consider a range of strategic objectives, beyond the primary objective of dismantling the facility, such that any potential negative impacts from the project are minimized.

4.2.1. Political and scientific aims

The development of fusion energy facilities, and planning for their eventual decommissioning, takes place against a background of significant political and public interest, with current implementation and decommissioning programmes often relying to a significant extent on public funding sources, in accordance with their research mission. Accordingly, when considering the decommissioning of a fusion facility it is advisable to consider the technical details of the programme in the wider context of the political and scientific landscape at that time, e.g. the dismantling of a fusion facility provides a unique opportunity to conduct destructive sampling and analysis of materials, adding to current scientific understanding.

A corollary of the above is that decommissioning plans may evolve significantly over time in line with the evolution of the political landscape in general, and the evolution of fusion facilities, e.g. assets of value may add benefit if repurposed to align to new goals. The pace of funding provision for the programme may be impacted positively by enhancing stakeholder support through demonstrating relevant strategic aims and benefits. Once strategic objectives have been defined, a variety of options are typically assessed and scored to ascertain which option will deliver the most benefit. The cost and risk associated with each option needs to be thoroughly assessed as part of the process of developing a detailed decommissioning and associated expenditure profiles.

4.2.2. Financing framework

As is the case with fission plants, it will be important that arrangements are made prior to construction to ensure that costs associated with the future decommissioning of the plant and of management of the associated materials are covered.

Accurate cost estimation is challenging due to uncertainties associated with key factors which influence decommissioning (type of installation, design, age, radiological inventory, waste management, decommissioning strategy, incidents/accidents, site specific conditions, etc.). General guidance using a standardized structure for developing a cost estimate for decommissioning nuclear facilities is proposed in [31]. No similar guidance is available for fusion facilities, though the general methodological approach is likely to be applicable even if the detailed approach will be different.

The decommissioning activities likely to be most challenging are related to fuel and nuclear material management in the context of the tritium waste inventory, fuel storage, and overall material activation management. Risk quantification can be difficult on fusion projects as they are often first of a kind. At the early planning stages, pending the development of a detailed decommissioning strategy, a reference class forecast (RCF) approach may, for example, be used to quantify risk associated with the cost estimate. As more cost data relevant to fusion decommissioning becomes available over time, consideration may be given to developing a bespoke reference class forecast for wider use to support the development of future facilities, including the provision of pertinent information to policy makers and other stakeholders, including investors.

4.2.3. Public and staff communication

Implementing dialogue and engagement with local communities, e.g. through public outreach programmes, is important throughout the facility lifecycle is important, as is the case of fissile facilities [32]. Communication strategies are typically established to obtain timely support from staff and public on upcoming decommissioning activities. These need to take into account the likelihood that stakeholders may have very limited general understanding of the technical issues associated with fusion facility decommissioning, especially given that very little experience of this exists worldwide. Local communities will generally be directly affected by the decommissioning strategy adopted and its pace of implementation. The intended future use of the site is also typically an important consideration associated with a good communication with the public in a structured and positive way. Various approaches may be adopted for engaging with local communities, taking account of national customs and legal requirements, e.g. establishment of local information centres or citizens advisory panels.

Staff communication is also a key to the smooth transition from operation to decommissioning. In cases where the decommissioning organization wishes to retain staff from operations to undertake decommissioning activities (e.g. to take advantage of their in-depth knowledge of the facility and its operating history), this needs to be planned and clearly communicated. Some operational staff may be reluctant to transfer to a decommissioning project and are likely to search for alternative employment. Counteracting such a tendency — and thereby retaining key knowledge needed to decommission a facility — relies on an approach based on reassurance and providing a positive vision of the decommissioning project. Organizational knowledge is best retained when sudden changes in organizational structure are avoided.

4.2.4. End states

The desired end state of the decommissioning activity needs to be defined at an early stage, when approvals are being sought to construct and subsequently to operate a facility. In the event of a change to the overall strategic objectives, resulting in a change to the previously advised end state, early communication with stakeholders is advisable, with the decommissioning plan being updated to align with the new strategic aims. Large fusion facilities will generally be located on sites with significant industrial infrastructure, including connectivity to the electricity grid. Where buildings, plant and land could be repurposed for a new fusion experiment or power generation project this may fit well with national overall aims for energy generation and/or other scientific endeavours. Such an approach may sometimes be preferable, in terms of overall resource usage, to the work needed to enable site release for unrestricted future use or restricted future use, due to the resource value of land with power grid connections, shielded buildings and associated permits.

4.2.5. Sustainability

Sustainability is generally considered to comprise three interrelated pillars: environment, economic, and societal and, accordingly, the safety of human health and environment is a prerequisite to achieving the sustainable decommissioning of fusion facilities. Experience from the decommissioning of facilities using fission suggests there are important potential impacts on local economy, employment, habitats in the surrounding area [33]. The decommissioning of fusion facilities may benefit from experience from

such facilities in terms of the consideration of environmental, economic and societal aspects. Environmental, economic and societal perspectives of stakeholders can inform political interest, being an important aspect of decision-making for decommissioning and which goes beyond safety considerations. Such stakeholder perspectives can be understood and incorporated through stakeholder communication and engagement activities as described in 4.2.3.

Sustainability may also include consideration of circular economy principles as regards the recycling and reuse of materials, components, and building in line with the regulation on waste management and clearance as covered in 4.1.2. The future sustainable land use is interrelated to the decommissioning strategy and the end state as mentioned in 4.2.4. Even in cases where sustainability is not explicitly covered in regulations, national regulatory requirements associated with the management of radioactive waste may require — explicitly or implicitly — that sustainability considerations be addressed as a metric in optioneering exercises, and as a topic to be considered during design activities. Ensuring the safety of the decommissioning of fusion facilities through regulatory processes is a prerequisite of sustainability.

4.3. MAIN PHASES OF DECOMMISSIONING OF FUSION FACILITIES

The administrative steps for decommissioning a fusion facility are determined by the applicable regulatory framework, which in turn depends on the applicable country-specific legal framework (see Section 4.1). Subject to these differences, the following are the main distinct phases and associated administrative step, as illustrated in Fig. 8:

- Post operation or post-shutdown activities¹⁷;
- Safe enclosure or delay phase (if applied);
- Dismantling or final dismantling phase;
- Termination of the authorization or delicensing.

Generation of materials and waste, including radioactive waste, occurs during implementation of all the active dismantling operations indicated below.



FIG. 8. Indicative of major decommissioning stages after operating a fusion facility.

¹⁷ This stage is also sometimes referred to as POCO (post-operational clean out), especially in the United Kingdom.

4.3.1. Preparation phase

The preparation face for dismantling will usually begin a few years prior to the permanent shutdown of the facility and will continue afterwards while post-shutdown activities are undertaken. In some jurisdictions, the operator of a fusion facility is required to declare to the relevant national regulatory authorities an intention to permanently shut down the facility a few years (e.g. two years in France) prior to the date of the planned shutdown (see Section 4.1.1).

Once permanent shutdown has occurred the facility needs to be put in a safe state — pending its immediate or deferred dismantling — involving the removal of tritium from the facility, draining and detritiation of remaining fluids in the circuits. It may also involve removal of some activated dust from the in-vessel components. This phase is typically referred to as a post operational phase.

Activities undertaken during the post-operation phase are normally covered by the same regulatory authorization (or licence) applicable to the operational phase of the facility. Together with the physical changes being made to the facility, an important activity during this phase, the detailed decommissioning plan and other documents are prepared for submission to the regulatory authorities.

Having completed the post operational activities, the level of hazard associated with the facility is reduced and the authorization regime may be modified in line with the conditions outlined in the detailed decommissioning plan and associated documents. In many regulatory regimes this involves the granting of a specific licence for decommissioning, or otherwise a significant modification to the operating licence.

4.3.2. Safe enclosure or delay phase

Where a deferred dismantling strategy is adopted, involving a safe enclosure or delay phase, the state of the vessel is monitored at a predefined frequency. During this phase the dose rate inside the vessel diminishes through radioactive decay of the activated components. This phase may last from a few to several decades, when applied, after which the main vessel components are dismantled, including the vacuum vessel, magnets, cryostat and thermal shields. Where possible, the dismantled components that cannot be recycled or released from regulatory control (through a clearance procedure) are emplaced in a suitable disposal facility or are placed in a storage.

4.3.3. Final dismantling and termination of the authorization

The final phase of decommissioning project may involve the demolition of buildings (depending on the chosen end state) and measurement of any residual activity on the site of the facility, to such an extent that the achievement of the agreed site release criteria are met to the satisfaction of the regulatory authorities. By way of example, the UK approach for the STEP facility is described in [9].

5. PLANNING AND IMPLEMENTATION OF DECOMMISSIONING

Key considerations in terms of planning the decommissioning of a fusion facility, and which inform the development of the preliminary decommissioning plan, include:

- Incorporate decommissioning requirements from the beginning of the design of the fusion facility:
 - Careful selection of material relevant to fusion needs, e.g. material with low levels of impurities in relation to activation (Ni, Co, Nb);
 - Design solutions to manage tritium inventory in radioactive waste and structures, systems, and components;

- Identify the envisaged safe state of the facility at shutdown (analysis of radiological, chemical and physical hazards);
 - Propose design options to facilitate decommissioning (to the extent possible).
- Implement systematic configuration management:
- Maintain records of design level changes through the lifecycle of the facilities;
 - Maintain knowledge transfer to the decommissioning operators and management.
- Implement decommissioning procedures during the licensing phase, maintaining close engagement with regulators;
- Identify waste routing options during the lifecycle of the fusion facility;
- Developing a risk management plan (risk analysis, cost and schedule projection, and mitigation strategies) of the decommissioning project.

5.1. GENERAL DECOMMISSIONING CONSIDERATIONS AT THE FACILITY DESIGN STAGE

By integrating decommissioning considerations into the facility design stage, nuclear facilities can be designed with a focus on optimizing the decommissioning process, minimizing costs, and ensuring safety and regulatory compliance. Incorporating lessons learned from previous decommissioning programmes will assist this process, e.g. JET decommissioning will provide relevant feedback and will help to improve the design of future facilities. Although the decommissioning strategy or plan can differ significantly depending on the specific fusion facility, it is always necessary to integrate different considerations such as safety, environmental impact, sustainability (including economical and sociological aspects), technical feasibility and cost optimization.

Important considerations to facilitate effective future decommissioning during the facility design stage and along the lifecycle include:

- Design for decommissioning: incorporate design features and considerations that facilitate future decommissioning. This includes minimizing the overall size of the facility and the likely future quantities of different classes of radioactive waste, use of modular designs, easy access to components, and minimizing the use of materials that may complicate dismantling. It is also good practice to design facilities for ease of decontamination and waste management, including application of standardization and compatibility with existing decommissioning techniques;
- Minimize the use of structural materials likely to be subject to significant levels of activation, e.g. activation analysis performed as part of the design process for the planned ARC reactor in the USA led to the selection of slow-activation vanadium alloy for use in the fabrication of the tokamak. Use of the new alloy will result in higher safety performance, as well as reduced overall cost – see Annex I-4;
- Depending on the decommissioning strategy, an option for managing tritium during decommissioning is by detritiation, especially for tritium recovery and reducing the classification of radioactive waste. Considerations for detritiation of the most tritiated materials from a fusion facility are best made at the design stage. This may be in terms of material selection to be compatible with detritiation treatments, the design of components to fit in a detritiation furnace, or the ability to segregate materials from components easily for materials with different detritiation treatment parameters. An alternative approach to management of tritium during decommissioning could involve the development and use of confining packages;
- Stakeholder engagement: involve relevant stakeholders, including operators, regulators, and potential decommissioning contractors, in the design process. Their input and expertise can provide valuable insights into the decommissioning requirements, technological advancements, and best practices for fusion;

- Training and knowledge transfer: invest in training programmes and knowledge transfer initiatives to ensure that personnel involved in the facility design are familiar with decommissioning principles, methodologies, and requirements. This will help them make informed decisions and incorporate appropriate decommissioning considerations into the design process;
- Regulatory compliance: understand and ensure compliance with relevant regulatory requirements and guidelines for decommissioning. This includes considering the specific decommissioning criteria and regulations set by the regulatory authorities in the relevant jurisdiction;
- Data management plan: design of data systems (such as for material characterization, health physics surveys, construction information and material information at build) to hold data through decommissioning;
- Decommissioning plan: develop a decommissioning plan as an integral part of the facility design. This plan typically outlines the steps, methodologies, and resources required for the decommissioning process and includes waste management, radiation protection, and environmental considerations. For instance, anticipated releases of tritium have to be taken into account with the aim of avoiding contamination of the site;
- Material characterization and inventory: establish a comprehensive inventory of materials and components used in the facility, including their properties, radiation levels, and potential waste streams. This information is critical for planning and estimating decommissioning activities, as well as for waste classification for disposal, recycling and clearance;
- Life cycle cost analysis: conduct a thorough life cycle cost analysis that includes decommissioning costs. This analysis typically considers various decommissioning strategies and their associated costs, including labour, equipment, waste management, and site restoration or repurposing. It will help in evaluating different design options and selecting the most cost-effective approach;
- Documentation and information management: establish a comprehensive system for documenting and managing design information, including specific decommissioning considerations (i.e. 3D scans and full 3D model of installation including modelling tools for cutting). This ensures that critical information is easily accessible and maintained throughout the facility's life cycle and during decommissioning.

Relevant examples of the implementation of these provisions are detailed in Annexes III-1 (ITER), III-2 (Japanese DEMO), and III-3 (ARIES).

5.2. PROJECT MANAGEMENT

Decommissioning a fusion facility constitutes a significant industrial project usually involving a large workforce from several different organizations. This will require a management team with a combination of competencies in fields which include safety, contract management, financial management, environmental protection, quality assurance, human resource management and communications.

5.2.1. Management of project risk

Project risk management will play a pivotal role during the decommissioning of fusion facilities. This provides the facility management with the ability to anticipate events that can delay or accelerate the execution of the decommissioning schedule or increase cost. A Risk Management Plan (RMP) is typically developed at an early stage of the process, defining the overall methodology and risk management process – see Fig. 9. In the context of project risk management, a risk may be defined as an unplanned activity triggered by an event that is foreseeable with a likelihood and consequence associated to it.



FIG. 9. Risk Management Process to Decommissioning Fusion Facilities.

Development of a detailed activity schedule and associated cost estimates follows a similar approach as with other decommissioning projects, albeit with higher levels of uncertainty concerning the duration and cost of specific tasks due to limited project experience. Such uncertainties are diminished as greater experience is gained in implementing similar activities. Current good practice for complex projects suggests that an integrated activity schedule be developed, including associated costs. A quantitative (Monte Carlo) risk assessment may be undertaken to identify confidence levels in the overall schedule and cost estimate, and in the achievement of key milestones. This also enables discrete risks to be ranked in accordance with their overall importance and assessment of the potential benefit of mitigation actions to reduce or eliminate major risks.

5.2.2. Knowledge management

Facilitating knowledge management within a project involves a combination of strategies, processes and tools to effectively capture, store, organize and distribute knowledge. Strong management support is important to ensure that knowledge management approaches are fit for purpose.

Knowledge management is an ongoing process that ideally is considered from the design stage, including the approach to gather information during the operating phase. Associated with this is the need to gain support from staff whose primary activities are on task implementation rather than knowledge collection. A continuous improvement approach may be adopted to ensure that knowledge organization systems are fit for purpose and provide accurate project information as needed to support the project management activities.

Important actions to facilitate knowledge management may include:

- Creation of a knowledge management strategy which describes the approach, goals, and tactics for managing knowledge within the project;
- Identify sources of valuable knowledge within the organization, including tacit knowledge (knowledge held by individuals) and explicit knowledge (documented knowledge). Develop

processes to capture this knowledge, whether through documentation, interviews, or other means;

- Procure and maintain knowledge management tools and software that can help organize and categorize knowledge effectively, e.g. processes are regularly reviewed and updated where necessary;
- Develop a classification system (e.g. metadata) to categorize and organize knowledge and develop guidelines for effectively creating, storing and sharing knowledge;
- Ensure that knowledge is stored securely, protecting sensitive information while making it accessible to those who need it, e.g. through access controls and encryption where necessary;
- Foster a culture of knowledge sharing within the project and provide training to employees on how to effectively use knowledge management tools;
- Utilize external sources of knowledge and integrate relevant external information to the extent possible;
- Establish a feedback loop to gather continuous employee feedback on the effectiveness of the knowledge management processes and implement improvements as necessary.

5.3. PREPARATION PHASE FOR DECOMMISSIONING A FUSION FACILITY

After the end of operations, several actions are needed to put the facility into a safe state and to prepare for dismantling works¹⁸:

- The tritium source term captured inside the vacuum vessel structure and components can be reduced by baking or glow discharge cleaning. Isotopic exchange is also possible by dedicating a last campaign to H-He plasmas;
- Regeneration of the cryopumps can then ensure the tritium fuel is transferred up to the storage getters. Purging of tritiated gases from the tritium plant process can also be performed by flushing with hydrogen and local heating. This tritium extracted from the materials is not to be considered as waste but as a valuable and accountable material that may be reused as fuel in other fusion facilities;
- Collection of dust inside the vacuum vessel using the techniques developed during operation;
- Full drainage and processing of cooling water primary loops is also a preparatory action that could take time depending on available liquid waste treatment facilities and discharge limits;
- Removal of activated in-vessel components by using the remote handling already available during operation will significantly reduce the source term on site provided waste treatment and off-site routing options are available.

A functional analysis needs to be performed to assess what would then be the dismantling sequence depending on the design of the facility, the need for supporting treatment facilities and decay requirements.

5.4. TECHNOLOGY AND TOOLS SELECTION

Technology selection for the decommissioning of a fusion facility is driven primarily by the project objectives and strategy such as cost, duration, or end state as defined by the project sponsor. The selection can also be supported by the assessment of the Technology Readiness Level between the different available technologies [34].

A graded approach and defence in depth is applied in undertaking the decommissioning safety assessment, facilitating the selection of decommissioning technologies that will mitigate radiological or chemical hazards and help maintain a defence in depth (redundant systems) approach to safety. It is important to note that many of the technologies used to maintain safety during fusion operation can be

¹⁸ Depending on the facility design, some or all of these activities may also be undertaken during the operational phase.

adapted, or extended into the facility, to support decommissioning phase. Recovery technology may be readily applied to mechanical systems, subject to consideration of the ageing of the materials necessary for sealing (joints) and to the condition of cabling (whose sheathing will have hardened and may therefore need to be replaced to avoid potential short circuiting).

The decommissioning of tokamak systems is extremely complex, not only due to the complicated construction of such devices, but also because of the associated radioactive and hazardous properties or materials, e.g. requiring the use of tritium abatement systems to address potential environmental impact (see Section 6). Although much of the equipment used in large scale demolition activities can be deployed for tokamak dismantling, the need to retrieve materials in a controlled manner results in specific equipment requirements.

5.4.1. Remote handling systems

Remote handling plays a key role in the decommissioning of fusion facilities because of the high dose rate of in-vessel components and structures. Accordingly, hot cells and active handling infrastructure are generally designed to enable their use also during the initial stages of decommissioning (e.g. for retrieval of in-vessel components for placement in storage). In addition, for components located away from the plasma, remote handling will help reduce personnel exposure and unplanned releases to the work environment.

The remote handling technology is designed to be able to withstand the high radiation levels or contamination by activated dust and tritium constituting the source term that can be mobilized in the fusion facility. The maintenance of such large-scale facilities requires a large hot cell / active handling facility. Once their operation is complete, these large-scale facilities will also need to be decommissioned. Different tasks require remote handling systems: from automated-remote sample collection, inspection of areas, dismantling, cutting, to handling and movement of large pieces of components.

The removal of materials inside a tokamak vacuum vessel is currently undertaken using remote handling equipment. During operation, this equipment enables the installation and removal of materials in tokamak systems without exposing personnel to elevated dose rates from activation, or tritiated atmospheres as a result of D-T fuelling. These systems may also be used during decommissioning to remove in-vessel components with the greatest tritium loadings and levels of activation. For example, at JET, the remote handling system has successfully been used throughout the operational lifetime of the system to install and remove in-vessel components such as plasma-facing tiles, to perform inspections, and to calibrate diagnostic instrumentation. It is planned to deploy the same system to remove in-vessel materials during the preliminary stage of decommissioning where person access to the JET vacuum vessel will be prohibited – see Fig. 10, showing the remote handling system used for inspection, calibration of diagnostic equipment, and removal / replacement of tiles and internal components.



FIG. 10. MASCOT manipulator inside JET (Courtesy of UKAEA).

Figure 11 illustrates one of the complex remote handling tools under development for ITER. It will be used for maintenance and removal of in-vessel components with maximal payload of about 800 kg in a radiation environment that can reach a few hundred Sv/h.

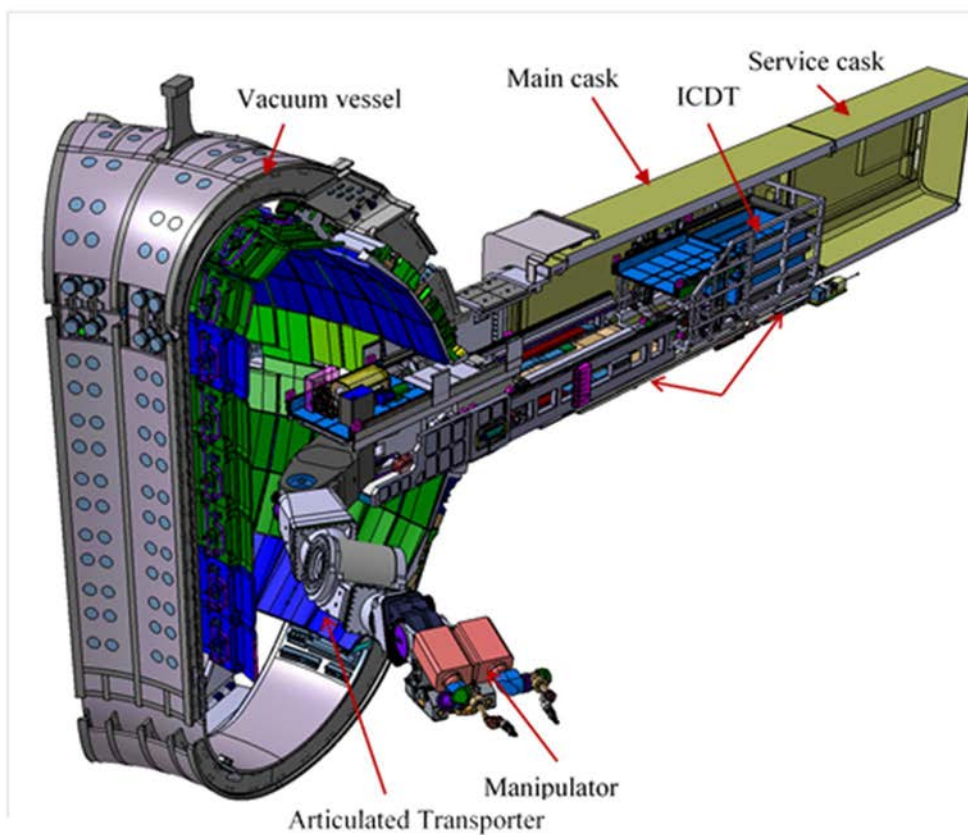


FIG.11. ITER Remote Handling System (Courtesy of ITER Organisation).

It may be necessary to modify and upgrade remote handling systems throughout their lifetimes, including for the increased burden imposed by decommissioning activities. Current fusion facilities have had sufficiently low levels of activation that radiation tolerant remote handling systems have not been a firm requirement. With the move to larger scale facilities and the resultant higher levels of activation, radiation tolerant remote handling systems will need to be designed for such purposes. In addition, with greater levels of tritium retention, there will likely be a requirement for remote handling outside of tokamaks to support maintenance and waste processing. For instance, to manage the maintenance operation, radiological waste streams, and during the dismantling, it is generally beneficial to have available a hot cell, with heavy reliance on remote handling for material processing activities.

5.4.2. Radiological monitoring equipment

The deployment of equipment to measure the radiological properties of an area or item is crucial in nuclear settings to ensure the ongoing safety of personnel, including early detection of fault scenarios. Fusion facilities will require a significant number of radiation monitoring devices to underpin safe operations, especially in terms of tritium monitoring and dose rate measurement. The need for such equipment will be even greater during decommissioning due to multiple parallel activities being undertaken. Furthermore, breaches of containment may be expected to be more frequent, which will require monitoring to ensure safety and control of releases. There is a significant requirement for the monitoring of airborne tritium concentration due to the high mobility of tritiated species, e.g. as suggested by radiological monitoring experience at JET. Upon the breaching of systems, movement of materials in sealed containers, or the processing of material to release tritium, frequent deployment of airborne tritium monitoring is needed to measure discharges, assess safe operating parameters, and as a form of verification of any decontamination activities over time. The expansion of tritium inventories, together with the overall volume resulting from decommissioning, suggests that this will have even greater relevance in future fusion facilities.

When tritium contaminated systems were breached during TFTR decommissioning both the breached area and the breathing zone of workers were analyzed for tritium using portable monitors. Additionally, each area in the experimental complex was equipped with tritium area monitors to warn workers if tritium levels rose to pre-determined levels. If tritium levels in the main processing area reached high levels, the area HVAC would shut down and a room de-tritiation system would automatically start to ensure the released tritium was cleaned up. The HVAC exhaust from each area in the experimental complex was also monitored for tritium before the effluent was released via the facility stack. If tritium levels on the HVAC exhaust reached high levels the associated area HVAC system would shutdown to ensure the air was not released to the environment.

The radiological monitoring of tritium is a key technical challenge and frequently relies on gaseous monitoring once tritium is already present in an area. When monitoring tritium retention within materials, destructive sampling is necessary to provide a robust estimate of tritium retained in a material (see Section 6.2.3). This is not a sustainable methodology for monitoring of larger scale fusion facilities with significant tritium usage due to the time and cost involved to achieve adequate monitoring. Recent advances in the use of laser-based analytical methods (such as laser induced breakdown spectroscopy or laser induced desorption methods) have shown promise in quickly detecting tritium retained on material surfaces and slightly below material surfaces. With further advancement, this may be relevant to future decommissioning and akin to the conventional radiological monitoring for radionuclide contamination within fission and healthcare nuclear applications.

5.4.3. Tritium handling infrastructure

The high mobility of tritium has significant implications in terms of containment and contamination spread. Currently, most tritium systems are constructed with multiple containment layers or barriers. During decommissioning it is commonplace to break these containment layers as components are removed, systems are inspected / characterized, etc. The use of mobile containment and ventilation via localized systems has been employed successfully at JET. For example, for some components, tritium

release rates of GBq/hour have been observed from components of significant size. The ability to locally isolate these materials with containment structures and ventilate or abate them to a relevant discharge point is crucial for controlling contamination spread. It is expected that such infrastructure will be used extensively as the JET tokamak is dismantled and tritium containment is lost.

During TFTR decommissioning flexible hoses connected to the experimental area negative pressure HVAC system were employed to remove/ventilate tritiated air from breached systems and exhaust the air to the facility stack. TFTR referred to these hoses as ‘elephant trunks’ and they played a critical role in minimizing worker uptakes and tritium contamination of the experimental areas during breaks into the tritium contaminated systems. Elephant trunk extraction arms for capturing aerosols and radioactive gas are currently used locally for remote handling tools maintenance.

5.4.4. Size reduction capabilities

An inherent aspect of decommissioning is disassembling and size reduction of systems such that they can be appropriately managed. With fusion facilities being inherently constructed of large metallic items, and relying on significant bioshield structures, the size reduction of fusion facilities is a key consideration within fusion decommissioning projects. There is extensive learning from the fission and particle accelerator communities with regard to size reduction of radioactive components from nuclear settings. TFTR displayed size reduction activities at large-scale during segmentation of the primary vacuum vessel – see Annex I-2.

For future fusion facilities at a power-production scale, there is likely to be significantly less personnel access to tokamak structures than there has been with systems to date. It is therefore likely that more remotely operated size reduction capability will be required during decommissioning.

At smaller scales, fusion facilities frequently use, or expect to use, rare materials due to their nuclear, magnetic, and plasma properties. Examples include breeding blankets (lithium, beryllium, lead), plasma facing tiles (tungsten, molybdenum, inconel, others), and magnetic coils (various options). The varied quantity, together with the chemical, activation, and tritium retention behaviour of these materials, may necessitate differing waste management routes. This may include requirements to segregate / separate materials for different treatments, decay periods and disposal sites. Accordingly, the size reduction capabilities may need to be sufficiently precise to enable material segregation / separation to facilitate further waste handling and/or processing. This is a unique property of materials arising from decommissioning of a fusion facility due to the inherent value, size and hazardous properties of materials, being analogous in the fission domain to separation of fuel rods into their constituent parts to enable various waste management routes.

The likelihood that significant size reduction infrastructure will be required to support operations suggests a need for development and deployment of enhanced size reduction capability to support the large throughputs and component sizes associated with decommissioning activities.

TFTR used a diamond wire cutting to segment the vacuum vessel – see Figs. 12 and 13, which show an evolution from the initial arrangement of the equipment (Fig. 12) to near the end of the process (Fig. 13). Note that many of the components were covered during the process to limit personnel exposure to potential airborne particulates due to radioactive activated particulates, and hazardous materials (chromium and other metal particulates).



FIG. 12. TFTR Vacuum Vessel Diamond Wire Cutting Arrangement (Courtesy of Princeton Plasma Physics Laboratory).

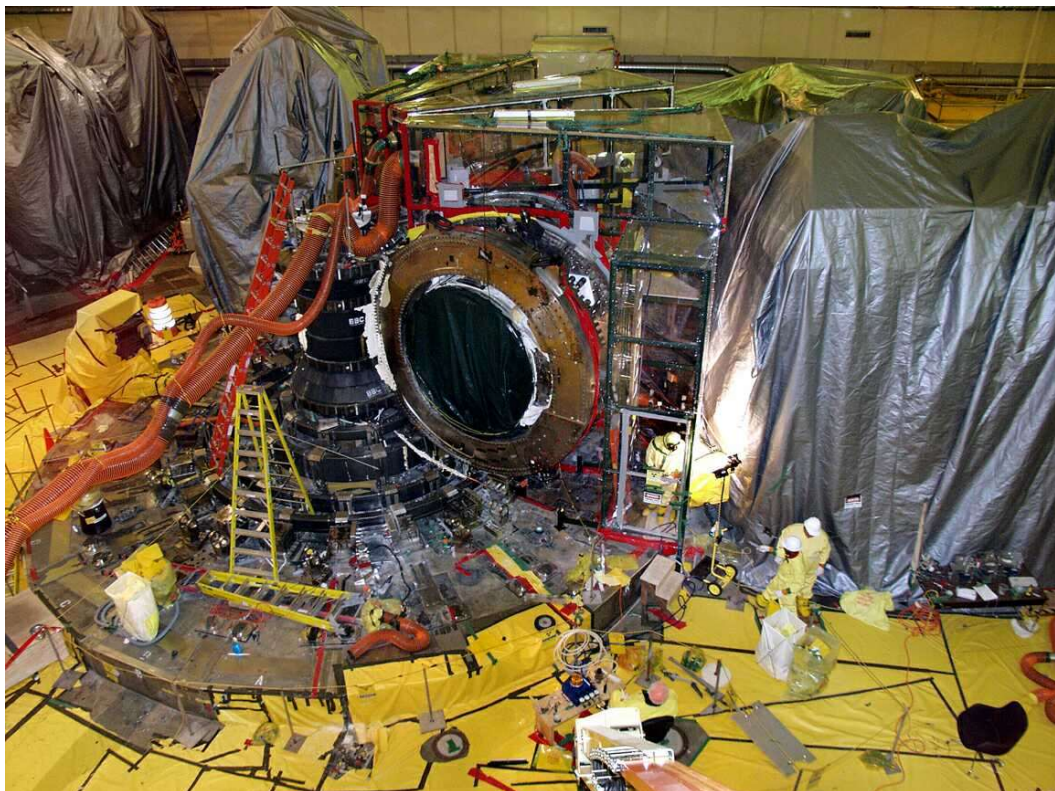


FIG. 13. TFTR Vacuum Vessel near the end of the Diamond Wire Cutting Process (Courtesy of Princeton Plasma Physics Laboratory).

5.4.5. Decontamination techniques

Decontamination enables the reduction of the amount of radioactive waste generated during decommissioning, with important feedback from the decommissioning of facilities in the fission domain being available regarding steel and concrete decontamination techniques. Their applicability to European DEMO materials is assessed in Annex I-3.

The choice of an appropriate technology is possible after determination of the type of contaminants and the degree of contamination of activated / contaminated steel and concrete structures. This requires a characterization plan for the whole facility – see Section 5.1.

5.4.6. Characterization capabilities

The characterization of waste generated in fusion facilities is complex and requires multiple independent measurements to underpin a full description of radionuclide content (see Section 6.2.3). The extent of tritium analysis capability in the supply chain is currently rare, and the combination of activated, tritiated and hazardous materials (such as JET beryllium) generally proves challenging for laboratory acceptance criteria. For emerging large-scale fusion facilities and their increased waste complexities in terms of activation, tritium loadings, and component size, characterization of materials will be even more challenging.

It is hence imperative that fusion facilities / sites have a dedicated characterization capability for their materials. This capability is needed to support operational activities, but the requirement will be even greater to support decommissioning. Experience from JET, and associated benchmarking with numerous other UK nuclear operators, has indicated that it is optimal to perform large-scale and high throughput analytical tasks on a reactor site as routine business (Fig. 14). There are likely to be significant benefits from augmenting the available capability with supply chain capacity to manage periods of excessive demand or equipment maintenance, though noting that some characterization activities such as detailed radionuclide fingerprinting is a specialist skill best undertaken by specialist laboratories.



FIG. 14. Left: high resolution gamma spectroscopy equipment at the UKAEA RADLab used for measurement of activation products. Right: One of the UKAEA pyrolysis capabilities used for bulk tritium measurements of small sub-samples of JET materials (Courtesy of UKAEA).

A key characterization activity which has been deployed in the early stages of JET decommissioning is the use of detailed activation modelling at component level. This has indicated expected dose rates, activation fingerprints, and decay profiles of materials. This information has been essential for designing characterization schemes for materials to align with waste acceptance criteria and other regulatory requirements.

As noted in Section 4.3.2, there is likely to be a period of delay between permanent shutdown and the commencement of dismantling of essential equipment, e.g. to allow a reduction in radioactivity levels

inside the facility and thus dose levels to workers due to radioactive decay¹⁹. In this situation special attention needs to be given to the final characterization of waste before departure from the facility, as easy-to-measure nuclides may already have decayed significantly. Where measuring a tracer radionuclide is difficult or even impossible (value below the detection limit) uncertainties in overall radioactivity levels, including due to hard-to-measure radionuclides, are likely to be increased.

Data quality objective methods can provide a statistically robust method for selecting the right number and representative location of samples to be taken, which will be particularly useful for large areas and structures such as concrete buildings or shielding where tritium may have penetrated.

The measurement of easy-to-measure radionuclides is key to verifying scaling factors and using key nuclides to support waste assay. Gamma spectroscopy can be used to identify Mn-54, Co-60 and Nb-94, however elements such as Fe-55 require samples, radiochemical separation followed by detection increasing time, effort and potential doses to workers. The use of gamma spectroscopy and x-ray spectrometry [35] is an alternative to intrusive sampling for activated materials. The combination of both gamma and x-ray spectrometry may offer a fast tool for characterization in a decommissioning setting. Further research for x-ray spectrometry will concentrate on the use of different detectors, challenges of non-radiological components in waste streams and also comparative work with current techniques such as Liquid Scintillation Analysis (LSA).

5.4.7. Recycling capabilities

The potential to recycle materials from the decommissioning of fusion facilities and from the replacement of vacuum vessels and in-vessel components is highly dependent on the national legal and regulatory framework and on the availability of recycling facilities – see Section 3.2.3. Economic considerations for recycling are discussed in Annex V-6.

5.4.8. Identification of R&D perspectives

Important technological challenges during decommissioning of fusion facilities, and which are likely to benefit from R&D activities, include:

- Development of remote handling cutting techniques that can handle the complexity of dismantling the tokamak core, while reducing the spread of particulate contaminants to other parts of the facility and thereby minimize secondary waste;
- Development of waste packages that can efficiently confine tritium and reduce outgassing: specific matrices and leak tight sealings. The package ought to be confining enough so that tritium diffusion characteristic time is significantly longer than that for natural tritium decay. Accordingly, the confining package design could rely on optimization of package design (size, material choice and thickness of package walls), an efficient sealing joint, and a confining cement matrix with potential incorporation of a tritium getter material;
- Development of radiological characterization tools (spectrometry, sampling) adapted to large scale activated components and to determine tritium content with less intrusive methods than destructive sampling;
- Development of calculation tools or models to estimate tritium diffusion into materials. This ought to take into account the potential impact of impurities contained in the materials. Industrialize waste detritiation techniques (baking/melting) adapted to the important amounts of materials from vacuum vessel components and the presence various radionuclides (see

¹⁹ Similar delays occur during outages for vacuum vessel replacement.

Section 6). Another key R&D objective of such developments is to limit tritiated water production²⁰;

- Technologies that are easier to maintain over time and easily understood by future generations.

For some of the proposed next step magnetic confinement fusion facilities, the in-vessel component structure will generally be larger than current examples (blanket system, divertor, vacuum vessel)²¹, requiring the development of large hot cells and high capacities remote handling tools. Since the radiological inventory of materials is likely to increase substantially with power-plant scale fusion facilities, it is imperative that analytical capability is developed to effectively measure tritiation and activation severity in materials.

It is also important that there is increased research and development of tritium removal techniques in materials for fusion facilities. The removal of tritium from polymeric and solid waste has been tried but is not at a scale commensurate with future fusion facilities. The increased tritium exposures of materials, and the use of tritium breeding materials, presents a key challenge for detritiation.

In terms of radiometric modelling, frequently referred to as ‘neutronics’, there is a key development in coupling the digital design files of future systems with activation codes. This has been applied at JET, but due to the age of the facility the design files are retrospective and do not represent a complete description of the tokamak. With future systems, such as ITER, DEMOs, power plants, digital information is used as standard as part of the design process and ought to be integrated with activation modelling software as soon as practicable.

5.4.9. Use of physical or digital mock-ups to achieve high technical readiness levels

Mock-ups may be physical, digital or a combination of both. As decommissioning activities are performed in a complex and hazardous environment, mock-ups support the safe performance of activities and improvement of technical readiness levels (TRLs) for the implemented technology.

For the purpose of this document, both types of mock-ups share the following objectives:

- Feasibility;
- Maintenance;
- Decommissioning;
- Training, for operations, maintenance and accidental scenarios.

The use of 3D models helps when physical mock-ups are not feasible (for example because of space and/or costs constraints), as demonstrated by recent dismantling projects carried out by industrial companies working in the framework of nuclear facilities dismantling. These projects have validated the effectiveness of digital models for various purposes.

To optimize the benefits of 3D models, gateways enable the integration of 3D data into calculation software, allowing anticipation of doses for maintenance and dismantling operators. The challenge lies in maintaining a 3D model (with updated as-built data) throughout the installation’s lifespan. This requires significant budgets and unwavering rigour from the outset of the projects. A technical prescription ought to be established to ensure the continuity of the models, with large means and formats kept in the public domain to avoid dependence evolutions or company changes, especially as the

²⁰ This is subject to the process limitations associated with the tritium concentration in the water, together with the requirement to have an operational water detritiation facility beyond the operational lifetime of the facility (the water detritiation system is usually coupled to the isotope separation system within the tritium fuel cycle).

²¹ Some machines currently under development by private sector organizations are much smaller than current generation machines.

lifetime of specific software may be much less than the lifetime of a major fusion facility. Regular reassessments are typically performed to ensure the model's quality over time.

A more traditional approach to the upkeep of data during the lifecycle involves the printing all the drawings and keeping them archived until dismantling. Subsequently, the models can be reconstituted from these drawings. Such an approach is becoming progressively less efficient with ongoing improvements to digital knowledge management systems.

It is important that the mock-ups be validated to ensure that they are representative of the current reality. Mock-ups are helpful tools which can be used for many purposes. There are challenges related to keeping the digital mock-ups up to date and they ought to be, as much as possible, representative of the current reality. Currently, digital mock-ups are used mainly for representation of the physical configuration of the facility, in order to design remote operations. In future, it is becoming increasingly likely that such models will generally incorporate the radiological and chemical characteristics of the facility environment. Such mock-ups could also be used to improve communication with the stakeholders involved, to ensure they understand the current configuration.

5.5. IDENTIFICATION OF CURRENT GOOD PRACTICES AND HIGHLIGHTS FROM PRACTICAL DECOMMISSIONING ACTIVITIES

Research scale fusion reactors built before the 2000s had limited design-level consideration for decommissioning, related to ease of dismantling, material selection, supporting facilities and systems. Priority was given to scientific experiments, and long-term end of cycle facility activities were not considered at first. In such devices the extent of activation was sufficiently low that there were rarely significant radiation protection implications in decommissioning and waste management. There was also low tritium loading and limited radiological risks.

5.5.1. General considerations

The complexity of decommissioning activities increases significantly when moving from H-H or D-D to D-T plasmas. For instance, facilities operating with H-H plasma or D-D plasma have lower activation and tritium inventory, allowing for decommissioning with fewer constraints (e.g. human access, hands-on for H-H plasma). The Frascati Tokamak Upgrade (FTU), which operated with H-H and D-D plasmas, radiological characterization measurements on components located inside the cryostat demonstrated that activation levels are lower than clearance levels for recycling and reuse as had been expected from historical data and semi-empirical analysis, without any constraints being required for their management and final destination – see Annex IV-3).

For facilities using D-T plasma, additional constraints are relevant, e.g. operation in the vacuum vessel (VV) is achieved through remote handling means at JET. Human access is not totally forbidden, if needed, but when longer operations of D-T plasmas occur, as foreseen in the ITER facility, human access to the VV after first production of fast neutrons is expected to be forbidden. In the case of ITER, with its foreseen neutron fluence during D-T experiments, even after some decades after the end of operation of the facility, access to all in-bioshield components, will still be forbidden.

From a technical standpoint, activation calculation plays a critical role in determining the radiological inventory and waste classification at the end of operation of the facility. Activation calculations can be developed and refined at various stages of design and procurement. They are frequently used to demonstrate regulatory compliance (or predicted regulatory compliance) for a given exposure. In addition, procurement of components and materials relies on specifications resulting from the activation calculations. There can be cases in which trade-off decisions will be needed when there is a scarcity of raw materials that meet the specification (material mechanical strength, cost, availability). It is good practice to avoid or eliminate paints as much as possible, in turn reducing the need for their sampling as part of the characterization process.

Finally, as part of the periodic activation assessments, sampling structural material can be performed as needed (using stainless steel coupons or concrete pouring). The coupons are then placed in representative environments so that they will face the same environmental conditions and ageing as the components of the facility during operation. Sampling and analysis are carried out at regular intervals to correlate the calculations with the observed radiation induced degradation phenomena. This procedure is currently underway during the construction phase of ITER.

Tritium migration in concrete, in particular where there is no liner, can also be problematic to manage at the end of dismantling. Indeed, depending on the levels of decontamination to be achieved, there is a risk of causing structural degradation of affected buildings, leading to complete demolition or to restrictions in further use. For this reason, it may be necessary to carry out destructive concrete analyses before defining the final state associated with these structures. As a first step, experience gained from dismantling of tritium contaminated facilities may be relevant. This issue may also be a topic for research and development.

Relevant examples of the implementation of such principles are presented in Annexes IV-I (China), IV-2 (ITER, France), IV-3 (Italy), IV-4 (Japan) and IV-5 (Russian Federation).

5.5.2. Calculation of levels of activation

There are important cost as well as safety benefits from having accurate knowledge of activation levels as the level of activation represents an important cost driver. Current computational systems and activation codes enable multiple scenarios to be modelled and refined efficiently. 3D CAD models will be readily available for future facilities (DEMOs, pilot plants, as well as fusion power plants), as they are already available for JET and ITER.

Current three-dimensional radiation transport codes (such as MCNP and OpenMC) are able to generate the neutron flux throughout the system. Several versions of software (such as DAGMC in the USA, McCAD in Germany, and MCAM in China) were developed to couple the CAD with 3D transport codes allowing relevant design details to be added to the neutronics model to estimate the neutron flux and energy spectrum profile with high fidelity. The neutron flux and energy spectrum is then integrated with the activation code (such as ALARA or FISPACT-II) to estimate the specific activity, radioactive inventory, source term for safety analysis and activation responses for licensing.

Performing such calculations enables a good estimate to be made of the shutdown dose rates applicable to maintenance operators during shutdowns, as well as having the time-dependent activity of all components to be consigned as waste, decay heat deposition, disposal classifications, recycling dose rate, and clearance index. According to the duration of irradiation, a good evaluation of the radioisotopes present in the radioactive waste over time is obtained.

Available feedback suggests that a comprehensive radiological characterization program of the facility is essential to effectively address the remaining uncertainties. Specific analysis capacities and capabilities are required to complete this process within a reasonable timeframe. On this basis, the decommissioning scenario can be aligned with the radwaste classification and its management route, prioritizing the development of dedicated storage or disposal facilities if needed and favouring clearance and recycling when possible. National waste policy and available storage or disposal options can impact the decommissioning scenario.

5.5.3. Tokamak Fusion Test Reactor

The TFTR at the Princeton Plasma Physics Laboratory (PPPL) was built in the early 1980s and performed D-D fusion experiments through the 1980s and into the early 1990s. Around 1992 tritium was brought on site and TFTR began its D-T experiments which continued through 1997 when the facility was shutdown. Approximately 100 grams of tritium were processed by TFTR. Decommissioning of TFTR began in 1999 and was completed in 2002. This marked the first

decommissioning of a D-T fusion facility and provided some valuable lessons learned and novel approaches.

5.5.3.1. Decommissioning cost

The cost estimate for TFTR decommissioning in October of 1999 was 40.3M USD. The project was completed in September of 2002, ahead of schedule, for a final cost of 36.6M USD.

5.5.3.2. Detritiation

In preparation for TFTR decommissioning PPPL developed a drum bubbler system designed to remove tritium oxide from a process stream. Drum bubblers were used to vent and detritiate highly contaminated components such as the TFTR vacuum vessel and large neutral beam enclosures by passing a contaminated gas stream through diffusers in a drum filled with water. The gas then bubbled to the surface of the water where it was 'dried' by a refrigeration unit before releasing. The bubblers were more than 90% effective in removing tritium oxide. Once the tritium content of the bubbler reached the maximum tritium concentration allowed for disposal the bubbler was removed from service. The residual water was treated with polymers which increased its viscosity until it formed a solid gel (i.e. solidification). This step was required for residual water acceptance into the waste storage facility.

5.5.3.3. Disassembly of tritium contaminated systems

TFTR was not designed with decommissioning in mind. To disassemble the tritium contaminated TFTR systems the project employed various tooling approaches to make the job easier and to minimize worker uptake of tritium. One of the more effective tools used was a pneumatic pipe crimping device. The crimping device was used to crimp contaminated piping and essentially seal the pipe before another pneumatic cutter was used to cut the pipe. A series of crimps were made on a pipe then the cutter was used to cut the pipe in the middle of the series or crimps. Once cut the piping ends could be sealed and prepared for disposal.

To minimize worker exposure to tritium while removing tritium contaminated piping during TFTR decommissioning, negative pressure trunks or 'elephant trunks' were used to remove tritium released at openings made in process piping. These trunks were connected to the area HVAC system that maintained negative pressure in their respective rooms. The gas was then discharged out the exhaust stack of the facility.

5.5.3.4. Segmentation of major components

The original plan was to segment the TFTR vacuum vessel using plasma torch cutting; however, the dose rate on the vacuum vessel was approximately 50 mrem/hr (0.5 mSv/hr) and torch cutting of stainless-steel presented potential hexavalent chromium exposure for workers. To mitigate these personnel safety concerns remote cutting of the vessel was performed using diamond wire. At the time, this was a novel approach as diamond wire cutting was being used mostly for concrete and stone cutting. To facilitate diamond wire cutting of the vessel required certain preparatory activities to be first undertaken. First, a team of workers entered the vacuum vessel in bubble suits and removed the carbon tiles from the areas where the vessel would be cut. Then the vacuum vessel was filled with a lightweight concrete to essentially seal the vessel before cutting. The diamond wire was operated at a distance from the vessel to minimize exposure. The vessel was cut into ten segments. This ensured that the tritium inventory in each segment was low enough to meet waste acceptance requirements.

Approximately 55 000 cubic feet (1 560 cubic metres) of radioactive waste was generated during the decommissioning of TFTR. Additionally, there were 400 tons of concrete shielding removed and about 200 tons of lead was removed and stored for future use. Concerning radionuclide handling concerns, the only issue concerned tritium content, as activation levels were not significant.

5.5.4. Joint European Torus

JET was permanently shut down in December 2023 and is currently in the preparatory phase of decommissioning. Main concerns include the distribution of tritium, particularly the tritium that remains trapped in the tokamak wall and in dust that has been accumulated in-vessel. Using available techniques such as vacuum vessel baking, glow discharge cleaning or isotopic exchange has been demonstrated to reduce tritium inventories. Remote handling activities of components are planned, and significant scientific information (e.g. measurement of activation as a neutronics code validation) is planned following tritium reduction. The remote handling (RH) tools are being designed to operate at elevated dose rates for an extended period of time.

JET had an initial total neutron budget (fluence) prior to operations with D-T beginning. This total budget was incrementally reviewed during system operation due to the research style of JET operations, modifications to the JET system, and the uncertainty associated with the neutron generation of experimental campaigns. However, the fluence levels were always retained within the neutron budget to limit the activation of JET systems, structures, and components. This was to prevent severe activation which would preclude some human maintenance scenarios for JET. A full summary of the evolution of the JET neutron budget is provided in [36]

JET is currently in the early stages of decommissioning, so a significant amount of learning has been gained at the planning-stages of decommissioning. The undertaking of decommissioning operations will be a unique opportunity to attain further knowledge of D-T tokamak decommissioning.

5.5.4.1. Management of plant configuration information

JET was designed in the 1970s and constructed in the early 1980s, hence there was little digital design information for JET. A significant amount of time and resource has been spent developing a digital model of JET from design drawings and plant surveys. This can now be used for detailed structural planning, activation modelling, and decommissioning sequencing but is still incomplete in some areas. Current design methods allow the use of digital technology from the outset and facilitate the updating of digital models throughout the facility lifecycle, such that they are ‘current’ when decommissioning needs to be planned.

A key finding from JET is that the exact material specifications for tokamak materials were not always recorded in a future-proof way. Hence there has been significant effort in determining what materials are made of to feed into activation modelling exercises. There is experience of material impurities at the part-per-million scale resulting in drastically different radiological characteristics than anticipated at JET during neutron exposures. It is therefore important that, for any future fusion facility with high neutron fluxes, material samples are kept in a secure fashion until decommissioning is underway.

5.5.4.2. Use of a digital twin to support decommissioning planning

Throughout the decommissioning progress, materials will be incrementally removed from systems. In fusion facilities which are highly complex, it is very beneficial to be able to display how systems will change over time. Due to the age of JET, and the lack of a detailed digital twin, there is limited opportunity to display how tokamak dismantling will progress outside of the vacuum vessel without significant work creating a digital twin. However, inside JET there is an existing digital ‘twin’ of the tokamak for rehearsing of remote handling tasks. Due to this, it has been possible to precisely articulate the stages of in-vessel component removal which are planned in the early stages of decommissioning. This has been very useful for articulating detailed waste arisings, the need for bespoke tooling and the resulting design, and for increasing staff understanding of operations.

5.5.4.3. Detritiation

UKAEA has developed a detritiation capability to facilitate the management of materials which were installed in the JET vacuum vessel but have subsequently been removed and replaced. These materials were highly tritiated and presented significant issues in terms of eventual disposal to a geological facility due to their ILW classification. However, operation of detritiation capability has reduced this operational waste liability to the lowest UK classification of radioactive waste. During detritiation, the released tritiated species are oxidized and captured in water as tritiated water (HTO). UKAEA is currently commissioning a water detritiation system which will convert HTO into gaseous tritium for use in experiments. Further information on detritiation can be found in Annex V-1.

An important advantage of having a detritiation facility during tokamak operations is that staff become highly competent in detritiation operations by the time decommissioning begins and the amount of waste processing via detritiation rises dramatically. There is also the opportunity to perform research and development into the optimization of detritiation activities – this has been a key activity in the early stages of JET decommissioning. There is also the opportunity to experiment on previously untested materials with this capability (such as solid beryllium in the case of JET) to aid with tokamak component design prior to installing potentially problematic materials in a tokamak.

5.5.4.4. Tritium accountancy

The JET system has historically proved challenging in terms of tritium behaviour. Following the first D-T experimental campaign in the late 1990s multiple grams of tritium were missing when accountancy checks were performed at the cessation of tritium operations. It was discovered that a few grams of tritium remained in the JET carbon wall tiles and was slowly released over time and clean-up operations. Detritiation trials on JET carbon tiles have demonstrated significant tritium removal from material – see Annex V-3. However, an amount of tritium (a few grams) was never released from the vessel and has been attributed to highly tritiated dust deposits which are below in-vessel structures and hence shielded from much of the clean-up and heating operations [36].

With further JET tritium campaigns, then with the ITER-like wall, there was a degree of uncertainty in tritium retention which may be present at the end of operations. This makes planning challenging, especially for non-tile materials in the vessel which still contribute a significant volume of waste. In practice, the tritium ‘missing’ from accountancy systems in the gas handling facilities is attributed to tokamak retention. Calculating this value is challenging and relies on multiple separate measures across several systems. Hence, to aid with future decommissioning planning, good practice suggests that multiple tritium accountancy systems are deployed, and that these have a good coverage of the entire tritium system such that waste management can be planned. This may include in-vessel techniques such as a laser induced spectroscopy methods, gaseous measurements via ion chambers, gas accounting calculations, or the use of coupons / sample mediums inserted into systems and routinely analysed for tritium content.

Since JET has been operated in incremental funding cycles, there have been several occasions in which JET has been scheduled for shutdown and decommissioning. Due to this a significant decommissioning baseline for JET was compiled in 2006–2007. This included a breakdown of JET disassembly, some activation modelling, a comprehensive waste survey of JET infrastructure, and engagement with regulators and government waste disposal sites. Though JET has surpassed the operational timelines outlined in the 2006–2007 baseline, the availability of this information has served as an analogue to a fission-style decommissioning plan for JET. This has meant that there is a large amount of relevant information available at the start of JET decommissioning which requires simple verification and update rather than starting from scratch. It is hence suggested that generating this kind of information during operations of a fusion facility saves significant time in decommissioning. There is also a major benefit of personnel being available to input into this information who have significant operational experience of the facility. If a decommissioning plan is not compiled at a detailed level until after operations have

ceased, there are likely to be key members of the facility workforce who have moved to other organizations, and a wealth of operational knowledge will have been lost.

5.5.4.5. Radiological characterization

A key finding with the onset of JET decommissioning is the forecasted rise in waste arisings and hence characterization required to support this. In a D-T system, radioactive waste arises due to activation from neutrons and from tritium contaminating materials. It has become apparent that, though UKAEA has existing tritium analytical capability, this will be stretched by forecasted decommissioning demands. Engagement with the UK supply chain and other nuclear operators who require tritium characterization suggested that there is a lack of capability for tritium characterization, especially when samples may also be activated and hazardous material (beryllium tiles for instance).

The same is also true for detailed radionuclide analysis to establish radionuclides in activation fingerprints which are challenging to detect. Though there are laboratories which can handle these analyses, very few can accept highly tritiated, moderately activated, and hazardous materials. This issue is only likely to worsen as fusion facilities deploy greater tritium loadings and generate greater exposures to 14 MeV neutrons than JET. Key learning from JET fore decommissioning is therefore that having a reasonable on-site capability for characterization of material is highly beneficial, firstly because the supply chain may not have current capability, and secondly because the costs and timescale implications of relying heavily on the supply chain are problematic.

5.5.5. ITER

Following the decision to build ITER, a decommissioning plan including the waste amounts and costs was established. Decommissioning experts are regularly invited to participate in the systems design review in order to provide suggestions that would ease dismantling, including definition of the risk management process. These considerations also have a positive impact on maintenance and Occupational Radiation Exposure (ORE) see Annex IV-2. As part of the periodic activation assessments, sampling structural material procedures is currently underway during the construction phase.

ITER radioactive waste inventory assessment, associated challenges and feedback for future fusion facilities are presented in Annex III-5. Reassessment of operation and decommissioning waste amounts consistent with the maintenance plan is performed frequently throughout the lifetime of the plant, from design to construction to operation and decommissioning. Secondary housekeeping waste generated in nuclear buildings supporting the fuel cycle or maintenance is not neglected as this will represent the main waste flow starting from the beginning of operations.

A methodology to assess radioactive waste from fusion facilities is under consideration, to enable the identification of specific materials that may not be accepted by the French repository for short lived radioactive waste, as well as defining the requirement for waste process and hot cell design. R&D efforts are needed on modelling tritium permeation into materials. In order to guarantee routing to waste disposal, the use of new materials may lead to development of specific waste immobilization matrices to deal with possible reactivity of fusion specific materials.

6. MANAGEMENT OF RADIOACTIVE AND HAZARDOUS MATERIALS AND WASTE FROM DECOMMISSIONING OF FUSION FACILITIES

In addition to technical feasibility and cost considerations, other objectives which have significant impact on decommissioning of fusion facilities need to be considered at the design stage and in advance planning for decommissioning. Specific design choices to minimize radioactive material and waste volumes and reduce environmental impact may include:

- Selection of materials such that decay timelines for activated waste are manageable, and waste can be safely managed and easily characterized (and potentially reused after a short decay period);
- The selection of materials and design of systems such that they provide minimal tritium retention or can be treated to remove tritium.

Several options may need to be analyzed to recover valuable elements (e.g. tritium), separate hazardous radionuclides (i.e. C-14, Nb-94, and/or Mo-99), in order to define an optimal set of process conditions depending on the envisaged final destination of the materials and waste (clearance, recycling or disposal). Specific materials from the tritium fuel cycle need also to be considered as well as others from housekeeping of the nuclear facilities.

In preparation for decommissioning, due attention is typically given to:

- Planning and allocation of adequate space, upgrade of waste handling infrastructure, etc., for decommissioning at the start of fusion projects;
- Early engagement with regulatory organizations to clarify regulatory issues related to waste processing, storage, disposal or reuse of materials;
- The development of treatment processes and eventual use of decay storage, such that waste from fusion can be reused or recycled (even if in other radioactive settings / scenarios) rather than disposed of.

6.1. OVERVIEW OF THE MATERIALS AND WASTE TYPES

D-T fusion facilities will typically contain large quantities of radioactive material generated through neutron activation and tritiation, with the extent of activated material being closely linked with the specific materials selection used for the plant design. Fusion facilities may also make significant use of non-radioactive, but nevertheless hazardous, materials that will require special consideration during decommissioning.

6.1.1. Radioactive materials

As discussed in Section 3.2, the main physical phenomena leading to the production of radioactive waste from fusion, are activation of materials and tritiation of the facility systems, structures and components.

6.1.1.1. Neutron activation

The primary neutron field, results from the $D + T \rightarrow {}^4\text{He} + n$ fusion reaction occurring in the tokamak vacuum chamber. The 14 MeV neutrons produced by this reaction collide with atomic nuclei of nearby materials, generating new unstable nuclear states that will undergo radioactive decay until they regain their stability. These radionuclides formed from neutron interaction will be generated in various components, starting with those located in the plasma facing region and then extending away from the centre of the facility. This activation process is directly linked to the neutron fluence, affecting the structures of the vacuum chamber as well as the various circuits containing fluids such as coolants and gases. Experience from fission reactors shows that corroded residues within pipework can preferentially activate and spread around cooling circuits.

The effect of activation is seen throughout the tokamak structures and to a lesser extent in cooling circuits and peripheral infrastructure such as bioshields and diagnostic systems. The radioactive decay of radionuclides formed from activation results in external dose rates from materials, which is a key hazard in a large-scale D-T fusion facility. Mobile activated materials form corrosion products, such as dusts formed from plasma-material interactions, and debris from cutting operations (during dismantling). These can form radioactive contamination during maintenance and decommissioning

operations. External radiological exposure from activation products after shut down is primarily due to cobalt-60 and cobalt-58 as gamma emitters, and to a lesser extent, beta emitters.

A neutron activation and waste assessment study carried for a EU-DEMO model is discussed – see Annex III-4. This modelling predicts the severity of activation in key components, generations of dominant radionuclides, and End-of-Life (EOL) waste classification for planned EU-DEMO operations. The paper also suggests some of the methods for waste minimization.

For the JET facility [37], modelling activities were undertaken at a component scale and these determined the predicted volumes and severity against the UK waste categories for such items via use of computational tools including MCNP, FISPACT-II and MCR2S methodology. These predictions were compared against measurements using high-resolution gamma spectrometry of components following neutron exposure.

Based on the results of an activation analysis for CFETR and the radioactive waste classification criteria in China, a preliminary radioactive waste classification for the major components of CFETR has been developed, together with advice on reducing the quantities of radioactive waste generated from the CFETR in-vessel components [38].

6.1.2.2. Tritium contamination from fuelling

A large fraction of the tritium injected to achieve D-T fusion is not consumed; and is instead recycled. However, energetic tritium ions may be implanted directly into plasma facing surfaces, and tritium atoms can permeate into solid structures. In both cases the tritium may be trapped in the material, a phenomenon that increases with neutron radiation damage in the materials. Significant tritium inventories may therefore exist in fusion reactor materials, beginning with those located in the plasma facing components. At JET, in-tokamak plasma facing components have displayed tritium retention on the order of GBq tritium per kg of material²² such that materials are classified as ILW by UK regulations, though would become LLW after a long period of decay storage (see Section 3.3). In future reactors, permeation may result in tritium inventories in ex-vessel components (e.g. those in the heat transfer system and elsewhere) that are not neutron-activated, thereby making these materials also radioactive.

At the end of operations, tritium can remain trapped within the fusion system materials. During operation, variations in temperature and pressure, absorption or degassing involving the materials occurs and thus tritium will migrate through systems and barriers over time. This can be controlled using secondary containment, forced ventilation and tritium permeation barriers.

6.1.2. Hazardous materials

Hazardous materials are substances that have the potential to cause harm to living organisms. These materials can be in various forms, such as solids, loose powders, liquids or gases. The hazardous properties are often attributed to the ability to biologically or chemically interact with organisms, or the ability to alter environmental conditions significantly. In the case of fusion several hazardous materials have been used or proposed for use including beryllium, lithium, lead, cooling fluids, asbestos and getter materials (such as depleted uranium for tritium storage).

The use of beryllium as a tokamak material (for example as a neutron multiplier in the blanket, first wall or wall conditioning material or diagnostic window) requires particular consideration due to its toxicity; small amounts of beryllium, if inhaled, can provoke chronic beryllium disease in susceptible individuals. In addition to its toxicity, beryllium possesses unique indirect long-term radiological risks due to the pollution of the material by uranium impurities. The ore used to manufacture beryllium

²² The quantity of activated waste is closely correlated to the selected structural materials used in the facility.

contains varying degrees of uranium, up to 150 wppm, some breeding blankets may exhibit impurity levels around 100 wppm or in some cases lower, e.g. the ITER requirement value is ~30 wppm and Japan Demonstration Fusion Reactor (JA DEMO) requires a significant reduction from this (see Annex IV-4).

Evaluation of the amount of the transuranic elements produced by nuclear transmutation is presented in Annex IV-4: in the case of use of Be12Ti, its uranium content needs to be controlled, so the total concentration of α radionuclides, such as Np, Pu, Am, and Cm, produced by uranium in Be12Ti, could be minimized.

Lead is used as a neutron multiplier, and lead-lithium can be used as a tritium breeding material in fusion facilities, e.g. [28] summarizes that impurities in lead-lithium tritium breeding materials also display the formation of chemotoxic isotopes such as ^{210}Po via activation of bismuth impurities in lead. This includes suggestions for filtration and purification of lead-lithium and also strict control of the bismuth content in lead to limit the formation of ^{210}Po .

6.1.3. Unique material morphologies

In addition to inventories of solid radioactive material described above, certain other activated material morphologies exist that present unique exposure hazards. In particular, during operation, activated and tritiated dust will be created due to plasma wall interactions, which erode some of the solid material facing the plasma. The material composition of dust is inherently dependent on the materials of tokamak plasma-facing components, this is hence highly system dependent. Operational experience suggests that dust size ranges from the order of nanometres to tens of microns. The tritium inventory of dusts is directly linked to the dust properties, especially if dust particles are covered by an oxide insulating layer or contain elements which can readily react with hydrogen species. In any case, dust presents a significant waste management challenge unique to fusion. Dust is conventionally collected via specialized means and after collection, is confined to avoid any spreading into the environment. Dust can also be generated by cutting operations undertaken during decommissioning or waste management.

Following tritium operations with carbon based plasma facing components at JET in the late 1990s, ~kg of highly tritiated dusts were found. When a sample of this material was analyzed, it was found to contain significant tritium inventory of TBq tritium per gram of dust, orders of magnitude more than plasma facing components [39, 40].

6.1.4. Activated materials

In order to reduce the decay time of activated material and reduce its overall environmental impact, reduced-activation materials are under development specifically for fusion applications. To reduce the activity, their composition has been tailored to reduce or eliminate elements (including impurities) creating significant amounts of long-lived radionuclides (refer to Annex III-1).

The alloy SS316LN-IG is being used as a construction material for ITER components because of extensive experience from fission reactors (e.g. use in the Superphénix reactor in France). SS316LN-IG limits certain impurities that lead to radionuclide formation, which is minimized relative to conventional SS316. The presence of nickel, molybdenum and niobium (impurity) in this alloy, however, still leads to the generation of the long-lived radioisotopes Ni-63 (100 yrs), Nb-91 (680 yrs), Nb-94 (20 400 yrs), and Tc-99 (211 000 years) that will preclude its classification as low-level waste in future, continuously operating D-T reactors. Cobalt impurities may also be problematic for surface disposal as it will generate Co-60. Other conventional alloys such as Inconel 718 and D9 steel that contain these elements have the same shortcoming for fusion applications (see Annex III-1).

With current and previous fusion facilities being at research scale, together with the availability of disposal sites which can readily accept materials, there were only limited requirements to limit activation due to low neutron fluences. However, in the future when systems have increased D-T

neutron exposures and will be significant in size, the use of low-activation alloys will be imperative to achieve sustainable waste management and to facilitate material recycling or reuse.

Avoidance of nickel and molybdenum activation products that would preclude classification as LLW was a primary motivation for the development of reduced activation ferritic/martensitic (RAFM) steels such as F82H (used in US and JA fusion designs) and EUROFER97 (used in EU designs). RAFM steels are a modification of grade 91 steel that uses tungsten and tantalum rather than molybdenum and niobium as minor alloying elements. These will still require a degree of impurity control to satisfy LLW disposal criteria; the composition of several RAFM steels is compared to SS316LN-IG in Table 4.

A study undertaken for EU-DEMO, based on analysis of the different waste management options depending on national disposals acceptance criteria for waste disposal, suggests that controlling the impurities leading to production Nb-94, Ni-63 and C-14 in EUROFER could satisfy the acceptance criteria for surface disposals. A list of dominant nuclides for EUROFER in EU-DEMO can be found in Annex III-4. EUROFER B3 is thus being developed with reduced content of Nb, N and Ni, as shown in Table 4. These materials will still be highly activated requiring remote handling and decay.

TABLE 4. COMPARISON BETWEEN VARIOUS LOW ACTIVATION STRUCTURAL MATERIALS IN TERMS OF CHEMICAL COMPOSITION (wt%)

Element	Weight %			
	SS316LN-IG	EUROFER	EUROFER B3	F82H
Fe	63.684	88.698	89.06933	89.3312
Al	-	0.002	0.0018	0.01
Ag	-	-	-	0.002
As	-	0.05	0.0005	0.002
B	0.001	0.002	0.001	0.0003
C	0.03	0.11	0.099	0.1
Co	0.05	0.01	0.0018	0.005
Cr	18.0	9.0	8.78	8
Cu	0.3	0.01	0.0013	0.01
Mn	2.0	0.4	0.544	0.1
Mo	2.7	0.005	0.0019	0.001
N	0.08	0.03	0.0285	0.005
Nb	0.01	0.005	0.00047	0.00005
Ni	12.5	0.01	0.0085	0.03
O	-	0.01	-	0.005
P	0.025	0.005	0.005	0.005
S	0.01	0.005	0.0017	0.002
Sb	-	0.05	0.01	0.0005
Si	0.5	0.05	0.024	0.1
Sn	-	0.05	0.0001	0.001
Ta	0.01	0.12	0.111	0.04
Ti	0.1	0.02	0.0001	0.05
V	-	0.2	0.195	0.2
W	-	1.1	1.105	2
Zr	-	0.05	0.01	-

Careful control of key impurities ensures ease in management of waste materials, e.g. [29] shows the maximum permissible niobium impurity concentration in the F82H structural alloy such that this material is not classified as ‘Greater than Class C (GTCC)²³’ radioactive waste according to the US waste classification system (see also Fig. 15).

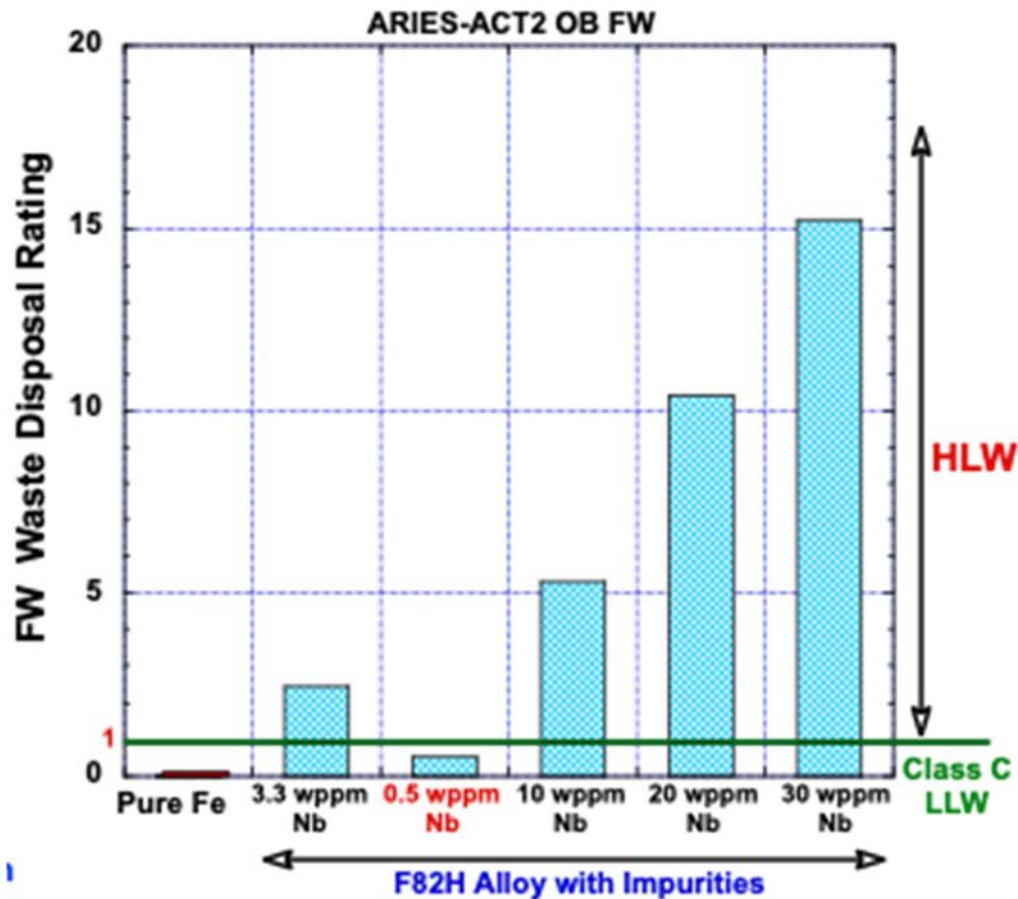


FIG. 15. Activation of F82H First Wall with varying Nb impurity levels, adapted from [29]²⁴.

Some modelling work has been undertaken on alternative designs of the EU-DEMO blankets, considering silicon carbide (SiC) as structural material [29]. The impurity content in SiC composition plays an important role in classifying waste and fulfilling disposal requirements.

6.1.5. Liquid and gaseous effluents

There are numerous effluents which may be generated outside of the immediate tokamak system. Coolants may corrode neutron activated structures and transport the resultant Activated Corrosion Products (ACP) throughout coolant circuits, leading to contamination of adjacent systems, such as coolant pipes. The ACPs are distributed throughout the circuit and structures, resulting in external dose rates from cooling infrastructure and potential contamination during maintenance operations, caused by

²³ GTCC radioactive waste exceeds radionuclide concentration limits associated with waste that may be emplaced in near surface disposal facilities, i.e. it requires to be disposed of in a geological repository.

²⁴ The waste disposal ratings shown in the figure distinguish waste suitable for near surface disposal (Class C LLW) and waste which requires geological disposal (HLW), the later typically being referred to as GTCC waste in the US waste classification system.

the release of ACPs as dust. External radiological exposure is primarily due to gamma emitters for ACP, with Co-60 being the main gamma emitter, and to a lesser extent, beta emitters.

The draining of the tokamak primary coolant loops needs to be performed at the beginning of decommissioning. The radionuclides resulting from water activation (if water is used as a primary coolant) include H-3, C-14, N-16 and N-17. Though the nitrogen isotopes provide gamma emissions, these are very short lived (7.13 and 4.17 seconds respectively) and thus are only a concern during operations and not during decommissioning [38]. Considering decommissioning, the resultant effluents have important ACP and tritium content, the latter due to permeation from the in-vessel components. In the case of ITER around 1 500 m³ of effluents have to be managed considering flushing of the circuit. Several cubic meters of spent resins have also to be discharged and managed after the facility shutdown. They are tritiated and are assumed to contain ACP. Tritiated effluents²⁵ are also generated by the Air Detritiation System (ADS) that will need to be maintained during the long period associated with decommissioning. All such waste is not compatible with release to the environment and may need to be treated by a specific process (see Section 6.3). As an example, tritiated fluids have been treated by incineration in UK. Other coolants may present unique radiological and chemical hazards, e.g. due to radiolysis.

There is experience of the formation of tritiated oils at JET. This has been especially prevalent in the active gas handling system when pumps have required maintenance or replacement (the use of dry pumps is implemented whenever possible to avoid management of tritiated oil). In general, these fluids have been sent for incineration due to the large tritium discharge limits for the UK incinerator facilities. The developers of future facilities, including ITER, may be expected to consider the use of dry pumps.

In terms of gaseous releases from decommissioning activities, there are numerous release pathways to be considered during decommissioning. Gas discharges were subject to ongoing monitoring during the operation of JET, in terms of tritium release as well as in terms of activated particulates which may have been generated. This was achieved via stack monitoring but also via biological and environmental sampling in the surrounding areas. Such activities are expected to be standard practice at other D-T fusion facilities and are required throughout operations and decommissioning. With the onset of decommissioning, and the breaching of systems and undertaking of size reduction activities, it is expected that there will be more challenging gaseous releases. This is partly due to the resuspension of existing loose particulate and tritiated species (such as tokamak dusts, active corrosion products, tritiated volatile compounds). This is also due to the generation of more active particulate and enhanced tritium release from materials during processing activities such as size reduction (cutting and grinding for example) and activities such as ventilated storage, detritiation, or surface treatments of waste.

The activation of gaseous species was accounted for during operation at JET, though this did not produce extensive quantities of highly activated gases. In the future, and especially with prolonged operations, significant volumes of activated gases may arise with sufficient half-life that they require storage or abatement. In particular, it was found that activation of the JET nitrogen heating loop results in the generation of short-lived N-17 and N-16 but also generated longer-lived isotopes such as tritium and C-14. Management of these gases in future systems is uncertain however, e.g. if gas systems are closed-loop there may be a build-up of radionuclides such as C-14 over prolonged operations, which will require special management at the decommissioning phase.

6.2. PROCESSING OF RADIOACTIVE MATERIALS AND WASTE

It is frequently the case that waste arisings during operation of facilities using radioactive material are a fraction of those which arise during decommissioning. In the case of current fusion facilities, operational arisings chiefly consist of redundant equipment, secondary waste from maintenance

²⁵ Including any tritiated condensate produced by abatement systems such as ADS.

activities, and waste arisings from system modifications (e.g. in-vessel tile replacement). Upon decommissioning, the significant tokamak infrastructure will become redundant and require processing, characterization, packing for disposal, and periods of storage. It is rare that waste facilities which support operations have sufficient capacity and capability for decommissioning challenges. Given the likelihood that waste from future systems will be significantly more complex, it is important that facilities are designed to accommodate decommissioning from the outset. This may be as simple as leaving sufficient space for decommissioning activities, or the ability to incrementally upgrade and extend supporting facilities as required. It may be the case that conventional buildings belonging to the perimeter of the facility can be reconverted to accommodate the required buffer storage capacity for radioactive waste from decommissioning, once they are no longer required to support operations.

Dealing with materials with enhanced levels of tritium represents a specific challenge, for which provisions need to be established prior to operation of the facility, e.g. the MKIIA divertor at JET, used for power and fuel exhaust, became activated to levels that precluded its management as LLW. Tritium was successfully removed and the used component managed as LLW – see Annex V-3. For reference purposes only, an illustration of the strategy for waste management of the Russian hybrid fission/fusion project is presented in Annex IV-5; such facilities are considered to be outside the scope of this publication.

6.2.1. General considerations

The management of radioactive materials refers to a series of activities involved in collecting, manual and remote handling, sorting, characterising, treating, and conditioning substances / components that contain, or may contain, radioactive isotopes. Depending on relevant national regulations, some materials (if of economic value) containing traces of radioisotopes and, coming from the fusion industry, can follow a dedicated process to be recycled either in the nuclear industry or even release for civil use if authorized by national regulation. In other cases, they are considered to have no further useful purpose and become radioactive waste.

For all these activities, it is crucial to follow strict protocols and safety measures to ensure the protection of the workers, the environment, and the public from potential radiation hazards and to take into account the potential impact for future generations.

Key steps involved in the processing of radioactive materials include:

- Radiological status (including tritium contamination) of the facility after shutdown: the knowledge of the radiological status of the facility after shutdown is a key issue conditioning the dismantling scenario;
- Radioactive material identification and characterization: radioactive materials are identified based on their nature (liquid, gas, solid, metal, soft housekeeping material, concrete.) as well as their specific isotopes and radioactivity levels. This identification is crucial for the identification of appropriate handling and processing methods;
- Radioactive material handling and sorting: specialized tools and equipment, such as lead-lined containers, remote handling devices, specific tools to limit the spread of contamination and shielding materials, are used to safely handle radioactive materials. These are example of measures to minimize radiation exposure to workers and prevent contamination. In addition to these measures, radiation protection can be ensured by the use of hot cells or glove boxes;
- Contamination control: contamination control measures (collective monitoring and personal hand/foot monitor) are implemented to identify the spread of radioactive substances;
- Contamination protection: this includes use of appropriate personal protective equipment (PPE), containment systems, and implementation of strict decontamination procedures;
- Radiation monitoring: regular collective and personal monitoring of radiation levels (dosimeter, radiation detectors) is conducted throughout the whole process to ensure that radiation exposure remains within acceptable limits;

- Radiation protection: this includes the use of appropriate personal protective equipment (e.g. lead apron), containment systems (e.g. hot cells, shielded glove boxes) and implementation of strict operational procedures;
- Radioactive material transportation: when radioactive materials need to be moved from one location to another, specific transportation regulations and guidelines have to be followed. Specialized containers, vehicles, and drivers are used to ensure the safe transport of these materials;
- Radioactive waste management: radioactive waste has to be managed and disposed of in accordance with regulatory requirements. Different types of waste, such as VLLW, LLW and ILW, may require different disposal methods, including surface, near surface and geological disposal facilities. The identification of physio-chemical characteristics and exotic chemical content is required to consolidate the waste management options;
- Safety and security: stringent safety and security measures are implemented to prevent unauthorized access to radioactive materials and mitigate potential risks. These measures include facility design, access controls, personnel training, and emergency response plans. These two aspects could be separated: ‘safety’ being related to the management of physical risks, whereas ‘security’ refers to the potential for human disruption.

6.2.2. Thermal treatment of materials

Thermal treatment can cover annealing or melting process, both techniques with proven benefits for material recycling, clearance, transportation, storage or disposal. In particular, annealing treatments are shown to be promising for tritium removal (see Annex V-3). Treatment by melting allows extraction of various isotopes such as tritium, C-14, and other activated species from the bulk matrix (see Annex V-3, Annex V-6 and [22]).

The option for future recycling of metallic materials from EU DEMO was considered in a melting study described in Annex V-6. This study considers the reduction in carbon content of steels to lower the impact of activation of materials (focused on long-lived C-14 and its impact on disposals). Experimental trials have been undertaken along with estimation of secondary waste arising from thermal treatment process to remove carbon.

A fusion reactor is simulated in [22], considering activation of materials on completion of operation, with recycling of core materials by melting being simulated, including separation of radionuclides during recycling. Results show a decrease in the dose rate of several orders of magnitude as a function of time in the metal, slag and dust. This phenomenon is correlated with a combination of decay and decrease of activated isotopes in the metal (such as Nb-94). It also noted that, for the recycled metal the dose rate remains almost constant as a function of the number of recycling times.

The scale of future thermal treatment processes will likely be significantly larger than any systems currently deployed. The size of components, potential dose rates, heating profiles required, and the potentially large tritium retention within materials will necessitate carefully designed facilities for thermal treatment of materials from fusion facilities. Most of the currently developed processes will produce tritiated water which has to be managed by a specific process of water detritiation or specific storage. Investigation of processes allowing recovery of tritium in a reduced form may be expected to be subject of further research and development.

6.2.3. Material characterization

Characterization of fusion components from D-T fusion facilities relies on multiple techniques and robustly underpinned sampling plans. This arises due to the distinct radionuclide behaviour of materials in fusion facilities. Characterization via destructive and non-destructive techniques (summarised throughout this section) are detailed in [30] with regards to fusion specific radionuclides and materials.

6.2.3.1. Tritium

Tritium is a low-energy beta emitting radionuclide which requires analysis via Liquid Scintillation Counting (LSC) for quantitative measurement of tritium trapped in materials. This necessitates sub-sampling of materials and capture of tritium in fluid. This is commonly achieved via pyrolysis (thermal degradation of sub-samples) or via acid dissolution (chemical degradation of sub-samples) before capturing products in fluid and LSC analysis. Tritium outgassing from materials can be in the form of tritiated hydrogen gas (HT) or HTO and may also be a crude characterization technique for highly tritiated materials.

6.2.3.2. Activation radionuclides (gamma emitting)

Many activation radionuclides emit gamma rays during decay and can hence be assessed via gamma spectroscopy. This is common practice for radionuclide analysis and can quantitatively assess the content of nuclides such as Co-60, Co-57, Mn-54, Zn-65. Due to the large size and composite nature of fusion components, this measurement process is often more effective after significant pretreatment of waste such as size reduction and segregation.

6.2.3.3. Activation radionuclides (beta emitting)

Some activation radionuclides decay via emission of beta particles only, or with weak associated gamma emissions. These particular radionuclides may be more problematic to uniquely detect and measure as compared to gamma emitting radionuclides. Beta emitting radionuclides can be characterized by a host of specialist laboratory techniques including gross-beta counting, LSC, mass spectrometry, and beta spectroscopy. This characterisation can be more time consuming than gamma spectrometry measurements.

Once an activation fingerprint has been suitably underpinned via beta and gamma measurements, the ratio of activity of easy-to-measure gamma emitters and difficult-to-measure beta emitters may be determined. This ‘validated’ approach can then be used to infer beta activity present from measurement of gamma ray emissions. When coupled with tritium measurements, this enables characterization of materials from fusion facilities against relevant waste classes, waste acceptance criteria, clearance levels, etc.

6.2.3.4. Experience

Pre-characterization measurements of components, based on Italian experience, have been carried out using both non-destructive and destructive analyses, respectively, for gamma and beta emitters determination. For this purpose, a transportable detection system equipped with a High-Purity Germanium (HPGe) detector was deployed for in situ gamma spectrometry (non-destructive analysis) and liquid scintillation counting (destructive analysis) was performed in laboratory.

In all laboratories, the measurement of beta emitters, such as tritium, is in fact carried out by the LSC technique. This technique is more complex to perform respect to gamma spectrometry because it requires a sampling and a dissolution with physical and chemical pre-treatment on the sample, which can be performed only in an authorized radiochemical laboratory. As a result, the preparation and measurement times become significant, and consequently, the characterization analyses are sometimes on the critical path of dismantling. Furthermore, due to tritium contamination heterogeneity, the sampling step is of key importance.

It may be anticipated that, for future fusion facilities, all the tritium measurement techniques have to be performed remotely on activated samples, thus requiring hot laboratories.

6.2.4. Specificities of tritiated material management

Tritiated waste from D-T fusion facilities provide a host of unique challenges compared with radioactive waste generated from other nuclear applications. A key factor in the management of waste containing

large quantities of tritium is its mobility. This results in the release of tritium from waste to the immediate environment, a process often referred to as off gassing. The off-gassing nature of tritiated waste requires careful consideration of a multitude of factors:

- What are the chemical species containing tritium (in particular HT or HTO)?
- How will off-gassing rates be managed to keep personnel safe?
- How will off-gassing rates be managed such that they limit cross contamination of areas?
- How will off-gassing rates be managed with regards to specific acceptance criteria (e.g. transport regulation leakage rates)?
- How will systems be decommissioned / dismantled and breaches in tritium containment be managed?

In most cases, the management of highly tritiated materials is achieved via the deployment of dedicated confinement systems. These may be double containment structures, the use of sealed containers with a high leak-tightness, or the use of surface coatings as barriers which are less permeable for tritium. There is also key control of any tritium releases via ventilation systems. These reduce the airborne concentration of tritiated species and hence limit personnel exposures and the severity of cross contamination. In some applications, releases may be sufficiently large that tritium abatement and capture systems are required for any ventilation. These have been previously deployed to limit environmental tritium discharges when storing or processing waste materials.

Regarding the disposal of tritiated waste, final repositories may impose stringent waste acceptance criteria for tritium so that the waste cannot be emplaced without a prior management step aiming at reducing the tritium amount in the waste (placement in storage for sufficient time to allow tritium decay and/or detritiation treatment) and/or confining tritium in the waste package (package designed so that tritium decay characteristic time is higher than the tritium permeation characteristic time).

Implementation of waste detritiation processes is best considered as an opportunity to ease waste management, reduce environmental impact of tritium release, recycle material for nuclear application (circular economy), and enable reuse of extracted tritium. There are several tritium removal techniques for materials, these include methods for removal of tritium from surfaces and methods for removal from the bulk of materials. The choice of the best available techniques relies on the assessment of several criteria such as safety and environmental issues, the required tritium removal efficiency, the physical properties of materials being treated, the technical readiness level of the process, the amount of secondary waste produced, the costs (investment, operating and dismantling costs), and public / regulator acceptability. The reduction of the overall waste volume also provides an opportunity for deployment of destructive detritiation methods, together with the potential preliminary cutting step required prior to waste processing (see Annex V-1).

Tritium accountancy in fusion reactors will likely be challenging but will be required for safety and security reasons. There are many sampling activities and measurements required to understand tritium behaviour, and there are still significant uncertainties in its retention. The main technical challenges according to JET feedback are as follows:

- Tritium is challenging to manage due to its mobility and limitations in real-time detection, leading to tritiated waste. It is difficult to estimate how much tritium resides within a system and how much could outgas during dismantling / processing, in the waste management facilities, and during disposal;
- Destructive sampling is required to deduce tritium content in materials, which necessitates waste handling and sampling. This could be improved via the use of sacrificial components being installed into systems at commissioning and being routinely retrieved and sampled to infer system-level tritiation;
- Refuelling systems for a D-T tokamak receive significant tritium contamination of internal structures through diffusion. Limited experience of the waste resulting from these systems

shows that they are difficult to analyze and treat without prior isolation before dismantling, and that they require numerous destructive analyses to appropriately characterize systems.

The quantity of tritium required to support continuous fusion reactor operation is likely to be approximately a few kilograms, distributed across the storage beds (located in the tritium building), the tokamak, the hot cells, and in storage of radioactive waste.

The current radioactive waste disposal landscape has been understandably refined for fission applications, hence current disposal and treatment facilities / organizations are rarely equipped or experienced with the management of highly tritiated materials. With the increasing focus on fusion, and the regulatory distinction of fusion mentioned in this text, it is probable that tritiated waste from fusion are likely to be catered for more holistically in the future. Fortunately, the relatively short half-life of tritium, and the potential for detritiation techniques to be implemented, enables effective management of highly tritiated materials without the necessity for disposal strategies such as those used for fission products or actinides found in other radioactive waste.

6.2.5. Detritiation processes and associated releases

During detritiation processes, tritium is desorbed from the materials being processed and is trapped in its oxidized form, HTO. Similarly, final clean-up systems of tritium plants (e.g. the Exhaust Detritiation System (EDS) at JET and Atmospheric Detritiation System (ADS) at ITER) process gases and convert the tritiated part of exhaust gases collected from the facility and the tritium plant into tritiated water that can be trapped or stored. This oxidation stage is achieved by either scrubbing columns (ADS) or catalytic oxidation (EDS). In any case, tritium is captured in water as HTO in current detritiation processes.

Water detritiation facilities have been developed in order to recover the trace amounts of tritium from detritiation fluids, but more importantly, to reduce ongoing waste management liabilities. Some proposed processes combine electrolysis with Liquid Phase Catalytic Exchange (LPCE) columns and a final treatment in cryo-distillation columns. In the case of JET, dedicated cryo-distillations columns have been installed, which enable the LPCE phase to be skipped, and concentrate the tritium from parts per billion levels to a percent level fraction. The tritium recovered is then processed by pre-existing Active Gas Handling systems (AGHS) whereas the tritium depleted stream is safely released to the environment (see Annex V-3).

PPPL made an agreement with the local governmental authorities that the TFTR facility would not release more than 500 curies of tritium (gaseous release) in one calendar year. Whenever a tritium contaminated system needed to be breached the system was pumped and purged with room air to gas holding tanks until the tritium concentration was as low as practicable. Then, as mentioned previously, negative pressure trunks, or ‘elephant trunks’, were used at the line break location to remove the released tritium in order to minimize worker uptake. This gas stream was analyzed for tritium concentration and then released through the facility stack via the respective areas HVAC system (see Annex I-2).

TFTR generated very little liquid effluent. The water systems supporting TFTR never reached levels of concern for tritium. The bulk of the liquid effluent from TFTR was generated by condensation from HVAC systems which was sent to liquid effluent collection tanks which were sampled for tritium and released assuming tritium concentration was low enough. During TFTR decommissioning some liquid waste was generated using drum bubblers; however, this liquid was solidified before the drums were shipped as radioactive waste.

ITER is investigating processes to manage its radioactive effluents during operation. This process will also be used intensively during decommissioning. The parameters taken into account for selection of the technology are:

- Decontamination factor to separate ACPs or solid particles;
- Process capacity to reduce treatment time and space for buffer tanks;
- Generation of secondary waste (spent resins or filters for example) and their compliance with disposal facilities;
- Compatibility of the treated water with Water Detritiation System available in Tritium Plant or industrial drainage acceptance limits (flow, chemistry, radioactivity)

Treatment of homogenous waste such as resins and concentrates consists of blocking with an immobilization matrix into a disposal container usually made of reinforced cement. Due to the important tritium concentration, the resins and concentrates generated by ITER are above the limits of existing incineration facilities. Benchmarking is being performed by ITER to ensure compatibility of the immobilization matrix with the chemical properties of the waste.

A comprehensive summary of tritiated effluent management from CANDU fission systems is presented in Annex V-5 and has a high degree of relevance to fusion facilities. The extraction of tritium from fluids and a comparison of fluid inventories for fission systems and ITER is provided displaying the similarities in scale. There is also a description of tritium emission pathways from future fusion facilities in comparison with current fission systems and a suggestion as to how such emissions may be controlled.

6.3. PROCESSING OF TOXIC MATERIAL AND REACTIVE CHEMICALS

As regards the management of toxic materials (Be, PbLi, Hg) within radioactive waste, it is crucial to implement proper protocols and precautions to ensure the safe handling, treatment, and disposal of these materials. The amounts of such materials may be high in fusion facilities. Due to their properties, several issues arise concerning acceptance for disposal of toxic materials in radioactive waste: impact on environment, possible reactivity with binding materials, instability over time, risks of explosion. Considerations for the processing of toxic materials in radioactive waste include:

- Identification and segregation: identify and segregate toxic materials present in radioactive waste to prevent cross-contamination. Conduct thorough waste characterization and employ appropriate analytical techniques to identify the toxic substances;
- Hazard assessment: perform a comprehensive hazard assessment to evaluate the potential risks associated with the toxic materials. Consider factors such as toxicity levels, physical properties, and potential interactions with radioactive components. This assessment will help determine the appropriate processing methods and safety measures to be implemented. Interaction between substances also needs to be evaluated, e.g. to determine potential production of gases, and prevent explosions by passivation if possible;
- Containment and engineering controls: implement containment measures to prevent the release of toxic materials during processing. Use sealed and secured containers, glove boxes, or other engineering controls to ensure the isolation of toxic substances from the environment and personnel;
- Personal protective equipment (PPE): provide workers involved in the processing of toxic materials with suitable personal protective equipment, including gloves, masks, protective clothing, and respiratory protection, to minimize their exposure and ensure their safety;
- Treatment and neutralization: employ appropriate treatment methods to neutralize or stabilize the hazardous materials within the radioactive waste. This may involve chemical processes, such as precipitation, filtration, or ion exchange, to remove or reduce the toxicity of the substances;
- Decontamination and cleaning: prioritize thorough decontamination and cleaning procedures to remove any residual toxic materials from equipment, containers, and work areas. This helps prevent contamination and minimize the potential for exposure to the toxic substances;
- Waste packaging and labelling: properly package and label the processed radioactive waste containing toxic materials to ensure safe transportation, storage, and disposal. Follow relevant

regulations and guidelines for waste packaging and labelling to inform of the presence of toxic substances and facilitate proper handling and management;

- Regulatory compliance: adhere to applicable regulations and guidelines governing the processing and disposal of radioactive waste with toxic materials. These regulations may include specific requirements for the treatment, storage, transportation (e.g. ADR) and disposal of hazardous or toxic waste components;
- Storage and disposal compliance: whenever needed, implement specific treatment (passivation, chemical treatment, immobilization) to ensure compliance of waste with storage and disposal waste acceptance criteria. This may require research and development studies (e.g. assessment of interaction between beryllium dust and cement matrix in the final waste package and new cement matrix development, or of the pyrophoricity of certain getter materials).

It is crucial to conduct these activities under the supervision of qualified personnel with expertise in both radioactive waste management and toxic material handling. Compliance with regulatory requirements and adherence to best practices is essential to protect human health, the environment, and to ensure the overall safety of the process.

The complexity of the measures to be taken aims to bring about a very significant reduction of toxic materials and impurities present in the components used. In particular, beryllium, which can be used as a plasma facing material or neutron multiplier in the blanket, is classified as a toxic material. Though beryllium has advantageous neutronic properties, and limited impact on plasmas in terms of radiative losses when poisoning a plasma, it has hazardous properties akin to asbestos.

At JET, precautions for beryllium contamination are equally as relevant as those for radiological contamination. This has meant that a rare skillset has been developed in terms of beryllium characterization, decontamination, control, and handling. This is unique to JET currently and discussions with waste disposal sites in the UK have indicated that disposal of radioactive beryllium are novel in the quantities associated with fusion facilities. The experience gained at JET was an important driver in a decision that beryllium would not be used in the ITER facility.

7. CONCLUSIONS

This publication reports on experience gained and current good practice regarding planning and implementation of decommissioning of D-T toroidal magnetic confinement fusion facilities, including aspects relevant to the management of the resulting materials and radioactive waste. It addresses a growing recognition, given current levels of interest internationally in the development of industrial-scale D-T fusion facilities like ITER, that consideration needs to be given to the decommissioning aspects and the management of radioactive waste from the earliest stages of the design process.

Globally, approaches to regulating the commissioning, operation, and decommissioning of D-T fusion facilities vary widely, reflecting the early stage of development of fusion energy and the application of a graded approach to safety when assessing the risk from early facilities. Regulatory considerations will become increasingly important as fusion facilities grow in complexity and fusion technology moves closer to deployment. While some Member States are currently developing regulatory frameworks for the operation and eventual decommissioning of such facilities, global experience with regulating D-T fusion facilities remains limited.

Even if there are differences between the various national regulations, decommissioning planning will need to consider the following main phases and associated administrative steps:

- Pre-shutdown and post-shutdown phases: these phases relate to activities performed as the facility approaches permanent shutdown and the period thereafter during which the facility is prepared for dismantling. An important activity during this period is the submission of the

decommissioning plan to the appropriate regulatory authorities. Depending on the decommissioning strategy, the facility prepares for:

- Immediate dismantling: the approach in which decontamination, dismantling, demolition and cleanup activities begin as soon as reasonably practicable following the facility's permanent shutdown.
 - Deferred dismantling: placing the facility in a safe and stable condition for an extended period before dismantling activities begin. During this interim phase, efforts focus on minimizing the mobilizable source term and addressing other hazards present at the time of shutdown. Once this period concludes, dismantling proceeds in a manner similar to that of immediate dismantling.
- Termination of the authorization: conducting a final survey, which is then verified by the regulatory authority. The results of this survey determine whether any restrictions need to be applied to the future use of the site or if it can be released for unrestricted use.

Current global experience in regulating fusion-generated radioactive waste comes primarily from experimental fusion facilities such as TFTR (USA), JET (UK), and others. These facilities present unique challenges due to the presence of tritium and the nature of the resulting waste, which may exhibit both radiological and chemical hazards. The quantity, radioactivity, and toxicity of such waste require tailored management strategies. Approaches to radioactive waste management vary according by Member State, depending on national legislation (e.g. radioactive waste classification, treatment, storage, disposal, or potentially clearance and/or recycling), alongside safety requirements and operator responsibilities.

Experience from the decommissioning of experimental D-T magnetic confinement fusion facilities has shown that such programmes can be lengthy and complex, primarily due to the scale and technical intricacies of the facilities. As a result, funding mechanisms for decommissioning and waste management need to be resilient to uncertainties and associated risks. These mechanisms need to be established prior to the start of operation, ensuring that adequate financial resources are allocated throughout the facility's operating lifetime.

Clear and transparent communication with the general public and other stakeholders regarding anticipated quantities of radioactive waste, other materials, decommissioning plans, and the intended end state of the facility is essential. Therefore, arrangements need to be established to communicate project objectives and strategies effectively to a broad range of stakeholder groups, including the plant workforce.

Decommissioning considerations for fusion facilities need to be integrated from the design stage to ensure that the management of radioactive waste and the eventual decommissioning can be carried out safely and efficiently. Key aspects include:

- Designing for decommissioning, including modularization of plant components, facilitating easy access to systems, structures and components, and minimizing the use of materials that may complicate dismantling.
- Selecting structural material with low impurities so that the extent of future activation is minimized (both in terms of volume and overall radiological activity).
- Design solutions need to aim to manage tritium inventory in radioactive waste and in structures, systems, and components. This include selecting materials compatible with detritiation treatments, designing components to fit within detritiation furnaces, and easy enabling segregation of materials with different detritiation requirements.

Key technical lessons from the decommissioning of TFTR and JET include the use of specialized processes such as remote handling, robotics, volume reduction techniques and detritiation ovens to recover and reduce tritium in materials while minimizing radiation exposure to personnel. It is widely

recognized that the development of future fusion plants will rely more heavily on digital design tools. This evolution may facilitate the decommissioning planning and execution, for example through the use of digital twin models that support accurate identification of system dimensions, materials, and properties.

For managing radioactive waste from fusion facilities, particularly those using D-T plasma, key considerations include:

- Pursuing the potential for decontamination, reuse, or clearance of mildly radioactive materials as a means of minimizing, to the extent practicable, the volume of waste requiring disposal in radioactive waste repositories.
- Undertaking research to assess the feasibility of recycling highly radioactive metal-based components, along with cost-benefit analyses comparing options with direct disposal. In parallel, the development of storage and disposal solutions tailored to the specific characteristics of fusion-generated waste needs to be considered.
- Materials selection and design optimization: careful selection of materials with low impurity levels can significantly reduce the volume and activity of radioactive waste as well as environmental impacts such as liquid and gaseous discharges. Neutron activation calculations play a key role in optimizing material choices. Reducing overall facility size is another important design strategy to support waste minimization and decommissioning:
 - Use low-activation materials: choosing materials that minimize the production of long-lived isotopes (e.g. Nb-94, Ni-63, C-14) is essential for meeting radiological acceptance criteria at existing disposal sites.
 - Management of hazardous activated materials: fusion facilities are expected to generate hazardous waste streams containing materials such as beryllium and lead-lithium. Upon activation, impurities in these materials can lead to the formation of chemotoxic radionuclides (e.g. Am-241, Po-210), requiring the development of long-term management and disposal strategies.
- Processing radioactive waste to remove or reduce specific radionuclides (e.g. C-14, H-3) can help reduce overall waste volume and activity, improving the prospects for acceptance at disposal sites.
- Research is ongoing into techniques such as annealing and melting, which aim to remove tritium and reduce carbon-based activation products in contaminated or activated materials.
- While existing fusion facilities have demonstrated good practices for tritium characterization and storage, larger-scale installations such as DEMO and future plants will face greater challenges, particularly in managing secondary waste, such as tritiated water.
- In fusion, much of the external radiological hazard at shutdown (e.g. from isotopes such as Co-60, Co-57, Co-58, Mn-54) can be effectively managed through decay storage or deferred dismantling over a period from 5 years to many decades. However, this approach carries the risk of knowledge loss over time, which needs to be mitigated through robust documentation and information management strategies.

As designated solutions for managing waste from the operation and decommissioning of fusion facilities, including storage and disposal, need to align with national legislation, it is important that such legislation adequately addresses the specific needs of these facilities. This may include considerations on how the waste acceptance criteria of existing repositories can be met, particularly in cases where the quantity or types of radionuclides differ significantly from those anticipated in the original safety cases for those repository facilities.

Finally, several areas of active research and development have been identified as critical to enhancing future decommissioning efforts and addressing radioactive waste management needs. These include:

- Remote handling cutting techniques capable of addressing the complexity of dismantling the tokamak core, with a focus on minimizing the spread of particulate contaminants within the facility and reducing the generation of secondary waste.
- Waste packages need to be developed that can efficiently confine tritium and reduce outgassing: specific matrices and leak tight sealings.
- Radiological characterization tools (spectrometry, sampling) need to be developed and adapted to activated large scale components and be capable of determining tritium content with less intensive methods than destructive sampling.
- Calculation tools or models need to be developed to estimate tritium diffusion into materials.
- Industrialization of waste detritiation techniques such as baking or melting need to be developed and adapted to handle large volumes and sizes of materials from vacuum vessel components.
- Technologies ought to be developed with long-term maintainability in mind and designed to be easily understood by future generations of operators and stakeholders.

APPENDIX I.

IAEA STANDARDS AND GUIDES FOR NUCLEAR FACILITY DECOMMISSIONING AND RADIOACTIVE WASTE MANAGEMENT

I.1. SAFETY FUNDAMENTALS

This publication is concerned with the decommissioning and associated radioactive material and waste considerations for D-T Magnetic Confinement Fusion (MCF) facilities having reached the end of their useful life reached its end of lifecycle, as well as their release from regulatory control.

At the time of writing this report, there is no definition on any IAEA safety requirements or guides about fusion system, or similar terminology. The authors of this report will use the term “D-T MCF” as a fusion system for which safety has to be taken into consideration to protect the people and the environment for radiation risk as results of conducting the practice of decommissioning and termination of such facilities.

Hence independently of the naming convention, a D-T MCF facility will require decommissioning and waste management to be performed in alignment with IAEA Fundamental Safety Principles [11]. In the context of decommissioning and waste management, “justification of facilities and activities” and “protection of present and future generations” [41] principles are very important because shows commitment to make fusion designs with consideration for safely decommissioning and management of radioactive waste.

A facility is defined in GSR Part 4, Rev.1 as (item g) “any other places where radioactive materials are produced, processed, used, handled or stored.” In this regard, a facility in which a fusion system operates will produce radioactive materials, handles and stored them through its operation and upon decommissioning.

I.2. DECOMMISSIONING SAFETY CONSIDERATIONS

In term of decommissioning safety, the applicable IAEA safety standard is in GSR Part 6 [41] and establishes the safety requirements recommended toward decommissioning. GSR Part 6 defines decommissioning as: “the administrative and technical actions taken to allow the removal of some or all the regulatory controls from a facility” [41].

It is important to note that GSR Part 6 states that the planning for decommissioning begins at the design stage and continues throughout the lifetime of the facility (Requirement 10, GSR Part 6 [41]). Also to implement the principle of graded approach during the practice of decommissioning. This means that the level of safety assessment needed to be commensurate with the magnitude of the radiation risks associated with the facility (Requirement 2, GSR Part 6 [41]).

Lastly, another key aspect for consideration is to ensure adequate financial funds for decommissioning (Requirement 9, GSR Part 6 [41]). Financial provision starts early on and is typically a requirement before starting operation of a facility. It is also recommended that facilities update the financial contributions accordingly throughout the lifetime of the facility to make sure funds remain adequate and available when needed (in case of sudden decisions to shut down or accidents)

Currently, there is no specific guidance for implementing the requirements outlined in the Decommissioning of Facilities (GSR Part 6) [41] for fusion facilities. However, existing safety guides such as the Decommissioning of Nuclear Power Plants, Research Reactors, and Other Nuclear Fuel Cycle Facilities (SSG-47) [32], and the Decommissioning of Medical, Industrial, and Research Facilities (SSG-49) [42] provide guidance that could be adapted to the decommissioning needs of fusion facilities using a graded approach.

It is important to note that during the practice of decommissioning of fusion facilities all exposures are considered to be a planned exposure situation, as defined in Radiation Protection and Safety of Radiation Sources GSR Part 3 [17] of the IAEA safety standards. These means application of the principles of justification, optimization of protection and application of dose limits.

One of the most important safety guidance from WS-G-5.1 [43] is site termination requirements. At the end of a fusion facility lifecycle, the goal is for unrestricted release of the site. Figure 16 shows the options to meet the dose constraint criteria [43]. To release the site from regulatory control unrestricted, “a final survey report demonstrate that the dose is less than 300 $\mu\text{Sv}/\text{year}$ without any control in place”. The guide also specifies that if “this dose constraint cannot be met, then controls will be put in place to meet the dose constraint criteria”. If such control fails, “the dose limit of 1 mSv/year should not be exceeded” [43]. In this scenario, the site is released as for restricted use.

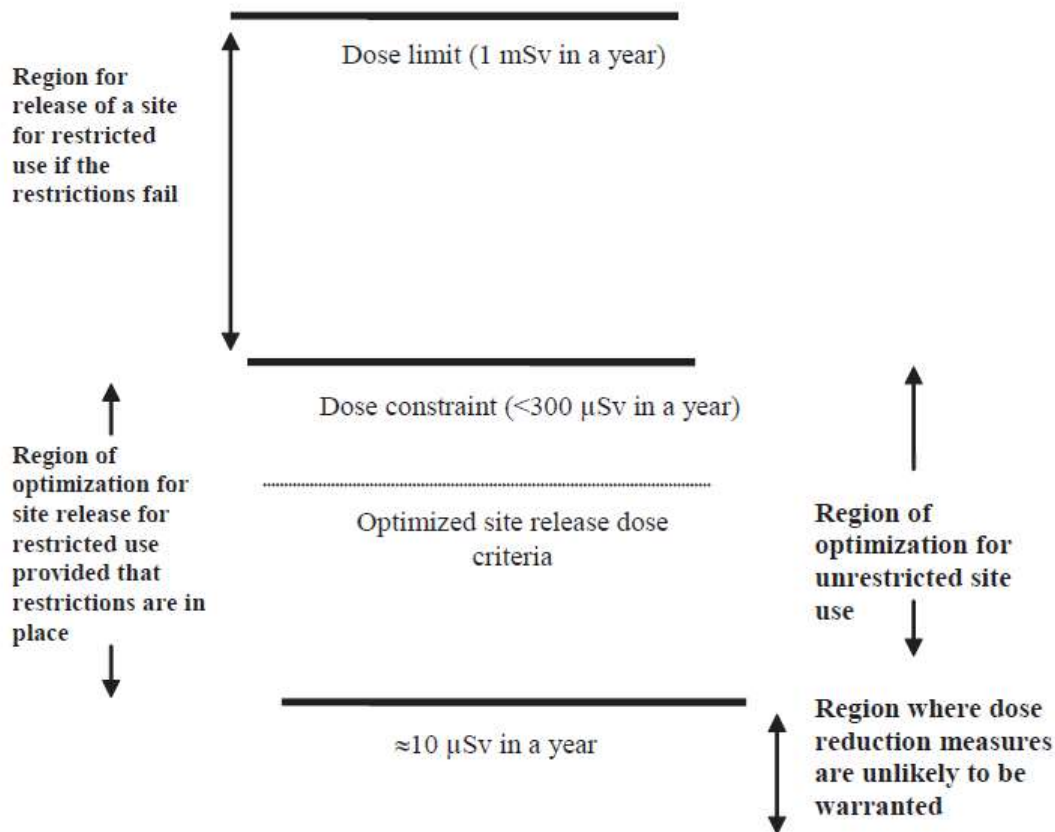


FIG. 16. Constrained optimization and regions of effective dose for members of the critical group (reproduced from [44]).

I.3. RADIATION PROTECTION CONSIDERATIONS DURING DECOMMISSIONING OF D-T FUSION FACILITIES

During the decommissioning of fusion facilities, the safety of the workers and the protection of the environment is a priority. Many changes will happen to the facility as the dismantling of structures, systems, and components happens. As a result, the workers will face industrial and radiological risks that need to be well characterized and managed throughout the evolution of decommissioning. The details of decommissioning execution are captured in the decommissioning plan. This plan usually adopts the concepts of graded approach [41] and defence in-depth [45] to protect the workers, the public, and the environment during the conduct of decommissioning actions.

According to the international standards on radiation protection [17] exposures to workers and to the public due to conduct of decommissioning actions fall under a planned exposure situation. Also, some unplanned events during decommissioning may lead to emergency exposure situations. These are the two exposure situations to be considered in the decommissioning plan and in its supporting documents, such as the safety assessment, and a compliance with the related requirements of [41] has to be demonstrated during the decommissioning process.

For decommissioning of nuclear facilities, the radiation protection relies on the principles of justification, optimization, and application of dose limits [41, 43]. These principles ensure through the establishment of administrative and physical controls that the dose to personnel will remain as low as reasonably achievable (ALARA), and the dose limits values not to be exceeded.

GSR Part 6 [41] provides the two decommissioning strategies adopted by member states for nuclear facilities. They are immediate and deferred dismantling. In principle, these two possible decommissioning strategies (or a combination of both) are applicable for fusion facilities. From a radiological protection standpoint, either option has its pros and cons. For example, if immediately dismantle post shutdown is chosen, then knowledge of plant operation, and overall personnel experience is readily available, but the radiological inventory is at its peak (with minimum time post shutdown for nuclides to decay) and the program may need to stand up additional administrative and physical controls to reduce personnel exposure. On the other hand, if deferred dismantle is chosen, then facility operation and inventory knowledge maybe lost or not accurate but allows time for the radiological inventory to decay. The goal of radiation protection is to support the decommissioning strategy that is most suitable to the member states.

Key factors influencing the selection of the decommissioning strategy from the perspective of occupational radiation protection are included in [46]. They are:

- Radiological inventory within the facility;
- Local availability of decommissioning experiences and related technology;
- Availability of experienced radiation protection and other decommissioning personnel;
- Intended level of involvement of contractors for decommissioning work;
- Availability of radioactive waste processing and storage facility.

Another radiation protection key aspect is the implementation of the requirement of graded approach as stipulated in [41]:

- “A graded approach shall be applied in all aspects of decommissioning in determining the scope and level of detail for any particular facility, consistent with the magnitude of the possible radiation risks arising from the decommissioning.”

For a fusion facility, this means assessment of risks to set controls in place to implement the radiological protection principles as the facility is dismantled. The systemic graded approach is applied to both radiological and on-radiological (industrial) risks.

Once radiological hazards are identified, decommissioning will be conducted following a defence in depth approach as recommended in Safety Assessment for the Decommissioning of Facilities Using Radioactive Materials [47]. Implementation of this approach will ensure that safety measures are appropriate to the degree of radiological risks. This will set layers of protection to mitigate risks to levels that are acceptable for the member state regulatory authority, and/or entity overseeing the decommissioning work.

The International Nuclear Safety Advisory Group [45] defines five levels of defence in depth:

- Level 1: Prevention of abnormal operation and failures;
- Level 2: Control of abnormal operation and detection of failures;

- Level 3: Control of accidents within the design basis;
- Level 4: Control of severe plant conditions, including prevention of accident progression and mitigation of the consequences of severe accidents;
- Level 5: Mitigation of radiological consequences of significant releases of radioactive material.

These levels of defence in depth are applicable when decommissioning a fusion facility and are recommended to be followed.

In summary, during decommissioning, the facility is changing as decommissioning activities take places. These will generate radiological and chemical hazards that are different from operation and will change overtime. Through this process it is important to establish a robust radiological protection culture integrated into the decommissioning plan and all activities involved. Also following the IAEA decommissioning principles and guidelines available as applicable to each decommissioning scenario.

I.4. RADIOACTIVE WASTE MANAGEMENT SAFETY CONSIDERATIONS.

As results of commissioning, operation and during the practice of decommissioning, radioactive waste will be generated by D-T MCF facilities. The main applicable IAEA safety guide and requirements are as follows:

- Classification of Radioactive Waste, GSG-1 [7];
- Predisposal Management of Radioactive Waste General Safety Requirements, GSR Part 5 [25];
- Disposal of Radioactive Waste, Specific Safety Requirements No. SSR-5 [48].

GSR Part 5 deals with all the steps in the management of radioactive waste from its generation up to disposal, including processing (pretreatment, treatment and conditioning), storage and transport [25].

Key safety considerations from GSR Part 5 regarding safety of radioactive waste management at the design stage of a nuclear facility are outlined below:

- Fusion designs are encouraged to select materials that minimize generation of radioactive waste by both volume and activity (Requirement 8, GSR Part 5 [25]). This consideration applies throughout the facility lifetime, including decommissioning and waste associated with decommissioning.
- Options for reuse and recycling of materials need to be considered during the design process, such that generation of radioactive material is minimized (Requirement 8, GSR Part 5 [25]).
- The waste streams generated through the commissioning, operation and decommissioning of a fusion facility needs to be classified in terms of quantity, physical and chemical form, for their future disposal (if declared as waste) or recycling (Requirement 9 GSR Part 5). Also to consider how these waste streams meet the waste acceptance criteria of established disposal facilities in the national legislation of Member States.
- Processing and storage of waste streams and/or source terms to ensure effluent discharges and clearance are in accordance with the national legislation in the Member State and in alignment with the IAEA safety standards. (Requirements 10 and 11, GSR Part 5 [25]).

Figure 17 shows the guidance for radioactive waste classification. The specific values are set by each national regulatory body [7].

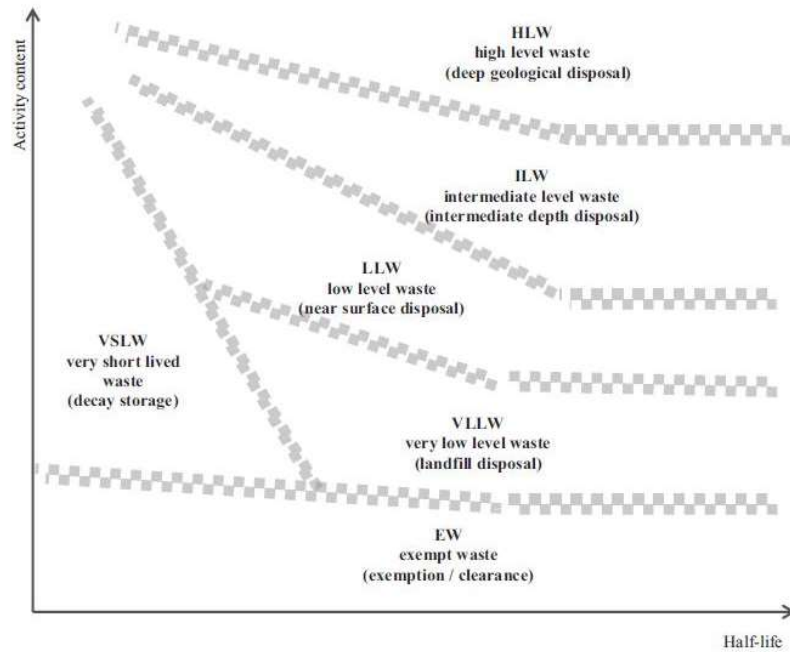


FIG. 17. Conceptual Classification of Radioactive Waste (reproduced from [7]).

SSR No. 5 deals with the safety consideration associated with the disposal of the radioactive waste. A disposal is defined as “the emplacement of radioactive waste into a facility or a location with no intention of retrieving the waste”. “Even if retrieval is not intended, it does not mean that retrieval is not possible” [48]. The disposal option appropriate to each waste stream class is shown on Fig. 18.

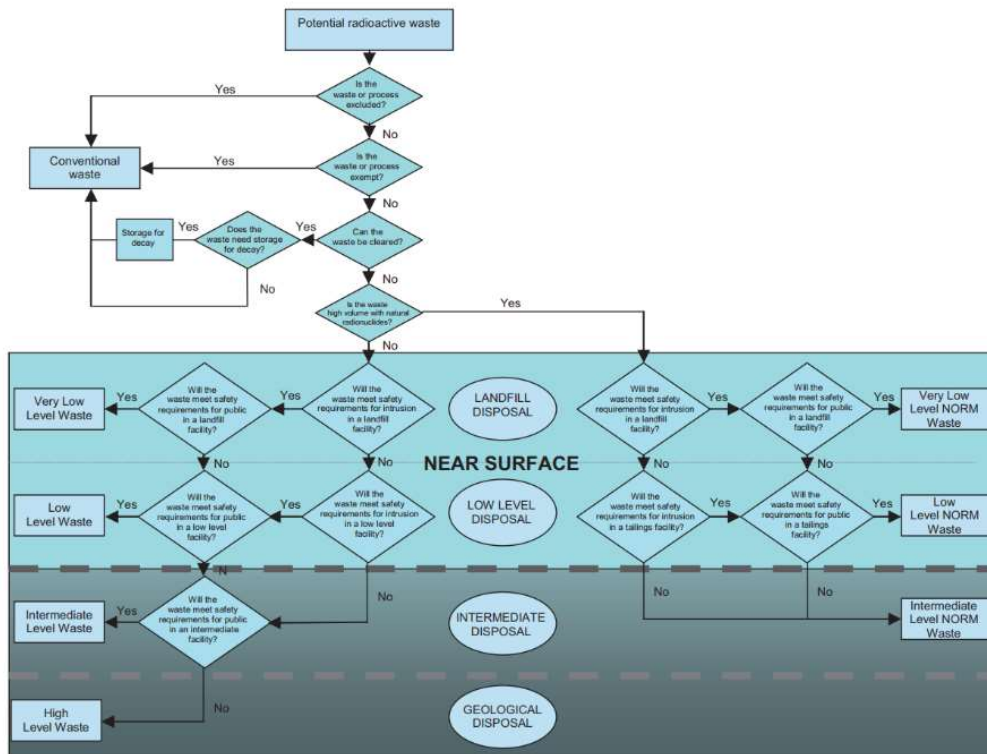


FIG. 18. Waste Disposal Options (reproduced from [7]).

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LIST OF ABBREVIATIONS

ADR	Agreement concerning the International Carriage of Dangerous Goods by Road (UN Treaty signed on 30 September 1957).
ADS	Atmospheric detritiation system
AIF	Agency ITER France
AISI	American Iron and Steel Institute
ASN	Autorité de Sureté Nucléaire (French Safety Authority)
CEA	Commissariat à l'Energie Atomique et aux énergies alternatives (French Alternative Energies and Atomic Energy Commission)
CFETR	China Fusion Engineering Test Reactor
D-D	Deuterium deuterium plasma
DEMO	Demonstration power plant
DOE	Department of Energy (United States)
D-T	Deuterium tritium plasma
EC	Electron capture
EDF	Electricité de France (French electricity utility)
EDS	Exhaust detritiation system
EOL	End of life
EU-DEMO	European Demonstration Power Plant
EUROFER	European Fusion Development Agreement (EFDA) Ferritic-Martensitic
H-H	Hydrogen-hydrogen plasma
HLW	High-level waste
HPGe	High purity germanium
HT	Tritiated hydrogen gas
HTO	Tritiated water
IAEA	International Atomic Energy Agency
INB	Installation nucléaire de base (basic nuclear installation in France)
IFMIF-DONES	International Fusion Materials Irradiation Facility

ILW	Intermediate level radioactive Waste
ITER	International Thermonuclear Experimental Reactor
IVC	In-vessel component
JA DEMO	Japan Demonstration Fusion Reactor
JET	Joint European Torus
JT-60	Japan Torus 60
JT-60SA	JT-60 Super Advanced
JT-60U	JT-60 Upgrade
LCA	Life cycle assessment
LCPE	Liquid phase catalytic exchange
LLW	Low-level radioactive Waste
LSC	Liquid Scintillation Counting
MCF	Magnetic confinement fusion
NNSA	National Nuclear Safety Administration (USA)
NPP	Nuclear power plant
NRC	Nuclear Regulatory Commission (USA)
NRC-NMSS	U.S. Nuclear Regulatory Commission's Office of Nuclear Material Safety and Safeguards
PFC	Plasma facing component
PPE	Personal protective Equipment
PPPL	Princeton Plasma Physics Laboratory
QRA	Quantitative risk analysis
QST	National Institute for Science and Technology (Japan)
R&D	Research and development
RAFM	Reduced activation ferritic/martensitic (steel)
RCF	Reference class forecast
RF	Radio frequency
RH	Remote handling

SS316	Stainless steel 316
TFC	Toroidal field coil
TFTR	Tokamak Fusion Test Reactor
TPL	Tokyo Power Laboratory
TRL	TRL Technology Readiness Level
TSDD	Tritium Systems Demolition and Disposal
UKAEA	United Kingdom Atomic Energy Authority
VV	Vacuum vessel
WCCB	Water-Cooled Ceramic Breeder

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Technical Meetings

Vienna (IAEA, Austria), 6–8 October 2021
Cadaroche (ITER site, France), 6–10 February 2023
Vienna (IAEA, Austria), 19–23 February 2024

ANNEX I.

IMPLEMENTATION OF DECOMMISSIONING ACTIVITIES

I-1. JAPANESE EXPERIENCE FROM DECOMMISSIONING JT-60

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Replacement of activated fusion core components in JT-60

JT-60U is one of two plasma generators in Japan which is regulated by Radiation Hazard Prevention Act (RI Regulation Act). After completing its missions, JT-60U completed its operation and experiments in 2008. After the shutdown, in order to upgrade JT-60U to JT-60SA, the main structure, whose weight is ~5,400 tons (~13,000 components), was dismantled, and the main structures of JT-60SA were newly installed in original JT-60U space, with superconducting coils. The main structures of JT-60U have been activated by 1.5×10^{20} neutrons generated by the D-D reaction during the 18 years of D-D discharge experiments. It was the first experience to dismantle an activated fusion facility in Japan. The JT-60U torus was composed of following systems and components. Those are 45 poloidal field coils (PFCs), 18 toroidal field coils (TFCs), a vacuum vessel (VV), Neutral Beam (NB) heating system, radiofrequency (RF) heating system (RF), diagnostics, tokamak ancillary system, and supporting structures as shown in Figure I-1.1. All main components of the JT-60U and its ancillary were required to be disassembled. Among them, only NB heating system was planned to be reused. Therefore, the disassembly of the NB heating system needs to be carefully handled by considering reassembly on JT-60SA.

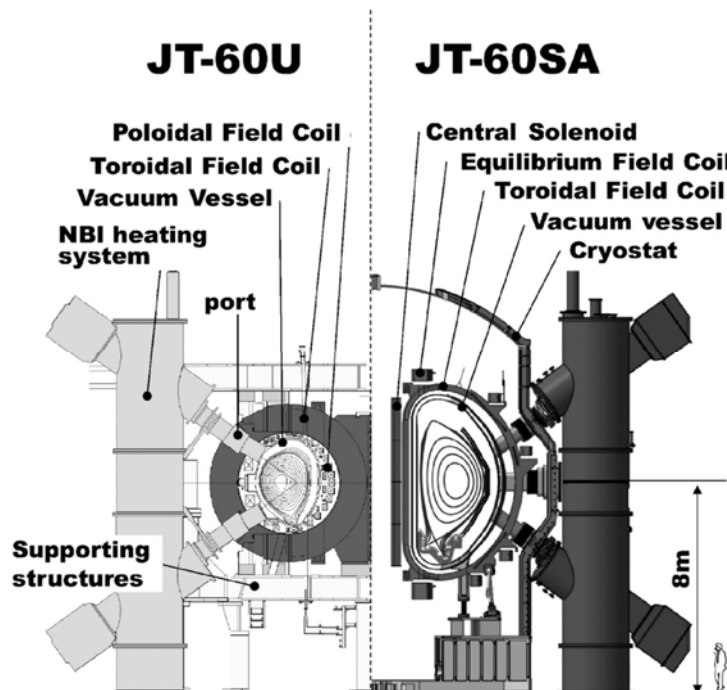


FIG. I-1.1 JT-60U and JT-60SA.

Requirements for dismantling of JT-60U were 1) dismantling period, 2) preparation of storage area of dismantled components, 3) efficient and safe dismantling (reduce radiation exposure of workers) and 4) classification of removed components.

1) Dismantling period

The JT-60U was shut down in August 2008, and an application for dismantling was submitted to the regulator and approved in August 2009. Dismantling work had to be completed by October 2012, since the equipment was scheduled to be delivered from the EU in early 2013. Therefore, the dismantling had to be carried out within a three-year time limit.

2) Preparation of storage area of dismantled components

Prior to the dismantling of JT-60U, the pre-assessment of the radioactivation level of the systems and components induced by neutrons was carried out in order to classify the radioactive components and to design the storage facility of the dismantled components. The Japanese radiation regulations require less than 1.3 mSv/three months at the radiation facilities boundary. Therefore, the storage facilities were to be isolated with public to keep this dose rate. In order to evaluate the necessary spaces for the storage facilities, the activation level and weights of the disassembled systems and components were roughly estimated by employing the area management corresponding to neutron fluence of each area. A storage building was built in order to store the large tokamak machine components such as TFCs, PFCs, the VV and supporting structures. Marine containers such as 20-feet or 40-feet were used for small structures packing. Marine containers were kept at storage yard in Naka-site, where JT-60U being constructed. Necessary space of storage yard was determined based on the results of pre-assessment. Components which are potentially tritium contaminated materials such as connected to the VV were sealed in metallic containers and stored in the storage facilities.

3) Efficient and safe dismantling

Figure I-1.2 shows the dismantling process of the JT-60U. First stage of the dismantling of the JT-60U was removal of the first wall tiles in the VV. Those first wall tiles contain tritium generated by DD reaction. Therefore, all first wall tiles were packed in the vacuum in order to prevent the tritium leak from the tiles. After the first wall tiles removal, ancillary facilities around the tokamak machine were removed. Next step was removal of TFCs, which was the critical pass of the dismantling. In the JT-60U, each TFCs were connected by welding to keep its structure against electronic magnetic force. To remove TFCs through the VV, the 20° torus segment of PFCs and the VV was partially cut and removed by lifting the TFCs through the removed section. After removal of all TFCs, the VV and PFCs was removed. Since the weight of the vacuum vessel exceeded the crane's capacity (250 tonnes), the VV was split in two and lifted and removed from the JT-60U frame. In this cutting process, a dry diamond wire saw was applied to cut the VV. Application of this saw was effective since it was able to cut several kinds of metals without temperature increase. This result clearly reduces the risk of fire due to temperature rise during cutting process.

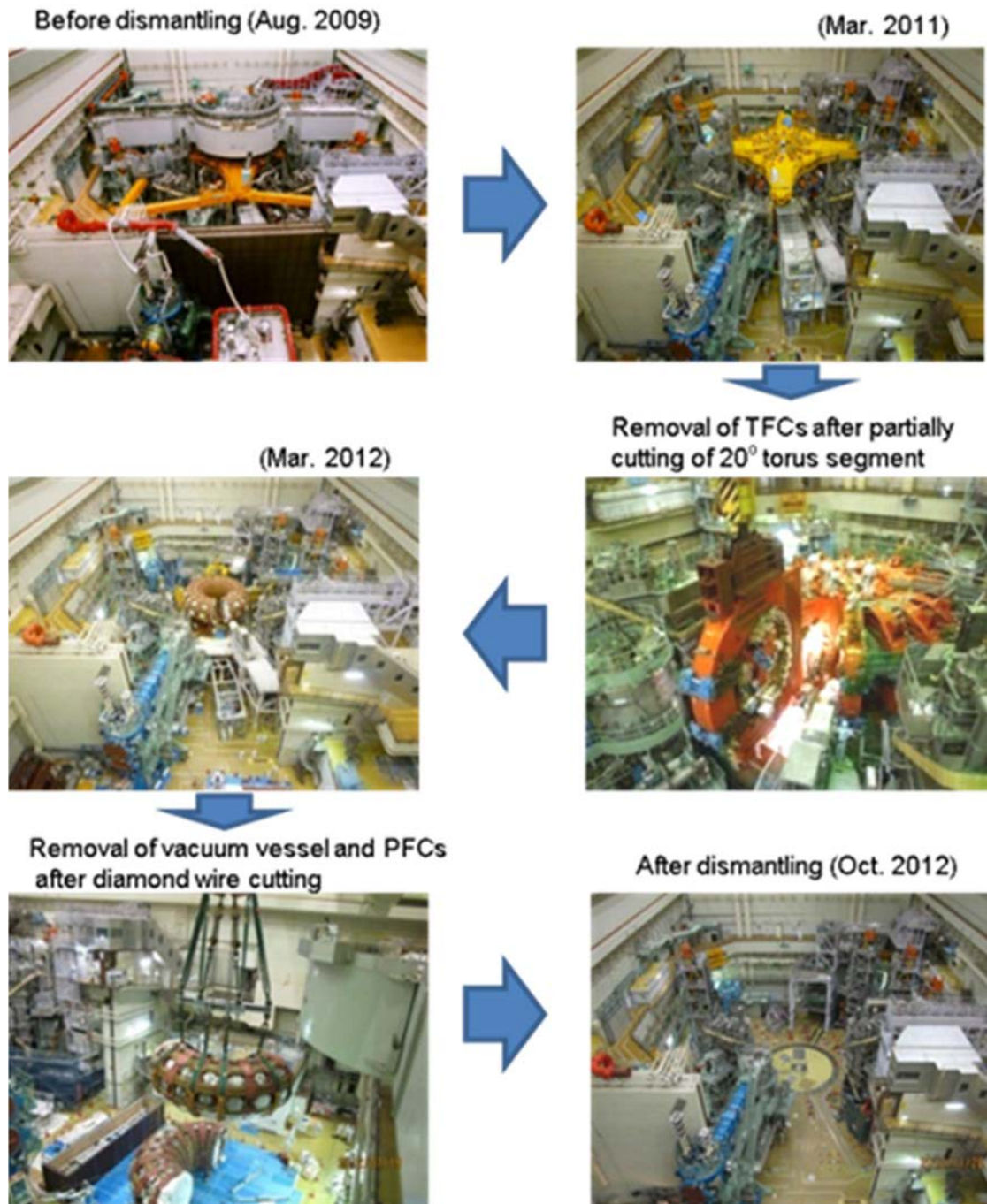


FIG. I-1.2. Dismantling process of the JT-60U.

Regarding the radiation control in dismantling of JT-60U, external exposure from activated materials and internal exposure from tritium were considered. Since JT60U was a D-D machine, and neutron yield is smaller than D-T machine such as JET, TFTP, external exposure from activated components was relatively low ($\sim 10\text{mSv/h}$ at VV surface). ^{60}Co in stainless steel was main radiation source. Regarding internal exposure to tritium, although the surface density of tritium was higher on the PFC surface ($\sim 10,000\text{ Bq/cm}^2$), the concentration in the air was kept low due to ventilation in the VV ($\sim 3,000\text{m}^3/\text{h}$) during the work in the VV. Radiation exposure of workers was within dose limit for radiation workers in Japan ($<100\text{mSv/5years}$).

4) Classification of removed components.

As the result of dismantling, ~13,000 pieces and 5,400 tonnes of components were removed as reported in [1]. Approximately 3,650 tonnes of large components such as the VV and TFCs were stored in the storage facility as mentioned 2). The metallic sealed containers with possible tritium contaminated components and stainless steel removed from high neutron fluence area were stored in the storage building (~200 tonnes). The containers without tritium contamination (~950 tonnes) were stored in the outside storage area. Weight of each container was range of 10 ~1000 kg.

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I-2. IMPLEMENTATION OF DECOMMISSIONING ACTIVITIES ON THE TOKAMAK FUSION TEST REACTOR (TFTR) AND THE TRITIUM SYSTEMS DEMOLITION AND DISPOSAL (TSDD) PROJECT

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Abstract

The Tokamak Fusion Test Reactor (TFTR) at Princeton Plasma Physics Laboratory (PPPL) performed D-T fusion experiments from 1993 to 1997. Approximately 100 grams of tritium was used supporting these experiments on TFTR. In October of 1999, several years after its operational campaign was completed, the TFTR Deactivation and Decommissioning (D&D) Project began. During project execution the TFTR vacuum vessel, coil systems, and supporting equipment from both the TFTR Test Cell and Test Cell Basement were removed leaving only the vacuum vessel pedestal behind. The Tritium Systems that supported TFTR were removed years later by the Tritium System Demolition and Disposal (TSDD) Project which was initiated in 2019. Many important lessons learned resulted from the activities implemented to safely complete these two projects.

INTRODUCTION

The Tokamak Fusion Test Reactor (TFTR) at the Princeton Plasma Physics Laboratory (PPPL) ran from 1982 through 1997. Up until 1993 only Deuterium-Deuterium (D-D) fusion experiments were conducted on TFTR. From 1993 through early 1997 TFTR performed Deuterium-Tritium (D-T) fusion experiments. During that time the supporting tritium systems sent approximately 100 grams of tritium (~1,000,000 Curies) to TFTR to support the D-T experimental campaign. After TFTR operations ceased in 1997, a long period of time was dedicated to activities aimed at lowering the tritium concentration in the TFTR vacuum vessel and supporting equipment. These activities included helium glow discharge cleaning, pulse discharge cleaning, vessel heating. And pump/purges with air.

In October 1999, the TFTR D&D Project initiated in 2000 several years after the completion of the D-T experimental campaign. The primary goal of the TFTR D&D Project was to remove contaminated and activated components from both the TFTR Test Cell and Test Cell Basement. The project was completed under budget and ahead of schedule in September of 2002 for an actual cost of 36.6M USD. More importantly the project was completed safely. Several novel approaches were incorporated into the project and serve as lessons learned for the fusion community.

The TSDD Project was initiated in 2019 with the mission of removing the legacy tritium system from PPPL. System removals began in early 2022 and were completed by spring of 2023. Many of the lessons learned from the TFTR D&D Project were incorporated into the TSDD Project.

MAJOR LESSONS LEARNED

Drum Bubbler for Removal of Tritium Oxide

PPPL developed and patented a drum bubbler system designed to provide long-term tritium removal in larger components such as the TFTR vacuum vessel and the Neutral Beam Box enclosures. This process was intended to reduce the quantity of radioactive waste generated during the project [1]. The Drum Bubbler system (Figure I-2.1) consists of “a 55-gallon drum containing water and an internal apparatus which aerates the water. The pump-purge gas process stream is introduced into the aeration process which removes the tritium oxide from the gas process stream. A vacuum pump located on a separate free-standing support cart provides the mechanism for process gas flow through the drum apparatus. The support cart also contains a refrigeration unit which condenses the moisture from the exhaust-gas stream and returns the condensate to the drum. This produces an exhaust stream containing less than 10% of the original tritium oxide. Overall, the Drum Bubbler system removes more than 90% of the tritium oxide from the process stream” [2].

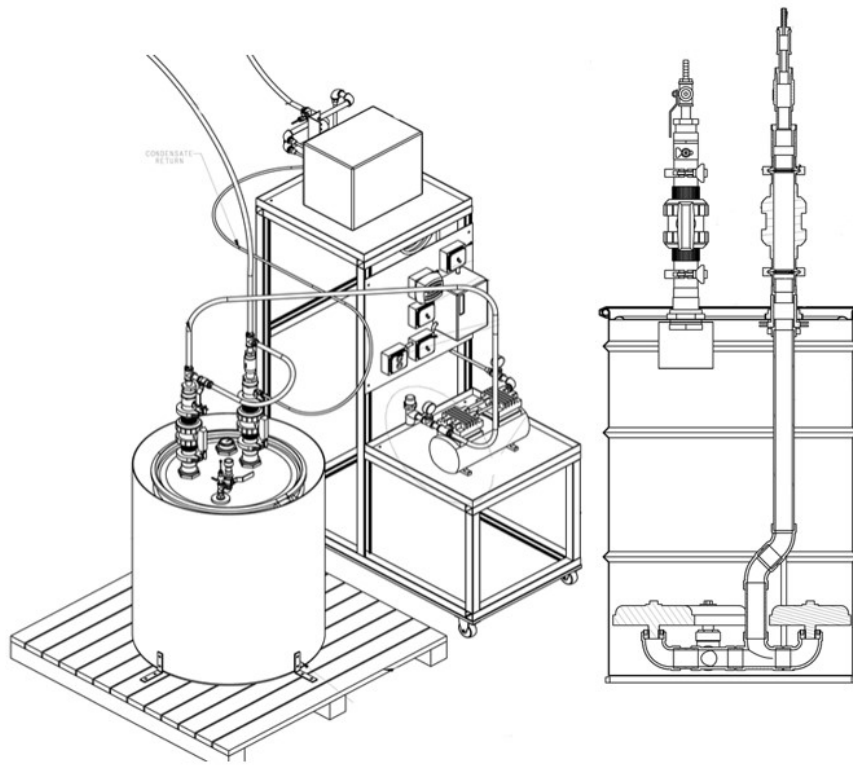


FIG. I-2.1 Drum Bubbler System.

Diamond Wire Cutting

The original TFTR D&D base plan was to use plasma torch cutting techniques to disassemble / segment the TFTR vacuum vessel; however, health concerns such as exposure to radiation, chromium, and metal particulate led the project to adopt diamond wire cutting technology instead where personnel could be at a distance to the cutting activities. At the time of the TFTR D&D Project diamond wire cutting was primarily used for concrete cutting and was still a novel approach for activities such as segmenting the TFTR vacuum vessel.

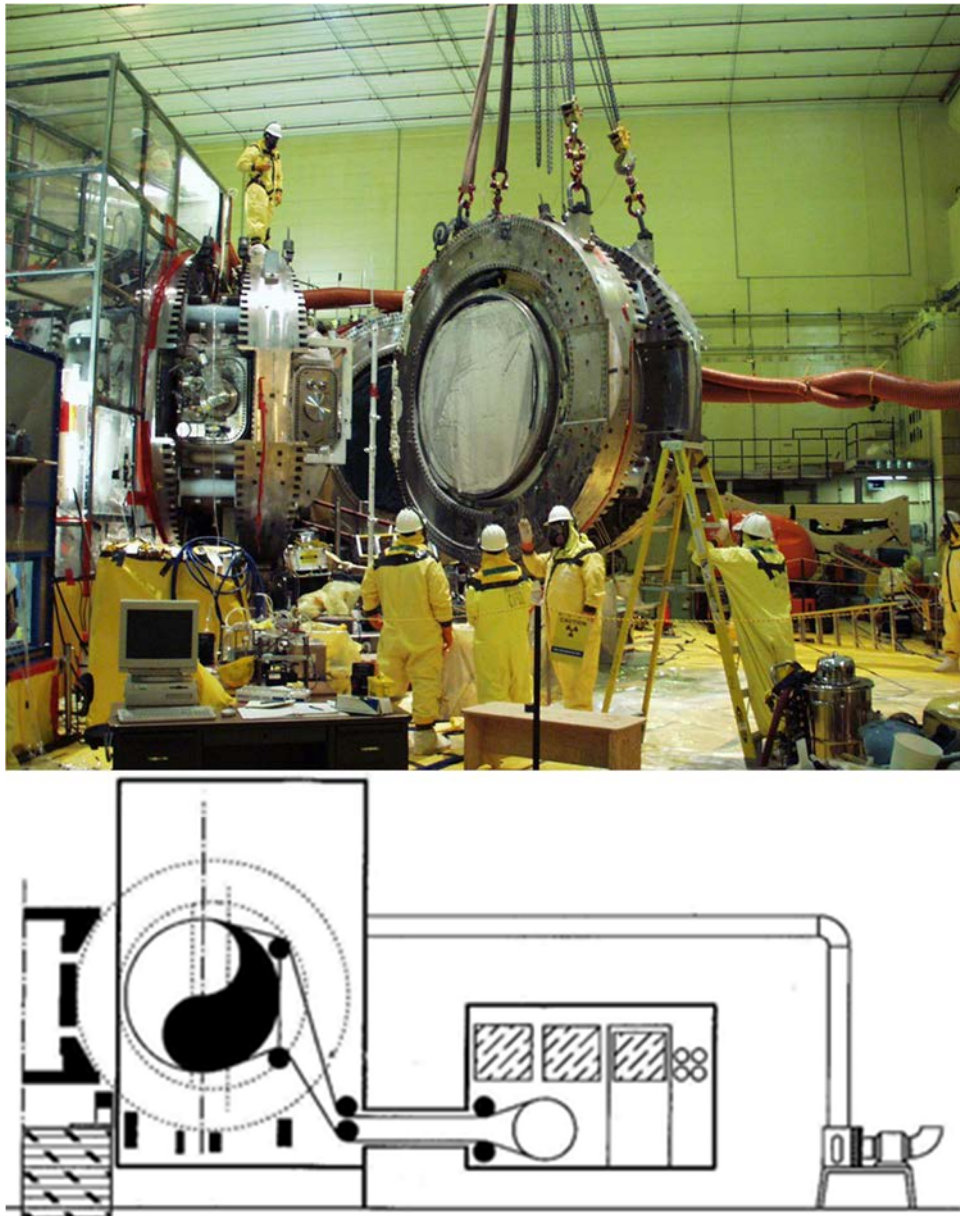


FIG. I-2.2 Vessel Segmenting with Diamond Wire Cutting.

Before the TFTR vacuum vessel was diamond wire cut the carbon tiles on the cut lines were removed, then it was filled with a lightweight concrete. The removal of the plasma facing tiles from the cut lines served to ensure highly tritiated carbon dust was not released when cutting a segment. The lightweight concrete fill ensured that the remaining tiles in the vessel were encapsulated and not exposed when vessel segments were cut and removed [3] (as indicated at Figure I-2.2).

Novel Tooling Approach

One of the more effective novel approaches used on both the TFTR D&D Project and the TSDD Project to remove piping with significant tritium contamination was crimping and cutting technology. Industrial crimpers were used to crimp piping in several locations on a segment of process piping then an industrial cutter, similar to what emergency responders use to extract people from motor vehicles after an accident, were used to cut the piping. The crimping served to virtually close the pipe before it was cut. This minimized the outgassing of tritium thereby minimizing personnel uptakes and releases of tritium to the environment [4].

Negative Pressure Exhaust Trunks

The Experimental Areas that contained and supported TFTR employed negative pressure zoning as an engineering control to ensure the areas with the highest likely tritium concentrations were maintained more negative in pressure with respect to the adjacent spaces. The HVAC exhaust systems that supported these negative pressure zones also had lines attached to the duct work which allowed for the connection of flexible exhaust ducts. The flexible exhaust ducts referred to as “elephant trunks” were used during breaks into tritium contaminated systems to ensure that any tritium outgassing that escaped from the piping being breached was exhausted through the system and to the exhaust stack after being analysed for tritium concentration by stack monitors. The employment of elephant trunks was instrumental in ensuring tritium uptakes to personnel were minimized.

SUMMARY

The TFTR D&D Project was safely and successfully completed in September of 2002 on schedule and under budget. Many important lessons learned can be gleaned from the novel techniques employed on the project. The TSDD Project safely completed the removal of the legacy tritium system in early 2023 and employed many of the same techniques that were developed during TFTR D&D ensuring that personnel uptakes of tritium and releases of tritium to the environment were minimized. The TSDD Project also completed the system removals on schedule and under budget.

ACKNOWLEDGEMENTS

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I-3. PRELIMINARY STUDY OF DECONTAMINATION OF STEEL AND CONCRETE FOR THE DECOMMISSIONING AND DISMANTLING OF EUROPEAN DEMO

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Abstract

This paper reports the preliminary study of decontamination and decommissioning techniques for steel and concrete of European DEMO, which have been investigated within the EUROfusion WPSAE project in cooperation with the industrial partner Framatome. The existing decontamination techniques in the nuclear industry have been assessed in order to perform the evaluation for DEMO applicability. A characterization plan is required to consider different characterization methods for activated / contaminated steel and concrete structures to determine the type of contaminants and the degree of contamination. The adopted methods are feasible to reduce the amount of nuclear waste during decommissioning. The existing detritiation techniques for different DEMO material types have been reviewed in the WPSAE. The detritiation techniques can be combined with the decontamination process either prior or as a subsequent step. New techniques specifically for DEMO will be indicated.

INTRODUCTION

The radioactive waste of a fusion power plant (FPP) have large quantities, huge amount of tritium, different radioisotopes compared to the nuclear power plant (NPP). Structural materials, such as steel (e.g. AISI 316L) for the vacuum vessel (VV), piping systems and ferritic martensitic steel (e.g. Eurofer) for the in-vessel components (IVCs, breeding blanket (BB) and divertor), are the dominant fusion waste. There is also a large quantity of concrete structures for the bioshield, containment and all rooms of different buildings (e.g. tokamak building, tritium building, and active maintenance facility). Regarding two design options of EU DEMO: the Helium Cooled Pebble Bed (HCPB) BB concept and the Water Cooled Lithium Lead (WCLL) BB concept, EUROFER mass of $3.69\text{E}+06$ kg and $4.19\text{E}+06$ kg respectively, stainless steel (SS) 316L mass of $2.17\text{E}+07$ kg are expected, and concrete mass of $2.50\text{E}+07$ kg [1]. The EU DEMO will produce a large quantity of radioactive metallic waste due to neutron activation and/or tritium contamination in operation. Handling and treatment of the metallic waste and concrete structures are mandatory for the decommissioning and dismantling of DEMO. Thus, the decontamination and decommissioning techniques for steel and concrete have been investigated within the EUROfusion WPSAE project. Existing techniques in the nuclear industry were first assessed in order to perform the evaluation for DEMO applicability.

ASSESSMENT OF EXISTING DECONTAMINATION TECHNIQUES

Decontamination techniques have been developed for conventional industry and adapted for the nuclear industry. Decontamination is essential in the decommissioning and dismantling phase. The main objectives of decontamination are [2]:

- to reduce radiation exposure of personnel and tools;

- to salvage equipment and materials;
- to reduce the volume of equipment and materials requiring storage and disposal in licensed disposal facilities;
- to restore the site and facility, or parts thereof, to an unconditional-use condition;
- to remove loose radioactive contaminants and fix the remaining contamination in place in preparation for protective storage or permanent disposal work activities; and
- to reduce the magnitude of the residual radioactive source in a protective storage mode for public health and safety reasons, to reduce the protective storage period or to minimise long-term monitoring and surveillance requirements.

Chemical and physical processes are used for decontamination. They are summarized in Table I-3.1. Chemical decontamination is primarily applied to clean metallic surfaces by removing contaminants such as fouling deposits and corrosion products like rust. In general, it is used in closed piping loops, where the chemical reagents are recirculated inside the piping system. Another application is the immersion of contaminated components inside a chemical decontamination bath, which is oftentimes accompanied by additional electrochemical or physical decontamination techniques such as an ultrasonic bath. In the nuclear industry several chemical processes / reactions have been established, differentiated by the used type of chemical agents [2]:

- **Chelation and Organic Acids:** binding of an organic chemical to a metal ion, enveloping the metal ion and bringing it into solution. The most common chelators used in decontamination are oxalic acid, citric acid, gluconic acid, ethylenediaminetetraacetic acid (EDTA), hydroxyethylenediaminetriacetic acid (HEDTA), Ethylenediaminedisuccinic acid (EDDS), Oxyethylidenediphosphonic acid (OEDPA), and Diethylenetriaminepentaacetic acid (DTPA);
- **Strong mineral acids and related material** that ionize nearly completely in aqueous solution: Hydrochloric acid (HCl), Nitric acid (HNO₃), Sulfuric acid (H₂SO₄) and Phosphoric acid (H₃PO₄);
- **Oxidizing and reducing (Redox) agents:** a metal oxide can be transformed into a different oxidation state, that might be more soluble with a redox reaction;
- **Electrochemical processes:** a chemical decontamination assisted by an electrical field.

Chemical decontamination is best suited for water carrying piping systems outside bioshield such as the WCLL and its primary heat transfer system (PHTS), PHTSs for the divertor (DIV) and vacuum vessel (VV), liquid waste treatment systems, and chemical and volume control system (CVCS) in DEMO.

Physical decontamination techniques can be divided into surface cleaning techniques and surface removal techniques, which are mechanical and manual decontamination. Surface cleaning techniques include brushing, wiping, flushing, vacuuming, ultrasonic (US) cleaning and strippable coatings – all techniques where the surface remains intact. In case the contamination has penetrated the surface through cracks or porous material, a diffusion technique is used, or at activation of the surface itself, a destructive surface removal technique is required:

Wet or dry abrasive blasting: different types of abrasive media are blasted onto the contaminated surface of a component or structure and remove the contamination or the top layer of the contaminated surface by abrasive forces with high decontamination factors. For dry abrasive systems, minerals (magnetite or sand) are blasted with a compressed air stream onto the contaminated surface [2]. In wet abrasive systems a high pressure water jet is used in combination with additional abrasive media and often also pressurized air. In general, wet and dry abrasive systems generate secondary waste which has to be collected and treated in a subsequent step.

Framatome has developed a specific process for decontamination of contaminated tubes with internal diameters of up to about 30 mm. The Mechanical Decontamination Process (SIVABLAST) has been developed primarily for steam generator tube decontamination with a closed circuit technique. This prevents blast media and dust from entering the adjoining areas. The recirculation blast unit (RBU) of

the blasting equipment can use a remotely controlled manipulator to connect each individual tube at the inlet side. At the outlet side, grit and scales are collected at a suction header or suction funnel and conveyed back to the blasting unit. The blasting grit is reused to minimize the generation of radioactive waste. A waste collection system is placed separately from the RBU and it is connected to the vacuum pump by hoses. A dust collector or waste drum is equipped with a filter which extract the dust from the air before it is exhausted to the station's ventilation system. The removed radioactive oxide scales will be collected in shielded waste drums. Considering possible airborne contamination, the waste collection system will be covered by a tent or cabin, which is connected to the power plant's exhaust air system. The whole process is controlled remotely.

- **Scarifying techniques** to remove the top layers of coated or uncoated concrete and steel surfaces down to the depth of uncontaminated surfaces, such as grinding / planning of surfaces, scaling, scabbling, milling / shaving. All scarifying techniques produce generally a large amount of dust.
- **Removal of concrete by drilling and spalling, jackhammer, (diamond wire) sawing or scarifying.** Explosives are not recommended due to potential release of tritium.
- **Further physical techniques** are microwaves, laser ablation, electro-hydraulic scabbling, bio-decontamination, electrokinetic decontamination, melting, plasma melting.
- **Combined physical / chemical processes** with foams, gels and pastes, and US cleaning. Hybrid technologies combining physical processes with one or more of the previously described chemical, electrochemical, biological, mechanical, laser or sonic methods are emerging technologies.

Most decontamination techniques are developed as manual techniques, which have been proven in the past that they are sufficient to adhere to the “as low as reasonably achievable” (ALARA) principle. Subsequently less effort has been made to develop remote-controlled or automatized versions of the mentioned techniques. Also the wide range of differing surfaces, materials and components appeared during the dismantling of a NPP leads to low degree of remote-handling and automatization of decontamination techniques. By far the larger volume of structures and components to be decontaminated and dismantled have a relatively low dose rate which allows manual operation either directly or with suitable shielding such as lead mats or screening walls. Highly activated or contaminated structures or components that prohibit manual techniques are generally of smaller volumes and dimensions that are dealt by automatized process regarding cost-effectiveness. Remote-controlled demolishing machines with robotic appendages, as well as other robotic applications following a Master-Slave-Control are available as well.

A matrix for the above decontamination techniques has been developed (Table I-3.1.) in the context of DEMO application and contains an evaluation with criteria such as handling, destructive, surface, material, blanket type, detritiation in combination, optimum recycling step, safety, nuclear waste minimisation, secondary waste, and advantages / disadvantages – see Table I-3.1. The existing detritiation techniques have been reviewed in [3]. The techniques such as thermal treatment at different temperature levels and pressure, laser treatment, melting, cold crucible, molten salt oxidation, surface abrasion, interim storage, evaporating and solidification line, and incineration are appropriate for short-/long-term reuse, recycling and disposal.

TABLE I-3.1. DECONTAMINATION CATEGORY AND EVALUATION MATRIX

Category	Process	Hand-ling	Destr-uctive	Sur-face	Eu-rofer	SS-316 L	Heavy-concrete	Ordinary-concrete	W-CL	H-CP	Detritiation	Optimum recycling step	Safety	Efficiency	Nuclear waste minimisation	2 nd Waste	Advantages [6]	Disadvantages [6]
Chemical	See above	Remote	no	Piping systems (internal), large areas	x	x	/	/	x	/	/	Direct reuse of system / component	o	o	-	Water, Chemicals, Reaction Products, IX resins	- “In situ” on closed systems - Reuse of components and systems - May remove contamination from internal and hidden surfaces - Minor problems regarding airborne cross-contamination	- Chemicals generate additional 2nd waste - Ineffective on porous surfaces - Treat corrosive and toxic reagents - Additional heating to 70-90 °C
Physical	Wet Abrasive Blasting	Manual, remote, automated	yes	Large areas	x	x	/	x	x	x	Surface abrasion or subsequent detritiation	Reuse of abrasive medium, disposal of debris	o	o	o	Water, Abrasive particles, Filter cartridges	- Low guiding & reset forces - Individual cutting forms - No vibrations	- Demand of water - Aerosols emissions - Very large amount of 2nd waste - Need of water/sludge treatment installation

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																		- Risks of cross-contamination
Physical	Dry Abrasive Blasting	Manual, remote, automated	yes	Large areas	x	x	/	x	x	x	Surface abrasion or subsequent detritiation	Reuse of abrasive medium, disposal of debris	o	o	o	Dust, Abrasive particles, Filter Cartridges	- High to very high yield - High performance - Highly versatile technique - hard to reach surfaces (air powered configuration) - Collection of dust and debris by default - Insensitive for metal inserts - rough surfaces - Low abrasive cost - Continuous recycling of abrasives	- 2nd waste - Risk of cross-contamination (abrasives recycling) - High personal safety requirements - High dust formation - Deep abrasion produces rather rough surface finish
Physical	Dry Abrasive Blasting	Remote	yes	Piping systems (internal),	x	x	/	/	x	/	/	Direct reuse of system /	+	+	o	Dust, Abrasive particles, Filter	- “In situ” on closed systems - Remote control	- 2nd waste - Risk of cross-contamination (abrasives recycling)

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
	(SIVABLA ST)			large areas								compon ent				Cartridg es	- Reuse of components and systems - Low 2 nd waste	- High dust formation - Not applied to tubes with large internal diameter
Physical	Kelly Process	Manual	yes	Large areas	x	x	/	x	x	x	Surface abrasion or subsequ ent detritiati on	/	o	+	o	Water, Filter Cartridg es	- Low 2 nd waste - Low guiding & reset forces - Individual cutting forms - No vibrations - Easy to use - Washed surfaces dry quickly - Good for large flat surfaces	- Demand of water - 2 nd waste - Need of water/sludge treatment installation - Higher energy consumption to generate pressurized superheated water - Not good for irregular surfaces - Not good for grease - Poor ergonomic design

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Carbon Dioxide Blasting	Manual, remote, automati zed	yes	Large areas	x	x	/	x	x	x	Surface abrasion or subsequ ent detritiati on	/	-	o	-	Dust, Filter Cartridg es	- Remove smearable contamination - Preserve substrate - No 2 nd waste (except HEPA filter)	- Not aggressive enough to strip coating or fixed contamination - High safety requirements - Risk of anoxia - Ventilation requirements
Physical	Ice Water Blasting	Manual, remote, automati zed	yes	Large areas	X	x	/	x	x	x	Surface abrasion or subsequ ent detritiati on	/	-	+	-	Water, Filter Cartridg es	- Remove smearable contamination - Preserve substrate	- Not aggressive enough to strip coating or fixed contamination - High safety requirements - Ventilation requirements

Category	Process	Hand-ling	Destr-uctive	Sur-face	Eu-rofer	SS 316 L	Heavy concrete	Ordinary concrete	W CL	H CP	Detritiation	Optimum recycling step	Safety	Efficiency	Nuclear waste minimisation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Liquid Nitrogen Blasting	Manual, remote, automated	yes	Large areas	x	x	/	x	x	x	Surface abrasion or subsequent detritiation	/	-	+	o	Dust, Filter Cartridges	- Versatile (coating stripping, removal of thick concrete layer, cutting) - High yield - Suitable for hard to reach areas - Suitable for rough surfaces - Insensitive for metal inserts - No 2nd waste (except HEPA filter)	- High safety requirements (personal & facility) - Ventilation requirements - High investment cost - Risk of anoxia - Complex technology - Process components implantation
Physical	Grinding (hand-held)	Manual (hand-held)	yes	Small area (thin layers), difficult to access	x	x	/	x	x	x	Surface abrasion or subsequent detritiation	Debris: disposal or plasma melt	-	-	-	Dust, Filter Cartridges	- Very good finishing - High yield - High performance - Moderate weight - Low vibration level - Collection of dust and debris by default	- Fine dust - Not suitable for rough surfaces - Sensitive to metal inserts - High consumable cost (disks)

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rofer	SS 316 L	Heavy concrete	Ordin ary concrete	W CL L	H CP B	Detriti- ation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																	- No 2 nd waste (except HEPA filter)	
Physical	Scaling (hand-held)	Manual (hand- held)	yes	Small area (thin layers) , difficu lt to access	/	/	/	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	-	-	-	/	- Flexible handling - hard to reach areas - Insensitive for metal inserts - Lightweight tool - No 2 nd waste	- High vibration level - Low yield (limited surface area coverage)
Physical	Scabbling	Manual (hand- held)	yes	Small area (thin layers) , difficu lt to access	/	/	/	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	-	-	o	/	- Flexible handling - Suitable for hard to reach areas - Lightweight tool - No 2 nd waste	- High vibration level - Rough finishing - Low yield (limited surface area coverage)
Physical	Scabbling	Remote, automati zed	yes	Large areas (thin layers)	/	/	/	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	o	-	+	Dust, Filter Cartridg es	- Large surface areas - Medium to high yield - No 2 nd waste (except HEPA filter)	- Require tailored interface with heavy duty - Carrier (vertical surfaces) - High vibration level

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																		- Rough finishing
Physical	Milling/Shaving (manual)	Manual (hand-held)	yes	Large areas	/	/	x	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	-	-	o	Dust, Water, Filter Cartridges	- Very good finishing - High yield - High performance - Moderate weight - Low vibration level - Collection of dust and debris by default	- Fine dust - Not for rough surfaces - Sensitive to metal inserts - High consumable cost (discs)
Physical	Milling/Shaving (remote/automatized)	Remote	yes	Large areas	/	/	x	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	o	-	+	Dust, Water, Filter Cartridges	- High yield - High performance - Collection of dust and debris by default - Insensitive for metal inserts– Moderate vibration level - Low consumable cost	- Require tailored interface with heavy duty carrier or engineered guiding system (vertical surfaces) - Fine dust

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rofer	SS 316 L	Heavy concrete	Ordin ary concrete	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Drilling Spalling	/ Manual, remote, automati zed	yes	Large areas (walls)	/	/	x	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	-	o	o	Dust, Filter Cartridg es	- No dust/slurry generation (besides drilling operations) - Applicable in hard to reach areas - Ease further hammering operations - Simple to use	- Pre-treatment - Need for further handling (hammering) - Reinforcemen t mostly requires additional cutting technique - Control of cracks spreading - Hazards related to damage to load-bearing structures
Physical	Jackhammer (manual)	Manual (hand-held)	yes	Large areas	/	/	x	x	x	x	Subsequ ent	Debris: disposal or plasma melt	-	o	o	Dust, Filter Cartridg es	- High yield - Very reliable - Insensitive to surface state - Insensitive to metal inserts	- Heavy equipment - dust generation - High vibration level - Reinforcemen t requires additional cutting technique - additional treatment(s)

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																		to reach adequate surface - finishing
Physical	Jackhammer (remote)	Remote	yes	Large areas	/	/	x	x	x	x	Subsequ ent	Debris: disposal or plasma melt	o	-	o	Dust, Filter Cartridg es	- High yield - Very reliable - Insensitive to surface state - Insensitive to metal inserts	- Dust generation - Reinforcemen t requires additional cutting technique - Additional treatment(s) to reach adequate surface - Finishing
Physical	Circular Sawing	Manual, remote, automati zed	yes	Large areas	/	/	x	x	x	x	Subsequ ent	Debris: disposal or plasma melt	-	o	o	Dust, Water, Filter Cartridg es	- No vibration - Precise cuts - Flush cutting - operated in dry conditions (at reduced cutting rate)	- Limited cutting depth - Hazards related to locking - Generation of dust/slurry - Elaborate installation - Risk of cross- contamination of hidden

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																		surface (wet conditions)
Physical	Diamond Wire Sawing (manual/remote)	Manual, remote	yes	Large areas (walls)	/	/	x	x	x	x	Subsequent	Debris: disposal or plasma melt	-	+	o	Dust, Water, Filter Cartridges	- Highly versatile technique - No limit on structure size - No vibration - Accurate cuts - operated in dry conditions	- Hazards related to rupture and locking of the wire - Dust/slurry generation - Risk of cross-contamination of hidden surface (wet conditions)
Physical	Microwave scabbling	Manual, remote	yes	Large areas	/	/	/	x	x	x	Surface abrasion or subsequent detritiation	Debris: disposal or plasma melt	-	-	o	/	- no introduction of media (water) - no equipment with wear parts - no 2nd waste	- Poor efficiency - Currently incompatible with constraints and safety requirements of a decommissioning project

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Laser Ablation	Remote, automati zed	yes	Large areas (flat surfac es), coatin g	/	/	/	x	x	x	Laser treatmen t or subsequ ent detritiati on	Disposal	+	+	o	Dust, Filter Cartridg es	- Low safety requirements - Low weight tool - Automation through low cost carrier - large surfaces - rough surfaces - Insensitive for metal inserts - Selective removal of coating - No 2nd waste (except HEPA filter)	- Low yield (very small surface area coverage) - No feedback yet available on long-term reliability and maintenance requirements
Physical	Electro- Hydraulic scabbling	Manual, remote	yes	Large areas	/	/	/	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	o	+	+	Dust, (Water), Filter Cartridg es	- Generate less 2nd waste than other technologies using water - Very efficient - Removes deep contamination	- Require a skilled operator - 2nd liquid waste - Information on performance data is scarce

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Electro kinetic decontamin ation	Manual, remote	yes	Large areas	/	/	/	x	x	x	Surface abrasion or subsequ ent detritiati on	Debris: disposal or plasma melt	o	o	+	Water, Chemic als, Reactio n Products , IX resins	- Great reductions in the amounts of 2nd waste generated - Minimal airborne releases - Minimal exposure of workers to radiation - Minimal damage to the concrete surface and reuse - Minimize the volume of 2nd waste - Achieve a solid waste form	- Require recycling of electrolyte pads and solulizer solutions to minimize 2nd waste - Require porose concrete structures - Not for coated structures; require previous removal of coating

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detriti- ation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Physical	Melting	Manual, remote, automati zed	yes	Deco mmiss ioned materi al	x	x	/	/	x	x	Melting and other Thermal treatmen t detritiati on techniqu es	as describe d in [4] and [5].	o	+	+	slag, filter dust	- Redistribution of radionuclides among ingots, slag and filter dust - Components with complex geometries with inaccessible surfaces - Homogenisati on of remaining radionuclides over ingot	- Complete destruction of the component - High energy consumption - Require high degree of automatizatio n for melting plant
Physical	Plasma Melting	Manual, remote, automati zed	yes	Deco mmiss ioned materi al	/	/	x	x	x	x	Thermal treatmen t detritiati on techniqu es	Plasma melt	o	+	+	slag, filter dust	- Redistribution of radionuclides among solidified plasma, slag and filter dust - Suitable as further treatment method of concrete debris - Homogenisati on of remaining	- Complete destruction of the component - High energy consumption - Require high degree of automatizatio n for melting plant

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
																	radionuclides over solidified plasma - Solidified plasma directly suitable for nuclear waste disposal	
Biologic al	Bio- Decontamin ation	Manual, remote	yes	Large areas	/	/	/	x	x	x	tritium can be a negative factor on the biologic al process	Biologic al waste treatmen t and disposal	+	o	-	Water, Chemic als, Biologic al Waste	- Depth of removal of surface can be controlled - “Hands-off” approach improving safety - Inexpensive - Elimination of airborne contamination	- Long time period to effect the decontaminati on - Constant monitoring of the slow process - Bacterial growth may be inhibited by surface components; detection of this effect is difficult

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Combin ed	Foams	Manual, remote	no	Large areas	x	x	/	/	x	x	/	Direct reuse of system / compon ent	o	+	-	Water, Chemic als, Reactio n Products	- Enables reuse of components and system - May remove contamination from internal and hidden surfaces - Minor problems regarding airborne cross-contamination	- Chemicals generate additional 2nd waste - Ineffective on porous surfaces - Treat corrosive and toxic reagents - Additional heating to 70-90 °C
Combin ed	Gels	manual, remote	no	Large areas	x	x	/	/	x	x	/	Direct reuse of system / compon ent	o	+	-	Water, Chemic als, Reactio n Products	- Enables reuse of components and systems - May remove contamination from internal and hidden surfaces - Minor problems regarding airborne cross-contamination	- Chemicals generate additional 2nd waste - Ineffective on porous surfaces - Treat corrosive and toxic reagents - Additional heating to 70-90 °C required

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Combin ed	Pastes	Manual, remote	no	Large areas	x	x	/	/	x	x	/	Direct reuse of system / compon ent	o	+	-	Water, Chemic als, Reactio n Products	- Enables reuse of components and systems - May remove contamination from internal and hidden surfaces - Minor problems regarding airborne cross-contamination	- Chemicals generate additional 2nd waste - Ineffective on porous surfaces - Treat corrosive and toxic reagents - Additional heating to 70-90 °C required
Combin ed	Ultrasonic Cleaning	Manual, remote, automati zed	no	Comp onents (size depen dent of the US- bath)	x	x	/	/	x	x	/	Direct reuse of system / compon ent	o	o	o	Water, Chemic als, Reactio n Products	- Established equipment - automatized cleaning process - good cleaning & decontamination results	- Component size boundaries (volume of the US-bath) - 2nd waste - Risk of cross-contamination

Category	Process	Hand- ling	Destr uctive	Sur- face	Eu rof er	SS 316 L	Heavy concre te	Ordin ary concre te	W CL L	H CP B	Detri- tiation	Optimu m recycling step	Saf ety	Effici ency	Nuclear waste minimi- sation	2 nd Waste	Advantages [6]	Disadvantages [6]
Combin ed	Electroche mical decontamin ation	Remote	no	Comp onents (size depend ent of the immer sion bath)	x	x	/	/	x	x	/	Direct reuse of system / compon ent	o	o	o	Water, Chemic als, Reactio n Products	- Enables reuse of components and systems - May remove contamination from internal and hidden surfaces - Minor problems regarding airborne cross-contamination - Used in combination with chemical decontamination to improve decont. effect	- Component size boundaries (volume of the immersion bath) - Electrolytes and water generate 2nd waste - Risk of cross-contamination - Require conducting surfaces - Inapplicable for complex geometries (pipes, valves, pumps)

Remarks:

- x: item can be treated with the selected process /: not applied 2nd waste: secondary waste
- Destructive: the process is considered as destructive / non-destructive.
- Safety: “+” safe process, “o” neutral process, “-” additional safety measures are required
- Efficiency: “+” high decontamination factor (DF), “o” average DF, “-” low DF
- Nuclear waste minimisation: “+” low volume, “o” average volume, “-” high volume
- IX: ion exchange

REQUIREMENT FOR A CHARACTERIZATION PLAN

Based on the DEMO design of Baseline 2017, four main types of contamination are available for steel components and structures and concrete structures:

- Structural activation of steel components and structures incl. permeation of tritium;
- Structural activation of concrete structures incl. permeation of tritium;
- Spread of loose contamination in steel components and piping systems through contaminated medium, e.g. activated corrosion products (ACPs), tungsten dust, tritium;
- Spread of loose contamination to concrete structures through leakages.

All of these contamination types are found in NPPs as well. Since fusion fuels are not solid, a high degree of contamination and distribution of radioactive components in the coolant due to defective fuel elements in a NPP are excluded for DEMO. For spread of contaminants in non-aqueous liquid solutions, the PbLi-loop for the WCLL blanket concept is designed to reduce the spreading of ACP and other impurities in the PbLi-Purification system.

Following the identified contamination types, a characterization plan can be developed, which is required as a part of facility decommissioning planning to define the nature, extent and location of contaminants, determine sampling locations and protocols, determine quality assurance objectives for characterization, and define documentation requirements. Following issues are considered [6]:

- “Determination of the areas and depth of radioactive and/or chemical contamination in concrete;
- The scope and volume of the materials to be managed by the routes established, depending on the initial values and the work and resources to be dedicated in attempts to apply other routes, e.g. if it is worthwhile to decontaminate a large volume of material to arrive at clearance levels;
- The scope and volume of material to be decontaminated or remediated, if this route is chosen for management prior to clearance;
- The scope and volume of radioactive material to be handled for final management as radioactive waste;
- Insight into the isotopes present and the possibility of assessing the source term,
- Evaluation of the key vector nuclides and acquisition of scaling factors for difficult to measure radionuclides compared to those easy to measure (gamma isotopes);
- Establishing the conditions required for adequate radiological protection in order to be able to carry out the works safely and using adequate confinement resources to prevent the dispersal of waste material; and
- Possibility of determining whether there have been migrations of contaminants towards neighbouring land, this being especially important in the case of groundwater and below the sub-soil.”
- Forecast of radioactive decay.

A characterisation study for activated concrete will consider the following aspects [6]:

- “Determination of areas irradiated with neutron radiation: mainly reactor vicinity;
- Analysis of building plans regarding non-homogeneous concrete (different types of concrete) and concrete structures (e.g. cable or pipeline penetrations or other potential locations for activity deposition);
- Analysis of operational effects influencing the apparent activity of the concrete (ingress of contaminated water or contaminated air): if there are indications that the activated concrete is also internally contaminated, the demolition of the concrete structure needs to be planned and performed differently as described below;
- Determination of project and construction data of the concrete structure, origin of the concrete and its additives;

- Analysis and reassessment of historic retained samples (e.g. test samples for concrete strength);
- Investigation of the chemical composition of recovered samples. The densities of all elements, contained in the concrete will be needed for calculations (analysis of the material from non-activated areas may be used without additional radiation protection effect);
- The radionuclides important for the demolition are to be specified on the basis of the chemical composition;
- Quantitative analysis of the activity composition of in-vessel samples for the purpose of determining a nuclide vector (or – if necessary – several nuclide vectors) for the mineral contents (including possibly contained metallic additives) and for the reinforcement steel. Therefore, the activities of the γ as well as the β -emitters needs to be determined;
- The use of scaling factors to determine hard-to-measure radionuclide activities for possible discrepancies in trace elements and impurities relative concentrations in the bulk material. Different concentrations distribution will result in isotope specific in depth activity distribution profiles.”

Generation of the characterization plan enables the determination of the necessity and applicability of decontamination (determination of nature and extent of contamination), type of decontamination method to be used and decontamination effectiveness after performance.

The main irradiation and contamination measurements are detailed in [7]. dose rate measurements, loose contamination measurements, surface counting, spectrometry measurements from which gamma spectrometry is the one most commonly used for concrete in situ measurement. Remote controlled or even fully automatized robots equipped with several kinds of measurements are the current research trend, e.g. gamma spectroscopy with video feed / 3D camera, laser scanning and position determination in combination with a layout drawing of the plant. This allows for very comprehensive information on the degree and type of contamination present in one room. Despite the in-situ measurements physical sampling and destructive assay remain necessary at the very early stages of the decommissioning programme [6]:

- “To list all radionuclides of significant importance for the decommissioning project;
- To establish scaling factors for hard-to-measure radionuclides;
- To establish radioactive waste inventories and define adequate waste evacuation routes;
- To perform environmental impact assessment studies;
- To discriminate isotopes migrating in the concrete and define contamination tracer(s);
- To evaluate contamination depth;
- To determine chemical composition of activated structures including trace elements with high activation cross section;
- To assess activation calculation (distribution profiles).”

Not only the most abundant radionuclides shortly after shutdown need to be measured, but also long lived radionuclides have to be determined, which might have a strong impact on final waste categorisation regarding their concentration.

Decommissioning and dismantling projects of NPPs have shown that information regarding the initial material composition of the used concrete, the activation of concrete and the spreading of contamination is not available at all times. Specifically in case of spilling of radioactive contaminated liquids that could lead to contamination ingress into small cracks in the concrete. Thus, a thorough and well documented collection of events is required. It is therefore highly recommended to establish a data collecting and documentation system during DEMO plant operation to record similar events in order to get data access for the dismantling phase. Additionally, detailed material specification and samples of the used material should be documented during the construction phase of DEMO.

At the current stage, a concrete characterization plan cannot be created for DEMO due to missing operational data and materials samples.

TECHNOLOGY GAPS AND NEW DEVELOPMENTS FOR DEMO

Based on the above study, technology gaps are identified for DEMO and new development is proposed:

- PbLi loop: the distribution and spreading of the ACPs in Pb-17Li of the PbLi loop should be analysed. After draining PbLi the loop should be cleaned and decontaminated to remove solidified PbLi, any loose contamination of ACPs, and reinstate the protective Al₂O₃ coating as recommended in [8] A corresponding chemical decontamination process shall be developed for the PbLi process;
- Tokamak dust: dust collection, treatment and disposal, as well as subsequent decontamination activities shall be investigated;
- Combined decontamination and detritiation: detritiation will have to be performed either prior to decontamination or subsequently. The latter and more likely scenario requires additional safety precautions due to the presence of tritium inside the structures to be decontaminated. It could therefore be a solution to once more analyse the detritiation matrix from [3] for compatibility with the mentioned decontamination techniques. New decontamination techniques e.g. multistage chemical decontamination with an additional detritiation step shall be developed;
- Prohibition of contamination: The best practice to increase decontamination factor is to prohibit contamination in the first place. Such areas and processes leading to contamination should be reassessed regarding the possibility to reduce the risk of contamination. For example, the cutting of the BB piping connection to the PHTS during its replacement shall be performed in such a way that no airborne contamination (sawing dust or aerosols from laser cutting) can occur. Otherwise, the VV has to be decontaminated before restarting operation of the FPP.

SUMMARY

The existing techniques in the nuclear industry have been assessed and a wide variety of decontamination techniques is available. Different characterization methods for activated / contaminated steel and concrete structures to determine the type of contaminants and the degree of contamination have been discussed. The adopted methods are feasible to reduce the amount of nuclear waste during decommissioning. The detritiation techniques can be combined with the decontamination either prior or as a subsequent step. New techniques specifically for DEMO have been indicated.

At the current stage, it is not possible to determine the best available practice for decontamination of steel and concrete structures within the DEMO project. The final characterization plan can only be created when operational data and materials samples of the DEMO/FPP (Fusion Power Plant) will be available. Because of the different types of fuel and neutron source between a NPP and a FPP, the decontamination paths of NPP are not directly transferrable to the FPP. For the WCLL and its PHTS, the DIV- and VV-PHTS with water coolant, the extensive experience from the light water reactor (LWR) in terms of contamination of piping systems through distribution of diluted activated corrosion products in cooling water is considerable for DEMO. Dust for the WCLL and HCPB concepts, and ACPs in PbLi loop have to be analysed and assessed.

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I-4. FUSION RADWASTE MANAGEMENT: WHERE WE ARE AND WHERE SHALL WE GO? APPLYING IEA'S FUSION WASTE MANAGEMENT STRATEGY TO THE ARC REACTOR

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Abstract

The management procedures for radioactive materials following the changeout of replaceable components and decommissioning Fusion Power Plants is “the back-end” of the fusion materials cycle. That approach followed routes leading to the minimization of waste, eventually disposed of as Low-Level Waste with no production of High-Level Waste, recycling of materials, and clearance. As business case, it is presented the waste management study, in the frame of the safety analysis done for the Affordable Robust Compact (ARC) reactor. ARC is a D-T Tokamak concept proposed by MIT (Massachusetts Institute of Technology). Since, the presence of highly energetic neutrons and the expected materials activation will be main concerns for safety studies. Neutronic transport and activation analyses have shown the inadequacy of Inconel-718 as VV inner and outer walls material aiming at clearance (CI as per NRC NUREG-1640 and/or Shallow Land Burials SLBs for LLW (reference limits to comply with Waste Disposal Rating <1). Other materials analyzed provided a better performance in respecting low activation criteria and V-based alloys were finally selected.

INTRODUCTION

The International Energy Agency (IEA) has started a global study on the waste produced by fusion energy. This study aims to look at the last part of the process that uses fusion materials. It's important for making sure fusion energy has the best possible positive effects on the environment. The activity began in the late 1990s and has involved experts from the EU, US, Russia, Japan, China, and South Korea [1-3].

The study looked at how to handle radioactive materials after replacing parts and shutting down fusion facilities. We call this the “back-end” of the fusion materials process. It covers all the steps needed to handle used radioactive materials from fusion facilities. This includes taking out parts from the device, reusing them through recycling or clearance, or getting rid of the waste in nearby sites as Low-Level Waste.

Over the years, the most important parameters that govern the back-end of the fusion materials cycle have been examined and quantified. A new fusion specific approach for the entire back-end cycle of fusion materials turns out to be necessary. The approach followed in the IEA study is comprehensive: it considers the evacuation routes for the waste and materials, the handling difficulties, as well as the critical issues and challenges facing all three main back-end solutions: recycling, clearance, and disposal. This plan needs to be improved, approved by national authorities, and supported by a special research and development program to tackle the important problems we found. It still allows for a full look at most of the factors involved in this complicated materials management system. It also helps to look into and compare different designs and materials to see how they affect the environment.

BACK-END OF THE ARC REACTOR MATERIALS CYCLE

Recent advancements in areas like High Temperature Superconductors (HTS), 3D printing, new testing methods, and new materials have created new possibilities and led to a second generation of designs for Nuclear Fusion Reactors. At MIT, researchers are working on a new type of fusion reactor called the Affordable Robust Compact (ARC) Fusion Reactor. They are doing this at the Plasma Science and Fusion Center (PSFC) [4-5].

It is a tokamak conceptual design proposed by Massachusetts Institute of Technology (MIT), the flagship of a new generation of fusion device – see Figures I-4.1 and I-4.2. A cheaper, smaller but even more powerful, faster way to achieve fusion energy. The goal of this conceptual design is to minimize the reactor size in order to reduce the cost of the plant. This will have as a benefit the reduction of the amount of potential radioactive waste. It has some new features, like using high-temperature superconductors. These allow for stronger magnetic fields and help make the tokamak smaller because of the way magnetic fields and size relate to each other. It also has a liquid blanket, which serves to cool the system and create more fuel. Another interesting thing is the design of a one-piece vacuum container, which can be easily replaced after a while. This is possible because of the high-temperature superconductors that can be separated, allowing the top part of the reactor to be taken off and the vacuum container to be swapped out [4].

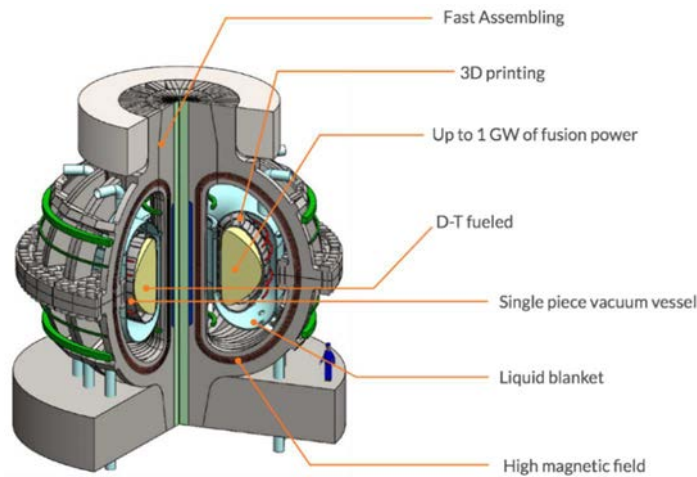


FIG. I-4.1. Scheme and main characteristics of the ARC reactor.

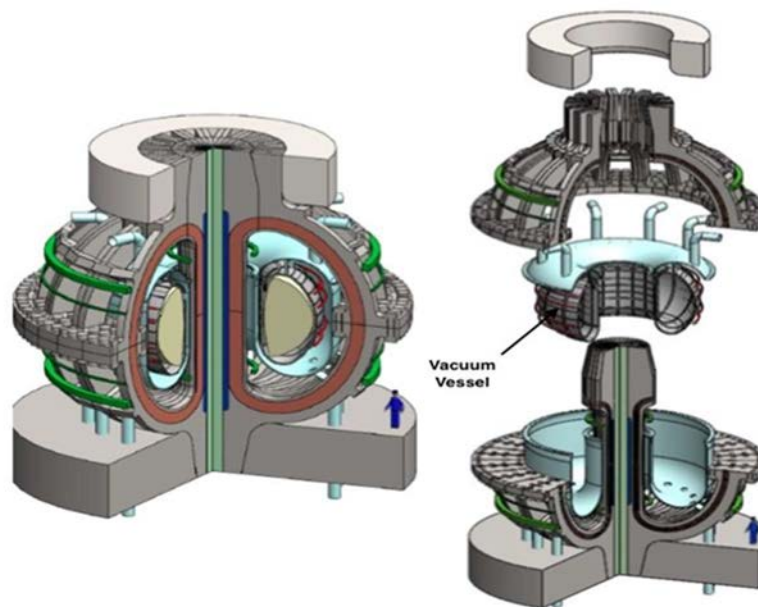


FIG. I-4.2. ARC: The demountable reactor.

The main purpose of this part is to analyze materials for the vacuum vessel that will be built as one piece. This vessel sits between the first wall and the tank that holds the breeding blanket made of Fluoride Lithium Beryllium molten salt (FLiBe). Because the vacuum vessel is located very close to the plasma chamber where plasma is created, it gets hit by a lot of neutrons. This can make the materials

inside it active and shorten the life of the component, leading to issues when it is time to remove it. The peculiarity of this technical solution is emphasised. In tokamak projects such as ITER or DEMO, radially, the sequence is plasma, FW, breeder blanket and Vacuum Vessel (VV). In the ARC design, the VV and breeder blanket are reversed along the radial axis.

The tools used for the low-activation analysis are the Monte Carlo N-Particle Transport code (MCNP) and FISPACT-II [6-10]. The first method looks at how many neutrons are in different parts, while the second method helps us measure how radioactive something is, how much radiation it gives off when touched, and how much heat it produces over time after the plant has stopped working. To pick the “best” material for the vacuum vessel, we looked at both its strength and how safe it is for the environment. Then, we compared the options. A chance that has been assessed before changing materials means using Isotopic Tailoring and Elemental substitution methods. This involves looking at the makeup of each alloy and removing or reducing isotopes or elements that cause issues for low activation. After making changes, new tests are run to see if those methods really improve the final results.

In this research, the neutron flux (actually, the incoming neutron current) was kept the same for all materials to ensure equal conditions for the analysis. The flow was calculated using the original materials of the ARC reactor as inputs for MCNP: pure FLiBe as liquid blanket and Inconel-718 for the vacuum-vessel. The final results for the neutron fluxes are $7.54\text{E}+014 \frac{\text{neutrons}}{\text{cm}^2 \cdot \text{s}}$ for the inner region of vacuum vessel, which is close to the plasma, and $5.02\text{E}+014 \frac{\text{neutrons}}{\text{cm}^2 \cdot \text{s}}$ for the region near the tank that contains the breeding blanket.

The expected lifespan of the vacuum vessel is about 2 years, so to make things realistic, we used a pulsed mode for the time. ARC is meant to have long pulses that can last at least 6 months. The main aim of the design is to reach a stable state. After this step, we can do the activation analysis using FISPACT-II.

The first material chosen for the initial design is Inconel-718. It has strong mechanical properties at high temperatures, with a yield stress close to 1000 MPa at 900K [5], which is the temperature planned for operation [4]. The main problem with that alloy has to do with its main part, which is nickel. The neutron flux created by the D-T reaction activates the material of the vacuum vessel. This is important for producing tritium and cannot be avoided. The only option is to choose specific alloys that help reduce this issue. Before we begin explaining the activation analysis, let's look at the criteria for low activation that were considered for this study.

To decide if a material can be cleaned, reused, or permanently discarded, we considered these points:

Recycling in the nuclear industry [3]: The important number is $10 \left[\frac{\text{microSv}}{\text{h}} \right]$. This is the maximum amount that the radiation level of materials may be so that workers, who have been exposed to radiation, can safely handle them without exceeding the annual limit of 20 mSv of radiation exposure for workers [13].

Recycling outside the nuclear industry: the value is $0.1 \left[\frac{\text{microSv}}{\text{h}} \right]$, that is the limit value for public exposure (1 mSv/y for 8760 h/y) [11].

Clearance Index (CI), which is an index that permits to classify the material as non-radioactive, above natural level, and defined by the Law [3]:

$$CI = \sum_{i=1}^A \frac{i}{L_i}$$

A_i and L_i refer to the specific activity and the clearance limit for the i -th nuclide in the material, and they are measured in $[\frac{Bq}{kg}]$. To check if the material can be recycled safely, this number needs to be less than 1, applying some safety limit due to uncertainties, after the reactor has cooled down for 100 calendar years. To assess the clearance index for each material chosen for the study, we used the clearance limit values [2] suggested by the IAEA.

Shallow Land Burial is a way to decide how to get rid of a slightly radioactive material, in case it can be classified in a – say – low-level waste according to US regulations. This decision is based on a score called the waste disposal rating (WDR), which is calculated using a specific formula.

$$WDR = \sum_{i=1}^{\#isotopes} \frac{C_i}{L_i} \quad (3.2)$$

C_i is the concentration of the i -th nuclide in the waste and L_i is the concentration limit value and they are expressed in $[\frac{Bq}{cm^3}]$ [9]. It is evaluated at 100 years after the shutdown of the reactor and it needs to be lower than 1 in order to classify the waste as a ‘Low-Level waste’ [3].

C_i , the amount of the i -th substance in the waste per unit of volume, is called its volumetric concentration, and L_i refers to the maximum allowed concentration level. Both of these are computed in $[\frac{Bq}{cm^3}]$ [9]. The WDR is checked 100 years after the reactor is turned off, and it needs to be less than 1 to be called “(American 10CFR6-like) Low-Level waste” [3].

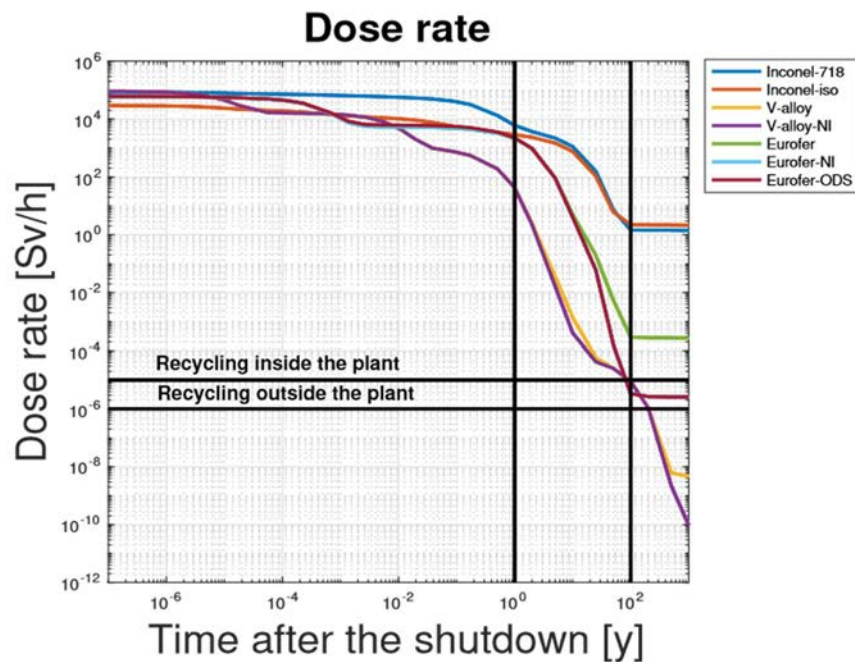


FIG. I-4.3. Activation of Inconel-718 vessel material

As expected, neutronic transport and activation analyses have shown the inadequacy of Inconel-718 as VV inner and outer walls material aiming at clearance [3] and/or Shallow Land Burials SLBs for LLW (reference limits to comply with Waste Disposal Rating <1). See Figure I-4.3 and Table I-4.1 [13]. Other materials need to be analyzed for a better performance in respecting low activation criteria.

TABLE I-4.1. CI AND SHALLOW LAND BURIAL INDEXES AFTER 100 YEARS FROM SHUTDOWN

	CI	WDR for SLB
Inconel-718 Inner wall VV	3.81E+007	4.15E+003
Inconel-718 Outer wall VV	3.29E+007	3.59E+003

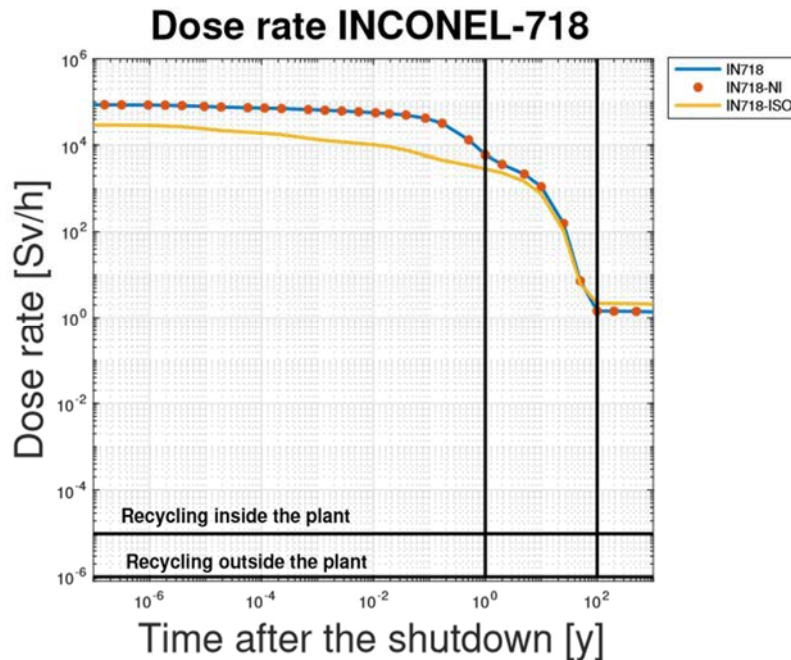


FIG. I-4.4. Comparison of dose rate in Inconel, V-alloy and Eurofer.

Both Vanadium-based V-15Cr-5Ti alloy and Eurofer steel have been considered as the main alternatives for Inconel-718 in the vessel. Figure I-4.4 shows the results in terms of dose rate [13]. Given the excellent results for the V-15Cr-5Ti alloy, also in terms of structural and thermo-mechanical properties, this material has been selected as the preferred alternative to Inconel-718, and a more detailed analysis has been carried out.

We first checked the clearance and SLB for the alloy that contains impurities to see if the standards are met or if the activation issue is linked to a small amount of a specific element. (Table I-4.2) [13].

TABLE I-4.2. CI AND SHALLOW LAND BURIAL INDEXES AFTER 100 YEARS FROM SHUTDOWN

	CI	WDR for SLB
V-15Cr-5Ti, with impurities, inner regions	1.48E+004	3.03E-002
V-15Cr-5Ti, with impurities, outer regions	3.29E+007	3.59E+003
V-15Cr-5Ti, without impurities, inner regions	5.86E+001	8.19E-006
V-15Cr-5Ti, without impurities, outer regions	2.41E+001	1.87E-006

The SLB rules are being followed, but unfortunately the clearance index is above one. To find out which isotopes make the Vanadium alloy still have high activity years after it was shut down, we check the percentage of each isotope's activity compared to the total. The main one responsible is C-14, which has a half-life of 5,708 years. It is created when N-14, a type of natural Nitrogen, gets activated through a reaction called (n,p). Nitrogen is a contaminant in the Vanadium alloy, and even if there is only a small amount, it affects how well the material performs over a long time. But even without any contribution from impurities, the CI is still above one.

Therefore, the outcomes of our activation analysis can be summarized in Table I-4.3 [13].

TABLE I-4.3. BACK-END OF ARC REACTOR: VESSEL ACTIVATION AND COMPLIANCE WITH CRITERIA

	CI	SLB	Recycling
Inconel	NO	NO	NO
V15Cr-5Ti	NO	YES	YES
Eurofer	NO	YES	NO
Eurofer-ODS	NO	YES	YES - inside the plant

SYNOPSIS

ARC has a unique design, and its vacuum container is placed very close to the area where plasma is made. This means it is always exposed to a strong flow of neutrons without any protection, which is quite different from how ITER and DEMO are designed.

The testing of different metal mixtures showed that when they are exposed to high levels of neutron radiation for a long time, for instance two years, it is really hard to get the radioactivity levels down, even after the plant has been closed for a long time.

The solutions we have explored during this research, from changing elements to using different isotopes showed some improvements, but they aren't enough, and the costs are quite high compared to the benefits they would bring.

The simulations done in this study identified a new potential material made from a Vanadium alloy that could be used in the construction of the ARC reactor. It also highlighted the importance of using a low-activation alloy to help make a tokamak that is cheaper, safer, and more efficient in every way.

This study pointed out to some of the ARC safety-oriented design and characteristics related to waste management and disposal: they will be included in a comprehensive safety analysis, aimed to demonstrate that ARC characteristics would easily permit the siting and licensing of this reactor in the US, according to that Nation's Federal regulations. It also confirmed the following:

- The adoption of low-activation materials to avoid the production of any High-Level Waste (HLW);
- The adoption of advanced materials to maintain a safety-oriented design, presenting a quite moderate radioactive inventory, and an even milder biological hazard potential.

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ANNEX II.

REGULATORY FRAMEWORKS

II-1. CHINA REGULATION STATUS FOR FUSION AND PROPOSED DEVELOPMENT PLAN

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Abstract

China establishes and improves a system of laws and regulations for nuclear facility. Especially, a series of detailed regulations have been established for fission nuclear power plant and research reactor. But China does not yet establish the regulation framework especially for fusion reactor, including fusion radwaste management. Some regulations adapted to the nuclear facility come from the safety requirements of fission reactor, which cannot be directly referred to fusion reactor on account of the obvious differences of safety characteristics between fusion and fission. Currently one task has been established to study the nuclear safety for China Fusion Engineering Test Reactor (CFETR) related to the regulatory framework in China, which is under implementation by Southwestern Institute of Physics (SWIP), Nuclear & Radiation Safety Centre (NRSC), China Nuclear Power Engineering Company (CNPE) and Institute of Plasma Physics of Chinese Academy of Sciences (ASIPP) now. The output of this task will be helpful for the development of regulations for fusion in China.

INTRODUCTION

China establishes and improves a system of laws and regulations for nuclear facility, covering radiological safety, conventional safety, radioactive pollution and environment protection, especially, a series of detailed regulations have been established for fission nuclear power plant and research reactor. But China does not yet establish the regulation framework especially for fusion reactor, also for fusion radwaste management. Now China nuclear safety regulator, the National Nuclear Safety Administration (NNSA), preliminarily recognizes the fusion reactor as one of the nuclear facilities according to its safety characteristics and has to comply with the laws and regulations applicable for the nuclear facility, which covers safety supervision of nuclear facility and nuclear fuel cycle facility, supervision of safety equipment, radioactive substance and radwaste administration, nuclear material control and so on. Besides, the conventional safety regulations, such as for work safety and occupational health, have to be complied.

But some regulations adapted to the nuclear facility come from the safety requirements of fission reactor, which cannot be directly referred to fusion reactor on account of the obvious differences of safety characteristics between fusion and fission. The NNSA encourages the fusion research institutions to develop the safety guideline together with the Nuclear & Radiation Safety Centre (NRSC) based on the safety characteristics, operational mechanism and principle of the fusion reactor. After that, the fusion regulations will be developed to cover all safety aspects of fusion reactor, such as site selection, licensing, design guideline, operation supervision and so on, which may partially take the fission regulations in reference.

Currently one task has been established to study the nuclear safety for China Fusion Engineering Test Reactor (CFETR) related to the regulatory framework in China, which is under implementation by Southwestern Institute of Physics (SWIP), NRSC, China Nuclear Power Engineering Company (CNPE) and Institute of Plasma Physics of Chinese Academy of Sciences (ASIPP) now. The output of this task will be helpful for the development of regulations for fusion in China.

CURRENT NUCLEAR REGULATORY FRAMEWORK

The China nuclear regulatory framework includes several levels, as shown in Figure I-2.1. The top level is nuclear laws issued by National People's Congress, including Law on Environmental Protection, Law on Nuclear Safety and Law on the Prevention and Control of Radioactive Pollution. The second level is nuclear safety control regulations (decrees) issued by the State Council, namely HAF series as listed in Table I-2.1. The third level is comprised of nuclear safety codes and details rules of implementation. The fourth level includes nuclear safety guides. The fifth level includes nuclear safety technical documents.

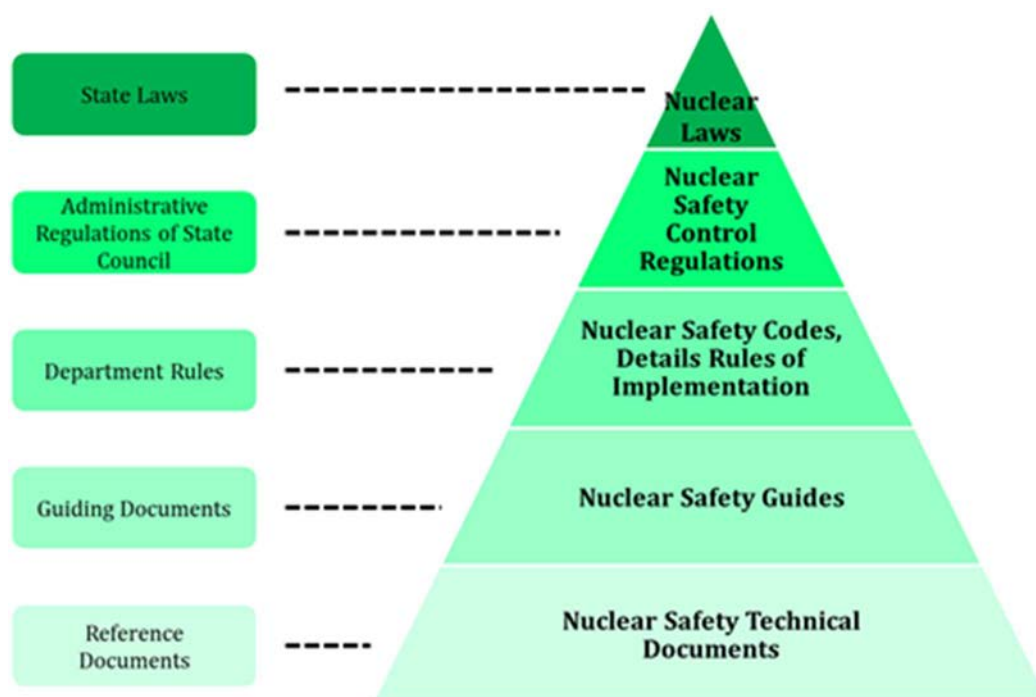


FIG. I-2.1. Current China nuclear regulatory framework.

TABLE I-2.1. NUCLEAR SAFETY CONTROL REGULATIONS (DECREEES) LIST

HAF series	Column 2
HAF001	Regulation on Safety Supervision and Administration of Civil Nuclear Installations
HAF002	Regulation on Emergency Management of Nuclear Accidents at Nuclear Power Plants
HAF400	Regulation on Safety Administration of Radioactive Substance
HAF501	Regulations on Nuclear Material Control
HAF600	Regulation on Supervision and Administration of Civil Nuclear Safety Equipment
HAF700	Regulation on Transportation Safety Administration of Radioactive Substance
HAF800	Regulations on Safety and Protection of Radioisotope and Radiation Installations

Currently the Tokamak experiment facilities, like HL-2A/M and EAST, are classified as radiation facility. But the classification for fusion reactor is not identified yet. In China, the nuclear facilities are divided into several classes, nuclear power plant, research reactor (including test reactor and critical facility), nuclear fuel cycle facility, radwaste treatment & disposal facility, and other nuclear facility requiring supervision. Considering the neutron fluence, decay heat and experience from ITER, it is preliminarily considered that the class of other nuclear facility requiring supervision for fusion reactor.

After that, the applicability analysis of China nuclear regulatory framework for China Fusion Engineering Test Reactor (CFETR) was preliminarily performed. It shows that the 3 laws (top level 1) and 7 decrees (top level 2) are generally applicable for CFETR. Due the regulations at top level 3 and 4 mostly focus on fission reactor or facility, so more detailed selection and analysis for fusion reactors is needed.

SAFETY FEATURES AND ASSESSMENT OF CFETR

CFETR is the DEMO concept proposed by Chinese fusion community to bridge the technical gap between International Thermonuclear Experimental Reactor (ITER) and the future prototype fusion power plant [1-3]. One of objectives of CFETR design activities is to assess its safety features.

Same as the magnetic confinement fusion reactor, the conditions of D-T fusion reaction in CFETR are strict, and the reaction can't occur if they are not satisfied. In this case, the fusion reactor does not have the issue of criticality feature, which are the fundamental difference between fusion reactor and fission reactor. The main resources of decay heat are blanket and divertor, which are inner components of vacuum vessel inside Tokamak. Based on the calculation, the decay heat of blanket and divertor is about 1% of full power after fusion power shutdown. Based on CFETR design (1GW) and operation strategy, the decay heat of blankets is about 8.0MW (10 years operation) and about 2.4MW for divertors (5 years operation). Even in the accident case, these decay heat of helium cooled in-vessel components can be removed passively by nature convection and thermal radiation, without the risk of melting. Comparing with fission reactor, the radioactive source of CFETR is different, including tritium, activated dust, and activated corrosion products (ACP). Especially a big mass of tritium, as the fuel of fusion reactor, will be sustained in the Tokamak, tritium plant and other systems. So, the radioactive containment is the most important considering the safety.

The main radioactive source of CFETR is including tritium, activated dust, and activated corrosion products (ACP), especially the tritium burning during CFETR operation. There will be a big mass of tritium sustained in the Tokamak plant and tritium plant. So, the radioactive containment is the most important part of acceptance criteria. In order to assess the safety of CFETR, the activation product, ACP and tritium have been estimated based on the preliminary design of CFETR [4]. The results show the activity inventory is 5.55×10^{19} Bq for blanket and 2.26×10^{19} Bq for divertor. The main radionuclide in water activation products is C-14 and short-lived N-16, N-17. During operation, the activity of water activation products in the primary heat transfer system is between $1 \times 10^{11} \sim 1 \times 10^{13}$ Bq/kg H₂O. ACP mainly come from the activation of Fe, W, Mn and other elements. At shutdown, the dose rate at 1m outside the main cooling pipe is about 8~50 μ Sv/h. It can be considered that The ACP of CFETR can be reduced by increasing coolant circulation purification capacity and controlling water chemical conditions. In order to minimize the tritium release, the confinement system and ventilation zoning has been considered in CFETR based on the experience from ITER and tritium laboratories.

Considering accidents, the initial event of CFETR mainly considers the following aspects that may cause damage and radioactive leakage to the system itself and fusion reactor (no consider the reactivity accident).

- Loss of power;
- Loss of cooling;
- Leakage of air, water or helium;
- Failure of magnet and system control;

- Failure of decay heat removal;
- Potential risk events during maintenance and transport.

Based on 16 major systems, the study includes initiating events, design basis accident, accident sequence, acceptance criteria and accident analysis result. In the radioactive consequence assessment for design basis accident (DBA), the preliminary analysis results show radioactive dose released to the environment after the accidents is under low, 4.32×10^{-7} Sv for pipe break accident of tritium extraction system, 2.16×10^{-8} Sv for ex-vessel LOCA accident of primary helium cooling system for costal site. In the radioactive consequence assessment for postulated siting accident (PSA) of break of all in-vessel components and break of all pipes of primary cooling system, the radioactive dose released to the environment is 1.5×10^{-3} Sv based on the following assumption:

The inventory of T in VV during operation is ~30g. During PSA, 9.3g T will be release to Tokamak building and 10% of that will be released to environment.

Atmosphere dispersion condition: based on the candidate coastal sites. After the accident, the short-term atmospheric dispersion factor was calculated according to the model of NRC regulatory guide RG 1.145. Based on current analysis of CFETR for DBAs and PSA accidents, the suggested 10mSv target can be satisfied. And expert also proposed that the offsite emergency requirements of CFETR can be simplified comparing to PWR power plants.

RADWASTE CLASSIFICATION AND MANAGEMENT

According to China Radwaste classification rule (Announcement No. 65 of 2017) [5], the radwaste is categorized as exempt Waste (EW), very short lived waste (VSLW), very low level waste (VLLW), low level waste (LLW), intermediate level waste (ILW) and high level waste (HLW) in China. Its classification criteria and relevant disposal options are shown in Table I-2.2.

TABLE I-2.2. RADWASTE CLASSIFICATION CRITERIA AND RELEVANT DISPOSAL OPTIONS

Radwaste type	Criteria	Disposal options
EW	personal effective dose $\leq 10 \mu\text{Sv/yr}$; specified radionuclides: $\sum_{i=1}^n \frac{C_i}{C_{i0}} \leq 1$	
VSLW	half-life of radionuclide < 100 day	clearance after storage and decay
VLLW	upper limit: 1~2 orders of magnitude high than clearance level	landfill disposal
LLW	specified radionuclides: $\sum_{i=1}^n \frac{C_i}{C_{i0}} \leq 1$; activity concentration of unspecified radionuclides $\leq 4 \times 10^{11} \text{ Bq/kg}$	near surface disposal
ILW	activity concentration $\leq 4 \times 10^{11} \text{ Bq/kg}$, and decay heat $\leq 2 \text{ kW/m}^3$	medium depth disposal
HLW	activity concentration $> 4 \times 10^{11} \text{ Bq/kg}$, or decay heat $> 2 \text{ kW/m}^3$	deep geological disposal

Based on the current assessment, some components of CFETR blanket and divertor are HLW waste at shutdown, while the rest components of CFETR are LLW waste. After 50 years of shutdown, all radwaste of CFETR are ILW or LLW waste. But some structural material of IVC stay at the ILW level up to 10000 years mainly due to the radionuclide Nb-94. The general options for radwaste management of CFETR have been proposed, clearance, recycling and reuse, disposal. The effort will be taken toward the clearance and recycling / reuse as much as possible.

SUMMARY

It shows that the 3 laws (top level 1) and 7 decrees (top level 2) are generally applicable for CFETR. Due the regulations at top level 3 and 4 mostly focus on fission reactor or facility, more detailed selection and analysis will be required for fusion reactor.

Currently, the preliminary applicability analysis of China nuclear regulatory framework of fusion shows that the top laws and decrees are generally applicable for CFETR, but the low level regulations, nuclear safety codes, details rules of implementation and guides, cannot imperfect applicable for CFETR at some aspects, such as fuel cycling management, radioactive source terms, radioactive effluents management, tritiated waste management, initial accident events and safety design principle, etc.

Based on the current situation, it was proposed to develop the special safety requirements for CFETR according to the safety features of fusion reactor, China domestic nuclear regulations and ITER experience with the collaboration with Nuclear & Radiation Safety Center. Depending on its maturity, it can be proposed to National Nuclear Safety Administration for review.

In the next step, the detail study and validation will be performed for the preliminary safety requirement and guideline for CFETR based on the China nuclear regulatory framework, also applicability study to other fusion reactors. Of course, the suitability assessment with involvement of nuclear regulator will be necessary.

ACKNOWLEDGEMENTS

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II-2. REGULATORY FRAMEWORK FOR FUSION FACILITIES IN FRANCE

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French nuclear law

Facilities using radiative materials are subject to regulations covering two aspects that are considered inseparable: radiation protection and nuclear safety. Radiation protection requirements are established by the Public Health Code and the Labour Code, whereas the regulatory framework for nuclear safety is established by the Environment Code (*Code de l'environnement*) [1]: “Installations in which radioactive elements are used are classified in three main categories: basic nuclear installations (*installations nucléaires de base - INB*), installations classified for environmental protection purposes (*installations classées pour la protection de l'environnement - ICPE*) in which activities using radioactive substances are carried out and defence-related nuclear installations and activities (*installations et activités nucléaires intéressant la défense - IANID*)”. INB and ICPE facilities are distinguished by the Environment Code in terms of activity and specific activity. The classification relies mainly on the inventory of the different radionuclides present in the perimeter of the facility (INB allows to manage higher amount of radionuclides). In accordance with this categorization, a fusion facility such as ITER is classified as an INB. Thus, the development here below refers to the INB case.

The 2006 Act on nuclear transparency and safety (*La loi relative à la transparence et à la sécurité en matière nucléaire*, TSN Act), established the legislative framework for all installations classified as INB, including establishing the Nuclear Safety Authority (*Autorité de sûreté nucléaire - ASN*) to the regulatory authority responsible for oversight of these installations. All the legislative and regulatory provisions that specifically concern INB are codified in the Environment Code [1].

The general technical regulations, provided for by article L.593-4 of the Environment Code, include all the texts of general scope setting technical rules for nuclear safety, whether ministerial orders or ASN regulatory decisions. It is supplemented by circulars, fundamental safety rules (RFS) and ASN guides, which are non-binding [2]. Among these general technical regulations, the order of February 7, 2012, defining the general rules for INB can be specifically mentioned concerning decommissioning activities (especially articles 6.1 to 6.8 and 8.3.1 to 8.3.4).

Decommissioning and permanent shutdown of a basic nuclear installation

The operator of an INB wishing to cease permanently operation of that installation has to inform the Ministers responsible for nuclear safety and the ASN [1]. The operator needs to declare the permanent shutdown date at least two years before the effective permanent shutdown (an update of the decommissioning plan is sent with this declaration). Then, at most 2 years later after the declaration of the permanent shutdown date, decommissioning authorisation application has to be submitted to the Ministries accompanied by a supporting data package (including a decommissioning file with the ASN in copy). Then the examination of the application cannot exceed more than three years, except special cases (target date for decommissioning decree obtaining). The operator has to send a copy to the ASN. The permanent shutdown and decommissioning authorisation application is subject to the same procedure as the creation authorisation application – see Fig. II-2.1.

A fusion facility classified as an INB (such as ITER) is subject to the same overall regulatory framework as a fission reactor. In accordance with the above, the key milestones for the purposes of regulatory control are:

- Authorization to construct the facility (INB creation decree);
- Authorisation to commission the facility (Notification by operator of intention to permanently shut down the facility (at least 2 years before permanent shutdown);

- Authorization to proceed with decommissioning (INB decommissioning decree);
- Agreement by regulator to declassify the facility.

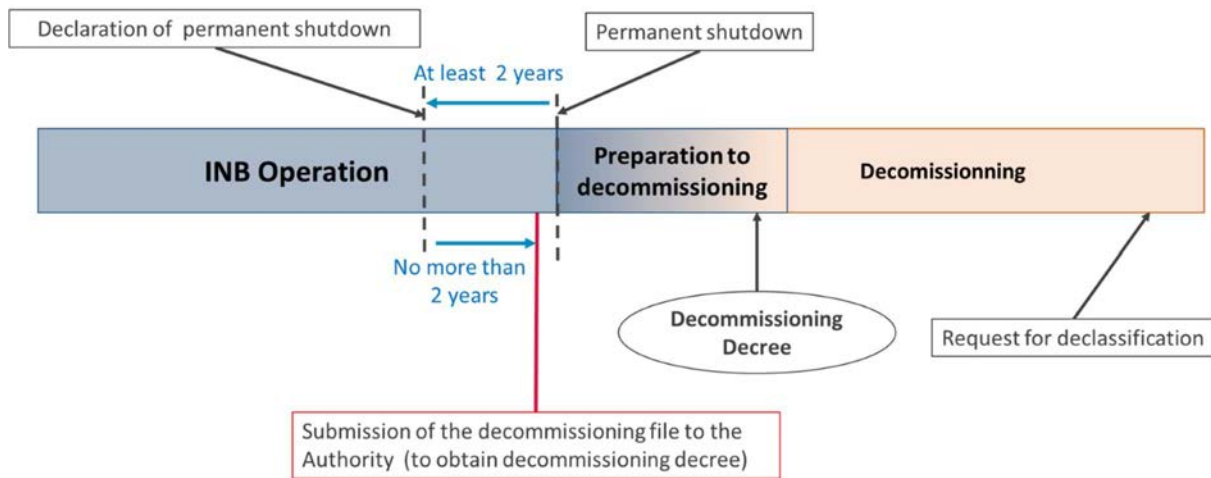


FIG II-2.1. Final shutdown and decommissioning procedure (reproduced from [3]).

Decommissioning plan

A key technical piece of the data package for decommissioning authorisation is the decommissioning plan. The main expected topics to be addressed in the decommissioning plan are the following ones:

- Presentation and justification of the retained decommissioning strategy;
- General considerations on the decommissioning, as for example:
 - Methodology, survey to be put in place after decommissioning;
 - Provisions made when designing the facility to facilitate dismantling;
 - Measures taken by the operator to guarantee the preservation of the history of the facility and the accessibility of the associated data;
 - Measures taken by the operator to guarantee maintenance of skills and knowledge of the installation;
 - Estimates of the quantities and methods of management of waste resulting from dismantling, taking into account existing or planned management solutions, developed within the framework of the National Plan for the Management of Radioactive Materials and Waste (PNGMDR) (see next paragraph);
 - Studies and any research and development work to be carried out.
- Decommissioning process:
 - Description and justification of the foreseen initial state;
 - Main steps;
 - Associated schedule;
 - Description of the activities;
 - Equipment to build;
 - Safety objectives, radiation protection and environmental protection measures;
 - Waste assessment;
 - Presentation of the main Protection Important Elements and Protection Important Activities;
 - Description of the methodology to decontaminate (soils, buildings);

- Organization description;
- Justification of the technical choices.

— End State:

- Presentation and justification of the retained end state;
- Foreseen use of the site;
- Impact assessment of the facility and the site after the end state is reached.

Radioactive waste management

The general regulations concerning waste are laid down in the Environment Code and the 2006 Act on the sustainable management of radioactive materials and waste (*La loi n°2006-739 du 28 juin 2006 de programme relative à la gestion durable des matières et déchets radioactifs*, Waste Act). Radioactive waste management policy is implemented through the National Radioactive Materials and Waste Management Plan (PNGMDR), updated every three years in the past and now five years, and which in turn is drafted on the basis of the National Inventory of radioactive materials and waste. The radioactive waste management framework also defines the role of public agency devoted to the management of radioactive waste, especially with radioactive waste disposal facilities, Andra (French National Radioactive Waste Management Agency).

Activities related to radioactive waste require an official licence and are subject in particular to the provisions of the Environment, Public Health and Employment Codes. The Environment Code [1] sets out the main principles governing the management of radioactive waste:

- “the sustainable management of any radioactive materials and waste, resulting notably from the operation and dismantling of installations using radioactive sources or materials, shall be carried out with a concern to protect human health, safety and the environment;
- the research and the implementation of the means necessary for the final security of radioactive waste are undertaken in order to prevent or limit the costs that will be borne by future generations;
- relevant means to ensure the final safety of radioactive waste shall be developed or implemented with a view to preventing or limiting the responsibilities to be borne by future generations;
- any producer of spent fuel and of radioactive waste shall be liable for those substances, without any prejudice to the liability of their holders as people responsible for nuclear activities.”

The development and operation of disposal sites in France for materials classified as radioactive waste is the responsibility of ANDRA, which currently operates or is in the process of establishing repositories for all classes of such waste in France. Accordingly, waste producers need to interact closely with ANDRA to ensure that waste is conditioned in a form suitable for disposal in the national repositories.

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II-3. MAIN REGULATORY DRIVERS IN JAPAN

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Nuclear / Radiation Regulation in Japan

In Japan, the main legislation applicable to nuclear / radiation facilities comprise:

- The Act on the Regulation of Nuclear Source Material, Nuclear Fuel Material and Reactors (Reactor Regulation Act) for nuclear reactor or nuclear materials regulation for the purpose of prevention of nuclear disaster;
- Act on the Regulation of Radioisotopes, etc. (RI Regulation Act) for radioisotopes and accelerators regulation for the purpose of prevention of radiation hazards;
- The Ordinance on Prevention of Ionizing Radiation Hazards (Radiation Ordinance) in Industrial Safety and Health Act for ionization radiation generators regulation for the purpose of ensuring the radiation worker's safety and health.

The Reactor Regulation Act and the RI Regulation Act are under the responsibility of Nuclear Regulation Authority of Japan, while Radiation Ordinance is under the responsibility of Ministry of Health, Labour and Welfare. The requirements established by the Reactor Regulation Act, which are applied to nuclear facilities, are generally more demanding than those established by the RI Regulation Act, which in turn are more demanding than the Radiation Ordinance. Nuclear fusion facilities are regulated by the above laws according to their potential risks. For example, the fusion facilities with H-H discharge, which generate X-ray, are regulated by the Radiation Ordinance through registration as the ionization radiation generators.

Fusion facilities with D-D discharge, which generate neutrons and tritium are regulated by the RI regulation Act as the radiation generators or tritium handling facilities. JT-60U (QST) and LHD(NIFS) are only two fusion facilities regulated by RI regulation Act as radiation generators. The regulatory framework for ITER, which is D-T fusion facility with large tritium inventory (~kg T), was discussed when the ITER was promoted to Japan (2000~2005). An Expert Panel was established to consider ITER regulation based on the safety characteristics of ITER such as the fact that ITER handles large amount of tritium, ITER has a potential energy (electric, magnetic) to break confinement barriers provided by the ITER technical basis document¹⁾. As a result, the panel recommended that ITER be regulated as an extension of the RI Regulation Act for the purpose of prevention of radiation hazards, not by the Reactor Regulation Act. They also recommended accidental analysis for ITER, while RI regulation Act does not require to the radiation facilities. Therefore, they recommended that ITER be regulated by new ITER regulation Act, which was a reinforced RI regulation Act.

The Expert Panel of Japanese Cabinet Office recommended Fusion innovation strategy to be adopted in April 2023. In this strategy, they recommended enhancement of discussion on safety feature of fusion DEMO reactor. Discussions on fundamental safety characteristics of fusion DEMO reactor and possible regulatory framework will be initiated soon.

Decommissioning Regulation in Japan

Decommissioning process is defined by the RI Regulation Act and the Reactor Regulation Act, respectively as shown in Fig. II-3.1. Here, ‘reactor’ means Reactor regulation Act and (RI) means RI regulation Act, respectively in Fig. II-3.1. The decommissioning processes of nuclear facilities and RI handling facilities under both laws are similar. The decommissioning processes consist of preparation of decommissioning, implementation of decommissioning, and waste management.

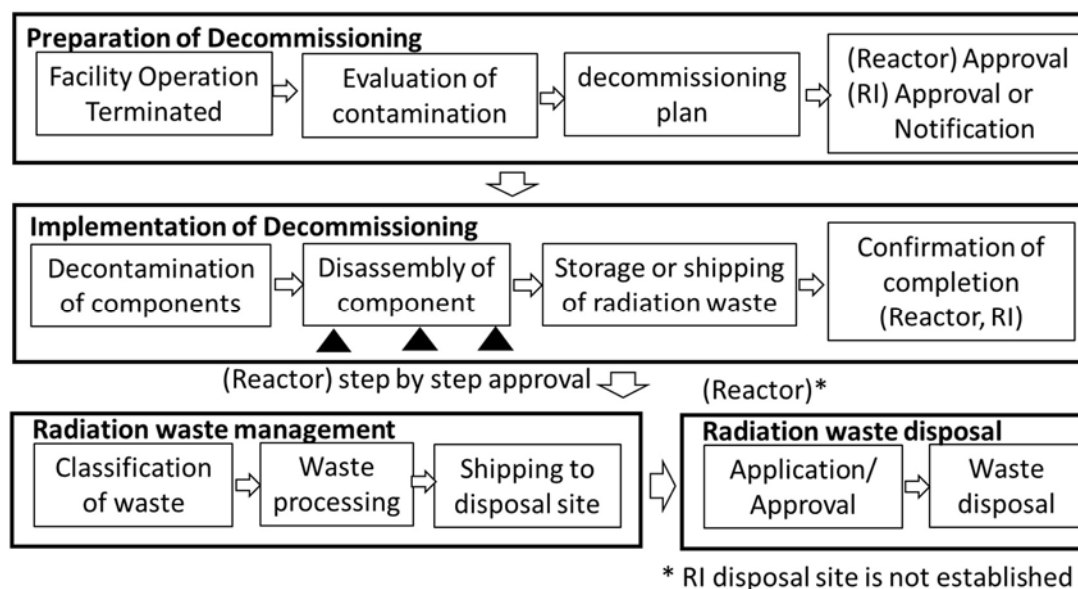


FIG. II-3.1. Decommissioning and waste management process of nuclear facilities and RI handling facilities in Japan.

While the decommissioning processes are similar between RI regulation Act and Reactor regulation Act, approval process is different in some processes. Reactor regulation Act has more permission processes than RI regulation Act in the implementation of decommissioning process. Reactor Regulation Act requires step by step approval of disassembly of the components in the implementation stage of the decommissioning,

Radiation waste management and disposal

The Reactor Regulation Law classifies radioactive waste into the following categories:

- High-level waste (HLW), which is vitrified waste generated in nuclear fuel reprocessing;
- Radioactive waste containing transuranic nuclides (TRU);
- Uranium waste;
- Intermediate level radioactive waste (ILW) containing control rods and in-vessel structures;
- Low level radioactive waste (LLW), containing filters, scrapped vessels, liquid waste, etc.;
- Very low-level radioactive waste (VLLW) containing concrete and metals.

The classification of radioactive waste generated at facilities regulated by the RI Regulation Act is to apply mutatis mutandis to the classification of waste under the Reactor Regulation Act. Since RI handling facilities do not generate HLW, TRU, and uranium waste, criteria of ILW, LLW, and VLLW are applied to radioactive waste are generated at RI handling facilities.

Regarding the radioactive waste disposal site in Japan, a near surface disposal (pit) facility exists at Rokkasho/Aomori for LLW generated at nuclear power plants. However, no disposal facilities exist for radioactive waste generated at RI handling facilities. Some organizations collect and store radioactive

waste generated at RI handling facilities such as institutes or universities. The disposal site for radioactive waste from RI handling facilities will be constructed in near future in Japan.

Regarding decommissioning of fusion facilities, the dismantling of JT-60U is the only one decommissioning experience of fusion facility regulated by RI Regulation Act in Japan ⁴⁾. In this case, the dismantled components were stored as reusable items or expected clearance materials, and radioactive waste was not generated except for combustible materials, etc.

As mentioned above, regulations for fusion power reactors have not yet been established in Japan. Therefore, regulations for decommissioning and waste management have also not been determined. However, decommissioning of fusion facilities will be similar to the decommissioning process in Fig. I-3.1. Regarding the radiation waste management, since nuclear fusion will not generate HLW, TRU and uranium waste, waste management will be similar to that regulated by the RI regulation Act.

II-4. REGULATORY DRIVERS FOR FUSION FACILITIES IN THE RUSSIAN FEDERATION

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Regulatory infrastructure for decommissioning nuclear facilities

Regulatory requirements to the process of decommissioning nuclear facilities in the Russian Federation are set by the Federal law № 170-FZ *On Atomic Energy Use* and laid out in further detail in regulations (federal norms and rules) endorsed by the national regulator, Rostekhnadzor.

Federal Law № 170-FZ *On Atomic Energy Use* (Article 33) stipulates that the procedure and measures providing safe nuclear facilities decommissioning (closure) has to be provided for in the facility design.

Overall requirements to decommissioning process for nuclear facilities are described in *NP-091-14 Decommissioning of Nuclear Facilities*. Additionally, all documents specifying general safety provisions for different categories of nuclear facilities (NPPs, RR, radiation sources, fuel cycle facilities, nuclear powered ships, etc.), e.g. *NP-001-15 General safety provisions for nuclear power plants*, *NP-016-05 General safety provisions for Fuel Cycle Facilities*, *NP-033-11 general safety provisions for Research Facilities* include requirements to decommissioning activities.

There are a number of additional regulations covering specific decommissioning safety requirements for specific types of facilities, e.g. *NP-012-16 Decommissioning safety provisions for NPP*, *NP-057-17 Decommissioning safety provisions for fuel cycle facilities*, *NP-007-17 Decommissioning safety provisions for Industrial Reactors*, etc.

Planning decommissioning is required at all stages of nuclear facility life cycle preceding its decommissioning (closure), including siting, design development, construction and operation, decommissioning (closure). Decommissioning (closure) is to be done in accordance with the established decommissioning program and relevant design documents.

Comprehensive engineering and radiation survey of a facility needs to precede decommissioning in order to evaluate technical and radiological status of components, equipment, building structures and territories.

Facilities undergoing decommissioning (closure) have to be properly staffed; the employees (personnel) need to have appropriate qualifications and permits. During facility operation, operating organization has to maintain all relevant records and information required for safe decommissioning.

Regulatory framework for fusion facilities

The current edition of the Federal Law № 170-FZ *On Atomic Energy Use* explicitly specifies that it is applicable to three general types of facilities - nuclear installations (which are identified as facilities containing nuclear reactors or fissile materials), radiation sources, i.e. facilities containing radioactive materials or generating ionizing radiation, and radioactive waste storage facilities. The law does not mention fusion reactions or fusion facilities and hence may be deemed to be inapplicable for future fusion facilities. No regulations specific to fusion facilities has been issued by any of the regulatory bodies, which contrasts with the detailed set of regulations and guides for each type of nuclear facilities.

Development of an adequate regulatory framework for fusion was one of the directions of the Federal Project “Controlled fusion and plasma technologies”, which started in 2021. The first step in development of regulatory infrastructure was to find a way to include fusion in the legislation. Several options were initially considered:

- Development of a separate law “On fusion energy”;
- Introduction of a term “Fusion facility” into the Federal Law № 170-FZ On Atomic Energy Use;
- Amendments to the definitions of “nuclear facility” and “radiation source” in the Federal Law № 170-FZ On Atomic Energy Use;
- Regulation of fusion facilities in accordance with the current legislation as radiation sources.

All of the options were discussed by experts, representatives of regulatory authorities, scientific and engineering community in 2021-2022. Option 3 was considered to be the most efficient as it allowed to introduce only minor changes to the legislation while introducing the term of fusion facility at the highest legislative level. Most of safety requirements would be specified in regulations issued by the regulatory body. Additionally, by specifying that fusion facilities may be classified as both nuclear installations and radiation sources it provides the necessary legal basis for a graded approach to regulation of fusion facilities. It was considered inappropriate to place all fusion facilities in a single category as most facilities would not meet the level of hazard corresponding to that of a nuclear installation, while some facilities, such as hybrid fission-fusion installations or large fusion plants with large tritium and activation products inventory have a much higher risk profile.

The selected approach was formulated in a draft law, which was agreed upon between all stakeholders during 2022 and has been introduced into the Government of the Russian Federation for further submission to the Parliament. The preparatory works have continued through several iterations in 2023–24. The draft law has been submitted to the State Duma (Russian Parliament) in early 2025 and has so far been passed in the first reading (of three). The Government decided to postpone the entry into force of the amendments to the Federal Law for two years. A set of regulations for fusion facilities in support of the Federal Law is expected to be drafted and approved during this period. It is thus expected that in 2025 the term of ‘fusion facility’ will be introduced into the legislation and relevant facilities will be explicitly categorized as regulated by Federal Law № 170-FZ on Atomic Energy Use. Relevant amendments, as well as regulations concerning the safety of fusion facilities, are currently being drafted by the national regulator with participation of the fusion community and will be issued by the regulator and enter into force within two years after passing of the law.

Considerations for fusion facility decommissioning

Specific regulations regarding safety of fusion facilities have yet to be drafted and enacted. This also applies to the guidance on fusion facility decommissioning. At the same time, the norms and rules *NP-091-14 Decommissioning of Nuclear Facilities* would still apply to fusion facilities. Specific guidance on fusion facility decommissioning, both as regulations and safety guides, may be expected in the future.

There are currently no fusion facilities with noticeable contamination awaiting decommissioning in Russia. At the same time, there are near term plans to construct and commission a number of fusion facilities with significant neutron fluxes. The relevant projects are at the design stage. *NP-091-14* has several requirements to the design, siting and construction stages relevant to decommissioning. Decommissioning concept needs to be available already at the design stage, describing decommissioning options considered for the facility, estimating the potential radioactive waste generation and appropriate waste management approaches, as well as procedures for maintaining records of all operational information relevant to decommissioning.

The document also requires the designer to make arrangements in the design to ensure safe decommissioning of the facility by:

- Using materials minimizing radiation hazards and waste generation at decommissioning stages;
- Applying design solutions facilitating decommissioning works and minimizing personnel exposure during decommissioning;
- Ensuring integrity of the facility for the period of operation and decommissioning,

- Ensuring availability of supporting systems required for decommissioning (i.e. their life expectancy encompasses the decommissioning period);
- Ensuring minimization of surface contamination;
- Planning decontamination and waste removal works.

II-5. RUSSIAN NUCLEAR SUPERVISORY AGENCIES AND NECESSITY TO HAVE STANDARDS AND REGULATIONS ACCOUNTING SPECIFIC NATURE OF FUSION AND HYBRID NUCLEAR FACILITIES

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Abstract

The status of the Federal service for environmental, technological and nuclear supervision (Rostekhnadzor) as nuclear regulator and the State Corporation Rosatom as the national operator is presented in the report as the main nuclear supervisory agencies in Russia. These agencies perform their functions based on the federal laws, rules and regulations and international safety standards for protection from ionizing radiation. They are responsible for licensing, state supervision of nuclear and radiation safety of civil installations using nuclear facilities on all stage of their life cycles.

Special attention is paid to safety related to radioactive waste and materials. Tritium, Deuterium and Lithium-6 important for fusion systems are not considered as nuclear materials, but as special non-nuclear materials that are also subject to strict accounting and control.

So far, in the absence of regulatory documents on thermonuclear and hybrid reactors, we have to rely on regulatory documents developed for fission nuclear power, whereas the reserve of priority documents for regulating the safety of thermonuclear reactors in the Russian Federation has been created since 1990.

The current priority is to finalize and approve regulatory documents on thermonuclear reactors in Russia, as well as to include the development of thermonuclear reactors among the tasks addressed by the Federal law "on the use of atomic energy", and to instruct Rostekhnadzor to take control of the development of thermonuclear and hybrid reactors.

SUPERVISORY AND CONTROL BODIES IN THE NUCLEAR INDUSTRY

In Russia, the Federal Service for Environmental, Technological and Nuclear Supervision (Rostekhnadzor) is responsible for the state supervision of nuclear and radiation safety of civil facilities using nuclear energy (nuclear facilities, NF) [1]. Similar supervision at military enterprises and facilities is carried out by the State Corporation Rosatom [2].

Rostekhnadzor in its activities relating to nuclear facilities is governed by the Federal laws of the Russian Federation (RF) as the lows:

- On the sanitary and epidemiological welfare of the population;
- On radiation safety of the population;
- On the use of atomic energy;
- The international safety standards for protection against ionizing radiation.

In addition to these documents, it also monitors the implementation of those related to its area of responsibility, such as:

- Decrees of the President of the Russian Federation;

- Resolutions of the Government of the Russian Federation [3];
- The Federal Norms [4] and Rules [5].

The regulation of safety of nuclear facilities is carried out in the following three directions:

- a) Development of regulatory legal documents, safety standard and rules;
- b) Licensing of Nuclear Facilities for all stages of their life cycle: placement, design, construction, manufacturing of equipment, construction, commissioning, operation, modernization, service life extension, and up to decommissioning, as well as storage, transportation, burial and disposal of radioactive equipment;
- c) Inspection of NF, monitoring the implementation of licenses, and, if necessary, the application of sanctions for violations.

NUCLEAR INSTALLATIONS AND MATERIALS

According to the law "on the use of atomic energy", facilities that use nuclear energy can be considered "Nuclear Installations" if they contain such materials as fissionable nuclear substances (Pu, U and Th, with the exception of U and Th contained in ore), as well as minor actinides (Np-237, Am-241, Am-243 and Cm-252).

If the quantity of nuclear materials does not exceed 300 g, then the facility is considered not as a Nuclear Installation, but as a Radiation Source. T, D and Li-6 are not considered as nuclear materials, but as special non-nuclear materials that are also subject to strict accounting and control.

NATIONAL OPERATOR

In Russia, in 2011, in accordance with the United Nation Joint Convention, a Unified State System for Radioactive Waste Management was developed. Also, the Federal Law on Rosatom as a National Operator for the management of radioactive waste (NO RAO) was adopted [2].

The task of the NO RAO, as a state production and environmental enterprise, is the mandatory and disposal (isolation) of all accumulated and disposed of radioactive waste produced in the country, taking into account any potential risks.

RADIOACTIVE WASTE CLASSIFICATION

Special attention is paid to the safety associated with radioactive waste and materials. Radioactive waste (RW) is any substance whose use is not provided for and whose radioactivity level does not allow free handling of them in accordance with current legislation.

By the Decree of the Government of the Russian Federation dated 19.10.2012, the concepts of special RW and removable RW are introduced. The special categories include RW for which the risks and costs associated with their removal from storage facilities and subsequent processing, aging, composting and subsequent disposal exceed the risks and costs associated with leaving such RW in its location. (These include, for example, RW after the Chernobyl accident, discharges into lake Karachay near the city of Ozersk, emergency submarine reactors). The rest of the RW are considered to be removed.

CRITERIA FOR THE ACCEPTABILITY OF RADIOACTIVE WASTE DISPOSAL AND RW CLASSES

The National Operator develops criteria and requirements for the radiation physical and chemical properties of radioactive waste and establishes compliance with these requirements, including packaging of radioactive waste for disposal, and packaging containers [6]. It ranks all RW into 6 classes (Table II-5.1) [7].

TABLE II-5.1. RADIOACTIVE WASTE CLASSIFICATION, SPECIFIC ACTIVITY, Bq/g

Category	HARW		MARW		LARW	MARW		LARW	
Class	1	2	2	3	3	4	5	5	6
^3H	$>10^{11}$		$10^8 - 10^{11}$		$10^7 - 10^8$	$10^4 - 10^8$		up to 10^4	Radioactive waste generated during the extraction and processing of uranium ores, as well as in the implementation of activities not related to the use of atomic
β^- (except ^3H)	$>10^7$		$10^4 - 10^7$		$10^3 - 10^4$	$10^3 - 10^7$		up to 10^3	
α -except transuranic	$>10^6$		$10^3 - 10^6$		$10^2 - 10^3$	$10^2 - 10^6$		up to 10^2	
transuranic	$>10^5$		$10^2 - 10^5$		$10^1 - 10^2$	$10^1 - 10^5$		10^1	

Class 1 includes “extremely dangerous” and “very dangerous” for humans highly active RW (HARW) and long-lived medium-active solid RW (MARW). After preliminary hold-up of waste, they are placed in containers and sent to deep burial points (to a depth of up to 500 m). However, such placement points are still under development in Russia, in particular in the Krasnoyarsk territory near Zheleznogorsk.

Class 2 includes solid waste and long-lived HARW that are subject to deep burial without prior hold-up the waste in order to reduce their heat deposition.

Class 3 includes short-lived solid MARW and long-lived low-level solid RW (LARW). The RW of this class can be isolated at near-surface disposal sites at depths of up to 100 m (in Novouralsk, Sverdlovsk region).

Class 4 includes short-lived solid LARW, including biological objects, and long-lived very low-level solid RW (VLARW). The RW of this class can also be buried in near-surface burial grounds.

Waste of Class 5 is liquid RW (organic and inorganic liquids, pulps, slurries). Their final isolation is carried out in burial grounds near the cities of Dimitrovgrad (Ulyanovsk region), Seversk (Tomsk region) and Zheleznogorsk (Krasnoyarsk territory).

There is also RW of Class 6. These are RW generated in the extraction and processing of uranium ores, as well as in the implementation of non-nuclear-related activities for the extraction and processing of mineral and organic raw materials with an increased content of natural radioisotopes. RW of this class is buried in near-surface burial mounds.

ABOUT THERMONUCLEAR AND HYBRID SYSTEMS

The Basic Sanitary Rules for Ensuring Radiation Safety of the Russian Federation prescribe that when deciding on the processing of radioactive materials (RM) for their return to the production cycle, they are guided by the principle of justification, namely, comparing the benefits obtained from the processing of RM with the risks, costs and yield of radioactivity during operations performed during their processing.

When developing thermonuclear fusion systems, the degree of radiation hazard of radioactive materials (RM) arising in the structure elements is usually assessed by the characteristic dose rate of gamma radiation (Sv/h), which accompanies the radioactive decay of elements formed in the system, and determines the conditions for handling them:

- When the gamma-dose rate is less than 0.025 mSv/h, the material is considered as an ordinary (non-radioactive) substance that does not require special tools (the level of manual treatment or “hands-on level”);
- At the gamma dose rate level in the range from 0.025 to 10.0 mSv/h, the PM can be reused (recycled) applying special tools (the level of remote “remote level” of treatment);

- At a dose rate level of more than 10.0 mSv/h, the RM cannot be recycled for reuse and are only subject to burial of varying complexity, depending on the class of radioactivity.

The purpose of the development is to provide residual radioactivity of Class 3 and below (see the next section).

CLASSES OF RADIOACTIVITY OF IRRADIATED MATERIALS

In the development of thermonuclear systems, the following classes of radioactivity of irradiated materials often appear, determined by their specific radioactivity (Bq/kg) [8]:

- Medium-Active RW containing beta-radiating nuclides (except tritium) with specific activity from 10^7 to 10^{10} Bq/kg;
- Low Level Long-Lived RW, containing radionuclides with a half-life of over 31 years, such as ^{14}C , ^{26}Al , ^{39}Ar , ^{53}Mn , ^{59}Ni , ^{63}Ni , ^{91}Nb , ^{92}Nb , ^{94}Nb , ^{93}Zr , ^{93}Mo , ^{99}Tc , and RW with a specific activity from 10^6 to 10^7 Bq/kg for containing b-emitting radionuclides (with the exception of tritium).

Also, this class includes waste that has a specific energy release of up to 10 W/m^3 , allowing for the possibility of radiochemical processing of irradiated materials. Since the dose rate and total specific activity are limiting factors, the decay time of up to 100 years seems appropriate for near-surface disposal and the use of dilution technology (clearance).

Another limiting factor is the presence of long-lived radionuclides and Tritium in the irradiated materials, such as: ^{94}Nb ($T_{1/2}=2 \times 10^4$ years, 1.2×10^5 Bq/kg), $^{108\text{m}}\text{Ag}$ ($T_{1/2}=418$ years, 1.4×10^6 Bq/kg), ^{10}Be ($T_{1/2}=1.6 \cdot 10^6$ years).

Of particular note is the use of PbLi eutectic as a liquid metal breeder and a coolant. Under the conditions of a fusion reactor, when the eutectic is activated, relatively short-lived radionuclides ^{203}Hg ($T_{1/2}=47$ days) and ^{210}Po ($T_{1/2}=138$ days) are formed, which have large radiological hazard potentials (inhalation dose factors) one hundred and one million times higher, respectively, than tritiated water. The risk of the spread of these volatile radionuclides needs to be taken into account.

CARBON-14

Special attention in the development of thermonuclear systems requires the formation of carbon-14, which has a long half-life ($T_{1/2} = 5700$ years, $E_{\beta^-} = 49.5$ keV) and has a high probability of entering the food chains of the biosphere.

Carbon-14 is formed in various reactor components mainly from nitrogen as a result of a nuclear reaction and $^{14}\text{N}(n,p)^{14}\text{C}$ and, in part, of oxygen-17 according to reaction $^{17}\text{O}(n, \alpha)^{14}\text{C}$. A comparison of the ^{14}C production rate in fission and fusion reactors was performed in [9, 10] (see Table II-5.2) [10].

TABLE II-5.2. CARBON-14 PRODUCTION RATE IN DIFFERENT TYPES OF FISSION AND FUSION REACTORS PER ELECTRICITY PRODUCTION UNIT (TBq/GW (el) per year

Fission Reactors		Fusion power blankets	
Light water	1–4	Water-cooled	7
Gas-cooled	6–18	Gas-cooled	16
Heavy water	10–12	—	—
Fast breeder	0.5–1	Lithium self-cooled	3

A typical ^{14}C -production rate from a light-water reactor is $\sim 0.2\text{--}0.4$ Bq/GW (electric) per year, or about 30% of the formation rate. Approximately the same level (~ 0.3 Bq/GW (electric) year) can be expected

for a fusion power reactor with a water coolant. At the same time, the total carbon-14 yield are small, since $1 \text{ TBq } (^{14}\text{C}) = \sim 6.0 \text{ g } ^{14}\text{C}$.

TRITIUM

In nuclear installations, tritium, formed during the so-called triple fission, accumulates in the fuel elements of the core and remains there until the end of the operation with these elements. Table II-5.3 shows estimates of tritium production ($T_{1/2} = 12.33 \text{ years}$, $E_{\beta^-} = 5.7 \text{ keV}$) in the main components of fission reactors of various types [9].

TABLE II-5.3. SPECIFIC ACTIVITY OF TRITIUM IN A FISSION POWER REACTOR NORMALIZED TO THE ELECTRICAL OUTPUT OF THE PLANT, TBq/GW(el) year.

Reactor Type	Fuel	Coolant	Moderator	Total
LWR-PWR	5.18×10^5	3.70×10^4	NA	5.55×10^5
LWR-BWR	5.18×10^5	Low	NA	5.18×10^5
HWR	5.18×10^5	1.85×10^6	5.18×10^7	5.42×10^7
GCR	5.18×10^5	Low	$(0-1.85) \times 10^5$	$(5.18-7.03) \times 10^5$
GCR-HTGR	5.18×10^5	1.85×10^5	$(0.18-7.40) \times 10^4$	$(5.2-5.9) \times 10^5$

NA: There is no rating.

Currently, the level of tritium activity of $\sim 10^8 - 10^9 \text{ Bq/kg}$ is considered to be acceptable. If we translate these activity values into grams, then the total amount of tritium appeared in fission reactors is only $\sim 1.5 - 2.3 \text{ g}$ per full power operation year.

The most significant difference is that in an energy thermonuclear reactor, tritium production is calculated in hundreds of grams per day, and the presence on the site and accumulation - in kilograms. At the same time, tritium produced in solid and liquid blankets, circulates in the fuel cycle paths, is sorbed in structural elements of the fusion reactor, is present in the sputtered materials of the plasma chamber, and in coolants.

THE RESERVE OF PRIORITY REGULATORY DOCUMENTS FOR THERMONUCLEAR SYSTEMS

A certain set of priority documents on regulating the safety of thermonuclear reactors in the Russian Federation has been created since 1990, when drafts of such documents as

- General safety provisions for thermonuclear installations;
- Requirements for the content of the safety justification report for thermonuclear installations with tokamak type reactors, and
- Requirements for the content of quality assurance programs for the construction of thermonuclear installations with tokamak-type reactors.

However, in 2009, the industry leadership lost interest in completing this work.

Due to the fact that the first thermonuclear installations will still have an experimental and research character, it is preferable to follow the existing regulatory framework for nuclear research installations, with reservations regarding the specifics of thermonuclear installations.

At the same time, it is advisable to use the existing backlog of regulatory documents developed earlier for fusion systems.

PRIORITIES FOR LEGISLATIVE JUSTIFICATION OF FUSION REACTORS DEVELOPMENT

As noted in [11, 12], preparations have begun to change the legislative framework in the form of amendments to the laws of the Russian Federation regulating the safe use of controlled thermonuclear fusion and plasma technologies.

Completing the development of regulatory documents on thermonuclear and hybrid reactors and their approval remains a priority and urgent task, as well as including the development of thermonuclear reactors among the tasks considered in the Federal law on the use of atomic energy and instructing Gostekhnadzor to take the development of thermonuclear reactors under its own control.

EPILOGUE

At the beginning of 2021, the Russian Federation approved and launched a comprehensive State Program for the development of technologies and scientific research in the field of nuclear energy use for the period up to 2024. The program will be implemented by the State Corporation Rosatom in close cooperation with the Kurchatov Institute as the main scientific organization, the Russian Academy of Sciences, universities and other scientific centres.

The Program includes not only fundamental scientific research on thermonuclear fusion. It also provides for the development of the legislative and regulatory framework in the field of nuclear energy use, including new types of nuclear installations, thermonuclear and hybrid systems. This work has already started.

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II-6. REGULATORY FRAMEWORK FOR FUSION DECOMMISSIONING IN U.S.A.

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Abstract

The U.S. Nuclear Regulatory Commission (NRC) is responsible for regulating all radioactive materials generated by commercial power plants. Other U.S. governmental agencies (such as the Department of Energy (DOE), the Environmental Protection Agency, the Department of Transportation, and the Department of Health and Human Services) also have roles in regulating the radioactive materials.

INTRODUCTION

The U.S. Nuclear Regulatory Commission (NRC) was created as an independent agency by Congress in 1974 to ensure the safe use of radioactive materials for beneficial civilian purposes while protecting people and the environment. The NRC regulates commercial nuclear power plants and other uses of nuclear materials, such as in nuclear medicine, through licensing, inspection, and enforcement of its requirements [1]. Until 1974 the Atomic Energy Commission (AEC) provided oversight of the nuclear industry; however, there was a perception that the AEC acted in manner that overly favoured the industry it was charged with regulating. Other U.S. governmental agencies (such as the Department of Energy (DOE), the Environmental Protection Agency, the Department of Transportation, and the Department of Health and Human Services) also have roles in regulating the radioactive materials.

The NRC's Office of Nuclear Material Safety and Safeguards (NMSS) oversees the licensing and regulation of nuclear facilities in the U.S. This regulatory function includes the processing, transport, and handling of nuclear materials. Another aspect of the NMSS's regulatory function includes facility decommissioning.

LAWS AND REGULATIONS

In the United States, the Code of Federal Regulations (CFR) provides the regulations that regulators, like the NRC, are required to uphold. The CFR provides 50 titles covering broad areas subject to regulation. Several of the regulations relate specifically to the nuclear industry; however, none of these CFRs specifically mention fusion energy systems. Some examples of the regulations that fusion facilities are subject to are Title 10 CFR 61 Licensing Requirements for Land Disposal of Radioactive Waste, Title 49 CFR Part 173 covering Transportation of Hazardous Materials, and Title 10 CFR 830 covering Nuclear Safety Management – see Fig. II-6.1.

FUTURE USNRC REGULATORY FRAMEWORK

The Nuclear Energy Innovation and Modernization Act (NEIMA) of 2018 required that licensing and regulating framework be developed and implemented for advance reactor designs, which includes fusion systems, by the end of 2027. In early 2023 several licensing and regulating options were presented to the NRC. The NRC approved the option for a byproduct material approach augmenting the framework in 10 CFR Part 30, "Rules of General Applicability to Domestic Licensing of Byproduct Material".

NRC staff are currently developing a new rule addressing both operations and decommissioning, together with a new volume of NUREG-1556 (Consolidated Guidance About Materials Licenses) [2]. This limited-scope rulemaking would amend the NRC's byproduct material regulations to establish regulatory framework requirements for fusion systems that are technology inclusive and supportive of a performance-based approach to regulation. The purpose of this proposed rulemaking is to respond to

NEIMA direction to establish a regulatory framework for fusion systems by 2027. The scope of this rulemaking is to augment NRC’s byproduct material framework to address near-term fusion systems by providing appropriate fusion-related definitions, develop content of application requirements of a licensing application, and determine other appropriate requirements.

DECOMMISSIONING OF NUCLEAR FACILITIES WITHIN THE DEPARTMENT OF ENERGY

In the U.S. DOE, the Infrastructure and Deactivation & Decommissioning Office, a branch of the Office of Environmental Management (EM), “develops policy for deactivation & decommissioning, real property asset management, sustainability, and energy management. The office leads the transfer of completed projects; manages the transfer of excess contaminated facilities and materials from National Nuclear Security Agency, Office of Science, and the Office of Nuclear Energy; and has the overriding responsibility to support field offices by enabling the effective execution of the mission” [3].

The decommissioning of DOE nuclear facilities is typically managed by EM; however, the Tokamak Fusion Test Reactor (TFTR) D&D Project and the PPPL Tritium Systems Demolition and Disposal (TSDD) Projects were funded and managed by the DOE’s Office of Science (SC). The Fusion Energy Sciences (FES) branch of SC was the funding source for TFTR’s experimental operations.

Projects within the DOE are executed in accordance with the requirements of DOE Order 413.3B: Program and Project Management for the Acquisition of Capital Assets. For nuclear facilities, this order implements U.S. Code of Federal Regulations, 10 CFR 830, Subpart B, which establishes safety basis requirements for Hazard Category 1, 2, and 3 nuclear facilities. A contractor has to perform work in accordance with the DOE-approved safety basis for a Hazard Category 1, 2, or 3 DOE nuclear facility and with the hazard controls that ensure adequate protection of workers, the public, and the environment [4]. This is a ‘cradle to grave’ requirement – see Figure II-6.1.

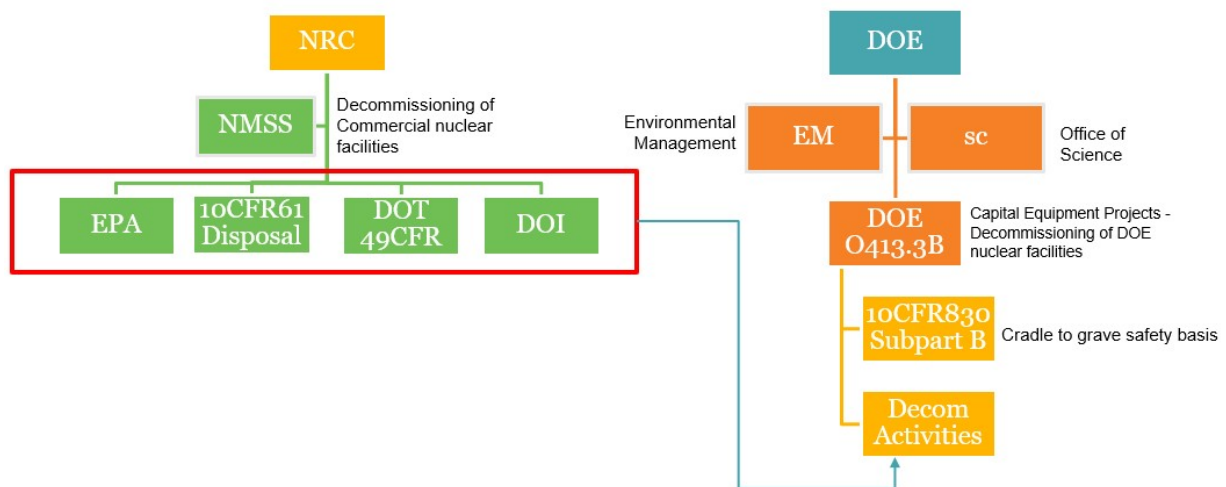


FIG. II-6.1. U.S. Regulatory Framework.

Since the early 1970s, both NRC and DOE have given some attention to the issue of reducing the amount of radioactive waste generated when decommissioning nuclear power plants [5]. The goal is to minimize the waste volumes requiring disposal through recycling and clearing as much of the activated materials as practical (see Section 4 of this report). This is highly desirable to reclaim the use of metal resources. The DOE provides a qualitative guide on releasing mildly radioactive materials for reuse in radiologically controlled facilities within the nuclear industries and at national laboratories. Several attempts have been made to clear or release the slightly radioactive materials from regulatory control [4]. In 2003, the NRC declared clearable materials with low concentrations of radioactivity can be deregulated and issued clearance limits for 115 radioisotopes of steel, copper, aluminium alloys, and concrete rubble from licensed nuclear facilities.

In recent years, the DOE and NRC held several public forums with fusion experts, offering an opportunity for different entities, both domestic and international, to share their thoughts about fusion regulations. This is a major step forward toward the development of regulatory framework that will be agnostic to all fusion concepts supported by the public and private sectors. In providing guidance specific to fusion power plants, it is hoped that the NRC will issue fusion specific decommissioning guidelines, recycling and clearance standards that include all radioisotopes encountered in fusion and develop fusion specific category for the remaining LLW and greater-than-Class C waste after recycling (see Section 4 of this report).

ACKNOWLEDGEMENTS

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ANNEX III.

PREPARATION OF THE DECOMMISSIONING PLAN – CONSIDERATIONS AT THE FACILITY DESIGN STAGE

III-1. ITER CASE STUDY

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Abstract

ITER is under design and construction phase. The design procedures and systems design integrate provisions to demonstrate the feasibility of future decommissioning. Recommendations and guidelines have been presented to the designers to facilitate future decommissioning. Decommissioning is one important driver of the Hot Cell Facility that will support machine maintenance and also decommissioning. Tracking of evolutions is mandatory to ensure compliance of the provisions for Decommissioning through the lifetime of the facility.

INTRODUCTION

ITER is a large-scale fusion experimental facility under design and construction in Cadarache, France. The Parties to the ITER Agreement (the ITER Members) are the People's Republic of China; the European Atomic Energy Community (Euratom); the Republic of India; Japan; the Republic of Korea; the Russian Federation; and the United States of America. Many ITER systems are provided as in-kind contribution. Most of those systems are first of a kind.

At the end of the experimental campaign, the ITER organization will have completed its mission and will be dissolved. As part of the ITER agreement the facility and remaining radioactive waste will be transferred to the host country France for final dismantling.

The goal of this paper is to present the example of the ITER Organization (IO) to illustrate how decommissioning is integrated into the design of the facility, and to provide some basis for requirements management to be followed along the entire project lifecycle.

CONSIDERATIONS ON DECOMMISSIONING IN THE SYSTEMS' DESIGN PROCESS

An ITER Decommissioning Officer has been nominated inside the ITER organization in order to deal with all technical matters related to the provisions for decommissioning to be taken into the design. The role of this transverse function is to define the approach, methods, procedures and organization necessary to coordinate, control and integrate all technical activities related to the ITER Decommissioning, from the early stage of design to the commissioning, operation phase and decommissioning phases.

For each ITER system passing the Preliminary Design Phase, a specific system's decommissioning plan has to be provided.

The system's decommissioning plan describes how to put the system in a safe state at the end of operations. The designers need also to identify and describe the techniques foreseen to perform dismantling of their system. For example, if hands-on decontamination or dismounting is possible or if use of remote handling is necessary. In that case, if these are expected to be tools available since they are needed during operational maintenance or if these are required to be developed for dismantling purposes only.

The document needs to include the expected list of radioactive waste to be generated during operation and decommissioning. This estimate is based on crosschecking the detailed system bill of material, the room-by-room breakdown, radiological data (activation, contamination) and the maintenance strategy (dismounting features, remote handling means, failure rates or replacement periodicity, etc.). The adequacy of the path to the radioactive waste treatment and storage systems in ITER has also to be assessed. One main concern is to identify radioactive waste that may not comply with the existing waste treatment and disposal facilities.

Furthermore, decommissioning experts are mandatory participants as panel members in system design reviews with the responsibility to give advice about decommissioning and raise issues whenever necessary.

The requirements derived from the System's Decommissioning Plan need to be included in the System's Requirement list update prepared and baselined in line with the design process and associated design reviews.

HOT CELL FACILITY REQUIREMENTS MANAGEMENT

When a complete set of Technical Requirements is correctly defined (e.g. using ranges, proper margins), including the associated decommissioning requirements, it is possible to have them tracked and fulfilled during all the phases of the fusion facility lifecycle. The following picture shows a simplified process for the requirements definition and their propagation.

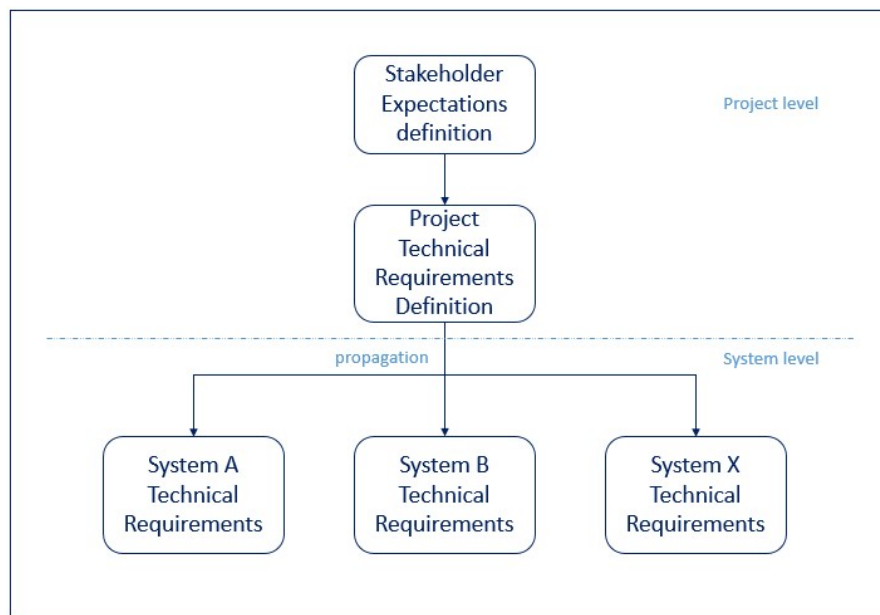


FIG. III-1.1. Simplified requirement management process.

The propagation of the requirements from the system level to the lower levels of the specification tree, needs to follow a process like the one shown in Figure III-1.2.

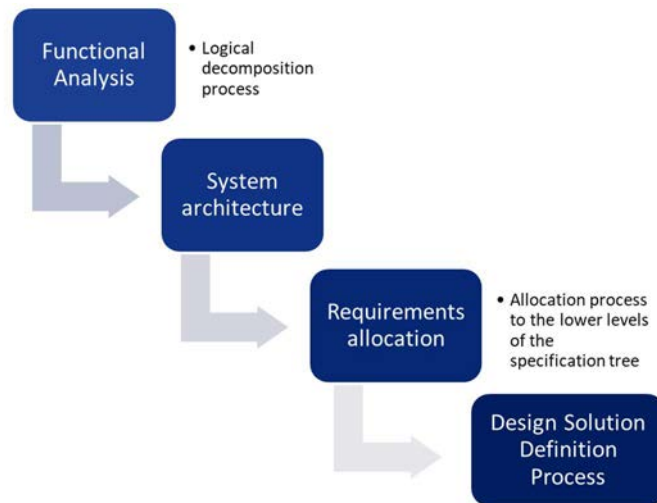


FIG. III-1.2 Simplified requirement allocation process.

As the design of the systems, structures and components belonging to the fusion facility evolves, the decommissioning scenarios would be updated, in order to converge to the proper design solution that can be implemented for the future decommissioning.

It is important to systematically follow a standardized approach for the requirements management along the full project life cycle, as for example the IAEA-TECDOC-1933 ‘Approaches to Management of Requirement Specifications for Nuclear Facilities throughout Their Life Cycle [2]. This is especially valid for:

- Relationship between requirements and design basis;
- Competences needed to correctly capture requirements;
- Ownership of requirements;
- Management of stakeholder interfaces.

Moreover, given the diversity of fusion technologies, the uncertainties associated to their research plans and as well as the legal framework in different countries, the decommissioning scenario needs to be kept as flexible as possible, thus the design of the decommissioning facilities (including the Hot Cell at ITER) needs to be modular and adaptable to the possible several phases of the ITER Project.

ITER DECOMMISSIONING FORUM

A joint forum has been organized in 2016 to give France’s (represented by CEA) and international feedback (participation of IAEA and OECD/NEA) regarding decommissioning to IO’s and Domestic agencies’ systems designers. See [3] for the communication made the on ITER Newslines.

Following this forum IO and CEA experts have established a common guideline for establishing a system decommissioning plan. Based upon this, generic recommendations given to ITER system designers to facilitate decommissioning include, for example, to:

- Facilitate the segmentation of large, contaminated or activated items for easy removal;
- Minimize the number of components in the design;
- Use modular design of systems to facilitate easy removal;
- Clad sumps and trenches in concrete floors with corrosion-resistant sheet steel, to protect them from contamination and to facilitate clean-up as well as to provide corrosion-resistant tanks to capture spills.

TRACKING OF EVOLUTIONS IMPACTING THE DECOMMISSIONING

According to article 6 of the ITER decree [1], “The Nuclear Operator must communicate each year:

- Information relating to changes in the system that may impact on waste disposal routes.
- Information relating to changes in the system could have a significant impact on the risks and inconvenience of dismantling.
- Generally, any useful information for setting the final shutdown and dismantling of the facility.”

The tracking of modifications is performed via the follow-up of Project Change Requests and Design reviews. A specific attention is paid on the possible evolutions of material specifications for components exposed to irradiation. Each modification can lead to evolutions in the amount and nature of waste, and dismantling operations time and resources.

Since 2013, IO has declared several design changes to the Regulator but not yet quantified the impact on decommissioning. An updated overall ITER Decommissioning Plan will be transmitted to the Regulator as part of the ongoing licensing process.

CONCLUSION

The ITER Hot Cell Facility will facilitate the radioactive waste treatment and storage process and thus will play an essential role during the ITER facility’s decommissioning. As the ITER design is still evolving, compliance with the whole set of requirements generated by the full completion of the decommissioning have to be closely monitored and updated in line with system design refinement.

Decommissioning requirements have been considered since the early stages of the project including licensing. Propagation of decommissioning requirements at the facility and systems levels has to be performed for the design of the Hot Cell, using a standardized approach.

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III-2. RADWASTE MANAGEMENT SCENARIO FOR THE JAPAN DEMO

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Abstract

The management scenario of a large amount of radwaste generated in every replacement of the used in-vessel components (IVCs) and the conceptual design of radwaste-related facilities such as Hot cell based on that scenario was proposed for Japan's DEMONstration fusion reactor (JA-DEMO). Even the maximum activation level of the blanket first wall (F82H steel) can be disposed of at the LLW (L2) and qualified for near-surface disposal. In addition, it was possible to reduce the amount by up to 40% compared to the case of once-through disposal by adopting reusing and recycling of the used IVCs.

INTRODUCTION

In the fusion plant, in-vessel components (IVCs) around the plasma are exposed to high energy neutron, so these IVCs need to be regularly replaced for the safety of the operation and these used IVCs become radioactive waste (radwaste). In order to reduce the volume of the radwaste generated in the fusion reactors, it is important to reuse and recycle these used IVCs. Therefore, an appropriate conceptual radwaste management design needs to be developed. In addition, radwaste classification is important from a social acceptance perspective.

WEIGHT OF JA DEMO AND VOLUME REDUCTION EFFECT BY RECYCLING AND REUSE

The current version of Japan's DEMONstration fusion reactor (JA-DEMO) has a major radius of 8.5 m, a fusion power of 1.5 GW [1], the estimated reactor volume of about 7,000 m³ [2], and a total weight of about 38,400 tons as shown in Figure III-2.1. The weight ratio of the blanket segment and diverter cassette at the time of initial loading is about 29 % of the total weight of the core components for the JA DEMO. However, if the plant life is 20 years, the weight ratio of the used IVCs will be increased to 75%. Here, it was found that the amount of radwaste at the decommissioning phase can be reduced by 40% when the used IVCs such as the blanket module, back plate, diverter mono-block unit, and diverter cassette is reused and recycled to the extent possible.

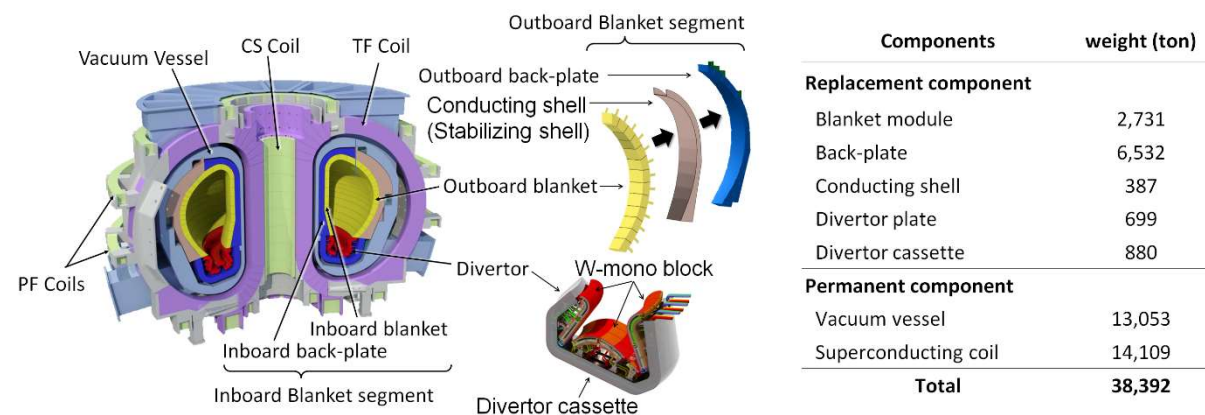


FIG. III-2.1 Conceptual design and weight of the components for the JA DEMO.

DESIGN OF THE HOT CELL BASED ON THE RADWASTE MANAGEMENT SCENARIO

The hot cell (Figure III.2.2) is designed based on the nuclear characteristic such as the dose rate and decay heat of the used IVCs. The used IVCs are treated as follows. In the disassembly area, the blanket modules and the divertor mono-blocks are removed from a back-plate and a divertor cassette body by the remote handling equipment after 7 years and 1.5 years of the cooling time in the temporal storage area, respectively. In this radiation environment, the disassembly area is estimated to be 100 Gy/h. The back plate and cassette body are moved to the reassembly building for reuse. The blanket modules and divertor plates are stored in the lower layer of the hot cell for 13 years and 5.5 years, respectively, until the surface dose rate decays to 10 Gy/h. Next, both the neutron multiplier and tritium breeder are recovered from each blanket module for recycling in the disassembling facility. Then, the empty blanket modules and divertor plates are cut into small pieces and stored in a container. After radwaste classification according to radioactivity, the radwaste is packed for disposal.

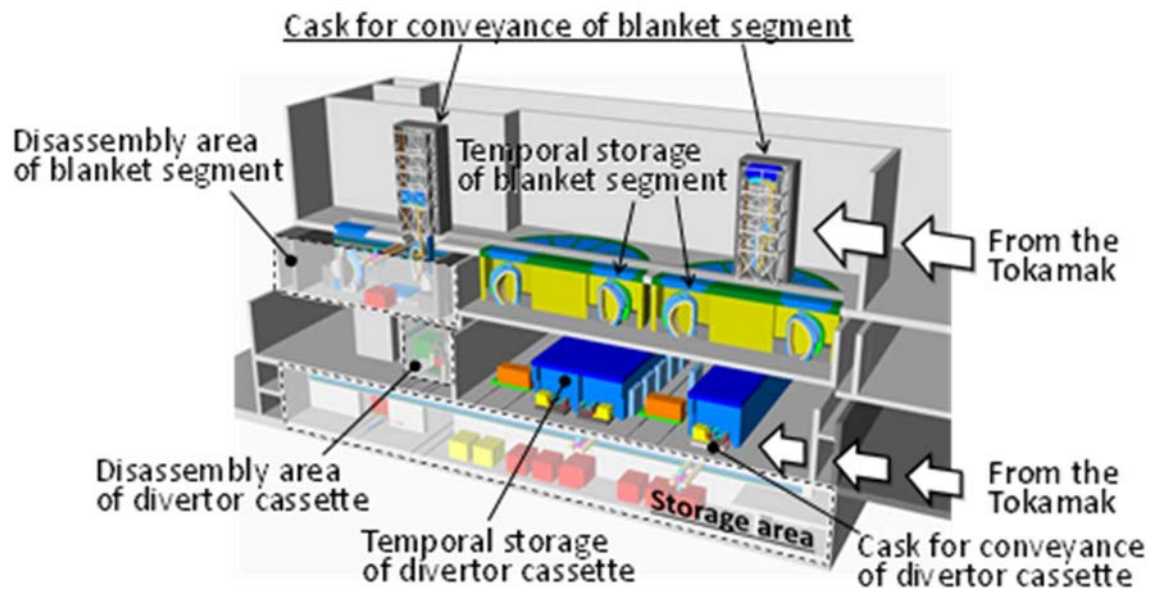


FIG. III-2.2 Conceptual design of Hot Cell for the JA DEMO.

CLASSIFICATION OF THE HIGHEST RADIOACTIVITY LEVEL GENERATED IN JA DEMO

The highest radioactivity level in the core components (which are blanket, divertor, shield, vacuum vessel (VV), magnet etc.) for the JA DEMO is estimated at F82H on the blanket first wall. Based on the results of nuclide transfer analysis in the disposal location, radioactive material of the blanket first wall can be disposed in a concrete pit (L2-level) with a depth of about 10 meters [3]. On the other hand, the biological shielding storing the core components was assessed to be at the clearance level based on the results of the activation calculation. Therefore, the radioactive material in the JA DEMO is limited to the core components. In addition, the uranium impurity in beryllium (which is a neutron multiplier loaded inside the blanket) has to be controlled within 0.85 wppm in order to fit within the L2-level. New purification processes have the ability to reduce uranium concentration in beryllium to allowable concentrations for L2-level (< 0.1 ppm) [4].

DECOMMISSIONING CONSIDERATIONS FOR JA DEMO

In order to reduce the amount of radioactivity at the time of decommissioning, it is essential to reduce the area of the building exposed to tritium as much as possible. In particular, it is important to handle tritium stored in the first wall for the IVCs. JA DEMO has adopted a policy of recovering tritium stored

in IVCs as much as possible in VV to facilitate handling during the transition to hot cells and hot cell storage. When replacing the IVC, assuming trouble such as getting stuck during movement, a one-month cooling period is secured in the JA DEMO so that the temperature of the IVCs does not rise even in such a trouble. During this cooling period before the replacing the IVCs, the decay heat is used to raise the temperature of the IVCs and especially recover tritium stored on the first wall of the IVCs [5].

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III-3. FUSION MATERIALS CONSIDERATION AT DESIGN STAGE TO SUPPORT DECOMMISSIONING

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An important distinction between fusion and fission involves the characteristics of radioactive waste products: activation level, quantity of waste, and final classification after decommissioning: land-based disposal, recycling, or clearance. To limit compounding the worldwide radioactive waste problem [1], fusion designers have developed a list of essential requirements for fusion materials that include: low-activation (to generate only low-level waste), radiation-resistance (to reduce the frequency of component replacement), good mechanical properties under 14 MeV neutron irradiation (to assure the structural integrity during operation), chemical compatibility with coolants and liquid breeders (to avoid the corrosion of structure), operation at high-temperature (to help enhance the thermal conversion efficiency and economics), and easy to fabricate complex configurations.

Regardless of the radioactivity level, all fusion materials (solids, liquids and gases) need to be treated and conditioned for further handling in a safe way during the decommissioning process with the main objective of permanent protection for humans and the environment. To avoid dealing with the highly radioactive by-products and to help reduce the disposal cost, the goal to generate only low-level waste was adopted by almost all fusion DEMOs and power plants. As such, reduced-activation steels, vanadium, tungsten and SiC/SiC composites are under development specifically for fusion applications. All fusion materials need to be carefully chosen to minimize (as much as practically possible) alloying with Al, N, Ni, C, Cu, Nb, Mo, Re, Ag, etc. that generate long-lived radionuclides, such as C-14, Ni-59, Nb-94, Mo-99, Re-186m, etc. Equally important, specific impurities (such as Nb, Mo, Ag, Re, etc.) need to be controlled to low level to help satisfy the low-level waste requirement [2]. Since most impurities are introduced during the fabrication process, manufacturers and suppliers of fusion materials in particular have to avoid introducing impurities during the production process, avoid using scraps, and rather employ highly pure *raw* materials.

Most DEMO and power plant designs developed to date employ the reduced-activation ferritic martensitic (RAFM) steel alloys – the most mature structure with extensive experience worldwide. Upon exposure to fusion neutrons, the steel becomes radioactive at different degrees, depending on the activation and decay characteristics of its constituents, the neutron flux level within the fusion design, and the service lifetime of individual components. Figure III.3.1 displays the activation of several steels – represented by the waste disposal rating (WDR) under ARIES-ACT2 operating conditions: outboard blanket; 1.55 MW/m² average neutron wall loading; ~10 y service lifetime [3]. Note that, according to the U.S. classifications of radwaste, the Japanese F82H steel and the European EUROFER97 steels could only generate Class C LLW (with WDR < 1) if the Nb and Mo impurities are reduced below 0.5 wppm and 5 wppm, respectively. Non-RAFM steels (SS316 of ITER and Inconel-718 and D9 of the ARC design) contain high Nb, Mo, and/or Ni alloying elements that would lead to a higher waste category than the Class C / LLW waste.

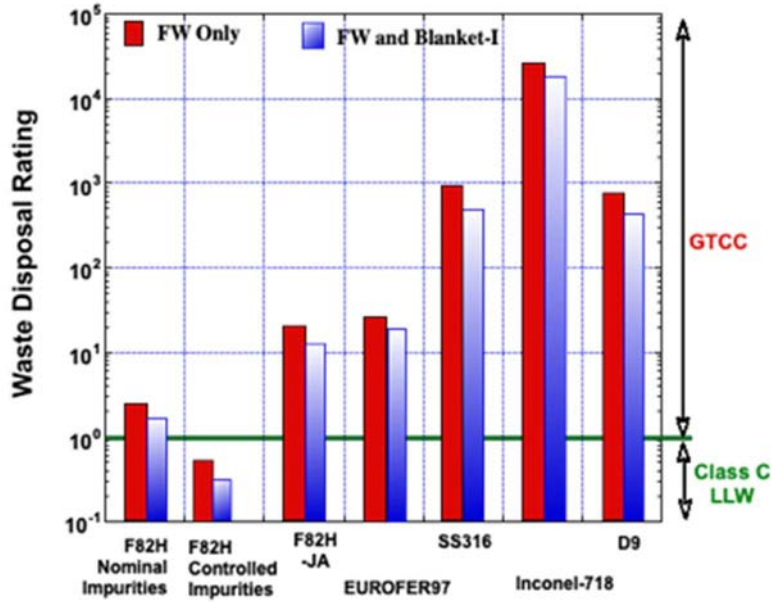


FIG. III-3.1. WDR of outboard ARIES-ACT2 first wall and blanket-I made of various steels (reproduced from [2]).

Another challenging material development issue relates to the divertor. At the present time, the best candidate structural material for the divertor is tungsten. Experts around the world developed several W alloys since the commercially available W-26Re fails to meet the Class C LLW requirement (Figure III-3.2) and is unsuitable for the challenging fusion environment. The W/W composites and W-TiC alloy are acceptable from the activation viewpoint and generate Class C LLW under the ARIES-ACT operating conditions [4].

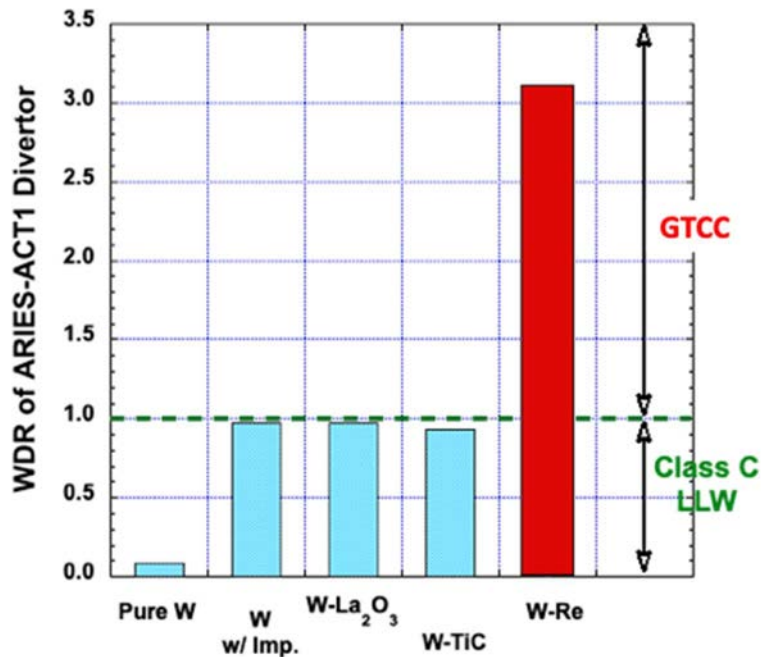


FIG. III-3.2. WDR of ARIES-ACT1 divertor made of various W alloys (reproduced from [4]).

Liquid and solid tritium breeding materials represent another source of large radioactive inventory (e.g. 17,000 tons of PbLi in ARIES-ACT2 [3]) that needs to be monitored closely for safe operation and decommissioning. Recycling of breeders has been investigated for containing lithium and beryllium that have limited resources. The flowing PbLi liquid breeder circulates in and out of the neutron radiation

zone in several separate flow paths, requiring time-dependent irradiation history to accurately estimate their activities and radioactive inventories. The primary radiological concern for PbLi is from its own radioactive by-products: Hg and Po [5]. It is essential to continuously filter and purify the PbLi online shortly after operation to avoid contaminating the ex-vessel sub-systems and assure safe operation. Ceramic (or solid) breeders require a significant amount of beryllium to satisfy the tritium breeding requirement [6]. As such, fusion power plants equipped with such blanket technologies will generate toxic waste from beryllium and lead. Beryllium is a rare element and ~200 times more toxic than lead. Unfortunately, to date, alternative neutron multipliers to Be and Pb have not been identified for D-T fusion blankets. Therefore, the chemical toxicity concerns necessitate the confinement of such materials during all operations to be done on these materials (from manufacturing, maintenance, up to waste management) and also during accidents. Moreover, the disposal strategy and facilities would have to accommodate the remaining Be and Pb toxic waste after breeder recycling.

While fusion was suggested to produce only Class C low-level waste (to avoid long term disposal and associated elevated handling and cost), its decommissioning is unique due to the large plant size and sizable tritiated materials [2]. A process that may take several years to complete. Procedures may include:

- Detailed timeline of the dismantling process (cutting, removal of components and equipment, and size reduction). The use of emerging technologies (artificial intelligence, advanced remote technologies (robotics), improved segmentation, and advanced techniques for cutting, volume reduction) is highly recommended;
- Segregation according to the type of radioactive materials (solid, liquid, gas);
- Classification for disposal, recycling, and/or clearance;
- Management of radioactive materials (preparation, characterization, packaging, transportation, licensing, and monitoring);
- Shipping to temporary storage or waste processing facility;
- Establishing “Citizens Advisory Panel” to address community concerns (such as radiation and tritium releases during decommissioning).

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III-4. EU-DEMO WASTE ASSESSMENT: SEVERITY, MASSES, AND ORIGIN – INFLUENCE OF IMPURITIES AND ALTERNATIVE MATERIALS

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Abstract

Accurate predictions of the severity and quantities of radioactive waste is an important part of the design, modelling, and engineering process of future fusion reactors. While fusion will not generate the same highly active spent fuel waste as fission, it will nonetheless create significant radioactive waste due to the interaction of neutrons with the materials of the reactor. For EU-DEMO, whole reactor waste assessment methodologies have been developed within the EUROfusion programme. The calculations involve performing detailed neutron transport (neutronics) calculations, followed by several thousand inventory simulations to predict the change in material composition (and hence activation) over time, both during reactor operation and afterwards, beyond end-of-life (EOL) due to radioactive decay. The inventory calculations are performed using FISPACT-II, a simulation platform developed at UKAEA over several decades and has been well validated for neutron-induced activation predictions. The activation-against-time profile for each material (component) is compared to waste classification limits (this work will focus primarily on the UK limits) to define the waste severity category. Here we will describe some of the highlights and key-issues from these calculations, showing that fusion reactors are likely to generate significant problematic waste streams that need to be mitigated by intelligent reactor design. We discuss some of the possible approaches to reducing the waste severity, including compositional adjustment, and waste extraction of impurities and problematic isotopes prior to storage.

INTRODUCTION

For the last several years, computational waste assessments of the conceptual designs for EU-DEMO have been performed by UKAEA as part of contributions towards the EUROfusion consortium work programme. These predictions of the likely waste severities and volumes arising from the future operation of DEMO are a critical part of the planning and design. As well as providing the inputs to the planning of decommissioning and waste handling strategies, the assessments provide feedback to where the design needs refinement in order to prevent or reduce some aspects of the radioactive waste that will unavoidably be generated during operation of any fusion reactor due to the interaction of neutrons with materials. Not only prevention and minimization of waste are at the top of the accepted waste hierarchy (i.e. prevent → minimize → reuse/recycle → dispose, from best to worst), but it would also allow fusion to maintain its claim as a ‘clean’ energy source.

The deuterium-tritium (D-T) reaction is the most common pathway being explored in fusion energy research and development (R&D). This reaction produces high energy (14.1 MeV) fast neutrons that move off in all directions from their point of generation (e.g. within the burning plasma of tokamak-style reactors like EU-DEMO) and interact, slow-down, and deposit heat within the surrounding containment structures and materials. These interactions allow the fusion energy to be recovered and then used to generate heat and electricity, but the nuclear reactions also cause changes to the isotopic composition of the irradiated materials. When radioisotopes are created in this process the material becomes ‘activated’. At the end of life or at the point of disposal, the amount of residual activation determines how waste material is treated, classified, and handled.

Classification of radioactive waste typically follows the three broad categories defined by the IAEA: high-level waste (HLW), intermediate-level waste (ILW) and low-level waste (LLW). In some countries / territories these classes are further subdivided, but for the purposes of judging whether a fusion reactor design meets the low waste severity and volume goal, these three are sufficient. The exact definition of each class also varies depending on the types of waste disposal facility available in each territory, which, particularly in Europe, makes it challenging to fully assess waste severity of a DEMO concept because its eventual location is unknown. However, previous work performed comparisons between different classification systems in Europe and worldwide, showing that the overall qualitative distribution of LLW, ILW, etc., does not vary significantly between repository type [1],[2]. Thus, for the present work, we consider only the UK limits for LLW because of its particular relevance to UKAEA and future fusion systems in the UK. In this system LLW is waste with alpha-particle-induced activity of less than 4 MBq/kg and combined beta/gamma activity of less than 12 MBq/kg. All waste above this limit is considered to be ILW at the point of disposal because it is reasonable to assume that any heat-generating material requiring cooling, which would be classified as HLW if sent for disposal, will be held on site for as long as required until no cooling is required – this is a key difference between fusion and fission as in the latter case the cooling times are hundreds or even thousands of years, while in fusion only a few years of cooling are likely to be needed (mostly due to the activation of tungsten [3]).

This paper briefly describes the methodology applied to performing waste assessments of conceptual EU-DEMO designs and then presents some of the latest results.

METHODOLOGY

The computational framework for EU-DEMO waste assessments starts from a reactor concept geometry that has been converted into the form required by a neutron transport simulation code. For the standard codes used for such simulations, including the stochastic Monte-Carlo code MCNP [4], [5] used in the studies performed at UKAEA, this entails converting a CAD geometry or similar into one in which the reactor regions or components are specified using volumes enclosed (defined) using mathematically describable surfaces, such as planes, spheres, cylinders, tori, etc. These volumes are then filled using homogeneous mixtures of the materials specified in the concept design. FIG. III-4.1 shows the cross section of the EU-DEMO design used for the results presented here; the 2017 baseline [6].

Figure III-4.1 also shows a total neutron flux map on a fine mesh-scale produced from the transport simulations with MCNP. The trajectories of 10^{10} 14.1 MeV source neutrons were simulated to produce this map. As well as the total flux in each region, the simulations also provide the energy distribution of neutrons corresponding to this flux. These energy spectra and total fluxes, along with the material compositions and reactor expected operational scenario²⁷, are the inputs required for the inventory simulations used to predict activation and hence waste classes.

²⁷ The operational scenario assumed for EU-DEMO is 30% availability during an approximately 22-year lifetime. The divertor is assumed to be replaced every 5 years and the tritium breeding blankets once, after the first 5 years of operation.

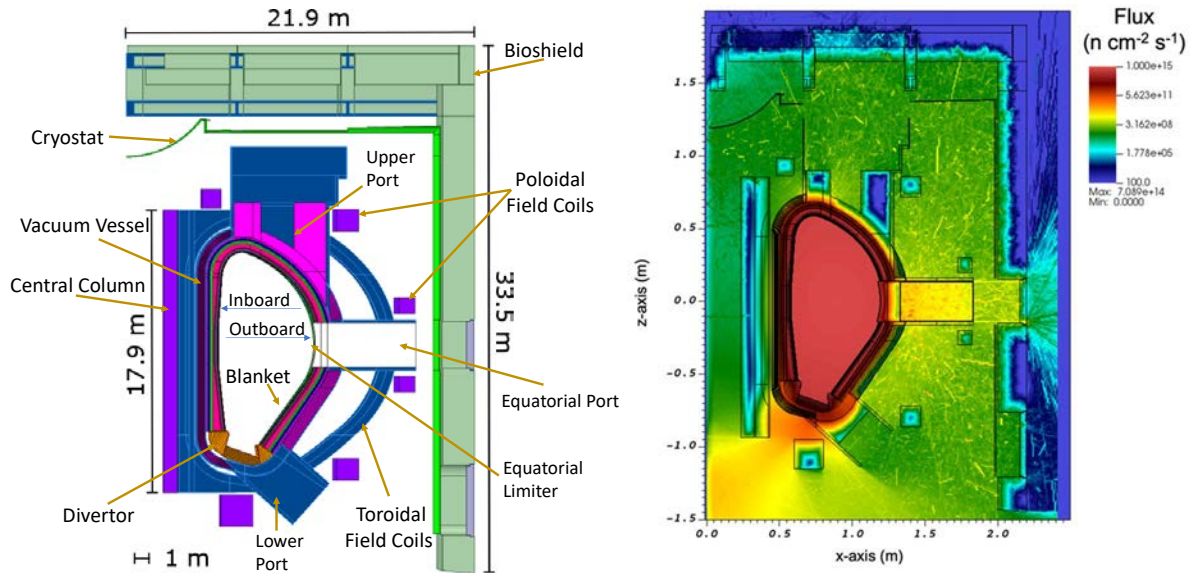


FIG. III-4.1: Left: toroidal cross section through the EU-DEMO neutron transport geometry; Right: neutron flux map obtained from transport simulations with MCNP. The neutron flux units are neutrons/cm²/s.

FISPACT-II inventory code

UKAEA uses FISPACT-II to perform transmutation and activation calculations. FISPACT-II is a simulation platform developed at UKAEA to predict the change in composition of materials exposed to irradiation. The code evolved from one originally developed for fission applications to FISPACT, which was maintained and improved during the 1990s and 2000s, before being refactored and modernized to FISPACT-II in the last decade. Written in modern-style Fortran and extensively validated, FISPACT-II is recognised as a standard tool for the prediction of transmutation and activation in neutron irradiated materials.

For a specific scenario of neutron environment, material and irradiation scenario, FISPACT-II combines energy dependent nuclear reaction cross sections with the neutron energy spectrum in the same, binned group energy-bin structure to define the effective total cross section s for each reaction. The reaction rate is then sf , where f is the total neutron flux. These reactions rates are then used in a set of coupled, ordinary differential (linear) equations – one equation for each possible isotope or nuclide – that include decay constants λ for radionuclides and which define the rate of change of concentration of nuclides in a material. FISPACT-II combines these ODEs in a matrix and solves numerically for each irradiation period (see [7] for more details).

FISPACT-II is continuously evolving and improving. For example, the latest release (version 5.0) now includes an Application Programming Interface (API), which allows a user to write integrated code to automatically run FISPACT-II and process results without the need for the traditional input/output file-write overheads. This API was employed for the results presented in this paper, resulting in significant reduction in computational time (although the majority of CPU time is spent on the transport simulations with MCNP).

The EU-DEMO waste assessment results shown here used TENDL-2017 [8] nuclear cross section data, which has been validated for fusion applications with FISPACT-II by UKAEA within the European programme [9][10].

WASTE ASSESSMENT RESULTS

For each material composition in each region of the reactor geometry the FISPACT-II results provided not only the evolution in activation during the EU-DEMO operational scenario, but also afterwards, due to radioactive decay alone and for up to 1000 years. At each point in time, the predicted activity was

compared to the UK-LLW limit to define whether the material would be classed as ILW or LLW at that time if sent for disposal. Activity was also compared to IAEA clearance levels to determine if the material could be considered as non-active or out-of-scope and thus not classed as radioactive, but those findings are not discussed here as they are typically only relevant for ex-vessel components and this does not alter the overall message concerning waste volumes and masses from the bulk of the radiation environment within EU-DEMO (and are also too conservative as they do not currently account for the likely tritium contamination that will be present throughout a wider fusion plant).

UKAEA has developed a number of different approaches to analysing and presenting the waste assessment data for EU-DEMO (see [1], 2, 11]). FIG. III-4.2 shows a typical example, where the complete activation response of one divertor, its constituent materials, and their dominant radionuclides is detailed. The top-left panel of the figure highlights that EUROFER is predicted to be the main contributor to the total average activity per unit mass in the divertor for several decades beyond DEMO end-of-life (EOL), before copper and copper alloys take-over. The combined contributions from these two materials – one, EUROFER, present in large quantities as the bulk structural material and the other, copper alloy, present in small amounts in cooling channels and pipes – leads to a prediction that the whole divertor will remain above the UK-LLW limit for at least 1000 years. In EUROFER, the top-right panel shows that the cause of the long-lived activity is ^{14}C (half-life of 5400 years), which is produced from (n,p) reactions in nitrogen [1]. In the copper-alloys, the activity is entirely dominated by ^{63}Ni , which has a half-life of only 100 years but is produced so readily in Cu that it is still present in significant concentrations at 1000 years. In reality it is reasonable to expect that the Cu-alloy pipes can be separated from the rest of the divertor (they are only mounted using small pins in the current design), and so while those few 100s of kg of piping is likely to remain ILW for more than a thousand years, processing of the remaining divertor materials – for example, decarburization of EUROFER to remove ^{14}C – is likely to allow much of that to be LLW-classifiable within 100-200 years (see the EUROFER plot in the figure).

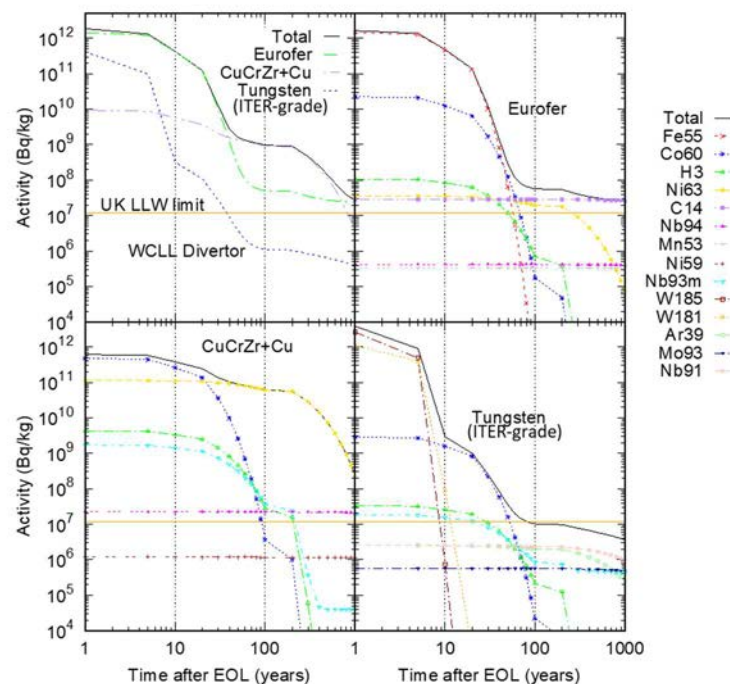


FIG. III-4.2: Activation profiles for the divertor of EU-DEMO²⁸.

²⁸ The upper left panel shows the relative contribution per unit mass to total activation from each material in the conceptual divertor design. The other panels use the nuclide legend on the right of the figure and show the activation profiles for each of

Waste mitigation

Over a number of years and iterations, the EU-DEMO programme as developed a number of mitigation assumptions for radioactive waste in response to the assessments produced by UKAEA. These include:

- Reduction of carbon in steels via decarburization to 1 weight parts per million (wppm) after operation to mitigate the radioactive caused by ^{14}C . Di Pace et al. [12] have developed and tested a system involving the use of oxygen blown over molten steel to remove carbon as CO gas before its conversion (via CO_2) and capture as calcium carbonate (CaCO_3). Despite the relative immaturity of the technique, it offers a potential solution, which could be extended to other impurities (such as Nb) and be developed further.
- 99% removal of any tritium in materials to reduce any disposal delay that might occur otherwise due to the decay of ^3H with its 12.3-year half-life. Tritium recovery will be an important part of the fuel cycle in fusion power plants to ensure self-sufficiency. While removal from waste after EOL is unlikely to contribute to the self-sufficiency, the detritiation techniques developed to support the refuelling are likely to provide solutions to maximize removal of ^3H from waste.
- Refinement of compositions of key fusion materials to include more optimistic / realistic levels of certain impurities that are otherwise predicted to create a long-lived waste problem. For EUROFER steel this includes reduction of Nb in the starting composition from 50 to 20 wppm, which is achievable since Nb is not a required element in the composition of the steel. Meanwhile, in beryllium, U content is considered to be 5 wppm or less to avoid the extreme alpha-activity that would be caused by higher U content; 100-150 wppm was the starting position based on standard Be grades, but the 5 wppm level is achievable with additional processing of Be during refinement from its ores. Selection of alternative materials. Previous work highlighted that the vacuum vessel (VV) of EU-DEMO could be one of the key sources of long-lived radioactive waste due to the design assumption of using 316 stainless steel. This austenitic steel as high nickel content (12.5% by weight) leading to the production of long-lived isotopes such as ^{63}Ni (100-year half-life) and ^{59}Ni (76000 years). Recognising that only the safety critical shell of the VV needs to be SS316 (due to its high standard of safety qualification and known performance characteristics), a design iteration replaced significant volumes of 316 in the VV with alternative steel grades including (and most significantly) reduced-activation ferritic-martensitic (RA/FM) EUROFER. This resulted in a factor 3 reduction in ILW from the VV at 100 years EOL [13]. FIG. III-4.3 shows a cross section through the DEMO geometry colour-coded according to when materials in some regions could be disposed of as LLW. The figure also shows (right image) the evolution in time in total mass of material classified as ILW – the histogram bars are subdivided into each component of the vessel as indicated by the legend colours (the listed components are further sub-divided using different patterns with the same base colour). The results in this figure include the mitigations described above – i.e. carbon removal, detritiation, initial composition refinement, and use of low-activation steels.

the materials in the divertor, showing how different radionuclides contribute to the activity from those materials. The results are the average response across all regions of the divertor.

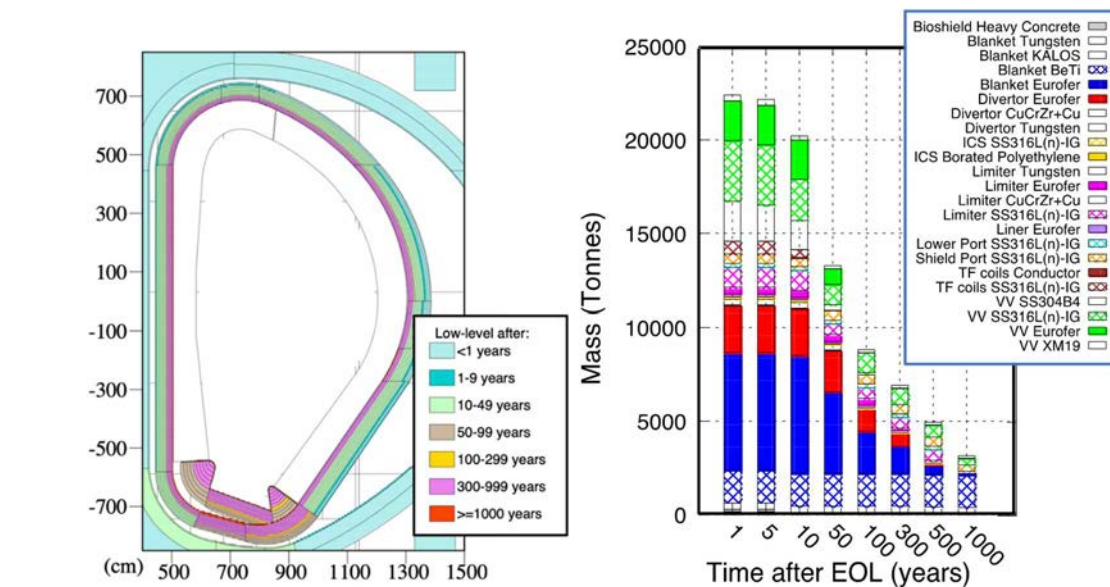


FIG. III-4.3: Left: cross section of EU-DEMO with the vacuum vessel, divertor, and ex-vessel regions (except the tritium breeding blanket)²⁹

FIG. III-4.3 demonstrates that, despite the mitigation approaches described above, there are still issues facing DEMO design: parts of the vacuum vessel, divertor and blanket are predicted to remain active enough to be classified as ILW for 100s of years beyond EOL, with some regions, particularly in the blanket, remaining at ILW activity levels for more than 1000 years. Combined with the relatively large mass and volume of the typical EU-DEMO design (and other fusion reactors), this results in a potential ILW legacy of several 1000 tonnes for centuries after operation (note that the results in the right image of figure III-4-3 include the mass produced from the single and triple replacements, respectively, of the blanket and divertor assumed for the current EU-DEMO lifetime plan).

DISCUSSION

Although there are still predicted to be waste disposal and decommissioning challenges for EU-DEMO, it is heartening to observe that fusion is developing approaches to prevent, mitigate and reduce waste even at the early conceptual design stage of commercial and near-commercial power plants. Fusion may not be able to live up to its billing as a truly clean energy source but considering waste handling at an early stage will enable the industry to convince governments, regulators, and the general public that, in contrast to the lessons from fission, fusion is tackling the waste problem up-front so as to significantly reduce the burden on future generations.

Fusion proponents need also to begin to discuss the relative risks associated with the waste that is predicted for the next generation of plants. While strict adherence to the activity limits associated with repositories and classification systems around the world paints a pessimistic picture, it is important to challenge the appropriateness of these limits since they may not be fit for purpose. For example, a consolidated block of steel from a fusion power plant might be classified as UK-ILW because its ¹⁴C concentration above the LLW limit by a factor of 2-3, but the radioactive carbon is static and embedded within the steel matrix. Such “ILW” poses a significantly lower risk to the environment than an equivalent mass of fission waste contaminated with spent fuel and other fission fragments, whose activity may be much longer-lived and orders of magnitude above the LLW limit. Current regulations in many countries,

²⁹ Coloured according to the time after end-of-life (EOL) after which the material in those regions could be disposed of as LLW. Right: Evolution in mass of UK-ILW classified material after EU-DEMO operation (beyond EOL). Note that the solid shading always refers to EUROFER, demonstrating its significant contribution to ILW.

including the UK, do not make any distinction between such waste and do not account for the relative differences in risk.

Figure III-4.4 illustrates the potential (positive) impact on the fusion waste landscape that a risk-based approach to classification and handling could have. It compares, for an earlier, 2015 EU-DEMO baseline, the theoretical waste classification of the steel (predominantly either RA/FM EUROFER or austenitic SS316) in each component under UK regulations to those for one of the near surface repositories in Japan. Japan's near surface repository for LLW [14] has higher limits for some of fusion's problematic radionuclides, including ^{14}C and ^{63}Ni (and contains no specific limit on ^{94}Nb – a nuclide that is predicted to prevent LLW classification of EUROFER in many European systems [1]), partially due to the characteristics of the 'trench' disposal approach that reduces the risk posed by radioactive waste. A future near-surface disposal (NSD) facility in the UK, which is currently being discussed, could change the picture here as well – only at-surface LLW and geological disposal is considered in the waste assessments.

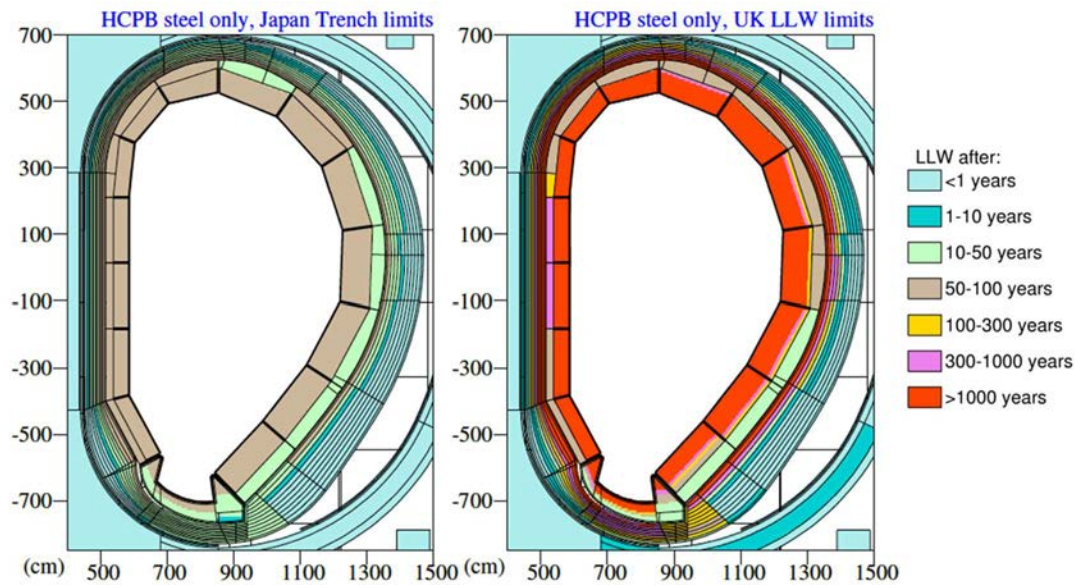


FIG. III-4.4: EU-DEMO (2015 baseline) toroidal slices coloured according to the time after which the steel in each component could be disposed of as LLW in Left: a near surface repository in Japan, and Right: in UK LLW repositories.

SUMMARY

This paper has presented a brief summary of the waste assessment predictions performed for the EU-DEMO fusion power plant concept. Even with various waste mitigation approaches and refinement of the assumptions around material compositions and choices, there are still predicted to be several 1000 tonnes of radioactive waste that would be difficult to dispose of without a deep geological facility (the requirement for ILW in most countries) for 100s or even thousands of years. However, fusion still has time to solve the waste issues, either by continued research and development into novel materials and waste treatment approaches, or by challenging the accepted view of risk around certain types of solid material structural (mainly steel) waste that, in contrast to fission which generates a severe spent fuel legacy, are the only major sources of radioactive waste that fusion expects to generate.

ACKNOWLEDGEMENTS

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number EP/T012250/1]. To obtain further information on the data and models underlying this paper please contact PublicationsManager@ukaea.uk.

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III-5. ITER RADIOACTIVE WASTE INVENTORY ASSESSMENT, ASSOCIATED CHALLENGES AND FEEDBACK FOR FUTURE FUSION FACILITIES

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Abstract

ITER Organization (IO) has developed a methodology to estimate the amount and characteristics of radioactive waste to be generated during operation and decommissioning. Radioactive waste will be generated by the replacement of component and housekeeping operations in all ITER nuclear buildings: tokamak, tritium plant and hot cell. This radioactive waste can be highly activated requiring remote handling treatment and also highly contaminated with dust and tritium. First of a kind treatment in hot cell and tritium decay facilities have to be developed to ensure the possibility for disposal of these waste in France. Feedback for the design of future fusion facilities can be already provided.

INTRODUCTION

ITER will produce radioactive waste during its operation (arising from the replacement of components and from process and housekeeping waste) and during decommissioning. The radioactive waste will be generated in the nuclear buildings inside the ITER facility (Figure III-5.1):

- The tokamak building and the tritium building, located in the tokamak complex; the tokamak building houses the tokamak machine; the tritium building mainly houses the fuel recycling system.
- The hot cell complex, in particular intended for in-vessel equipment maintenance activities, the treatment and storage of intermediate-level long-lived radioactive waste (ILLW) and purely tritiated waste, the treatment and export of Low-level and very low level solid and liquid radwaste from process and housekeeping operations. The Low level radioactive waste treatment building which is integrated in Hot Cell Complex.

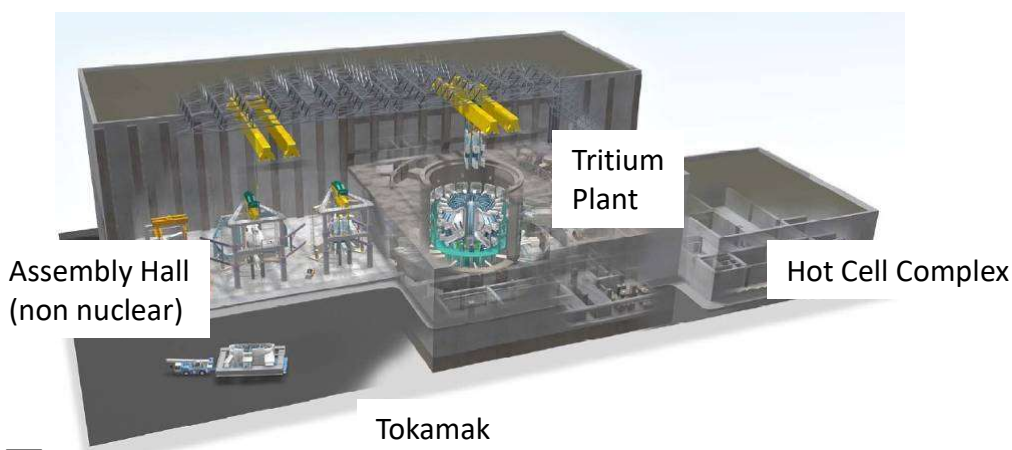


FIG. III-5.1. Overview of main ITER buildings.

During active operation, the walls of the tokamak will be heated by plasma, irradiated by the neutron fluxes generated by the fusion reactions and subjected to high concentrations of tritium. These materials, mainly of metallic type, will be loaded in tritium and can be strongly activated according to the nature and the purity of the materials.

The two phenomena related to radioactive waste producing activity are:

- Activation: The deuterium-tritium reaction generates 14.1 MeV neutrons whose interaction with all the materials surrounding the plasma causes the progressive activation of these materials. Activation levels decrease for the components located further away from the plasma.
- Contamination: Half of the fuel being tritium, it is found progressively trapped (phenomena of permeation, recombination, implantation etc) in the materials surrounding the plasma. Tritium is also present in the waste generated by the fuel processing equipment.

Dust particles are produced inside the vacuum vessel by various complex plasma–surface-interaction mechanisms. This dust becomes activated and tritiated which gives rise to surface contamination that propagates inside the tokamak during maintenance operations.

METHODOLOGY TO ESTIMATE THE AMOUNT AND CHARACTERISTICS OF RADIOACTIVE WASTE TO BE GENERATED DURING OPERATION AND DECOMMISSIONING

As the maturity of ITER nuclear buildings systems evolves, the maintenance requirements are refined more accurately which enables to assess more precisely the number of replacements or failures to be expected. During decommissioning all equipment located in rooms classified as nuclear waste zones have to be accounted for, including all piping, ventilation ducts and cable trays.

This can challenge the conservativeness of the early estimates performed at the early design phase and licensing for the creation of ITER basic nuclear facility. For this reason, ITER Organization (IO) has developed a methodology to estimate the amount and characteristics of radioactive waste to be generated during operation and decommissioning.

The goal is to provide up to date waste inventories and to identify specific materials that may not be accepted by the French Repository (ANDRA):

- Reactive waste will require specific passivation treatment (e.g. LiH, LiPb, Depleted Uranium);
- Fission chambers or radioactive sources require specific waste routing;
- Electronic waste, rare earths, etc.;
- Tritiated equipment that are not compatible with the acceptance limits without several decades of decay.

Each system reaching Preliminary Design phase has to provide a list of radioactive waste and characteristics. The template to fill is based on the systems bill of material for equipment located rooms classified as Nuclear Waste Zones. According to French regulations [1] a room is to be classified as Nuclear Waste Zone in case of added radioactivity (activation and contamination) without considering a minimum release limit. This zoning is defined as an input by the safety teams after analysis of environmental conditions and incidental scenarios. For each equipment part, it is requested to provide:

- Mass and size;
- Useful volume;
- Material, composition;
- Specific materials or chemicals identification;
- Existing radiological data during ITER lifetime (from first plasma to end of life);
- Activation;
- Contamination;
- Tritium content;
- Dose rate;
- Number of replacements or failures to be expected (zero means only decommissioning waste);
- Means of replacement (hands-on or using remote handling).

TABLE III-5.1. EXAMPLE OF RADIOACTIVE WASTE ASSESSMENT TRACKING TABLE

PBS	ZDN room (see 4.1)	Equipment part (see 4.1)	Mass of component (see 4.3)	Volume of component (see 4.3)	Material (See 4.4)	Composition (See 4.4)	Specific chemical Yes or no (if yes list and quantities) (See 4.4)	Form (solid, liquid,...) (See 4.4)	IDM reference for description of equipment	Radiological data (See 4.5 and appendix A)	Reference for radiological data	Number of replacement (see 4.6)	Means of replacement (see 4.7)	Transfer container (See 4.8)	Additional data
Xx	xx-L1-yy	Pipe xx	Xx tons estimated as...	Diam = xx Length = yy Thickness = zz	SS	Iter grade SS see ITER_D_...	no	solid	See ITER_D_...	Little activation But Highly contaminated No data	Radiological map ITER_D_ + component in red zone	0 (decommissioning waste)	TBD as attached to the structure	To be designed	
Xx	xx-L1-yy	Filter case	Xx g				Pb for shielding	Solid	BoM ITER_D_... DDD ITER_D_... Load spec... ITER D ...	ACP = xx Bq/g T = xx Bq/g	Estimation based on ...	3 during scheduled maintenance	Hands on replacement	200 L drum	
Xx	xx-L1-zz	filter	Xx tons		SS and paper			Solid			Activation calculation ITER_D_...	expected failure <1	RH class 3	Specific container described in ITER D ...	
yy	No equipment in ZDN room	/	/	/	/	/	/	/	/	/	/	/	/	/	

A re-estimation is to be performed at each design review or when introducing a change request. This can be the opportunity to study possible relocation of components to minimize the waste amounts, change of materials to limit activation or reduce toxicity. Similar approach is performed for liquid effluents. All this information needs to be placed under configuration control.

ITER WASTE INVENTORY REFINEMENT

ITER Organization (IO) is performing re-estimation of waste amounts to be generated during operation and decommissioning, considering the ramp up of activation from first plasma to full Deuterium Tritium Plasma considering the staged construction of the facility. The current waste types are described hereafter.

Intermediate-level and long-lived radioactive waste (ILW-LL) amounts and characteristics are relatively stable as the associated maintenance and replacement strategy is fixed by high level project requirements. The following large scale in-vessel components will become activated and tritiated leading to around 3500 tons of ILW-LL waste to be produced during operation and decommissioning phase [2]:

- Divertors (1 full replacement campaign during operation) (see Figure III-5.2);
- Blanket Modules First Walls (1 full replacement campaign during operation) (see Figure III-5.3);
- Blanket Modules Shield Blocks;
- All Port Plugs front part.

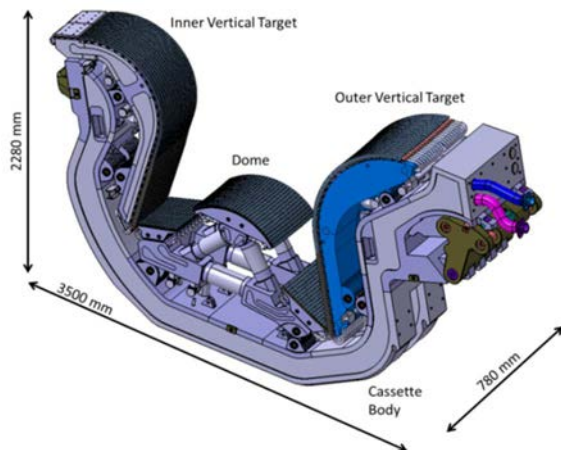


FIG. III-5.2. Divertor.

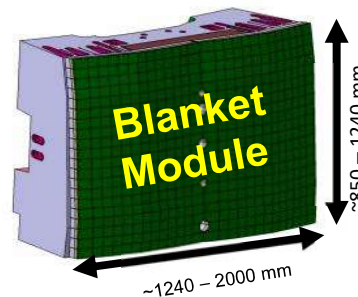


FIG. III-5.3. Blanket Modules.

Main neutron activation isotopes contributing to the ILW-LL classification are: ^3H , ^{60}Co , ^{10}Be , ^{63}Ni and ^{54}Nb . This was determined using 3D calculations based on activation at the end of the operation. A part of tritium permeation is also added. The waste classification will not be challenged by refinement of materials specifications during fabrication by the suppliers.

TABLE III-5.2. MAIN RADIONUCLIDES ACTIVITY OF ITER COMPONENTS COMPARED TO CSA REPOSITORY LIMITS (LMA), AT SHUTDOWN AND AFTER 50 YEARS DECAY (IN BOLD WHEN HIGHER)

CSA LMA (Bq/g)	Blanket Modules				Divertor				VV Dust	
	First Wall		Shield Block		First Wall		Cassette body		VV Dust	
	SD	50 yrs	SD	50 yrs	SD	50 yrs	SD	50 yrs	SD	
³ H	2.00E+05	4.61E+08	2.78E+07	1.08E+06	6.52E+04	4.66E+08	2.81E+07	1.01E+06	6.07E+04	3.41E+09
¹⁰ Be	5.10E+03	6.18E+04	6.18E+04	1.50E-02	1.50E-02	8.14E-03	8.14E-03	8.56E-04	8.56E-04	2.86E+02
⁶⁰ Co	1.30E+08	3.31E+08	4.65E+05	4.22E+07	5.93E+04	3.25E+07	4.56E+04	4.36E+06	6.12E+03	4.64E+05
⁶³ Ni	3.20E+06	4.55E+07	3.21E+07	4.91E+06	3.46E+06	4.17E+06	2.94E+06	7.39E+05	5.21E+05	1.20E+03
⁹⁴ Nb	1.20E+02	4.08E+03	4.07E+03	2.91E+03	2.91E+03	1.70E+03	1.70E+03	1.97E+02	1.97E+02	2.68E+01

Components further away from the plasma will be classified as Low-Level Waste as less exposed to neutron radiations but still with important dose rate of the back of port plugs: Equipment in Port Cells are located beyond bioshield plug have dose rate that needs to be lower than 100 microSv/h after few days of decay. Equipment in the Tokamak Galleries are much less activated but still to be considered as very low level waste. All these equipment can be contaminated at a lower extent to tritium and dust in case of breach of confinement during their maintenance / replacement. An effort is being performed with Host Country to refine the total amount of radioactive waste based on the methodology described previously.

A total of ~40000 tons of LIL-SL and ~20000 tons of VLL-SL waste can be considered conservatively for all operation and decommissioning: the figures will be reduced as some systems have not reached their final design yet. Estimation of the tritium permeation is also being performed. Mass and dimensions can also be important for components such as cryopumps showed in next table.

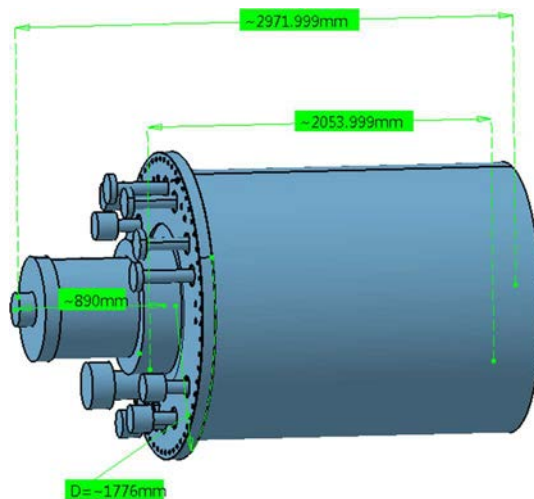


FIG. III-5.4: Cryopump weights about 7.4 tons and will be classified as LILW-SL.

Radioactive liquid effluents (~185 m³/y) are generated by the operation and maintenance of the Tokamak Cooling Water System: this effluent is contaminated with Activated Corrosion Products (ACPs) and tritium. In case of in-vessel Loss of Coolant Accident, the effluent will be also charged with the in-vessel dust source term inventory. Maintenance and decommissioning of the tokamak cooling water circuit will generate metallic waste also contaminated with ACPs Activated Corrosion Products (ACPs) and tritium that permeates the material (filters valves, pumps etc). Homogeneous waste such as resins and oils will also be generated.

It has also to be added ~1000 tons of purely tritiated waste arising from the tritium plant processes (i.e. only highly contaminated with tritium) for which at least 50 years decay or specific treatment is mandatory before reaching ANDRA CSA acceptance limits. Plastic housekeeping waste have also to be accounted for as purely tritiated waste. The tritium source term of these waste is being refined by IO, considering the relevant feedback from tritium facilities worldwide [3].

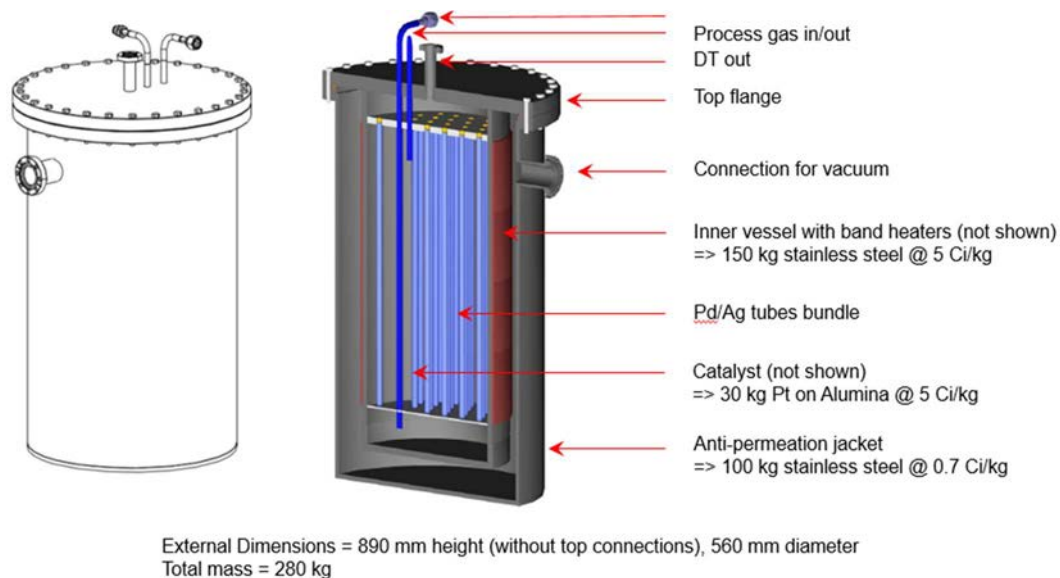


FIG. III-5.5 Fuel cycle equipment (getter bed) contaminated with tritium.

During operation, the main source of radioactive waste (185 m³/y) will be the secondary radioactive waste from housekeeping operations (plastic protections, clothes, gloves etc) [4]. IO is refining a re-estimation of the amount of housekeeping plastic waste considering the recent development of the tokamak maintenance strategy based on number of operators, type of interventions and type of protective equipment used. The tritium source term is also being refined using simulations of the permeation into the plastics as it needs to be lower than for tritium facilities.

CHALLENGES FOR ITER WASTE MANAGEMENT

The goal is for IO to ensure existing options for routing to the final French repository all foreseen radioactive waste to be generated during operation and decommissioning. Identifying proper radioactive waste inventories is mandatory to design the ITER hot cell and radwaste facilities that will be used for the treatment and interim storage of the waste, in order to demonstrate that all the ITER waste will be safely manageable with the existing outlets. This demonstration also needs to be provided to the French regulator.

Design, construction and operation of such radioactive waste facilities can represent one of the main investment items of a fusion facility. The functions need to be available depending on the ramp up of activation to be expected following the research plan.

It is emphasized that the nominal waste flow will not be generated from the start of plasma operations. The analysis of the activation of the in-vessel components show added radioactivity can be progressively measured, requiring nuclear waste zoning. Thus, the resulting waste need not be routed to waste disposals without proper administrative precautions and confirmation of their radioactive classification.

Once the key Deuterium Tritium nuclear plasma experiments will have been performed, the maintenance of the in-vessel components will generate radioactive waste that are too big to be transported in off the

shelf shielded transport containers. The size of components such as divertor and port plugs is too big to rely on conventional hot cells with off the shelf through the wall manipulators. Access to all sides of these components and tilting being required, dedicated heavy shielded robots have to be developed for the handling of such waste. This results in the ITER hot cell being a first of a kind with largest nuclear shielded red zones. Such large volumes require a specific air treatment system to collect the tritium outgassing from the waste as well as inerting due to the explosive atmosphere linked to the presence of dust and hydrogen isotopes.

Regarding the in-vessel waste (ILW-LL) treatment processes themselves, maximum tritium activity is estimated to be high thus reduction of the tritium content and outgassing of the waste is mandatory in addition to a maximal 50-year decay period. Feasibility of efficient reduction of the tritium source term in metallic waste has been demonstrated at laboratory's scale [5]. One of the challenges for ITER is now to develop a large scale thermal oven to be operated remotely after volume reduction of the waste. Volume reduction is mandatory to optimize the packing ratio and limit the storage space required in hot cell as further decay will be required to reach the drastic tritium limits for deep underground disposal.

Characterization of waste generated in fusion facilities requires proper attention as some activation products may be difficult to measure with off the shelf spectrometers. Tritium inventory is also challenging due to the large amounts put in place in the fuel cycle mass balance compared to the precision required for waste traceability and comparison with disposal limits. This implies taking frequently consequent number of representative samples and managing associated measurement in laboratories. In the case of in-vessel components, due to the dose rate, such sampling and analysis have to be performed in shielded cells.

Managing housekeeping waste generated during plant maintenance is also challenging due to potential high level of tritium contamination. Outsourcing the treatment of such waste to external melting or incineration facilities in France is not compatible with their releases authorizations. As decay storage facilities are not available and could represent an important financial effort to ongoing and future fusion projects, different options are being investigated to reduce the tritium inventory considering isotopic exchange, heating or melting.

The reactivity of dust with cement requires development of specific conditioning matrices to immobilize the waste prior to disposal. Addition of tritium trapping materials in the cement is also investigated in order to reduce the outgassing and releases to the environment at the disposal site. In case of wet waste such as resins, ITER Organization is investigating the possibility to use formulas compatible with existing matrices developed for nuclear fission industry.

Chemical stability is also analyzed for tritium fuel cycle waste that are new to waste disposals designed for fission plants radioactive waste. For these waste specific discussions will also be performed with ANDRA on the acceptance conditions based on their physical form as they contain materials that may be reactive with water such as depleted uranium, zeolites or Pd/Ag membranes.

FEEDBACK FOR FUTURE FUSION FACILITIES

The size of in-vessel components that will become activated is one of the biggest challenges related to fusion energy radwaste management as it requires large hot cells and customized remote handling tools. It is recommended to investigate the feasibility to dismount into smaller size sub-components that could be transported more easily for offsite treatment.

Establishing the overview of the overall radioactive waste flow that is in line with the maintenance strategy has to be performed from the start of the design and frequently reassessed. Secondary housekeeping waste generated in nuclear buildings supporting the fuel cycle or maintenance may not be neglected as they will represent the main waste flow starting from the beginning of operations. Waste that could be generated in case of incident also needs to be assessed. It is recommended to interface early with waste repositories and treatment service facilities in order to identify all routing options and start filling the missing gaps.

The use of new materials may lead to the development of specific waste immobilization matrices. This development may also take several years and could be a blocking point for waste disposal if not anticipated.

Provisions have to be taken before the start of nuclear operations with the design of the ITER facilities that will be used for the treatment and interim storage of the waste, in order to demonstrate that all the ITER waste will be safely manageable with the existing outlets. This demonstration also needs to be provided to the French regulator.

When performing activation calculations to assess the waste classification and dose rate constraints for treatment, it is important to consider not only end-of-life build-up but also the ramp-up corresponding to the experimental program for the components replacement due to maintenance or failure during operations: consider if exceeding LLW limits can be avoided by early replacement. New nuclear activation calculations have to be performed if the selected materials grade or components geometry evolve due to design improvements or refinements or fabrication: change in impurities contents can lead to different results in waste classification.

Modelling the tritium permeation into materials and spreading of contamination outside the vacuum vessel during maintenance is also a key item to establish waste classification, treatment and disposal options.

During operations, tracking of events and accidental contaminations and frequent reassessment of activation scenarios is mandatory to identify possible evolutions of the design parameters that enabled to establish radwaste production forecasts.

CONCLUSIONS

ITER radioactive waste management during operation and decommissioning has been identified from the beginning of the project as a key engineering issue requiring dedicated resources. Waste inventory and routing options are frequently discussed with the Regulator. Future fusion facilities will benefit from ITER feedback taking into account the specific regulations of each country. Safe, sustainable and cost effective radioactive waste management is a key factor for the public's acceptance of fusion energy.

ACKNOWLEDGEMENTS

Disclaimer: The views and opinions expressed herein do not necessarily reflect those of the ITER Organization. I would like to thank all the ITER radwaste team for their support.

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ANNEX IV.

DEVELOPING THE DECOMMISSIONING STRATEGY / PLAN

IV-1. DEVELOPING THE DECOMMISSIONING STRATEGY / PLAN APPROACH RETAINED BY CHINA

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The Experimental Advanced Superconducting Tokamak (EAST), the first fully superconducting tokamak, has operated in China for nearly 16 years since its first plasma obtained in September 2006. The category of the EAST device is a radiation facility because it will operate with the D-D plasma for the duration of 1-1000 s and the peak value of the prompt neutron production will be up to $1\text{E}+15$ n/s. The accumulative neutron yield reached $\sim 1\text{E}+18$ by 2022. With the application of high power auxiliary heating systems on the EAST machine, it is foreseen that the annual neutron yield will increase and may reach $\sim 5\text{E}+18$ towards shut down.

The measurement by 2022 has shown that the maximum shutdown dose rate and beta contamination level inside the vessel are respectively $\sim 2.3\text{mSv/h}$ and $\sim 3.6\text{ Bq/cm}^2$ at 10 days after shutdown. It is planned to cool for one year after the EOL. This is to allow the preparation of the decommissioning of the device including to liaise with the nuclear safety authority that will oversee the safe decommissioning of EAST device. In this duration, the radiation monitoring of X/g dose rate, beta contamination and radioactive isotopes will be conducted. The deduction shows that the corresponding dose rate and beta contamination might reach $\sim 0.7\text{ mSv/h}$ and $\sim 1.1\text{ Bq/cm}^2$, respectively, after cooling for a year. Based on the predicted activation level, it is anticipated that only small part of in-vessel components in the EAST device will fall in the category of very low level waste (VLLW), and the rest part of the main machine, the auxiliary systems and the building of the torus hall will be exempt waste (EW), i.e. clearance level. The radiation monitoring equipment will be in place to verify the level of radioactivity so that the dismantling activity will be planned to separate the components of VLLW for temporary storage before being transported to the permitted centre for further process or storage. The remained components and hardware will be recycled without being constrained for nuclear facilities.

On the roadmap of magnetic confinement fusion development in China, the China Fusion Engineering Test Reactor (CFETR) [i] has been proposed to serve as the bridging step. CFETR is designed to operate in a steady-state mode with D-T fusion power, ranging from 200MW to 1.5GW. It aims to fill in the physics and technology gaps from the international fusion experimental reactor ITER towards the future DEMOnstration fusion reactors. Two blanket concepts, i.e. helium cooled ceramic breeder blanket (HCCB) [ii] and water-cooled ceramic breeder blanket (WCCB) [iii], were proposed as main candidates of breeding blanket for CFETR. Dedicated study has been conducted to assess the radwaste management for the CFETR with the adoption of alternative blanket concept. Preliminary results showed that in-vessel components, including large part of the breeding blankets and part of divertor, would remain as intermediate-level waste (ILW) even after a 100-year cooling time, and the vacuum vessel itself and ex-vessel components would become low-level waste (LLW) with 1-year cooling after shutdown [iv-v].

It is noted that, if the impact of the tritiation of components is taken into account, the radwaste classification will need to be reassessed for all components in direct contact with tritium in both tokamak machine and process system in the tritium plant. Updated designs and advances in the tritium processing technologies will be applied to manage the tritium contaminated waste, general considerations include:

- Minimization of the tritiated waste production by the storage decay for tritiated waste or the employment of advance techniques, such as the advanced tritium extraction system, fast recycling loop, tritium permeation barrier, modular design, and recycling and reuse of purified materials.

- Tritium removal and detritiation before dismantling for process system. Detritiation operation may be performed before or during further treatment, such as recycling, if needed. Tritium needs to be removed to reach the clearance level (currently is $1\text{E}+5$ Bq/kg) or lower before reuse.
- Detritiation factor requirement for different management approaches have been preliminary assessed as the guidance for the development and implementation of detritiation techniques that undergo continuous development to fit for different conditions, including gas, liquid, solid components and surface contamination.

The applicability of existing regulatory framework for fusion applications needs to be carefully assessed. Therefore, we strongly recommend the early engagement with the Chinese nuclear regulator so as to develop the regulatory framework suitable for fusion facilities. It is necessary to optimise the parameters from the design phase, especially for those that may affect the decommissioning. For example, the irradiation scenario would determine the activation level after the EOL; the planned replacement of in-vessel and ex-vessel components would have impact on the amount of radwaste and the associated radioactivity; the temporary storage and transport related to the site selection would place requirements on on-site activities including monitoring and packaging etc. The R&D for key technologies, e.g. detritiation and remote handling, needs to be planned to pursue more advanced treatment of tritiated waste and highly activated large-size components. The approaches to assess the radwaste classification and the quantification of tritiated waste need to be further developed in order to aid the development of decommissioning plan.

IV-2. DEVELOPING THE DECOMMISSIONING STRATEGY / PLAN – ITER CASE STUDY

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Abstract

Provisions for ITER decommissioning have been included since the beginning of the project. The phenomena leading to radioactive waste production are identified and allow to determine the ITER source term. An initial and preliminary decommissioning scenario has been established. A working group has been created to identify all evolutions of the facility that could impact decommissioning, allow optimization of the various phases of ITER decommissioning and reassess the decommissioning fund.

INTRODUCTION

ITER is an experimental fusion facility under construction in Cadarache, France. Its goal is to demonstrate the feasibility of producing energy by fusion reaction of Deuterium and Tritium. The nuclear operator of the facility is ITER Organization (IO). As nuclear operator under French nuclear laws and regulations, the ITER Organization has to propose a decommissioning plan and make provisions for the decommissioning of its Nuclear Licensed Facility.

At the end of the 20 years of operation including a period of experimental campaigns and a period of deactivation activities, the ITER organization will have completed its mission and will be dissolved. As part of the ITER agreement, the facility and remaining radioactive waste will be transferred to the host country France for final dismantling. Host country mandated the CEA Agency ITER France organisation (AIF) to follow the evolutions of the project in view of this final dismantling phase.

Several technical and organizational challenges are being addressed to fully demonstrate the technical feasibility of fusion up to final dismantling. The goal of this paper is to present the example of the ITER case study for developing the decommissioning strategy / plan.

SOURCE TERM INSIDE ITER NUCLEAR BUILDINGS

The two phenomena leading to radioactive waste production inside the tokamak are:

- Activation: the deuterium-tritium reaction generates 14.1 MeV neutrons whose interaction with all the materials surrounding the plasma causes the progressive high activation of these metallic type materials (constituting elements and impurities). The main neutron activation isotopes contributing to the Intermediate Level Waste-Long Lived (ILW-LL) classification according to the national regulation are: ^3H , ^{60}Co , ^{10}Be , ^{63}Ni and ^{94}Nb . These were derived using 3D calculations based on activation at the end of the operation.
- Contamination: half of the fuel being tritium, it is found progressively trapped (phenomena of permeation, recombination, implantation etc.) in the materials surrounding the plasma and fuelling systems. Additional contamination can be derived from the dust produced in the vacuum chamber and the contamination by activated corrosion products mainly in the tokamak cooling water system. This gives rise to surface contamination propagated outside the vacuum vessel during maintenance in adjacent volumes around the tokamak and in particular the hot cell.

An important source term is also located outside the tokamak:

- In the tokamak cooling water system;
- In the tritium plant housing the fuel injection and recycling system and storage;
- In the hot cell, intended for in-vessel equipment maintenance activities and the treatment and storage of radioactive waste.

INITIAL PROVISIONS FOR ITER DECOMMISSIONING

The Final Design Report of 2001 specified that the ITER organization at the end of operation will be responsible for starting the machine decommissioning through a deactivation period after which the facility will be handed over to a new organization managed by the ITER host country. It is necessary to provide a feasible and flexible plan for the decommissioning of the ITER machine and associated active components. According to the ITER agreement of 21 November 2006 signed between the partners to the project, as well as to the headquarters agreement signed between France and the ITER Organization on 7 November 2007, ITER Organization has to generate a decommissioning fund suitable for the decommissioning of ITER facilities. Following the deactivation phase, this has to be transferred including the ITER facilities to France, after its acceptance.



FIG. IV-2.1. ITER Organisation / France sharing of responsibility for ITER decommissioning and initial estimation of the duration the main phases.

The aim of the Deactivation phase is to put the facility in a safe state by removal of all tritium, all activations products and dangerous materials that can be mobilized.

Furthermore, reduction of the source term has to be done and can be achieved by the following means:

- Reduction of tritium inventory from the in-vessel components and from vacuum vessel using established techniques available during tokamak operation such as baking, glow discharge cleaning and gas or steam flushing.
- Storage of tritium fuel in getter beds (removal from site to be foreseen for potential reuse).
- Draining, evaporation and detritiation of resulting fluids using the existing detritiation facilities in Tritium Plant and Hot Cell evaporators.
- Removal of any accessible and activated dust to the greatest possible extent from the in-vessels components and vacuum vessel using the existing remote handling equipment.
- During Deactivation phase, removal of in-vessel components using remote handling tools can then be achieved by reusing remote handling equipment designed for maintenance. It is followed by cask Transport from the Tokamak to the Hot Cell for remote processing. This requires sufficient space and customized remote handling tools and waste treatment processes to deal with the increased capacity requirements compared to the operational phase of the tokamak. The large size tokamak in-vessel components become activated and tritiated leading to around 3500 tons of Intermediate-level and long-lived radioactive (ILW-LL) waste being produced during operation and decommissioning phases. Note: disposal acceptance criteria are under discussion (detritiation, packaging, decay and interim storage).
- In order to reduce the dose rate inside the vacuum vessel a Decay Phase is assumed to last 23 years before starting dismantling of the vacuum vessel. During this phase, no site activities would be required except security and monitoring.
- The main activity during the Dismantling Phase is the removal and disposal of the vacuum vessel, magnets, cryostat and thermal shields. Dismantling of Vacuum Vessel is the most complex activity to be performed. It requires opening the bioshield and cryostat and re-establish plant systems in particular confinement, followed by remote cutting and removal of the vacuum vessel sectors.



FIG. IV-2.2. Illustration of ITER vacuum vessel remote dismantling.

ITER DECOMMISSIONING OPTIMIZATION WORKING GROUP

In 2017, ITER Organization and CEA have agreed on the technical reference state of the nuclear facility before change of nuclear operator. This agreement was the starting point for a joint working group to optimize technically the phases of ITER decommissioning. The decommissioning working group is composed of ITER Organization and CEA experts.

The first achievement of the joint working group was for ITER Organization and CEA to analyse the changes occurred since 2001 that impact the ITER decommissioning. The output is captured in a report approved in the middle of 2020 listing all changes in the assumptions regarding the radiological conditions, the functions and the design changes of ITER nuclear buildings since 2001. The report also analyses evolutions in terms of waste management.

The working group will also analyse legal evolutions that occurred since the signature of the ITER agreement which may impact the decommissioning scenario, such as new laws or regulations related to nuclear facilities as well as international standards.

IO and CEA/AIF have jointly established a list of issues derived from the 2001 decommissioning plan reference solution. The 2001 decommissioning plan reference solution is recalled and other solutions were discussed during working group meetings between CEA and ITER. The goal is to identify blocking points and issues related to the new design and nuclear regulation for the future decommissioning of the ITER facility.

The IO and CEA Joint Working Group has defined the following objective over the coming years:

- Perform functional analysis of all decommissioning phases;
- Reassess time and resources to perform all decommissioning phase;
- Reassess overall optimized decommissioning scenario based on current design and regulations;
- Reassess primary and secondary radioactive waste to be generated during decommissioning;
- Integrate staged approach configuration of ITER;
- Integrate best practices for change of nuclear operator;
- Integrate optimized tritiated waste management solutions;
- Reassess overall decommissioning costs following OECD/NEA costing methodology [1].

In parallel, based on the inputs from the joint Working Group, ITER Organization will establish the necessary documentation related to decommissioning for the licensing with the Regulator.

The methodology for tracking of evolutions impacting decommissioning is described in the ITER Case Study, Annex III-1 ‘Preparation of the Decommissioning Plan – Considerations at the Facility Design Stage’.

ITER WASTE ROUTING REVIEWS

A waste routing review was performed in 2018 involving IO, ANDRA and CEA/AIF management. It was based on description of the waste characteristics, flow and treatment process for each waste category. The goal was to secure good confidence in current waste management strategy and anticipate issues regarding acceptability of specific waste due to their toxicity, reactivity or high tritium content. The outcome was the definition of a short- and long-term action plan to develop processes, increase the knowledge by the repository of specific radioactive waste generated in fusion facilities and follow up of R&D programs.

In 2021, a specific task force was created involving experts from all ITER Members to refine detailed tritiated waste inventories and make recommendations in order to select the most adapted waste management solutions in particular waste detritiation, packaging and interim storage before disposal.

CONCLUSION

The decommissioning strategy of ITER Facility has been initiated at the early stage of the project. Before the launch of the construction of ITER Facility, an initial decommissioning plan was already established. During the design and construction processes, all the technical evolutions of the machine have to be followed and their impact on the decommissioning plan analysed. It will lead to the establishment of a more detailed, up to date and optimized scenario for the decommissioning plan. During the operation of the facility, specific processes will occur to adapt the decommissioning plan to the real state of the ITER facility and the best practices as well as the regulatory framework. To establish the decommissioning plan, the on-going studies are not taking into account future possible dismantling methods but they demonstrate the dismantling feasibility based on existing methods. In parallel to technical studies, the cost to be reported into the decommissioning fund will be regularly updated with regards to the different evolutions to be taken into account.

The ITER decommissioning studies will help demonstrating the sustainability of Fusion Energy regarding waste management and decommissioning by anticipating the definition of strategy that is optimized in terms of resources, time, and safety and incorporating international feedback.

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IV-3. ITALIAN EXPERIENCE CONCERNING THE DECOMMISSIONING OF THE FRASCATI TOKAMAK UPGRADE

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Frascati Tokamak Upgrade main parameters and its regulatory history

Frascati Tokamak Upgrade (FTU) is a compact, high-magnetic-field tokamak built following the FT (Frascati Tokamak) experiment in Frascati. The main parameters of FTU are listed in Table IV-3.1.

TABLE IV-3.1. FRASCATI TOKAMAK UPGRADE MAIN PARAMATERS

Maximum toroidal magnetic field	8 T
Maximum plasma current	1.6 MA
Major radius	0.93 m
Maximum installed additional power (per pulse)	8 MW
Maximum neutron production (theoretical)	6×10^{15} neutrons/shot
Feed gas for plasma production	Hydrogen, Deuterium

Construction of the FTU plant was authorized by the Ministry of Industry on Jan. 7th, 1986, and operation pursuant to art. 55 of the Presidential Decree 185/64 was authorized by the decree of the same Ministry no. XIII – 326 on Nov. 2nd, 1993. Operation began at the end of 1993.

FTU officially ended its experimental activity on Dec. 20th, 2019. Therefore, ENEA requested the closure of the practice and, pursuant to points 5.11, 5.12 and 5.13 of Annex IX to Legislative Decree 230/95, the plant decommissioning and related authorizations were requested. Pursuant to Legislative Decree 101/20, entered into force on Jul. 31st, 2020, which repeals the previous Decree 230/95, such a request included the Radiation Protection Expert Technical Report, the Radiological Characterization Plan for Radiometric Verification, and the Proposal for Technical Requirements to be applied during decommissioning operations.

The Ministry of Ecological Transition provided authorization to proceed with the FTU decommissioning on Apr. 2nd, 2021, with Technical Requirements to be respected.

According to the Radiological Characterization Plan, a radiological pre-characterization of the materials generated by the FTU decommissioning through Non-Destructive Analyses (NDA) and Destructive Analyses (DA) has been carried out to identify and quantify radionuclides coming from neutron activation processes. Results have been transmitted to the Italian Nuclear Regulatory Authority (ISIN, National Inspectorate for Nuclear Safety and Radiation Protection).

Given the absence of significant neutron activation phenomena on the analysed samples, a proposal for the release of all materials and interested areas was requested to remove any radiological constraint. The final notification from Regulatory Authority to Ministry concluding the dismantling activities is still awaited and expected in the next months.

Operational history and theoretical evaluation of neutron activation levels

As discussed in [1]:

“The analysis of the FTU logbook has shown that the effective average neutron flux and the total neutron fluence during the operational phase have been some orders of magnitude less than the design values.

Decommissioning of FTU considers only neutron activation processes as source of radioactivity without possibility of components contamination.

Based on historical data and semi-empirical analysis based on calculations (i.e. FISPACT simulations), a theoretical quantification of relevant radionuclides produced by neutron activation has been performed for the components located inside the cryostat, considering the effective total fluence and the energy spectrum of emitted neutrons characteristic of deuterium-deuterium fusion reaction. Activity concentrations <0.01 Bq/g for all the characteristic radionuclides resulted from this evaluation. The comparison of these results with the clearance levels pursuant to Table I-1B of Legislative Decree 101/20, showed that calculated activity concentrations are lower than clearance level. As an example, an activity concentration of $5 \cdot 10^{-4}$ Bq/g of ^{60}Co has been evaluated for the vacuum chamber stainless steel while the clearance level 0.1 Bq/g is set for this radionuclide. Free release is expected for most of the materials present in the FTU Plant without constraints for the final destination or limitations for subsequent recycling or reuse.

Radiological pre-characterization plan

The radiological pre-characterization plan aimed to define the type and quantity of radionuclides present in the plant, to enable planning and design of operational phases of dismantling, i.e. removal of components and structures, radiological characterization, management of waste and materials to be cleared and recycled / reused.

The decommissioning plan is structured in:

- Removing the cryostat and ancillary parts around the innermost vacuum chamber, through removal of flanges, screws and bolts (without cuts or desoldering);
- Removing the innermost structure as a whole (without cuts or desoldering), i.e. the vacuum chamber and surrounding electromagnets including the ports (the experimental “windows” of the vacuum chamber), by lifting and placing the whole structure on a trolley to be moved to the adjacent hall awaiting for successive utilization.

From calculations, only the structures contained inside the cryostat could have a radiological relevance about neutron activation, so that the materials dismantled by FTU were grouped into two macro-categories:

- Group a): materials subjected to neutron flux (inside the cryostat) and potentially activated;
- Group b): materials not subjected to neutron flux or subjected to neutron flux of intensity negligible for activation purposes (outside the cryostat) and presumably not activated.

Components and materials belonging to group a) were divided in homogeneous groups on the basis of elemental composition and location. According to this grouping, materials available for measurements in pre-characterization phase were, for example:

- Titanium-Zirconium-Molybdenum (TZM) alloy facing tiles of the limiters. The limiter is the cylindrical band on the inner surface of the vacuum chamber which the plasma faces to and gives off excess of heat. Its purpose was to preserve the integrity of the liner of the vacuum chamber;

- Support racks in AISI 316L steel on which the TZM cladding tiles of the toroidal and poloidal limiters were mounted;
- Copper components placed inside the ports of the vacuum chamber;
- Ferrous metal components of similar composition to the vacuum chamber including tubular structures, flanges, gates and larger diagnostics (currently non-removable).

According to the technical request of ISIN, to reuse or recycle a material coming from FTU, the measured concentration activity (both per unit surface and mass) has to be less than the Clearance Levels, and the measurement needs to reach a Minimum Detectable activity Concentration (MDC) equal to or less than 30% of the corresponding Clearance Level. In Table IV-3.2, the authorized Clearance Levels for reuse or recycle for each characteristic radionuclide are reported.

Radiological pre-characterization of materials coming from decommissioning of FTU involved non-destructive and destructive analyses. The former aimed to the identification and quantification of gamma emitters coming from neutron activation processes (^{54}Mn , ^{60}Co , ^{58}Co , $^{110\text{m}}\text{Ag}$) by means of in situ gamma spectrometry; the latter were focused on quantification of the activated products with no significant gamma emission (e.g. beta emitters) via Liquid Scintillation Counting (LSC). For each one of the homogeneous groups previously defined, a measurement campaign was carried out.”

TABLE IV-3.2. CLEARANCE LEVELS IN TERMS OF SURFACE AND MASS ACTIVITY CONCENTRATIONS FOR REUSE AND/OR RECYCLE PURPOSES

Radionuclide	Reuse	Recycle	Reuse/Recycle
	Surface activity concentration (Bq/cm ²)	Surface activity concentration (Bq/cm ²)	Mass activity concentration (Bq/g)
^{54}Mn	10	10	0.1
^{55}Fe	1000	10000	1
^{59}Ni	10000	10000	1
^{58}Co	10	10	1
^{60}Co	1	10	0.1
^{63}Ni	1000	10000	1
$^{110\text{m}}\text{Ag}$	1	10	0.1

More than 100 distinct items belonging to the different homogeneous groups have been characterized in-situ using a transportable detection system equipped with Germanium HPGe detector, ISOCS System (In-Situ Object Counting System) to quantify gamma emitters generated by neutron activation processes. Only few items, such as some support racks for TZM tiles of toroidal and poloidal limiters, have been found affected by detectable neutron activation phenomena. Even in those cases, levels of radioactivity determined are similar to background and below the authorized Clearance Levels. Activity values have been converted to surface activity concentration by using a few thousand cm² as averaging areas, instead of several hundred cm² advised for “Direct Reuse” in RP 89. As an example, two options for averaging areas have been evaluated for the TZM tiles homogeneous group (5300 cm² and 740 cm², corresponding respectively to 60 and 9 TZM tiles). Applying the fixed-sample strategy by UNICEN-189, a Noether number equal to 13 has been calculated in both cases. This means that residual activity concentrations are order of magnitudes below the corresponding Clearance Levels. An averaging area of 5300 cm² allowed to check the 100% of the waste stream even with a number of measurements less than 13 (total number of tiles is 360, divided in 60 tiles per sample, means 6 measurements). An averaging area of 740 cm² means to check only the 33% of the waste stream (13 measurements, 9 tiles each, means to check 117 tiles of 360). Being the radioactivity concentrations below the Clearance Levels, the setting of the averaging areas at few thousand cm² allows the achievement of the clearance objectives with the advantage to have radiological check on larger masses of waste stream with respect to the case of using smaller surfaces.

To quantify beta emitters (^{55}Fe , ^{59}Ni , and ^{63}Ni), some samples of AISI316L steel from chips obtained from one of the supports of FTU toroidal limiter have been physically and chemically treated in laboratory through acid digestion and chromatographic separation methods, to be characterized through LSC. Not-irradiated samples, made by the same type of steel, have been analysed following the same procedure for evaluating the blank conditions. The sample concentration in the LSC vials is strictly connected to the quenching, particularly to the colour quenching. To achieve the requested MDC in a reasonable time, a preliminary study has been carried out to select the best procedure and the best sample concentration in the LSC vial. As for the non-destructive gamma spectrometry, also LSC experimental determinations revealed activity concentration of each radionuclide searched for less than the MDC (0.3 Bq/g for each radionuclide) and lower than the corresponding Clearance Levels.

As discussed in [1]:

“Radiological pre-characterization results demonstrated that materials present in the FTU Plant can be cleared to unconditional released as expected from historical data and semi-empirical analysis, without any constraints for the final destination or limitations for subsequent recycling or reuse”.

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IV-4. MANAGEMENT STRATEGY ON RADIOACTIVE WASTE OF FUSION DEMO REACTOR OF JAPAN

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Abstract

As the considerable amount of radwaste could impact on the public acceptance of DEMO, an appropriate radwaste management scenario needs to be developed. A radwaste management scenario for Japan's DEMO (JA DEMO) was developed based on the characteristic residual heat and dose rate of used In-Vessel Components (IVCs) such as blanket segments and divertor cassettes. A single blanket segment comprises several blanket modules and a back-plate equipping the blanket modules. The lifetime of each IVC was set considering an allowable displacement per atom. The back-plate of the blanket segment and the cassette body of divertor cassettes are reused to reduce the amount of waste by arranging the re-welding points under neutron shielding. The disposal scenario was developed in consideration of dismantling IVCs, storing for decreasing the residual heat and dose rate before waste classification, and cutting them into small pieces for container storage. After radwaste classification in accordance with radioactivity, the radwaste is packed for disposal. The radwaste amounts to 892 containers of 1.6 m cubes for the blanket module and divertor. The analysis indicates that the induced radioactivity of the blanket first wall (F82H) is higher than that of the other IVCs such as the divertor and back-plate, and that long-lived radionuclides (e.g. ^{14}C , ^{99}Tc , and ^{63}Ni) are produced in high concentrations. Classification of radwaste disposal is determined by evaluating public radiation exposure following disposal, considering radionuclide transport via likely biosphere exposure pathways. The radiation exposure is as low as $3.4\text{ }\mu\text{Sv/year}$, lower than the shallow land burial criterion ($\leq 10\text{ }\mu\text{Sv/year}$), when the used blanket is cooled for 10 years and is disposed in a concrete pit near the land surface. From the assessment, all radwaste originating from used IVCs will be qualified as LLW and can be disposed of in a shallow land burial.

INTRODUCTION

Previous radioactive waste (radwaste) research has focused on decommissioning. There is a concern with regard to fusion reactors that replacements of in-vessel components (IVCs), such as blanket segments and divertor cassettes, generate a considerable amount of radwaste and represent serious environmental burdens. Therefore, we considered a strategy to reduce the radwaste generated by IVC replacements. Thus far, studies on fusion-activated material recycling have focused on reducing the amount of radwaste that requires disposal [1]. Owing to the necessity of replacing the IVCs, radwaste management is required during fusion reactor operation. Therefore, an appropriate conceptual radwaste management design needs to be developed. In addition, radwaste classification is important from a social acceptance perspective. In Japan, a fusion DEMO reactor generates only low-level radwaste. However, low-level radwaste is not always disposable. In Japan, the low-level radwaste classification system divides radwaste into three types: L1, L2, and L3 (Table 1 [2]). The L1 waste is disposed of in the sub-surface (50 to 100 m depth below the land surface) with the high-performance Engineered Barrier System (EBS). The other categories involve two kinds of near-surface ($<50\text{ m}$ depth below the land surface) disposal. One is trench disposal and does not use EBS, and the other is concrete pit disposal and uses EBS. So,

far, the L2 and L3 disposal methods have good track records. Therefore, limiting the amount of L1 waste produced from a fusion reactor is an advantage for public acceptance.

In this report we describe the development of a waste management plan for IVCs based on decay heat and dose predictions from Japan DEMO. We consider the likely classification of radwaste from Japan DEMO in Japan by assessing the likely biosphere impact of the main sources of radioactivity in disposal waste under scenarios where that waste is released. A final section also discusses the issues of uranium impurities in Be, which is the main moderator material in Japan DEMO.

TABLE IV-4.1. FRAMEWORK FOR CLASSIFICATION OF THE LOW-LEVEL WASTE

	L3 LLW	L2 LLW	L1 LLW
Disposal depth	Near-surface (< 50 m)		Sub-surface (50 m <)
Engineered barrier system	Without EBS Trench	With EBS Concrete pit	With High Performance EBS
Management period	~ 50 years		~ 300 – 400 years
Disposal site	Available		None

MANAGEMENT SCENARIO FOR RADIOACTIVE WASTE GENERATED IN SCHEDULED MAINTENANCE

Specifications of JA DEMO reactor

A conceptual design of Japan's fusion DEMO (JA DEMO [3, 4]) is shown in Fig. IV-4.1. The radwaste management scenario considers DEMO conditions with a major radius of 8.2 m, a fusion power of 1.5 GW, and a plant life of 20 years. The average neutron wall loads of the blanket area and divertor area are estimated to be 1.0 and 0.3 MW/m², respectively. Additionally, the maximum neutron wall loads of the blanket area and divertor are estimated to be 1.66 and 0.53 MW/m², respectively. The blanket segment comprises a number of blanket modules mounted to a back-plate, and the blanket module cooling system is separated from the back-plate. The structure material of the back-plate is assumed to be SS316LN. A conducting shell, composed of Cu alloy (CuCrZr), for plasma positional stability and high beta access is installed between the blanket modules and the back-plate. The blanket concept was developed with a mixed bed of Li₂TiO₃ and Be₁₂Ti pebbles, shown in Figure IV-4.1 [5]. The surface of the blanket modules is covered with a 0.5 mm tungsten coating (W). The structural material of the blanket modules is assumed to be F82H; reduced-activation ferritic/martensitic (RAFM) [6]. The total weight of the blanket segments is 9 260 tons. The divertor cassette comprises divertor plates and the cassette body. A divertor plate is constructed of W mono-block and RAFM (F82H) cooling pipes and substrates [7]. A Cu alloy (CuCrZr) is used for the cooling pipes near the high-heat-flux region (~10 MW/m²). The total weight of the divertor cassettes is 1,580 tons. The materials and weights of the IVCs are shown in Table IV-4.2.

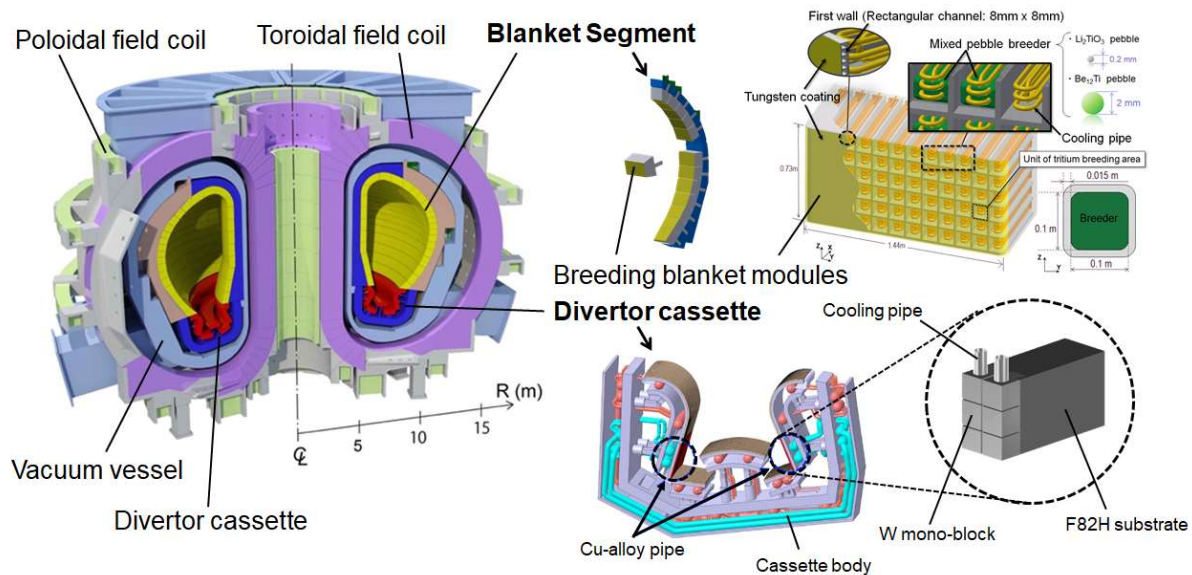


FIG. IV-4.1. Conceptual design of the JA DEMO reactor.

An MCNP-5 [8] neutronics analysis, using the FENDL-2.1 nuclear library [9], indicates 17.0 displacement per atom (DPA) per year for the first wall, 0.2 DPA/year for the back-plate, and 1.7 DPA/year for the Cu-alloy pipe. The irradiation times of the blanket modules, divertor cassette, and back-plate are assumed to be 2.8, 1.0, and 10 years, respectively, according to the allowable DPA for each IVC (Table 2). The plant availability for the JA DEMO is assumed to be 50%.

TABLE IV-4.2. WEIGHT AND DPA OF IN-VESEL COMPONENTS AND ALLOWABLE DPA VALUES.

Component	Material	Manage	Weight	(ton)	Max DPA (DPA/year)	Allowable (DPA)
Blanket module	W coating	Disposal	12	2,728	17.0 (First wall, F82H)	< 20 (F82H)
	F82H	Disposal	1,935			
	Li ₂ TiO ₃ , Be ₁₂ Ti	Recycling	781			
Divertor cassette	W monoblock + cooling pipe	Disposal	155	1,580	1.7 (Cooling pipe, Cu)	< 1 (CuCrZr)
	F82H substrate	Disposal	698		0.7	< 20
	F82H cassette	Reuse	727		(Cassette, F82H)	(F82H)
Back-plate	SUS316LN	Reuse	1,638	6,532	0.2	< 3
	F82H		4,894		(Case, SUS316LN)	(SUS316LN)
Total			10,840			

Assessment of the residual heat and contact dose rate for DEMO

The neutronics analyses for the blanket, divertor, and back-plate were performed using THIDA-2 code [10]. Figure III-4.2 shows that, for a DEMO reactor with 1.5 GW fusion power, the estimated residual heats immediately after shutdown of the blanket, divertor, and back-plate were 30.6, 8.6, and 1.1 MW, respectively. A month following shutdown, the total residual heat decreases to 5.5% of the initial value. The maximum residual heats immediately following shutdown and one month later are 40.3 and 2.2 MW, respectively.

The contact dose rates of the W coating, the first blanket wall, and the divertor W mono-block and substrate are shown in Fig. IV-4.3. After shutdown, the contact dose rates of the blanket and divertor decrease to 100 Sv/hr after 7 and 1.5 years, respectively, and to 10 Sv/hr after 20 and 7 years, respectively.

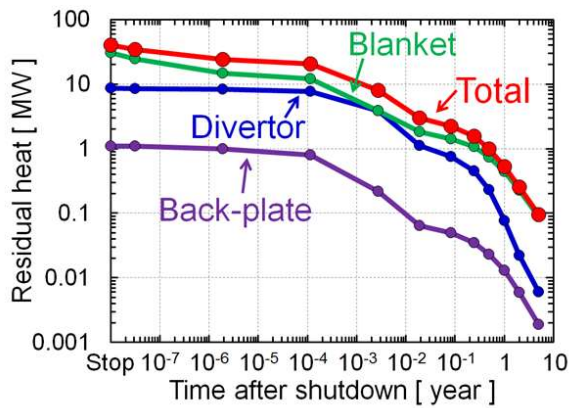


FIG. IV-4.2. Residual heat for DEMO.

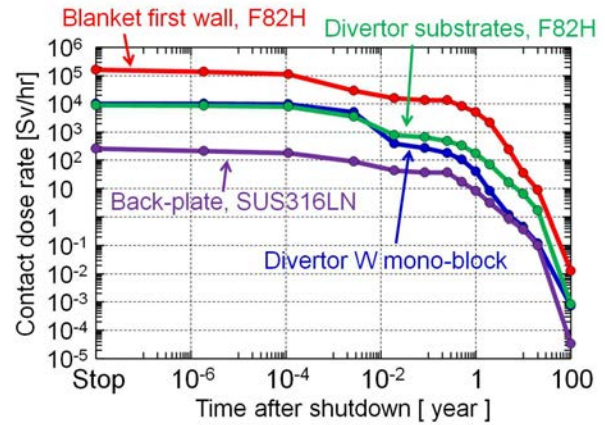


FIG. IV-4.3. Contact dose rate of the in-vessel components.

Radwaste management scenario

We developed a radwaste management scenario for DEMO based on the residual heat and dose rate of the used IVCs, such as blanket segments and divertor cassettes (Figure III-4.4). The IVCs need to be contained in a cask to deliver to the hot cell as the sector is contaminated with tritium and radioactive tungsten dust. When the IVCs are removed from the vacuum vessel (VV) after turning off the water supply and disconnecting from the cooling manifold, the other IVCs in the VV need to be actively cooled. The temperature of the removed IVCs rise during the replacement because of loss of a coolant. A possible measure to avoid such a temperature rise is to adopt a cool-down in VV for the decreasing the residual heat. In this situation, the IVCs can be cooled by natural convection.

In the hot cell, the used IVCs are handled as follows. Blanket modules are removed from a back-plate by remote handling equipment after 7 years of cooling. In the divertor cassette, the divertor plate W mono-block is removed from a cassette body, after 1.5 years of cooling, using remote handling equipment. The radiation environment in the hot cell is estimated to be 100 Gy/h. The dose rate of the in-vacuum vessel at ITER ('The Way' in Latin) during maintenance using remote handling equipment is limited to 250 Gy/h [11, 12]. The IVC dose rate of DEMO is one order of magnitude higher than that of ITER. The DEMO blanket and divertor also require 7 and 1.5 years, respectively, of cooling time to reduce their dose rates to ITER dose rate level. When the IVCs are naturally cooled by gas convection in the hot-cell storage area, the maximum temperatures of the blanket module, blanket segment back-plate, and divertor W mono-block are theoretically estimated at 274 °C, 190 °C, and 156 °C, respectively, 26 days for the blanket module and segment and 13 days for the divertor W mono-block after transfer to the hot cell (Fig. IV-4.5).

Next, the neutron multiplier and tritium breeder are recovered from each blanket module for recycling in the disassembling facility. After, the empty blanket modules and divertor plates are cut into small pieces and stored in a container. The radiation environment in the facility is estimated at 10 Gy/h, and the temporal storage is 13 and 5.5 years for the blanket and divertor, respectively.

After radwaste classification in accordance with radioactivity values, the radwaste is packed for disposal. During the disposal process, breaking up the F82H and W into small pieces reduces the radwaste volume, contributing to a reduction in storage space. Since the small pieces are packaged in containers with mortar for proper disposal, the residual heat needs to be removed by natural cooling to keep the temperature below 65 °C, which prevents water evaporation from the mortar. The Li_2TiO_3 and Be_{12}Ti pebbles are collected for recycling. We found that the small pieces can be kept in a temporary storage to meet the mortar temperature conditions during approximately 13 years for the blanket and 5.5 years for the divertor. The structural material of the blanket and divertor is cut into small pieces and packed in a suitable container. The total amount of low-level blanket module and divertor cassette radwaste is 892

for 1.6 m cube containers. This total includes 324 and 568 containers of blanket modules and divertor cassettes, respectively.

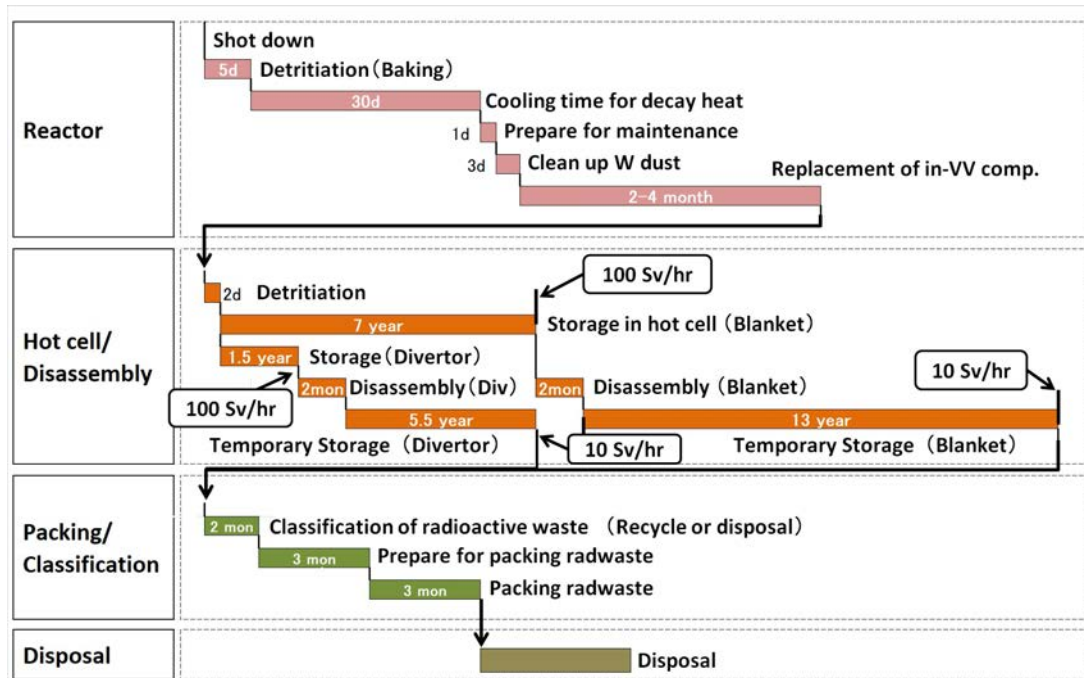


FIG. IV-4.4. Radioactive waste management scenario of the in-vessel components for DEMO.

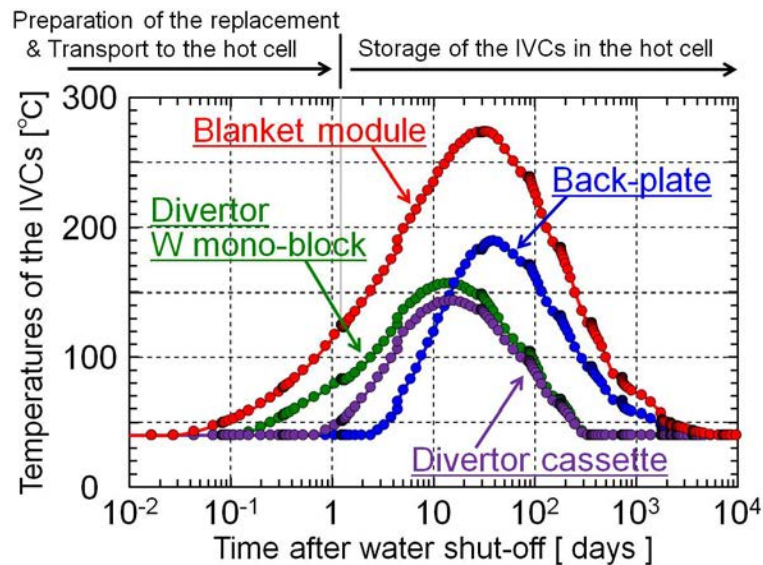


FIG. IV-4.5. Temperatures of the IVCs during replacement.

RADIOACTIVE WASTE CLASSIFICATION

Neutronics analysis for activation of IVCs, such as blanket segments, divertor cassettes, and back-plates, was carried out using THIDA-2 code. In this calculation, impurities in all IVC materials are taken into consideration. The material compositions with tungsten and SS316LN impurities, as well as data on F82H impurities, were taken from the literature [13,14]. Figure IV-4.6 shows that the induced radioactivity of the first blanket wall is higher than that of other IVCs, including the divertor cassette and back-plate, more than 1,000 different radionuclides are produced in each of the regions, and long-lived radionuclides (e.g. ^{14}C , ^{99}Tc , and ^{63}Ni) are produced in the first wall at high concentrations. For example,

the concentration of ^{14}C (half-life of 5730 years) is calculated to be 5.38×10^{10} Bq/ton after a continuous 2-year operation (2 FPY) of the reactor at the peak neutron wall load.

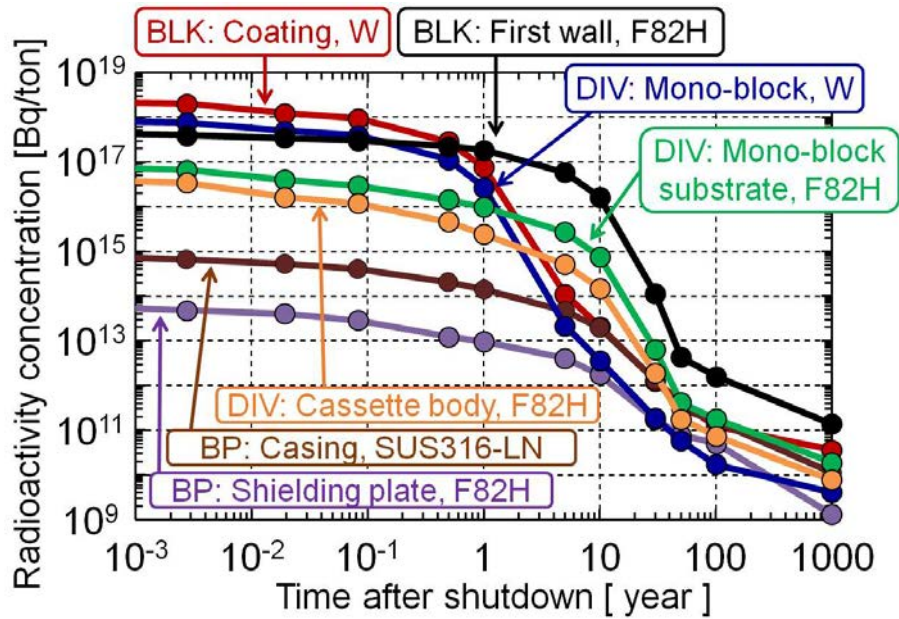


FIG. IV-4.6. Radioactive concentration of the in-vessel components.

Selection of radionuclides from each material

The approximately 1000 radionuclides generated for each material were narrowed down to approximately 100 radionuclides by the following method. The prerequisites for the screening method were assumed to be the following: (1) the radwaste contents will be destroyed after 10 years, and after 10 years, the radionuclides will leak; (2) radionuclides that permeate from the artificial barrier, such as a concrete pit, are instantly transported to the surface layer biosphere; and (3) the groundwater near the surface layer will be used immediately as a well, from which the water is subsequently withdrawn by an individual.

The radionuclide concentration in a well water is calculated using the following equation.

$$C = \frac{\left(\frac{A \times N}{T}\right)}{Q} \text{ (Bq/m}^3\text{)}, \quad (3.1-1)$$

where N (g), T (day), and Q (m³/year) are the weight of the radwaste, melt time of the radwaste, and flow velocity of the groundwater, respectively. N , T , and Q were assumed to be 1 g, 100 days, and 1.0×10^{10} m³/year [15] from the selection viewpoint with a relative priority order for radionuclides from the IVC materials. A (Bq/g) is the maximum radioactivity concentration of the IVC materials.

C_{limit} is defined as the maximum concentration of radionuclides within well water that satisfies Japanese regulations [16]. Therefore, more than 1,000 radionuclides generated for each material in the IVCs were selected as the top 100 radionuclides based on equation (3.1-2). For example, ^{14}C , ^{53}Mn , ^{63}Ni , ^{93}Mo , ^{94}Nb , ^{99}Tc and $^{186\text{m}}\text{Re}$ of C_{limit} (10^6 Bq/m³) are 2.0, 30.0, 6.0, 0.3, 0.5, 1.0 and 0.4, respectively.

$$\text{Index} = \frac{C}{C_{\text{limit}}}. \quad (3.1-2)$$

We found that the dominant radionuclides are ^{14}C (half-life of 5.70×10^3 years), ^{53}Mn (half-life of 3.7×10^6 years), ^{63}Ni (half-life of 100.1 years), ^{93}Mo (half-life of 4.0×10^3 years), ^{94}Nb (half-life of 2.03×10^4 years), ^{99}Tc (half-life of 2.111×10^5 years), and $^{186\text{m}}\text{Re}$ (half-life of 2.0×10^5 years).

Underground radionuclide migration analysis

In Japanese nuclear regulation, the dose limit of underground radionuclide migration in the biosphere needs to be estimated using the radwaste classification disposal method, which is based on the dose limit ($<10 \mu\text{Sv/year}$). The dose of the radionuclides in the biosphere is evaluated by the Generic safety assessment code for geologic disposal of radwaste Greater than Concentration Limit of shallow land disposal (GSA-GCL) [17] calculation code for disposal plan safety and to ensure low levels of radwaste in shallow land disposal. This methodology can be applied to the generic safety assessment, which evaluates the migration pathway from a repository to the biosphere using a groundwater flow code. The GSA-GCL code is based on a groundwater migration scenario and consists of source term, geosphere, and biosphere models.

Model calculation for underground radionuclide migration

For the radionuclide migration analysis following burial disposal using the GSA-GCL code, the disposal method was assumed to be the concrete pit method (L2). This analysis was based on the premise that groundwater flows on the bottom and each side of the disposal area, permeating the facility. Therefore, this study assumes that groundwater reaches the radwaste, and groundwater containing radionuclides permeates concrete pits and other low-permeability layers, which use a bentonite–sand mixture soil, and migrates via the soil and aquifer (Fig. IV-4.7) [18]. For a biosphere dose evaluation, a scenario with a high dose rate in every radionuclide was selected based on a comparison of two scenarios. The first scenario assumed that the soil region is recycled by digging. The other scenario assumed that humans are exposed to radiation through food/livestock and drinking water, following the migration of groundwater containing radionuclide to wells and rivers.

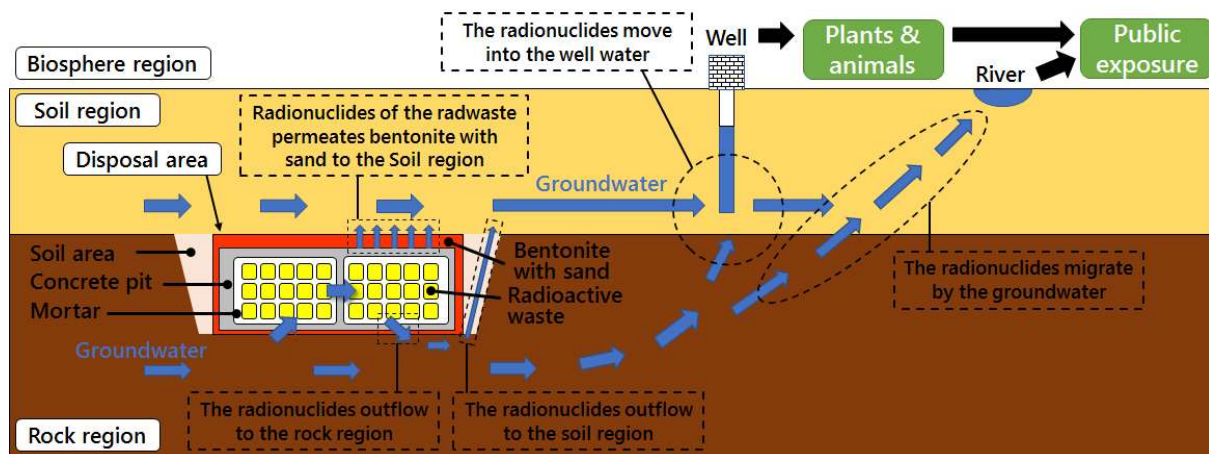


FIG. IV-4.7. Migration model for the radionuclides.

As a barrier to radionuclide migration, only a low-permeability layer of bentonite–sand mixture soil is considered because of the following three reasons. First, it is difficult for a concrete pit to maintain its waterproof function because of cracks caused by extra loads, such as earthquakes and radwaste corrosion and expansion. Second, the Nuclear Regulatory Committee suggests that an overdependence on the confinement function due to the facility's location and structure (both artificial and natural barriers) needs to be avoided [19]. Third, the Nuclear Regulatory Committee does not request that concrete pits have a waterproof function, but asks that a low-permeability layer using bentonite–sand mixture soil have an approximately 10^{-8} cm/s permeability coefficient [20].

The following three data are required for radionuclide migration analysis on the low-permeability layer using bentonite–sand mixture soil:

- ① Migration due to groundwater flow in the low-permeability layer (permeability coefficient).
- ② Migration due to the concentration differences in the low-permeability layer (diffusion coefficient).

- ③ An adsorption effect in the low-permeability layer (distribution coefficient: the ratio of radionuclide concentration in the solid phase to the liquid phase in pure water). The permeability coefficient of the low-permeability layer is assumed to be approximately 10^{-8} cm/s [20].

The low-permeability layer diffusion coefficient is evaluated according to the chemical form of each radionuclide, and the low-permeability layer pH was assumed to be approximately 11.5 [21]. Regarding dissolved oxygen in the low-permeability layer pore water, Eh, an index in water chemistry for oxidation–reduction potential, was assumed to be approximately 100 mV. This value is used because many of the materials in the radwaste layer are iron, and FeS₂ is a constituent mineral in the low-permeability layer bentonite and works as a reductant [22]. On the basis of the low-permeability layer pore water pH and Eh, a chemical form of radionuclides in bentonite was assumed.

The conditions of the selected radionuclides are estimated by the Eh-pH diagram [23, 24] at pH = 11.5 and Eh = 100 mV. As an assessment, each radionuclide was assumed to be a negative ion. Therefore, the diffusion coefficient of the negative ion in bentonite (with sand) was assumed to be 1.26×10^{-3} m²/year [21]. The chemical form of tritium was regarded as HTO, and the diffusion coefficient in the neutral chemical form was assumed to be 9.47×10^{-3} m²/year [21]. The diffusion coefficient of the positive ⁴¹Ca ion was assumed to be 1.26×10^{-3} m²/year.

The groundwater velocity in the low-permeability layer (v , m/year) was calculated using the following equation [15].

$$v = k \frac{i}{\varepsilon}, \quad (3.2-1)$$

where k (cm/s), i (–), and ε (–) are the permeability coefficient, hydraulic gradient, and voidage, respectively, in the low-permeability layer. The permeability coefficient and hydraulic gradient of the low-permeability layer are assumed to be approximately 10^{-8} cm/s and 0.03, respectively [18]. The voidage in the low-permeability layer was calculated using the following equation [22].

$$\varepsilon = 1 - \frac{\rho_d}{\rho_r}, \quad (3.2-2)$$

where ρ_d (Mg/m³) and ρ_r (Mg/m³) are the theoretical and dry densities, respectively, of the low-permeability layer. The theoretical and dry densities were assumed to be approximately 2,700 kg/m³ [22] and 2,000 kg/m³ [25], respectively. Therefore, the voidage in the low permeability was calculated as 0.259 using equation (3.2-2).

We found that the groundwater velocity in the low-permeability layer is 0.00036 m/year.

Dose rate in the biosphere

The classification for radwaste disposal is determined by the evaluation of radiation exposure to society after disposal and considers radionuclide transport via likely biosphere exposure pathways. In the case of the used blanket structural material, dominant radionuclides related to the low-level radwaste (LLW) classification include ⁹⁹Tc, ³⁶Cl, ^{186m}Re, ⁹³Mo, and ¹⁴C (Fig. IV-4.8). Our assessment using GSA-GCL code indicates that the radiation exposure is as low as 3.4 μSv/year, lower than the shallow land burial criterion (10 μSv/year), when the used blanket is cooled for 10 years and is disposed of in a concrete pit near the land surface. The situation is the same for the used divertor and back-plate of the blanket segment. From the assessment, we conclude that all radwaste originating from DEMO IVCs in every periodic replacement is classified as LLW and can be disposed of in a shallow land burial.

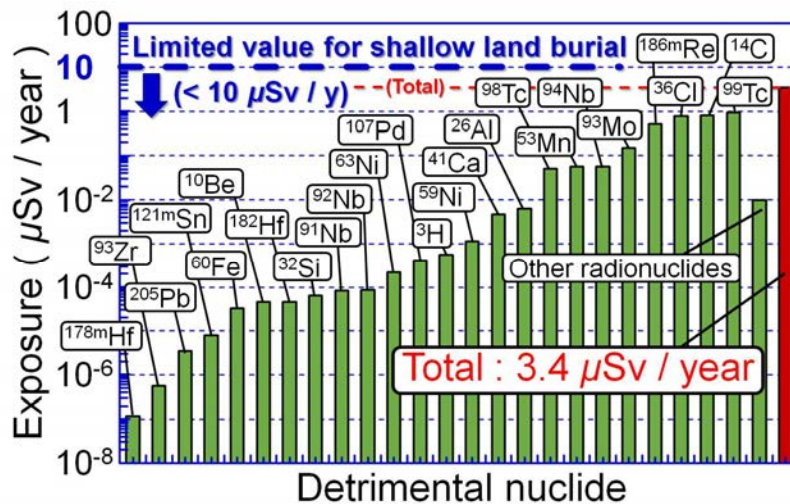


FIG. IV-4.8. Exposure anticipated in the shallow land burial of the in-vessel component of the blanket modules without mixed breeding materials.

Discussion of impurities in beryllide as neutron multipliers for shallow land disposal

Beryllium (Be) is used as a neutron multiplier in DEMO reactors to efficiently produce tritium as fuel. In current DEMO research, scarce Be resources are supposed to be recycled from used blankets because of their high-volume consumption. The blanket in the DEMO research uses Be_{12}Ti to reduce the volume of hydrogen produced in the event of a coolant pipe rupture in the blanket box. The total load of Be_{12}Ti in the reactor was approximately 500 tons. It is a discussion on the disposal for radioactive waste produced in the recycling process of Be_{12}Ti and the end of operation.

Uranium contained as an impure substance is a concern during the recycling or disposal process of activated Be_{12}Ti [26]. Be often exists near uranium mines, and a few wppm of uranium exists as an impure substance in Be. This uranium content is equivalent to the uranium content in the concrete used in the Fugen Nuclear Power Station [27] and Be does not need to be designated as uranium radwaste according to Japanese nuclear regulations. However, if all 500 tons of Be_{12}Ti in the blanket contained a few wppm of uranium, the total amount of uranium in the reactor would amount to a few kgs. Therefore, transuranium elements produced by nuclear transmutation need to be considered.

As a result of neutronics analysis, when the used neutron multiplier Be_{12}Ti in the blanket is cooled for 10 years, the uranium content of Be_{12}Ti needs to be less than approximately 0.85 wppm, so the total concentration of α radionuclides, such as Np, Pu, Am, and Cm, produced by uranium in Be_{12}Ti , is less than approximately 10^{10} Bq/ton. Assuming a DEMO reactor with an 8.2 m major radius and fusion output of 1.5 GW (Figure IV-4.9), this value is the acceptable level for shallow land disposal in Japan ($< 10^{10}$ Bq/ton) [2]. In Fig. III-4.9, Am concentrations increase through disintegration from ^{241}Pu to ^{241}Am after 100 years. However, this incremental value of the radioactivity is 3% of the total and thus not a large problem. Therefore, in the JA DEMO, a significant reduction from the ITER requirement value (~ 30 wppm) is required. Japan was challenging the number with a novel Be refining method developed. It found that new purification processes can reduce U concentration in Be to allowable concentrations for shallow land disposal (~ 0.1 wppm) [28].

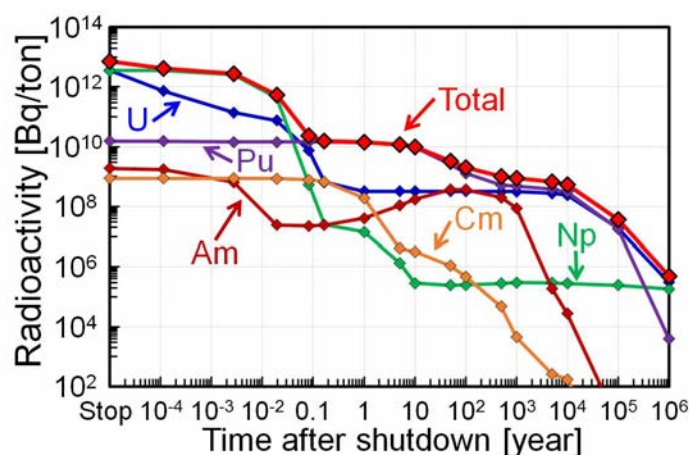


FIG. IV-4.9. Radioactivity of the beryllide with impurity for neutron multiplier.

SUMMARY

As the amount of radwaste can be considerable and impacts DEMO plant design, an appropriate conceptual radwaste management scenario design is needed. The main parameters of DEMO are a major radius of 8.2 m and a fusion power of 1.5 GW. We developed a radwaste management scenario for the DEMO based on the residual heat and dose rate of used IVCs. In the hot cell, the used IVCs were handled as follows. First, blanket modules are removed from a back-plate with remote handling equipment after 7 years of cooling. Next, both the neutron multiplier and tritium breeder are recovered from each blanket module for recycling in the disassembling facility. The divertor mono-blocks are also removed from a cassette body after 1.5 years of cooling, and the radiation environment in the hot cell is estimated to be 100 Gy/h. Then, the empty blanket modules and divertor plates are cut into small pieces and stored in a container. The radiation environment in the facility is estimated at 10 Gy/h, with a temporal storage of 13 and 5.5 years. After radwaste classification according to radioactivity, the radwaste is packed for disposal. The total amount of blanket module and divertor cassette low-level radwaste is 892 1.6 m cube-shaped containers.

Classification of the radwaste is also important from a social acceptance perspective. The analysis indicates that the induced radioactivity of the first blanket wall is higher than that of other IVCs, such as the divertor and back-plate, and that long-lived radionuclides are produced in the first wall at high concentrations. Classification of radwaste disposal is determined by evaluating public radiation exposure following disposal, considering radionuclide transport via likely biosphere exposure pathways. Our assessment indicates that the radiation exposure is as low as 3.4 $\mu\text{Sv/year}$, lower than the shallow land burial criterion (10 $\mu\text{Sv/year}$), when the used blanket is cooled for 10 years and is disposed in a concrete pit near the land surface. The situation is the same for the used divertor and back-plate of the blanket segment. According to the evaluation of DEMO radwaste classifications, we also found that the interim storage process may not be required.

ACKNOWLEDGEMENTS

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IV-5. HYBRID FACILITIES WASTE IN THE RUSSIAN FEDERATION

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Hybrid (fusion-fission) systems, a concept, which was first elaborated several decades ago, use abundant neutrons produced in a fusion reaction to initiate a fission reaction. A fusion-based neutron generating core with a subcritical fission-based blanket has several advantages over pure fusion reactors. First, a plant would be feasible with significantly smaller reactor dimensions. The fission blanket may be designed to implement a number of functions including producing nuclear fuel for traditional fission-based reactors, generating energy and neutrons for waste burning and may be complemented with a tritium breeding blanket to produce fusion fuel. A hybrid system has slightly lower requirements to fusion yield and thus might be introduced somewhat sooner than genuine fusion systems. It is also viewed as an alternative to fast reactors for nuclear waste transmutation and nuclear fuel breeding and as a testbed for materials and technology that could be later applied for fusion technology.

In the Russian Federation hybrid systems form an integral part of the national fusion development effort, with significant resources allocated to hybrid concept development. A number of conceptual designs have been produced since the late 1980-s, the most recent being DEMO-FNS developed by Kurchatov Institute [1].

It is yet unclear as to what specific design of a hybrid facility would be deployed first, and when will this happen. Still, the current level of maturity of the concept and the described advantages of the fusion-fission systems would suggest that it may happen within a few years of deployment of pure fusion systems or even earlier in the event that hybrids be prioritized in the national program. This calls for consideration of the problem of decommissioning such facilities and management of radioactive waste produced in process of operation and decommissioning.

With hybrid facilities combining both fusion-based core and a blanket zone where fission reactions take place, the produced waste would have species with features common to both fusion and fission:

- High tritium content;
- High content of neutron-activated materials;
- Transuranium nuclides;
- Fission products.

The fusion part of the waste is expected to be highly similar to what will be expected from a pure fusion facility. The fission part of radwaste would vary depending on the source nuclide selected for the fissile blanket – uranium or thorium. Preliminary calculations performed for depleted uranium, natural uranium and natural thorium metallic blankets show that spent blanket materials would have radionuclide profiles comparable to those of PWR spent fuel in terms of fission products. Content of trans uranium elements would have some differences, especially for thorium-based blankets.

Another problem with blanket material reprocessing, though not immediately connected with radwaste management, is poisoning of the target nuclide product (Pu-239 and U-233 for uranium and thorium options respectively) with even isotopes Pu-238 and U-232 and their decay products, leading to severe increase in radioactivity. It remains questionable whether these products would be loaded back to the hybrid or thermal reactors (which will require remote handling and additional radiation protection measures) or would need to be separated and treated as radioactive waste.

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ANNEX V.

MANAGEMENT OF FUSION SPECIFIC RADIOACTIVE AND TOXIC MATERIALS

V-1. DETRITIATION PROCESS SUMMARY AND OPPORTUNITIES: APPLICATION TO TRITIATED METALLIC WASTE

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Abstract

During its operating and dismantling phases, a fusion reactor using deuterium and tritium (DT) as fuel will generate tritiated components requiring maintenance, and different types of tritiated solid waste for which suitable and appropriate waste routes need to be provided. Depending on the relevant national waste management strategies, and waste acceptance criteria (WAC) of the final repositories, the management of this tritiated fusion waste often combines a series of technical processes. Implementation of detritiation processes is to be considered as an opportunity to improve waste management, reduce environmental impact of tritium release, recycle material for nuclear application (circular economy), and enable reuse of extracted tritium. Implementing an appropriate non-destructive detritiation process during operational phases of a fusion system also brings an opportunity to ease the maintenance of tritiated components via reducing the associated tritium hazard of materials. The correct balance between the benefits and drawbacks of using a detritiation process is to be assessed on a case-by-case basis. Though detritiation has proven promising in research and fusion applications, it has to be noted that most of detritiation processes still require an increase of Technology Readiness Level (TRL) to be confident for industrial-scale use. In particular, there is a current lack of R&D on fusion relevant materials and the behaviour of tritium in these materials. Future DT fusion systems are likely to implicitly rely on tritium reduction and recovery methodologies such as detritiation due to increased tritium use compared with current systems. Hence material detritiation behaviour is a key research area to support future fusion endeavours.

INTRODUCTION

During its operating and dismantling phases, a fusion reactor using deuterium and tritium (DT) as fuel will generate tritiated components requiring maintenance, and different types of tritiated metallic waste for which suitable and appropriate waste routes need to be provided. Depending on the relevant national waste management strategy, and waste acceptance criteria (WAC) of final repositories, the management of this tritiated waste often combines a series of processes. These processes are likely to include a cooling phase (where short-lived activation products in materials can decay), waste treatment(s), conditioning steps for disposal (such as immobilisation of materials requiring sometimes a preliminary cutting step), and interim storage for tritium decay. Following these processes, it is then likely that materials will be transferred to a final repository or possible reuse (circular economy) if waste has an activity below any relevant clearance levels. Implementation of detritiation processes is to be considered as an opportunity to ease waste management, reduce environmental impact of tritium release, recycle materials for nuclear application (circular economy), and enable the reuse of extracted tritium. Implementing an appropriate non-destructive detritiation process during operational phases of a fusion system also brings an opportunity to ease the maintenance of tritiated components via reducing the associated tritium hazard of materials.

TRITIUM REMOVAL FROM MATERIALS AS AN OPPORTUNITY FOR WASTE TREATMENT OR COMPONENT MAINTENANCE

Tritium removal from materials via detritiation processes is an opportunity for waste management and for the maintenance of components. With respect to detritiation aiding with maintenance of components,

such a process eases maintenance implementation by the reduction of tritium hazard. The reduction in tritium content via detritiation also offers the possibility of reuse of components rather than disposal [1]. Due to the harsh environmental conditions seen by plasma facing components, few in-vessel components will be suitable for short-term re-use, and where radiation damage occurs, longer term recycling may not be an option. On the contrary, components within the tritium plant, cooling plant, vacuum pumping plant or remote handling equipment that are earmarked for scheduled maintenance / refurbishment can be identified generically as candidates for detritiation prior to short term reuse. In this case, surface treatment or light thermal treatment are likely sufficient since these techniques are unlikely to degrade component physical properties.

In the field of waste management, a tritium removal process provides more flexibility for the safe handling of materials throughout waste processing, from production of waste to disposal activities. This follows since reduction in tritium content of materials reduces the rate of tritium release to air, the reduction in tritium permeation through Personal Protective Equipment (PPE), and the reduction in tritium cross contamination to other materials and the generation of secondary waste materials. The process also has the key benefit that materials can have a reduction in total radioactive content, and potential accident scenario severity. This eases compliance with transportation constraints and enables disposal sites of a lower radiological containment to be used – for example a near-surface disposal store rather than a deep underground repository. There is experience in the UK of materials from the Joint European Torus (JET) system being detritiated to the point where they can be disposed of in hazardous landfill conditions rather than in a deep underground repository. This has a variety of cost, efficiency, and public perception benefits.

Wherever interim storage of waste is required prior to a final repository, waste treatment could reduce the interim storage duration and surface storage infrastructure demands. In some cases, applying such a treatment could lead to a possible downgrading of the initial waste category (from Intermediate Level Waste down to Low Level Waste) and limit storage facility safety constraints (less complex ventilation / containment system). These benefits are potentially significant over the extended lifetime of a storage facility. Detritiation processes also enable key benefits in terms of radioactive material transport by reduction in the tritium gaseous releases from materials. The IAEA transport limit of 0.96 GBq/day/package of tritium release [2] may be exceeded by highly tritiated materials if transport is required between sites via public roads / railways etc.

The main drawbacks for detritiation arise from the management of the gaseous or liquid effluent which contains extracted tritium. Depending on the adopted detritiation process, all extracted tritium can become secondary waste which needs either be treated in view of reuse, stored, or disposed of. Indeed, a full detritiation process requires a detritiation unit adapted to the type of waste to be treated, together with a system designed to reduce the environmental impact of the gaseous effluents produced by the detritiation unit (see Fig. V-1.1).

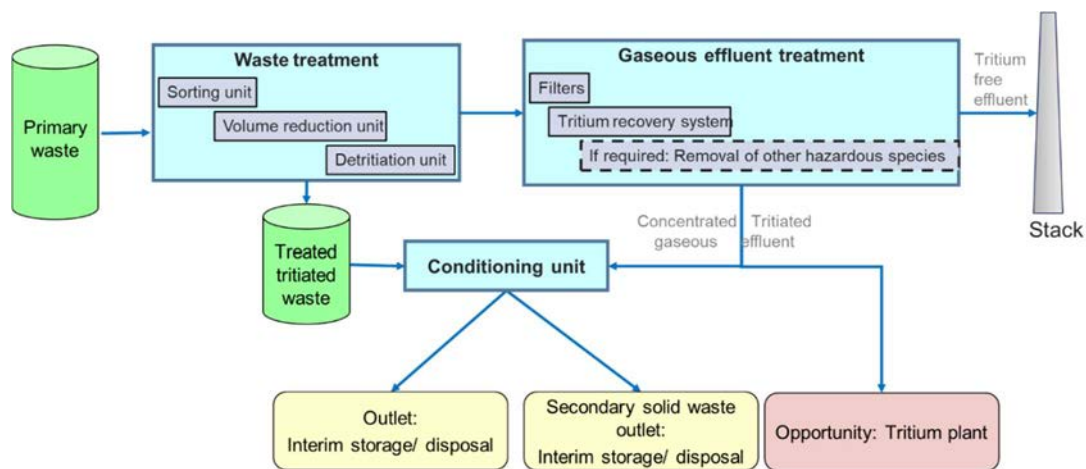


FIG. V-1.1. Detritiation process principle for waste materials.

As an example, UKAEA has developed the following waste management strategy for the JET solid tritiated Intermediate Level Waste that under UK regulations would require deep geological disposal at great expense and complexity (see Fig. V-1.2).

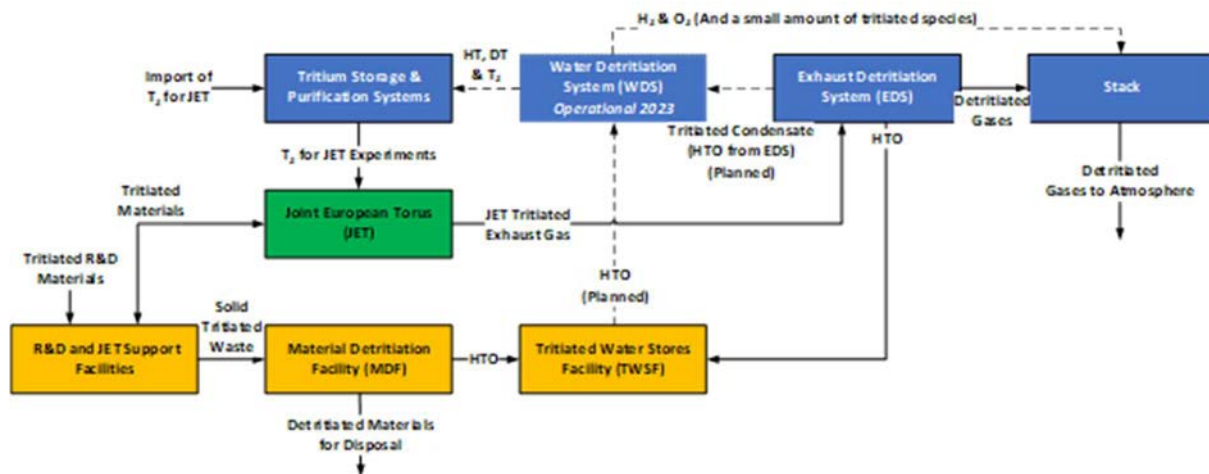


FIG. V-1.2. JET tritiated intermediate level waste management principle.

There are several tritium removal techniques for materials, these include methods for removal of tritium from surfaces and methods for removal from the bulk of materials. Surface efficient treatments include plasma or laser abrasion, leaching, chemical or electrochemical etching, polishing etc. Bulk detritiation techniques include thermal treatment in a vacuum or in a reducing or oxidising carrier gas. There is also experience of using a wide range of temperatures during thermal treatment depending on the materials being treated. UKAEA have frequently used temperatures in the order of 1000 °C to detritiate carbon fibre, Inconel, steel, and copper materials. There are decades of experience of melting materials to release tritium at CEA. The choice of the best available techniques relies on the assessment of several criteria such as safety and environmental issues, the required tritium removal efficiency, the physical properties of materials being treated, the technical readiness level of the process, the amount of secondary waste produced, the costs (investment, operating and dismantling costs), and public / regulator acceptability. The reduction of the overall waste volume is also an opportunity where destructive detritiation methods such as melting are deployed and by the potential preliminary cutting step required prior the waste processing. Generally speaking, surface treatment methods are unsuitable for waste with large tritium loading in which tritium will have permeated deep into materials, such as those expected from ITER.

FUSION MATERIALS FEATURES

In fusion facilities, plasma facing as well as structural materials need to meet specific requirements to ensure safe operation of facilities. Indeed, plasma facing materials are facing extreme thermal loads, repeated intense thermal shocks, repeated interactions with plasma ions, neutral particles, and neutrons. Exposure to these fusion conditions leads to damages that may not jeopardize the integrity of the materials. In addition, these materials are required to limit plasma heat loss and pollution, ease the cooling of systems, limit chemical interaction with hydrogen isotopes, present a poor toxicity, have a low tritium retention and low activation properties, and finally need to be industrially produced [3]. Similar but less stringent constraints exist for the structural materials which are not directly plasma facing (for example tokamak structural segments). Though not directly exposed to plasma, a large amount of structural materials are likely to become activated by fusion neutrons during operations and are also liable to become tritiated via tritium permeation around a fusion system. Both plasma facing and structural materials present key challenges in terms of hazard and in terms of waste disposal for future fusion facilities. As a consequence, in the recent years, many R&D efforts have focused on the development of new or improved materials for fusion applications, leading to the use of specific materials compared to the conventional materials used in fission applications.

Despite research efforts, the main issue for tritium removal treatments from fusion materials is the lack of literature / data currently available. In particular, there is uncertainty in the behaviour of fusion relevant materials towards tritium (permeation, off-gassing, tritium removal, plasma-material interactions and their impact on tritium retention, etc.). Indeed, most of available knowledge concerns the widely used material – stainless steel. It is a key area for development that more fusion relevant materials (such as metallics like tungsten, Eurofer and breeder blanket materials) are investigated with respect to their tritium behaviour in fusion system conditions.

FOCUS ON THERMAL TREATMENT

Among all possible materials that could be used within fusion facilities, only few have been subject to tritium diffusion experiments. Moreover, most of experiments have been performed on small and surface contaminated samples that were artificially loaded with tritium during a few hours under moderate temperature. It is frequently the case that such samples have very large surface area to volume ratios (S/V ratio) which may enhance tritium removal efficiency compared to larger scale materials likely to be present in a fusion system. There are also limited studies on materials which have received plasma-relevant damage such as neutron damage or ion implantation of tritium ions. As a consequence, scale up of thermal treatment performance obtained on surface contaminated samples is potentially not representative of reality, and needs to be validated by tests on relevant samples both in terms of tritium contamination and size. Nevertheless, based on extensive literature [4-14], and CEA and UKAEA results, the following main conclusions can be raised on stainless steel.

- Impact of sample size: The smaller and thinner the sample (producing a favourable S/V ratio), the more efficient the thermal detritiation process. Surface contamination is favourable for efficient thermal detritiation. Where materials require a carrier gas for tritium capture during detritiation, there is a clear assumption that surface exposed to gas flow, and hence have promoted isotopic exchange of tritium, are detritiated more efficiently. Hence sample geometry may be a key factor under certain detritiation conditions where carrier gases are used, or in scenarios where tritium diffusion through bulk materials is relevant.
- Impact of carrier gas: use of a reductive gas as a carrier modifies the HTO/HT ratio in the gaseous effluents to almost 1. This is clearly an opportunity wherever HT & HTO release authorization is differentiated (health and environmental impact of HT being lower than that of HTO) and for circular economy scenarios where direct recovery of HT is sought. Using an oxidising carrier gas leads to produce mainly HTO in the gaseous effluent. Furthermore, oxidising the surface may create a kind of permeation barrier that contributes to limit tritium release. The presence of steam in the carrier gas improves the detritiation factor even at low concentrations. Above a limit, the increase of water content in the carrier gas did not reveal any impact on the detritiation factor [11].
- Impact of treatment duration: measurement of the detritiation kinetics generally reveals a plateau in tritium release which appears more or less rapidly depending on the temperature, contamination, type and size of the part. It has been demonstrated that tritium release is concentrated early in thermal treatment cycle and that a prolonged duration at the same elevated temperature does not produce significant additional tritium release from materials.
- Impact of temperature: Depending on the temperature, the materials and the species under consideration, the permeability will be governed either by tritium species mobility or by solubility. Increasing the temperature induces the homogenization of tritium contamination (tritium migrates from the surface to the atmosphere, but also from the surface into the depth of the part) and the reduction in the surface contamination. Thermal treatment conducted at a too low temperature provokes migration of tritium into the depth of the part instead of removing it, and a temporary drop in the tritium desorption rate and the surface contamination. This effect eventually disappears with the gradual migration of tritium towards the surface over time. It seems that a treatment temperature in the order of or above 1000 °C is required to obtain a significant detritiation factor for most current materials which have been tested such as carbon fibre, Inconel, steels and copper. Materials such as aluminium with a low melting point have been detritiated at lower temperatures than 1000 °C such that they do not melt and compromise

current detritiation systems. Experience from the JET system has demonstrated a reduction in the tritium radioactivity by 3 orders of magnitude in carbon fibre and Inconel materials via thermal treatment under air at ~1000 °C [15]. However, melting of material is the most efficient treatment for deeply tritium contaminated stainless steel as detritiation factor of around 1000 is obtained compared to nearly 100 for thermal treatment at 1000°C. Melting also offers the opportunity of a significant volume and tritium out gassing reduction factor.

Similar conclusions could be raised for other fusion relevant materials such as different grades of Inconel, beryllium, copper, Eurofer, tungsten etc. Consideration is needed of material properties such as oxidation (tungsten and beryllium), melting points (very high for tungsten), and the tritium initial inventory, distribution, and permeation in materials.

CONCLUSION

Implementation of detritiation processes is to be considered as an opportunity to ease waste management, reduce environmental impact of tritium release, recycle material for nuclear application (circular economy), and promote reuse of extracted tritium. The correct balance between benefit and drawbacks is to be assessed by a case-by-case analysis due to the inherent complexity associated with material detritiation. If correctly implemented, the detritiation process can provide a unique fusion benefit in which radioactivity from waste materials is recycled for use in future fusion applications. There is also the opportunity to reduce the tritium hazard of materials prior to handling for maintenance, analysis, or disposal, and to reduce environmental impact in a general sense. These twin benefits have potentially significant cost and time implications for future fusion systems with large tritium loadings.

Most of the currently identified detritiation processes still require an increase of TRL to enable confidence for an industrial use. Furthermore, extensive R&D support is still needed as most of the supporting results for detritiation were obtained at lab scale with non-representative samples in terms of material nature, size, characteristics and tritium contamination. Extensive results exist on non-irradiated stainless steels but there is a lack of information in all but a few fusion relevant materials. In particular, specific attention needs to be paid when developing new materials which may make detritiation processes more problematic (or significantly easier). For example, some material impurities could contribute to tritium trapping (C, Ti) and reduce the detritiation treatment efficiency.

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V-2. MANAGEMENT OF FUSION SPECIFIC RADIOACTIVE AND TOXIC MATERIALS – EU DEMO

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European DEMO objectives

As discussed in [1³⁰], the European DEMO aims to demonstrate the production of net electricity, the feasibility of operation with a closed-tritium fuel cycle and the adoption of maintenance systems capable of achieving adequate plant availability while demonstrating the safety and environmental advantages of fusion power compared with fission designs. Therefore, a coherent strategy for dealing with the waste arising from the reactor needs to be part of the justification for realizing DEMO. This strategy is required to minimize the environmental impact of the reactor whilst demonstrating an economically feasible approach. In terms of decommissioning and waste, the following high level plant requirements have been established:

- The DEMO plant has to be designed for an acceptable decommissioning cost and duration;
- The DEMO plant has to minimize the amount of radioactive waste produced (e.g. by applying the ALARP criterion for material choices, appropriate radiation shielding, waste separation and recycling in Active Maintenance Facility, and long term waste immobilization);
- The DEMO plant has to minimize long-term geological disposal waste;
- The DEMO plant has to be able to detritiate efficiently structures, fluids and remote handling equipment (i.e. minimize spent component waste).

Considering these requirements ought to become an integral element of the design process and they need to be constantly monitored to minimize all waste, especially long-lived radioactive waste. EU DEMO is currently moving from the pre-conceptual to the conceptual phase and no host country is selected yet while waste management and decommissioning depend on national legal and regulatory framework. The on-going studies aim thus at integrating generic requirements into the design and identify any showstopper that would require to design changes [1].

Waste minimization strategy

The waste minimization strategy is based on 3 axes: reduction at source, consideration of recycling and reuse and waste management optimization. The reduction at source relies first on the choice of the materials considering in particular low activation materials such as the Reduced Activation Ferritic-Martensitic steel EUROFER97 as structural material for in-vessel components [2]. Work is also on going on impurities to get materials with better behaviour in terms of waste and dose rate while keeping the expected level of mechanical properties. Resistance to corrosion is also being studied considering anti permeation barriers that enable to prevent the corrosion of EUROFER by PbLi and minimize the tritium permeation rate from the breeder to the coolant and thus limit the inventories [3]. Nevertheless, attention is needed concerning the thickness of such barriers to limit waste as Al-based coating is foreseen for such barriers and its activation could be detrimental to waste classification [1]. Development of reliable technologies will also enable to minimize replacement and/or maintenance and thus limit the waste production at source. Optimization of the recycling and reuse opportunities are also being studied for EUROFER and stainless steel AISI 316 [4], tungsten and beryllium (Be12Ti is used in the Helium Cooled Pebble Bed blanket design). Waste management optimization relies in particular on the segregation of the various waste fluxes considering designs that will allow segmentation of component to prolong their service lifetime and enable a possible dismantling of elementary parts in order to separate them according

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to their radioactivity content (and reusability). Development of processes to reduce the inventories considering decontamination from dust and activated corrosion materials, detritiation (in-situ baking, heating and melting) and decarburization (to reduce ^{14}C inventories) area also being analysed.

Waste classification

In order to evaluate the impact of the design choices on the waste minimization strategy, evaluations of the waste classification are regularly performed. As there is no site selected and thus no clear waste acceptance criteria to be considered, waste classification is performed using different types of classifications: those based on global activity level (as used in UK or Germany) or those based on nuclide-specific limits (such as France or Spain) [1].

Such evaluations [5] have shown that regardless of the waste regulations, most of the in-vessel steel will not meet low-level waste (LLW) requirements on the 1000-year timescale. Results are slightly better for vacuum vessel, but inner layers do not meet LLW requirements. Depending on the regulations, ^3H , ^{14}C and ^{94}Nb are the main responsible isotopes for classification as Intermediate Level Waste (ILW). If detritiation and decarburization can be considered for ^3H and ^{14}C removal respectively, ^{94}Nb can't be easily reduced. Nevertheless, chemical analysis of the Nb content in EUROFER94 produced by steel makers showed that a maximum of 5 ppm could be achieved (instead of the 50 ppm in the current specifications). This new value could greatly improve the performance of the EUROFER in terms of waste management and could possibly lead to LLW classification in French disposal.

The optimization studied have focused up to now on the limitation of Intermediate Level Waste and the High Level Waste (HLW) was assumed to be zero. There is currently no disposal available in Europe for HLW so only preliminary waste acceptance criteria can be considered. In UK and France, there are criteria in particular on dose rate and heat output [1]. As a first assessment the following maximum heat output have been considered at the time of disposal:

- UK limits: 60 W for 2m box or 1.9 W/t of waste;
- French limits: 10 W/package or 2.5 W/t (considering a packing factor of 5t per 5 m³ package for massive components).

On this basis, parametric studies have been performed considering classification at material or component levels, impact of W self shielding¹ in FISPACT calculations, impact of U impurity level in Be₁₂Ti (between 1.4 and 105 ppm), impact of tritium removal (none, 95 and 100% efficiency). Assessments have been done for both European breeding blanket designs (Helium Cooled Pebble Bed and Water Cooled Lithium Lead) at the end of life when 50 dpa are expected to be reached. At material level, more than 100 years of radioactive decay would be necessary to be considered as LLW for both concepts. Tungsten self-shielding as no influence on the HLW/ILW classification. Depending on U amount in Be₁₂Ti, between 14 and 26 years of radioactive decay are needed to move from HLW to ILW. The most important factor is detritiation efficiency as for some materials such as KALOS (ceramic pebbles, consisting of Li₄SiO₄ and Li₂TiO₃, developed as tritium breeding material for ITER HCPB) 100 years of radioactive decay are needed to move from HLW to ILW without detritiation. This duration can be reduced to 45 years and 1 year with 95% and 100% detritiation respectively. On this basis, it appears that decay heat will have to be managed when waste is interim stored in DEMO and high efficiency detritiation of breeder materials and neutron multipliers needs to be achieved before sending waste to disposal. Considering the national classification at component level, about 20 years of radioactive decay is needed for HCPB blanket with 99% efficiency of detritiation to reach ILW classification. ^3H , ^{54}Mn , ^{94}Nb , ^{108}mAg prevent those blankets to become LLW at plasma shutdown and, even with 100 years of radioactive decay, ^{94}Nb remains above LLW acceptance criteria in France. The same issue is with vacuum vessel inboard and shell.

R&D program and conclusion

EUROfusion is funding R&D programmes in order to fulfil the high level plant requirements on waste and decommissioning. This concerns in particular the establishment of inventories for solid, liquid and gaseous waste stream and the treatments to be developed inside and outside the plant to facilitate their management for final disposal [1]. Experimental campaign of material detritiation using the Materials Detritiation Facility (see Section 5 of this report) operated by the UKAEA (thermal process) and W dust detritiation considering sintering in ENEA [6] and Molten Salt Oxidation and Induction Heating in a Cold Crucible processes in IPP.CR [7] are being performed. In the near future it is foreseen to test technologies for activated steel treatment including recycling, optimize the waste recycling process, develop the processing of ‘difficult to manage’ waste streams (mercury, PbLi (incl. waste from PbLi decontamination), pebbles, getter beds...) and characterization of tritium and «difficult to measure» nuclides, improve the packaged waste (blocking matrix, tritium outgassing) and outline a first decommissioning plan

The ultimate goal at conceptual phase is to identify any showstopper and identify needs for the Active Maintenance Facility and Waste Treatment Facility.

ACKNOWLEDGEMENTS

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V-3. MANAGEMENT OF RADIOACTIVE WASTE FROM FUSION – THE JET EXPERIENCE

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Abstract

The paper provides a brief introduction to JET at Culham, UK and the current operational programme. JET commenced operations in 1983 and has conducted 2 experimental deuterium-tritium (DT) campaigns which have currently used ~15% of the total 14 MeV neutron budget. The current JET operational programme, which is dedicated to ITER support, is due to complete in 2023; this will include a further DT campaign to fully exploit the remaining neutron budget. JET is funded by the European Commission through EUROfusion and operated by UKAEA. JET produces a range of waste types and activities in which tritium is the predominant radioisotope, however activated materials and finely divided Beryllium dust are also present in waste originating from ‘in-vessel’ components. UKAEA has developed processes and opened waste routes to seek to deliver ‘zero’ operational waste at the end of JET operations. However, twenty-seven tons of hard (metallic and carbon) Intermediate Level Waste (ILW) had been generated which did not meet current waste acceptance criteria for existing disposal facilities due to the level of tritium present. Therefore, deferred disposal in a future Geological Disposal Facility (GDF) was the baseline strategy. Following a research study, UKAEA has developed an innovative thermal detritiation treatment process and complimentary Water Detritiation System which will not only enable the down categorisation of waste but also facilitates the recovery of tritium for reuse; thereby closing the fusion fuel cycle for ILW for the first time. The UK Government is investing in a range of fusion related research facilities at Culham including the Hydrogen-3 Advance Technology (H3AT) facility. By partnering with others, H3AT will become a world centre of excellence for tritium (and waste) research at the heart of which will be an ITER relevant tritium processing loop.

INTRODUCTION

The Joint European Torus (JET) Tokamak is sited at Culham, Oxfordshire and funded by the European Commission through EUROfusion. It was established with the long-term objective of providing a safe, environmentally sound prototype for fusion reactors. The operation of JET has been carried since 2000 by the United Kingdom Atomic Energy Authority (UKAEA). The first plasma was produced in 1983 reaching 16 kiloamps (EUROfusion Consortium of Research Institutions, 2008). JET has operated for nearly 40 years with numerous experiments most notably the first tritium experiment (PTE1) in a Tokamak in 1991, and at the time the world record in fusion power production in 1997 producing 16 MW utilising deuterium and tritium (DTE1). The JET facility is currently operating the second deuterium-tritium experiment (DTE2) where the majority of the remaining neutron budget will be utilised. The use of higher energy neutrons will have a greater potential to activate the structural components of the experiment significantly changing the activity of radioactive waste, dose rates and hazard. The use of tritium as a fuel and will also create tritiated components above 12 GBq/t limit for Intermediate Level Waste (ILW) categorisation. A minor proportion of the tritium will be attributed from neutron activation of materials. In 2023 there is possibility of a further tritium experiment being planned.

Legislation and policy for radioactive substances in England

UK legislation for environment is devolved, with different regulations and regulators depending on the country. In England, where the JET is based, the operation of the site is covered under the Environmental Permitting (England & Wales) Regulations 2016 (EPR2016) and regulated by the Environment Agency (EA). The EA authorises activities related to radioactive substances through Permits. This includes the keeping and use of radioactive materials, the receipt of radioactive waste, accumulation of radioactive waste and also disposal of radioactive waste. The permit, authorised to the UKAEA for the operation of JET and related facilities, contains standard conditions and specific limits to the site. As part of the standard conditions there are a number of requirements to ensure the operator, UKAEA, applies Best Available Techniques (BAT). For example, the use of BAT to minimise the use of radioactive materials, use of BAT to minimise the generation of waste and minimise discharges in terms of public dose and environmental impact to name a few.

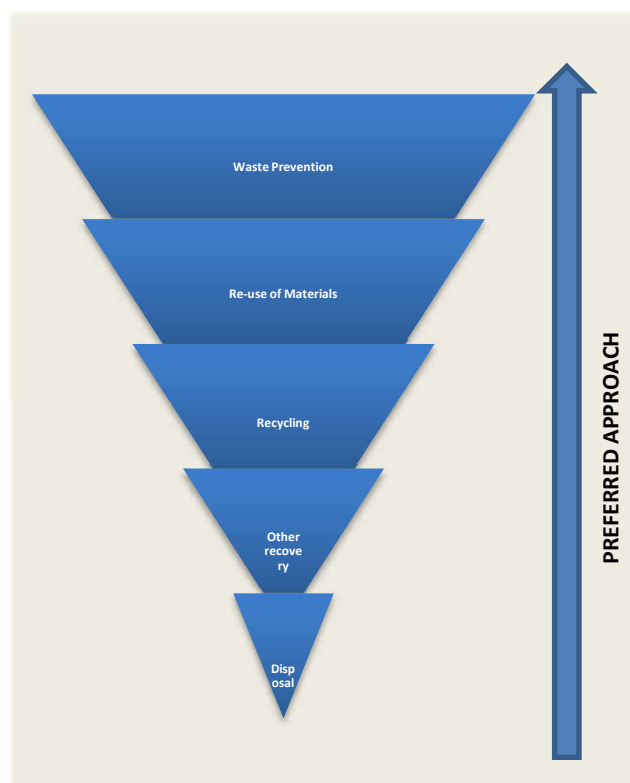


FIG. V-3.1. The Waste Hierarchy based on (UKAEA, 2020).

A basic principle of radioactive waste management in the UK is to implement the Waste Hierarchy (UKAEA, 2020) shown in Figure V-3.1. The Waste Hierarchy was initially set out through the European Directive 2008/98/EC on Waste and implemented in England through The Waste (England and Wales) Regulations 2011 for non-radioactive waste. The principles were adopted via the UK government policy on Low Level Waste (LLW) management (Defra, DTI and the Devolved Administrations, 2007) and regulatory expectations include the use of the hierarchy for prevention of radioactive waste as primary goal and with less preferred options being re-use of materials, recycling, other recovery and final last resort option of disposal.

UK waste is currently classified into High Level Waste (HLW), Intermediate Level Waste (ILW) or Low Level Waste (LLW) dependent on activity.

HLW and ILW are commonly grouped together and referred to as Higher Activity Waste. They are classified based on activity level and not radionuclide specific where the either the beta/gamma component is above 12 GBq/t and/or the alpha component is above 4 GBq/t. The differentiator between

HLW and ILW is that HLW generates heat which needs to be taken into account in the design, storage and disposal facilities for that waste. At Culham, no HLW is produced. The policy for Higher Activity Waste is devolved in the UK. In England, where no treatment route or current disposal route is available, waste is envisaged to be transferred to the Geological Disposal Facility (GDF). Waste is passively stored in interim storage facilities, preferably close to the production of the waste, to enable the decommissioning of facilities and reduction of hazard. Decisions regarding how waste is passivated, packaged and stored requires robust disposability assessment cases to ensure waste will be acceptable to the GDF. It is an expensive process requiring significant infrastructure to produce packages to high specifications and ensure there are significant efforts for records management of the waste stream.

LLW is where the activity is below 12 GBq/t for beta/gamma and/or 4 GBq/t for alpha and again is not radionuclide specific. Currently in the UK there is one national facility for the disposal of LLW which is the Low Level Waste Repository (LLWR) in Cumbria. The facility has a finite space and is utilised for radioactive substances disposals from all sectors including the nuclear industry, research facilities, universities and hospitals which meet the Waste Acceptance Criteria (WAC). The UK strategy for LLW is to avoid using the LLWR where possible by utilising the Waste Hierarchy and alternative techniques such as advanced waste characterisation, combustion, super compaction, metallic treatment and for problematic waste specific alternative treatment.

A sub-category of LLW is known as Low Activity LLW (LA-LLW), which comprises waste that can be safely disposed alongside commercial, municipal or industrial waste in specified engineered landfill sites. Each designated landfill has a WAC and can normally accept levels of 200 Bq/g averaged over a 10 tonnes consignment with a maximum specific activity of 1000 Bq/g.

In terms of radioactive waste management there are three further categories to consider: Out of Scope (OSR), Exempt and Clean. Out of Scope (OSR) is where the item/material is not radioactive for the purposes of the EPR2016 legislation and is no longer subject to the radioactive regulatory requirements, however it may still be required to be managed as a non-radioactive waste for management and disposal. The limit for whether an item is OSR is radionuclide specific and is derived from scenarios leading to a public dose of 10 μ Sv/y. The limits can be found in UK government guidance (BEIS, 2018) and specifically used for solid waste and liquids specified as 'relevant liquids' under EPR2016.

Waste production and composition at JET

Operations of the JET facility produces both radioactive and non-radioactive waste that need to be managed. Radioactive waste generated includes solid, gaseous, aqueous and other liquid waste which are regulated under the site Permit.

The experiment produces activated metals including INCONEL®³¹, tungsten, copper and stainless steel with key radionuclides such as Co-60, Fe-55 and Ni-63 identified within fingerprints. The activation of metals will also produce a small amount of tritium (H-3) however the majority of tritium within the facility will be from the use of tritium as a fuel and permeation into materials. Permeation of tritium is dependent on many factors but notably heat and porosity of materials. In 2010, the structure of the JET torus was changed from use of carbon composite tiles and replaced with an 'ITER-like' wall (Matthews & al., 2011). The 'ITER-like' wall utilises beryllium and tungsten which protects the primary vessel from extremely hot plasmas. However, in return beryllium dust is generated from erosion and deposition which is hazardous with a similar nature to asbestos. The increase of activation in materials will also lead to increased dose rates and therefore increased risk to workers.

For gaseous waste, the operation of the experiment produces tritium waste in form of tritiated water (HTO), elemental tritium (HT) and small proportion of tritiated methane (CH₃T), beta/ gamma particulate and also short-lived radionuclides in activated air. Discharges are monitored using instrumentation and exhausted via stacks where required. Site limits for gaseous discharges exist for

³¹ INCONEL is a registered trademark of the Special Metals Corporation group of companies

specific radionuclide groups within the Permit and where required facilities have internal action levels set as percentages of the overall limit to ensure that discharges are monitored and reviewed ensuring BAT is being applied to minimise discharges.

In accordance with NDA guidance on good practice³², “aqueous discharges are produced from various routes including:

- Aqueous waste generated within radioactive designated areas, such as hand-washings and/or floor-washings, scrubbers in ventilation systems, and drain down of cooling water systems;
- Stack condensates;
- Desludging and rinsing out of effluent tanks / bunds;
- Relatively small volume aqueous waste from process and decommissioning operations,
- Surface water (which includes rainwater) or groundwater that has entered and collected in potentially contaminated facilities, drainage networks, tanks, sumps and bunds.”

Aqueous discharges are collated into holding tanks where the tanks are re-circulated prior to sampling for gross beta, tritium, high resolution gamma spectroscopy and beryllium. The results are reviewed and on meeting the discharge limits the holding tanks are then discharged into the trade effluent system under the Permit. The trade effluent system is regulated under a separate Permit with requirements to monitor non-radiological components. This is then discharged into a local water course.

Solid waste are produced from maintenance, replacement and repair of the experiment including metals, concrete, plastics, dust and High Efficiency Particulate Air (HEPA) filters used for abatement of discharges to atmosphere for beta/gamma particulate as examples. Supplementary to this solid secondary waste is generated from personal protective equipment (PPE), housekeeping and analytical waste.

A range of techniques are utilised for waste characterisation at JET including x-ray fluorescence to detect material type (i.e. differentiate between Inconel-600 and Inconel-625), use of liquid scintillation for analysis of H-3, beryllium analysis using UV Florescence and high-resolution gamma spectroscopy to measure key radionuclides such as Co60. Where further analysis is required to support fingerprint samples are transferred to external laboratories for radiochemical separation and analysis.

A key operational target of JET is to ensure there are ‘zero’ operational waste remaining at the end of JET operations. For example, in 2019, 60,487 kg of radioactive waste arose and 58,926 kg were consigned.

JET WASTE INNOVATIONS AT CULHAM

UKAEA has developed processes and opened several waste routes to seek to deliver the ‘zero’ operational waste target. The paper will discuss two innovative processes to resolve waste management issues, treatment of waste tritiated to ILW levels and the recovery of H-3 from aqueous waste as a potential fuel.

Detritiation of the MkIIA divertor

The MkIIA divertor was used for power and fuel exhaust. It was installed in JET in 1996 and consists of an INCONEL support structure with large carbon-carbon fibre composite tiles. The structure was utilised in the 1997 DTE1 campaign and therefore became activated and also tritiated to ILW levels. The MkIIA divertor was removed in 1999 leading to the production of 5 tonnes of waste. An additional 22 tonnes of hard ILW inventory was also identified with similar H-3 levels. The current baseline for ILW from JET is to package the waste in a stainless steel box contained in a 6m³ robust concrete box. The waste will

³² See Bibliography (16)

then be co-located at a nearby nuclear licensed site, Harwell, with other ILW waste streams awaiting the availability of the Geological Disposal Facility.

Removal of tritium from the MkIIA divertor was proposed as a potential option to reduce activity levels from ILW to LLW (according to UK's classification) and allow availability of disposal routes (Whittaker, Newman, & Coombs, 2014). A trial was proposed utilising three carbon-carbon fibre composite tiles selecting a high, medium and low H-3 off-gassing and heating the tiles to 1000°C for 6 hours in air and capturing H-3 in a water bubbler system. Samples of the tiles were taken prior to treatment and post treatment. The treatment and characterisation were undertaken by Tradebe Inutec, United Kingdom and results shown in Table V-3.1.

TABLE V-3.1. CHARACTERISATION OF CARBON-CARBON FIBRE COMPOSITE TILES PRE AND POST HEAT [15]

Tile reference	Tritium Activity Pre- Bake-out (GBq)			Tritium Activity Post- Bake-out (GBq)			% of H-3 activity removed
	Surface H-3	Bulk H-3	Total	Surface H-3	Bulk H-3	Total	
2-OW-9	0.639	0.577	1.25	0.007	0.063	0.0742	94.08
12-IW-2	0.947	1.757	2.74	0.005	0.015	0.0232	99.15
14-BN-4	47.343	10.997	57.3	0.002	0.012	0.0146	99.97

Several observations were made from the trial. The use of high heat oxidised 10% of the carbon within the tiles. The majority of the tritium was found in the surface of the material rather than the bulk and the trial removed >94% of H-3 activity. Detritiation was more successful in the most highly active tile. For tiles 12-IW-2 and 14-BN-4, based on the weight of the tile and total H-3 remaining, the tiles would be classified as LLW, therefore successfully proving that waste could be detritiated from ILW to LLW according to the UK classification. Tile 2-OW-9 was marginally above the 12 GBq/t level.

The trial was used to develop a business case with European Commission support to design an industrial sized process that would treat 150kg of hard metallic waste in a thermal run. The thermal process utilises a bespoke retort furnace with a flowing air atmosphere. Exhaust gases from the facility are catalysed to convert elemental tritium to tritiated water and carbon monoxide to carbon dioxide. The exhaust gas is then fed through three industrial sized bubblers containing water to capture the H-3. The facility was installed at Culham and is known as the Materials Detritiation Facility (MDF).

The following are preliminary results from of the commencement of operations to January 2021 for all waste types through the facility and then broken down into specific materials such as copper, Inconel and stainless steel (Fig. V-3.2). Samples were randomly taken and prepared using pyrolysis followed by liquid scintillation counting (LSC) at UKAEA.

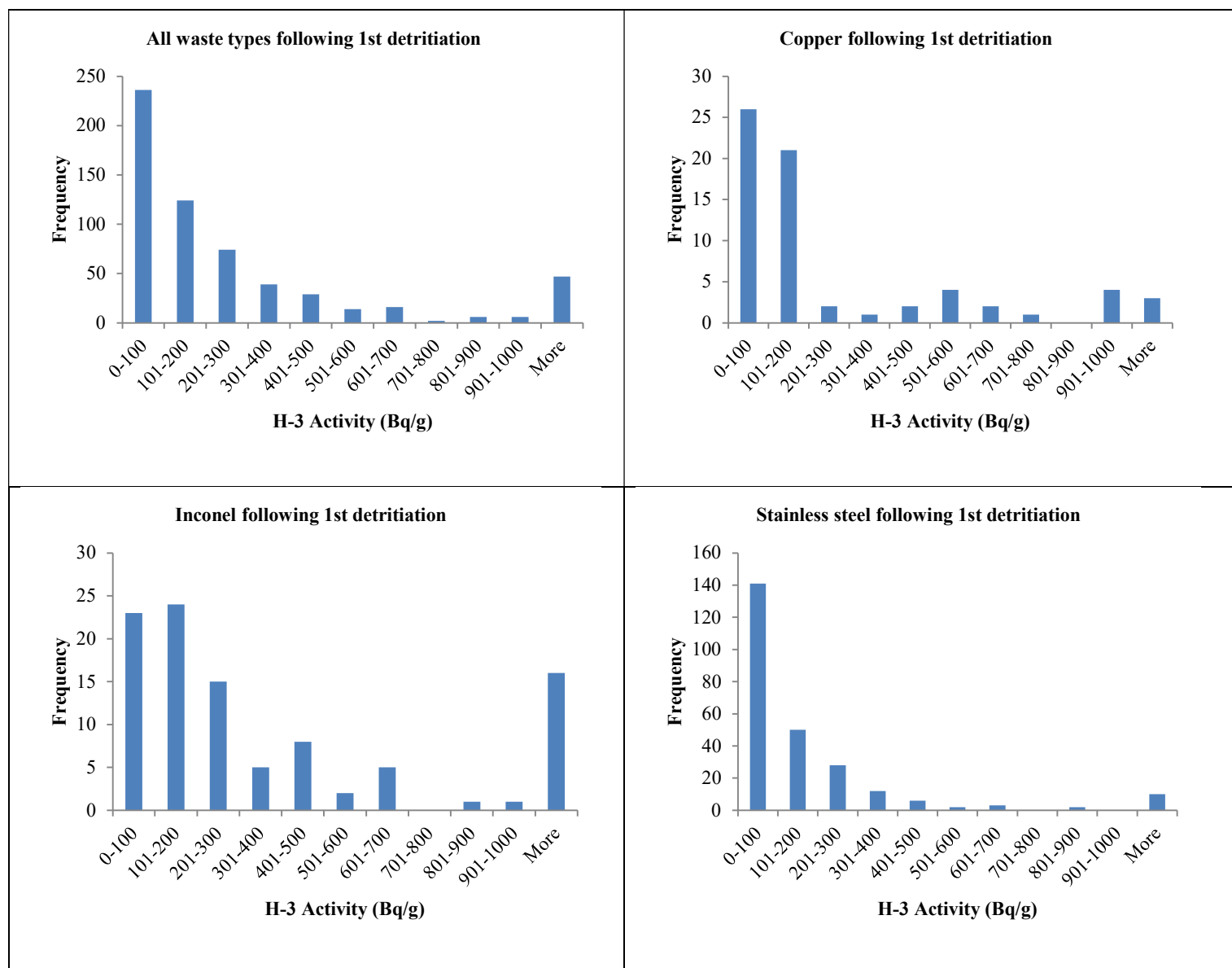


FIG. V-3.2. Results from MDF for tritium in waste post treatment.

The results have shown that post treatment, H-3 levels are notably observed between 0-100 Bq/g. The OSR limit for H-3 in UK legislation is 100 Bq/g and therefore significantly reducing the waste liability from ILW to LLW (according to UK's classification) and potential 'out of scope' of regulation. It is noted that the waste stream is activated and therefore other radionuclides such as Co-60, Fe-55 and Ni-63 are present in the waste that require management and suitable disposal routes to be considered. A number of samples showed results above 1000 Bq/g (the specific activity limit for specified landfill) and therefore items were treated for a second time. The possibility of in-transient H-3 was investigated by preparing further sub-set of samples with acid digestion followed by LSC which confirmed similar results to the original technique of pyrolysis and LSC.

Further work to refine parameters for material type will be made including temperature stages, run-times and number of samples to support disposal.

Detritiation of tritiated condensate

As part of the operation of the MDF and Torus, tritiated abatement condensate is produced which needs to be managed. The Active Gas Handling Facility (AGHS) was installed at JET to supply tritium and deuterium to the experiment but also receive the exhaust gases from JET for recover and to recycle

tritium. The Exhaust Detritiation System (EDS) which is part of the AGHS can remove >99.9% of tritium contents from the exhaust. This is carried out by oxidising the hydrogen isotopes before discharging gaseous emissions to a monitored stack. This minimises gaseous emissions of H-3 and dose to members of the public. In return the system converts the gaseous tritium into tritiated water. Historically this material has been sent to Ontario Power Generation in Canada for treatment and recovery of the H-3 with significant costs associated with storage, packaging, transport and treatment of the material. The requirements for acceptance for the treatment route became more stringent with the necessity for filtration and polishing to be utilised prior to dispatch.

The installation of a filtration and polishing system would require significant investment and a business case was developed to design and build a plant that would clean contaminated water and recover H-3 on-site utilising three stages; filtration, electrolysis and permeation, and cryogenic distillation and cooling. The system is known as the Water Detritiation System (WDS) which is designed to be an unmanned system which can take activity up to 185 GBq/l. The output of the system would be elemental tritium (HT) which is collected on and stored on uranium beds for re-use therefore enabling the site to recover and recycle tritium from operation of the Torus and also treatment of waste. The facility is due to enter full active operations in 2023.

FUTURE DEVELOPMENTS AT UKAEA

The UKAEA site has received significant UK government funding to support the research capabilities and diversification of the site to being new facilities linked to fusion and fission research, notably the Hydrogen-3 Advanced Technology (H3AT) centre. The diversification of the site's research capabilities brings societal benefits such as jobs directly and indirectly linked to the mission to 'lead the delivery of sustainable fusion energy and maximise scientific and economic benefit'.

Supplementary to diversification of site, JET operations will cease at the end of 2023 and the facility will move from an operational to a decommissioning phase.

Hydrogen-3 Advanced Technology Centre (H3AT)

The H3AT facility is a state-of-the-art facility which is UK Government funded in partnership with industry, academia, government organisations, international collaborators and national laboratories. The goal of the facility is to expand upon the tritium and waste management innovations at UKAEA and build a world class centre focused on tritium technology. A highlight of the facility is a scaled-down version of the ITER tritium fuel cycle which can operate in a closed loop. This will allow modifications to be proven safe on the closed system and therefore de-risk changes on the main ITER system. The facility will develop novel tritium processes to support DEMO and other commercial fusion ventures and will include; uranium bed storage and delivery, cryogenic distillation, water detritiation systems, impurity separation as well as test cells, gloveboxes, wet chemistry laboratories for smaller experiment / component testing. The facility is being futureproofed to support 25+ years of experiments and to support this a variation to the site Permit is being made. The facility is due to fully open in 2022 however small-scale experiments are possible with the use of Interim-H3AT in an alternative on-site facility.

JET Decommissioning

In 2023, JET experiments will cease and the decommissioning of JET will become a UK liability. As discussed earlier, the waste following DTE2 produced from the experiment will be significantly different in terms of activation, tritium levels and dose rates from increased neutron budget. It is estimated that the cost of decommissioning will be £400m. The advanced JET system itself contains significant assets and equipment which can be re-used and components which will support future research for ITER and DEMO and therefore decisions will be made on where items are for re-use or deemed waste. The change from operations to decommissioning will bring a significant change to the culture and processes required to manage the challenge.

To support JET decommissioning sophisticated models utilising FISPACT (M.Fleming, 2021) have been generated identifying activation and H-3 held within Torus components. The information will be used to develop characterisation plans and waste routes based on the activity and hazards such as dose and beryllium. The use of early modelling and categorisation of waste will enable the site to develop a waste-led decommissioning programme with purpose-built facilities.

SUMMARY

In summary this paper covers an overview of management of radioactive waste from JET in the UK and regulatory framework. Following the final DTE2 and potentially DTE3 experiments there will be significant changes to the waste stream producing activated and tritiated components.

UKAEA with support from the European Commission have successfully trialled and built a thermal treatment process which has reduced a significant and challenging ILW waste stream to LLW. Further to this the site has developed a tritium recovery system, WDS which will enable recycling of tritium and re-use as fuel.

The UKAEA site has expanded and currently building the H3AT facility to be opened in 2023 as a world-class H-3 centre. JET decommissioning will begin following cessation of experiments bringing new challenges in terms of waste streams, hazards and waste routes to be managed.

ACKNOWLEDGEMENTS

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We would like to acknowledge Tradebe Inutec for their support in the treatment and characterisation trials of the JET MkIIA Divertor. In addition, we would like to thank Ian Wilson, UKAEA, for the management and operation of the Materials Detritiation Facility and Dan Paley, UKAEA, for assisting with data manipulation for sampling results post treatment.

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V-4. USING SELECTIVE SMELTING TO IMPROVE THE WASTE CLASSIFICATION PROSPECTS OF ACTIVATED STEEL IN FUSION

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Abstract

The paper investigates a selective smelting regime where higher activity components are smelted with lower activity components with the aim to reduce the amount of ILW waste for disposal. This has been applied to the EU-DEMO Fusion Plant Helium Cooled Pebble Blanket (HCPB) 2015 baseline design waste assessment. The paper assesses the structural steels in the breeder blanket and the vacuum vessel. Other benefits of a smelting regime are considered, including decarburisation and detritiation. Key radioisotopes are identified for disposal in different countries – UK, Centre Stockage de l'Aube (CSA, France) and Italy – with comparisons drawn between them. For the UK disposal option large savings in the amount of EUROFER ILW can be achieved with decarburisation but minimal savings are achieved for SS316 ILW due to high levels of Ni63. For the CSA the reverse is true with Nb94 the key nuclide for both steels with no savings made in the amount of EUROFER ILW and significant savings made to SS316 ILW when Nb is reduced. The paper illustrates the potential benefits of international arrangements for waste disposal: individually the highest potential reduction in overall in-vessel ILW was 40% in the UK, while, with a suitable arrangement between the UK and France, a potential reduction of 84% could be achieved. The selective smelting regime may be applicable within the wider nuclear waste industry.

INTRODUCTION

Fusion Waste Goals

Zero legacy waste is a key goal for nuclear fusion within the European Union and will be key if fusion is to gain the support of the public. Zero Legacy waste equates to having virtually no ILW waste after 100yrs of plant shut down [1]. The public and subsequently political support will be vital to the commercial realisation of fusion in the future first through achieving the funding to build prototype / demonstration reactors and then to encourage investors to purchase / build their own fusion reactors for energy generation in the future. Fusion has historically been a clean source of energy with some statements claiming there is no radioactive waste produced. The statement is likely in context with fission facilities which produce significantly long-lived radioactive waste requiring long-term solutions in comparison to fusion which produces activated waste that can be managed in 100 years.

However, it has become clear to those involved with fusion that there will be large amounts of problematic activated waste produced. This has led to careful material selection to reduce the severity of the waste produced. A key example of this is the EUROFER97 steel, where several compositional choices have been developed to try and reduce the severity of the activation. These choices include swapping Mo to W and Nb to Ta. EUROFER97 is not the only option with several other fusion steels such as F82H-mod being potential candidates. EUROFER97 and F82H-mod are the leading two choices with EUROFER97 being preferred due to better mechanical properties [2, 3].

The other key area of work looking at meeting this goal concerns waste management and waste processing techniques used to reduce the problematic waste post activation. This includes investigating

potential heat treatments and for some materials chemical processes, such as chlorination, to separate out the most problematic isotopes. This would allow a far smaller mass of highly activated material to be disposed while the bulk material could then hopefully be recycled or at least disposed of in an acceptable way.

Highlighted Issues with Fusion Steels

Neutronics analysis carried by M. Gilbert *et.al* [4] (2017) highlighted that large amounts of the steel used within the Breeder Blanket (BB), Breeder Blanket Support Structure (BBSS) and Vacuum Vessel (VV) would be Intermediate Level Waste (ILW) at the end of reactor life even after 100yrs interim storage. This includes all the EUROFER97 within the BB which would remain ILW for greater than 1000yrs. This is of great concern in relation to the waste goal with there being over 1000 tonnes of ILW EUROFER97 from the BB alone. Material selection choices are highly unlikely to improve this particularly for EUROFER as EUROFER97 was designed with minimal activation issues as a key consideration therefore for fusion usage it is unlikely to be improved upon for activation properties in the near future. Other non-steel materials have been suggested in the past such as V alloys and SiC however they are not as developed as Reduced Activation Ferritic Martensitic (RAFM) steels along with other concerns. The primary concern for SiC is the generation of insoluble He under intense neutron activation which could damage the ceramic or composite structure. For V alloys the issues relate to high solubility levels of interstitial elements such as N, O and C which could cause catastrophic embrittlement problems [5–8]. As such the remaining option is to reduce the amount of ILW steel produced post activation, through waste processing techniques. It is planned that the first-generation Breeder Blanket (BB) will be replaced in EU-DEMO (DEMO) after 5 calendar years (CY) due to neutron damage while the improved second-generation is planned to last the remaining 15CY of planned DEMO operations.

ILW limits

Waste limits vary from country to country and even between individual disposal facilities. For the scope of this project key disposal limits at existing facilities in France and the UK have been investigated due to the timescales of DEMO. These limits may also inform the proposed location for the DEMO reactor as locating DEMO within a country / region where the resulting waste can be managed and disposed of would be adventitious in terms of decommissioning costs and environmental impact.

UK Limits

The first regulations to consider are the UK's as they have general acceptance limits for waste that cover all radioisotopes, although individual facilities still have overall inventory limits for different radionuclides based on a safety case for their location. The main limit of interest for the fusion steel waste is the UK Low Level Waste (LLW) limit of 12GBq/t (12MBq/kg) for β/γ [4]. Based on the neutronics data, Gilbert *et. al.*, (2017) [4] key elements that prevent the waste from reaching the LLW) limit can be identified. For EUROFER97 the key isotopes are those of C14 and H3 which between them can account for over 60% of the total activity after 100yrs with C14 accounting for around 50% of the total activity. As such if these could be removed from the steel post irradiation the activity could be lowered significantly, potentially meeting the LLW limit, and current steel making processes make this a realistic option (see section 2.3) For SS316 the key isotope is Ni63 as this accounts for more than 90% of the total activity for the stainless steel. This is not ideal as Ni would be very difficult to remove from the steel using conventional techniques [9, 10].

France Limits

The regulations in France for the Centre Stockage de l'Aube (CSA) are very different to those of the UK as they have waste package limits based on the individual isotopes themselves partially based on radiotoxicity, half-life and more properties [4]. Logically this would appear to be a better way of classifying waste however it does make it more complicated and requires far better knowledge of the material composition and a better developed characterisation schemes than generic limits as set in the UK. This radioisotope specific method in some cases can be beneficial to waste disposal as isotopes such

as C14 are given high levels compared to the UK as the radiotoxicity of C14 is low as it is a β emitter. The French limit for C14 is 92MBq/kg which is far higher than the 12MBq/kg of the UK which would appear to make disposal of EUROFER97 easier in France based on the key isotopes highlighted by the UK regulations [4]. The same could be said of SS316 based on the key isotope of Ni63 which in France would have a limit of 3.2GBq/kg over a factor of 100 greater than the limits in the UK. [4] However, there is one radioisotope in particular for both steels that challenges the regulations in France, Nb94. Unlike the C14 for EUROFER97 and the Ni63 for SS316, which both originate from elements that help achieve the respective properties within each steel, Nb94 is an unwanted impurity. The typical levels of Nb in EUROFER97 are around 0.001wt% yet even this small amount is enough to cause problems after activation [11]. The limits in France for Nb94 are 0.12MBq/kg. This also happens to be the same limit as for the El Cabril repository in Spain. [12, 13] All the EUROFER97 within the BB and BBSS come in well above this limit while the innermost areas of SS316 in the VV are also affected. Due to the very small amount of Nb within the steels it would be very difficult to remove at any stage either before or after activation through conventional processes. There are concerns with measuring Nb94 levels in active waste. Complicated measurement techniques such as ICP-MS have detection levels as low as 0.7wppb for Nb which could be used along with modelled activation data for Nb to measure the Nb94 content. Another cheaper way would be to use a scaling factor from an easy to measure isotope such as Co60. This has been achieved for graphite components [14] and something similar could be used to measure fusion steel waste.

Italy Limits

While this report does not investigate in detail at the regulations in Italy, they are worth considering in the context of changes to limits. The limits in Italy changed recently in 2015 with specific changes to certain key radionuclides such as Ni59 and Ni63. These Ni nuclides described as ‘long lived’ in the IAEA report [15] have had their limits increased from 3.7MBq/kg to 40MBq/kg [16]. However, for any other long-lived radioisotope such as Nb94 and C14, both of which have half-lives between those of Ni63 and Ni59, have had their limits reduced from 3.7MBq/kg to 0.4MBq/kg. The increase in the Ni limits may show the ability of the regulator to accept appropriate justifications for a particular radionuclide limit increase as these are problematic for the disposal of some activated stainless steels used in the fission industry as the bulk material for the reactor vessel in some cases.

SMELTING PROPOSAL

Smelting for Fusion

Smelting is highly likely to be a key component in the waste management strategy for DEMO. This is because smelting has many benefits for waste management purposes. It allows for the reactor components to be homogenized and cast into consolidated ingots. This allows for more efficient storing of the reactor material due to a significant volume reduction [17, 18]. The homogenization of the material allows for easier monitoring as now only a small piece of the smelted ingot needs to be tested as it is representative of that whole ingot whereas a part straight out of the reactor would have a heterogeneous activation profile in comparison. The act of smelting also has the effect of liberating any H-3 within the material including deeply embedded H3 which may not have been removed if a thermal desorption treatment was used instead. It is also possible that other compositional changes could be made during the smelting process such as C removal which would remove the C14 as well. This may also allow for the material in ingot form to be used as feedstock for future reactor components preventing the material from being ‘waste’. The smelting of steels is a very mature industry and there are many possible methods for achieving this.

Selective Smelting

It is proposed in this report that the benefits of homogenisation are used to reduce the amount of ILW waste. This would be achieved by smelting problematic areas of the reactor with lower activity areas with the aim that the averaged activity for the whole smelt is below the ILW limits. For EUROFER97 this would mean smelting the BB with the BBSS as the BBSS does meet LLW limit within 100yrs for

the UK limits. In the case that this method is not successful it is proposed that the most active sections are selectively removed before smelting. This could take the form of the First Wall (FW) being removed from the BB before the BB is smelted with the BBSS. This could be continued, with detailed neutronics analysis, to within the BB itself. This is shown in Fig. V-4.1. where if the whole BB (yellow) when melted with the BBSS (purple) is predicted to be above the LLW limits then the FW would be separated. The rest of BB (red) would then be melted with the BBSS and so on until a smelt that is predicted to come in under the limits is identified. A similar methodology can be used for the SS316 within the VV by segregating the VV into layers.

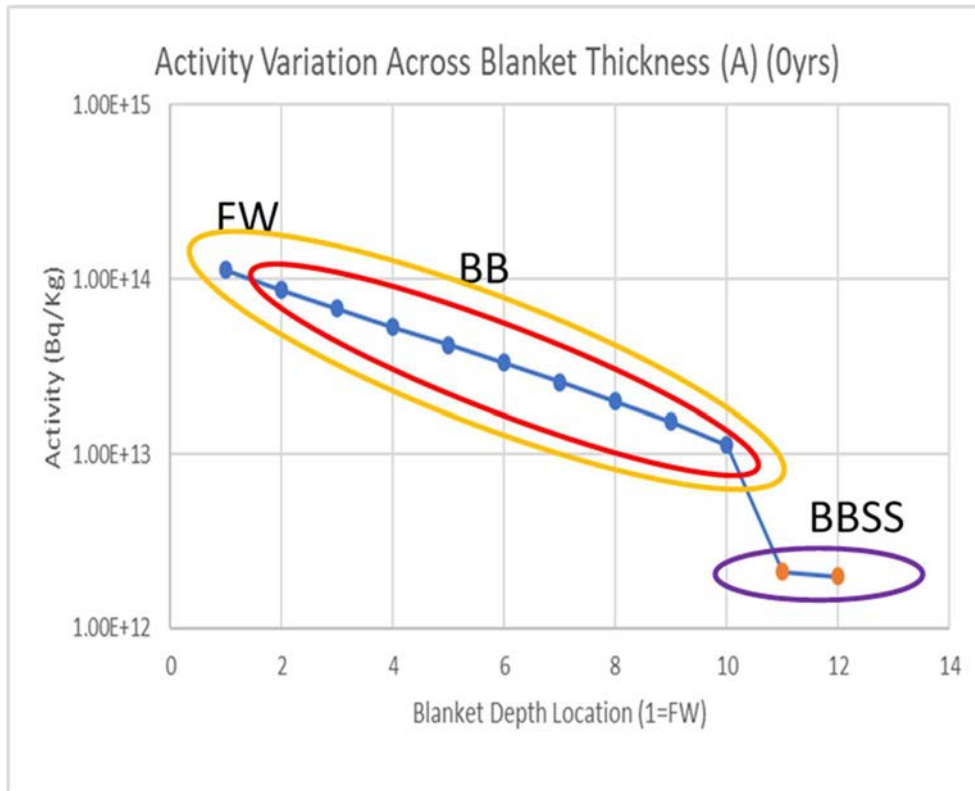


FIG. V-4.1. Graph showing the activity distribution across the blanket and support structure illustrating how separating out the FW may result in a lower activity.

H3 and C14 Recovery and Removal

Another benefit of smelting that is of interest is the ability to alter the composition of the melt. H3 removal from the steel will occur during smelting without any extra processes as it will leave in the gas phase. This will lead to a reduction in the activity of the smelt metal and allow for recovery of a key fusion resource [18, 19]. Theoretically it is also possible to remove C from the melt, a process which is done during regular steelmaking through several methods of which an O₂ blow is the most common. As the removal of C would include the removal of C14 which for UK limits is an isotope of concern this would be beneficial for reducing the bulk activity. [15] The historical method for removing C from steels is the Bessemer process which reduced C levels from pig iron (around 4% C) to levels below those in steels at that time on very short timescales (20mins of O₂ blow) [20]. As such it would initially seem likely that the C14 could be removed from the steels at least to some extent using even potentially current methods. Newer methods such as the Kawasaki Steel top oxygen blowing (KTB) method can produce very low carbon steels, 10ppm, in around 1h [21]. This method utilises an O₂ gas blow in a vacuum environment. This is now a mature technology in Japan and could be utilised for the treatment of active waste if the levels of contamination are commercially acceptable [22].

ANALYSIS METHODOLOGY

Modelled Data

The modelled data was produced using the Monte-Carlo neutron-particle transport (MCNP) and FISPACT-II codes, by Gilbert *et.al* (2017) [4], for a 10° toroidal extent based on the geometry of the “EU DEMO 2015” Helium Cooled Pebble Bed (HCPB) reactor design [24]. The reactor was broken down into 20 radial sections further broken down into 12 tangential sections. The BB is broken down into 14 radial sections are then broken down into 10 tangential subsections with the same orientation of the VV sub sections. The model does not consider any H3 impregnation from ‘external’ sources such as the plasma or the H3 bred within the BB. It only takes into account H3 produced through activation within the steels themselves. Subsequently the H3 levels in the data would be an underestimate of the true expected values. However, as the smelting process will liberate any H3 present including embedded H3 this is not envisaged as an issue for the purpose of this report. This data with regards to the BB refers to the 2nd generation BB as it is assumed that the 1st generation will be far easier to dispose of (only receiving about ¼ of the flux) and if this smelting method works for the 2nd generation it will for the 1st. The VV data is for its whole lifetime within the reactor (both sets of BB not just 2nd gen BB).

EUROFER97

For EUROFER97 the data described above was analysed For EUROFER97 the BB and BBSS were assessed utilising a tangential section (1/12th) for representation, and constituted the BB and BBSS which are primarily regions EUROFER97 is used within the reactor. The mass of EUROFER97 in each of these subsections and as such for the whole of the BB and BBSS where calculated. The Volume Correction Factor (VCF) used for the BB was 0.64 for the innermost subsection of each section (represents the FW) and 0.0985 for the rest of the BB. 0.58 was used for the BBSS sections. The VCF is required to account for voids within the blanket volume which would be occupied by coolants and breeder materials during operations. The density used for EUROFER97 was 7.75g/cm³ [23].

For each subsection the total activity, total dose, C14 activity, H3 activity and Nb94 activity were recorded and consolidated for analysis. C14, H3 and Nb94 were chosen as they are currently the key radioisotopes that impact the disposal regulations for EUROFER97 in the UK and France. This consolidated data was then used to predict the averaged activities of the BB sections and subsections with the BBSS sections and compared to the regulations at 0, 80, and 100yrs after reactor operations have ceased using the mass of each subsection to weight the average calculations appropriately and to determine the masses of ILW and LLW produced. This was also done with the H3 and C14 contributions removed to simulate the effect of removing them during the smelting process on the activity and ILW mass. Estimated reduction factors following detritiation and decarburisation are 10,000 for H3 and 1000 for C14 (although in practice this may be closer to 100 than 10000) [15] [24]. As the reduction factors are so high the total amounts of H3 and C14 were removed from the values to simplify the calculations after a representative calculation showed that even with a 100 x factor reduction in C14 the C14 activity contribution did not affect the results compared to a 1000x reduction or a full, 100%, removal. The Nb content assumed for EUROFER97 in this model is 0.005% which is higher than the measured composition of some EUROFER97 melts at 0.001%. Therefore, this model potentially indicates an overestimation of the waste problem caused by Nb94 for EUROFER97.

SS316

SS316 is found within the VV and the data described previously was analysed in the same way as for EUROFER97. Masses were calculated in the same way to EUROFER97. The VCF for the SS316 whole VV was 0.6 and the density for SS316 used was 7.99g/cm³. As with EUROFER97 the activity data for H3, C14 and Nb94 as well as the totals for activity and dose were consolidated. The decision was made not to include Ni63 within this as there is not at this time a viable way of removing it from the steel pre- or post- activation and individually is not a concern for the UK regulations. Ni63 in the UK just contributes to the overall activity and for the regulations in France even the highest Ni63 levels were far

below the limit of 3.2GBq/kg at around 1GBq/kg after 100 years. The total mass of SS316 in a DEMO reactor is not analysed in this report as only the primary in-vessel volumes were calculated. The outermost layers of SS316 considered will be immediately available for LLW disposal after DEMO operations and it is assumed the rest of the SS316 waste is not deemed to be of major concern at this time [4].

RESULTS

EUROFER97

France LLW limits are not met due to the BBSS having an activity up to 6x higher than the limit for Nb94. This means that even with the ‘true’ Nb levels within EUROFER97 the BB and BBSS would still all be ILW under the regulations in France.

The average activity after the smelting process, including H3 and C14 removal, is below the UK LLW limit but only by a small margin as shown in Fig. V-4.2. In practice some sections will need to have their FW removed to meet the UK LLW criteria creating a small amount of ILW level waste. This is illustrated in Fig. V-4.2. by the blanket depth location. The depth location number refers to the first layer of the blanket included in the smelt i.e. depth 1 is whole blanket, depth 3 is the whole blanket with layers 1 and 2 removed and treated separately.

Expected ILW mass before smelting was 1035t, after smelting 661t (36% reduction) and after smelting and H-3, C-14 removal 144t (86% reduction) as seen in Fig. V-4.3.

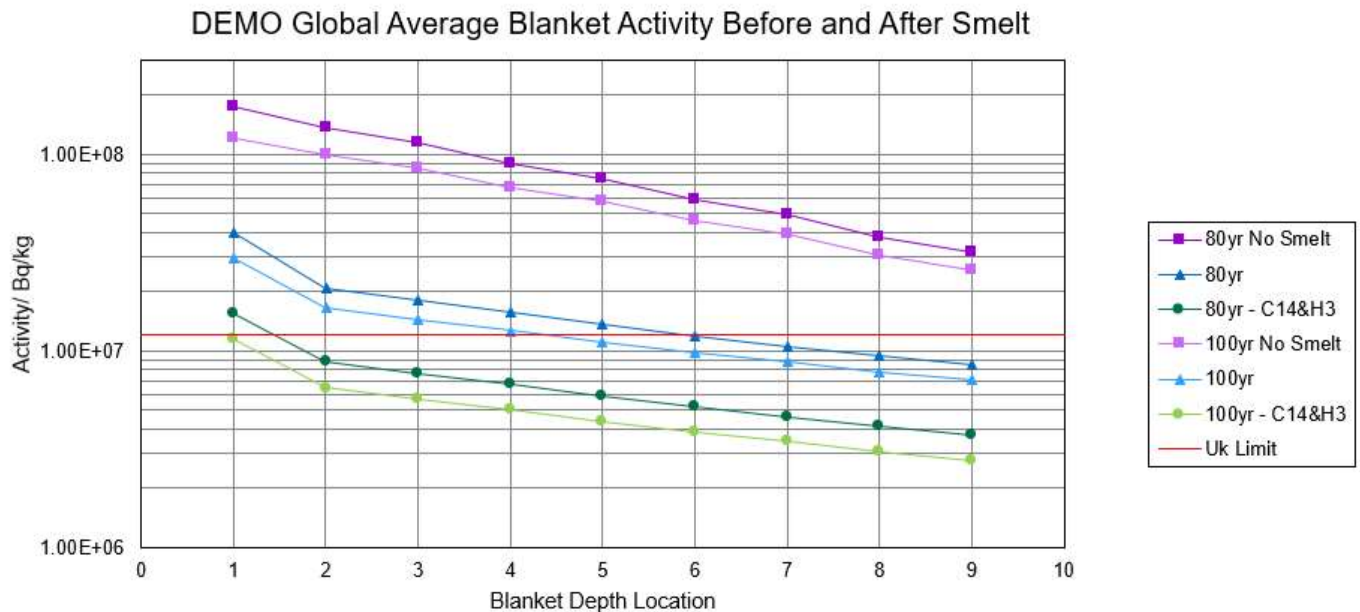


FIG. V-4.2. Graph showing the various smelting options after 80 and 100yr interim storage,³³.

³³ There are 3 options illustrated: Pink/Purple lines show the activity with no selective smelting, Blue lines show the activity after selective smelting has been carried out and the green lines show the activity after selective smelting and removal of H3 and C14. After 100yr with H3 and C14 removal the global activity for all the EUROFER97 drops below the UK LLW limit

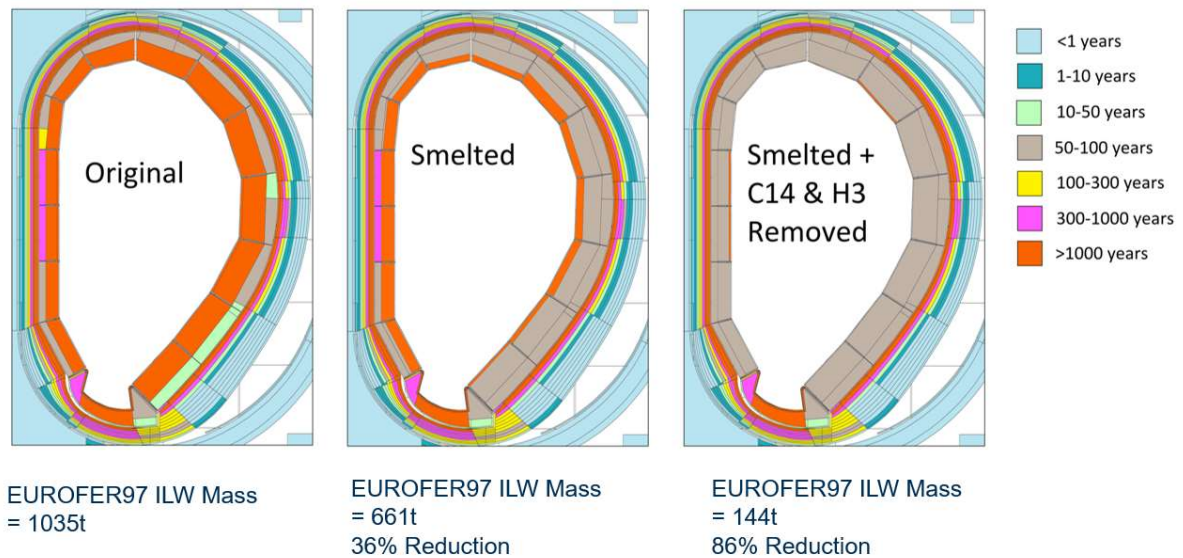


FIG. V-4.3. Schematic showing the reduction of EUROFER97 ILW with smelting and H3, C14 removal.

SS316

Smelting does lower the expected ILW under UK limits. However, this requires around 80 years of interim storage for what is a fairly minimal impact, with only 24.53% reduction in ILW under UK regulations (3371t to 2544t). It does not seem particularly beneficial compared to the other results seen in this report. As Ni-63 accounts for over 90% of the total activity, H-3 and C-14 removal have little to no impact on overall activity therefore C-14 removal may not be required. H-3 removal will still occur regardless and it would still be desirable to recover any H-3 so this still needs to be considered when identifying the technologies required for this process. French limits after smelting allow for the majority of the VV to be disposed of immediately after reactor operations cease. For some sections this would be the case before smelting as well, however smelting is still recommended for the other benefits it brings such as volume reduction. For a full selective smelting regime, a 73.66% reduction in ILW (966t to 254t) is achieved for the VV under French regulations as shown in Fig. V-4.4. (H-3 and C-14 removal have not been included as they have no impact in this scenario with Nb94 the key isotope when considered under French regulations).

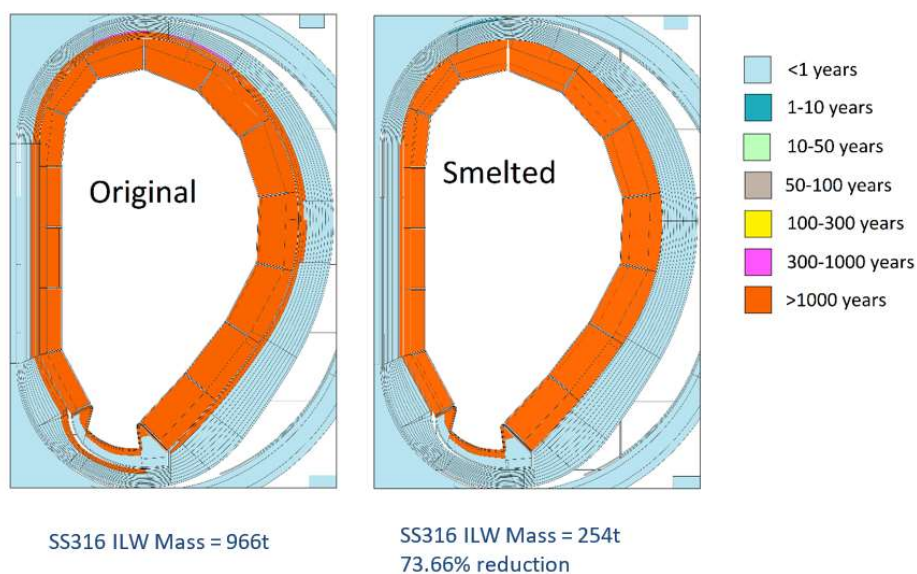


FIG. V-4.4. Diagram showing the reduction in ILW after 100 for the VV under French regulations following the selective smelting regime.

Utilisation of Regulatory Frameworks

Looking at the data combined there would appear to be one obvious solution in terms of the waste treatment routes currently available – the BB and BBSS to be sent to the UK and the VV to France. When sending the BB, BBSS and VV to different locations for disposal the results achieved show the lowest total amount of ILW witnessed in this report.

Fig. V-4.5 shows that combining regulations from UK and France gives an 81% reduction in expected ILW when EUROFER97 and SS316 considered with the EUROFER97 having the H3 and C-14 removed.

A negative of splitting the waste this way is that it would make transportation considerations more important as some long-distance non-road transportation will have to take place regardless of where DEMO is built. This would increase the environmental impact of the disposal through increased fuel usage for the transportation and the transportation costs will almost certainly be higher. Suitable authorisations will also be required for trans-frontier shipments of radioactive waste from the relevant competent authorities.

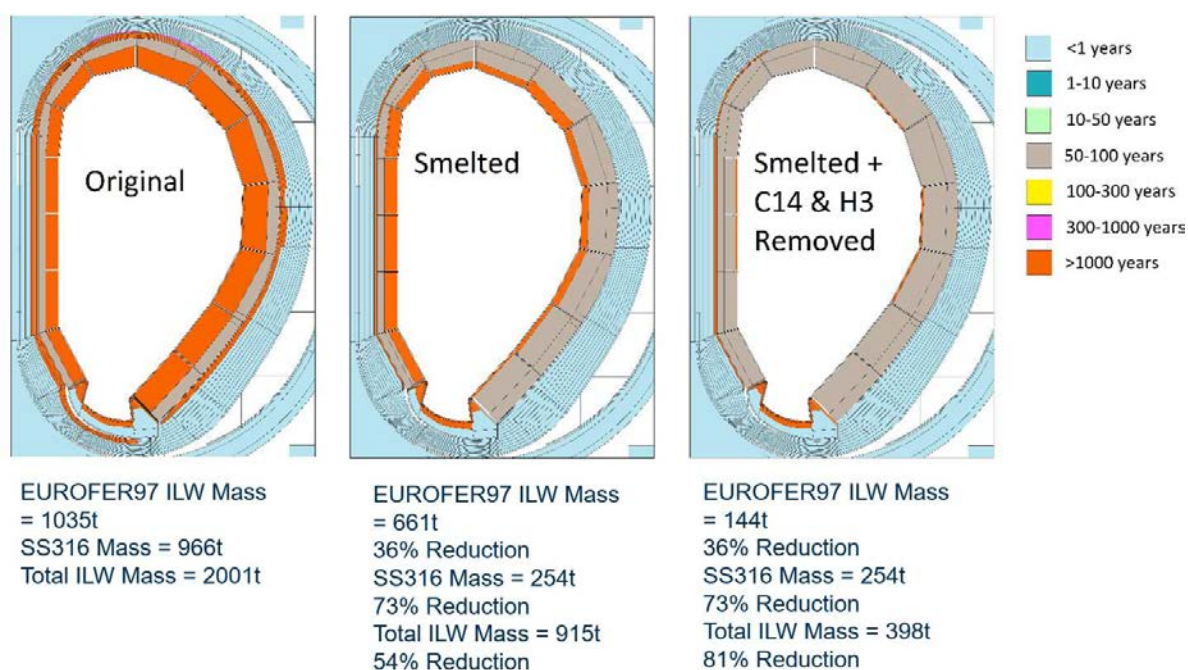


FIG. V-4.5. Diagram showing the effect of selective smelting on both the BB and VV if the waste can be disposed of in separate locations under different regulations. BB in UK and VV in France.

DISCUSSION

ILW reduction

The results show that the proposed smelting technique could be successful in reducing the amount of ILW produced and while it does not reduce the amount of ILW steel to zero, it does produce a significant reduction. As previously mentioned, smelting is highly likely to be part of the waste management strategy for DEMO regardless of this method of smelting and hopefully the C-14 removal could easily be incorporated into the waste management plan for DEMO. This analysis shows that selective smelting is a feasible way to lower the steel ILW content from DEMO. This would hopefully still be the case if there are design changes made to DEMO or the predictive modelling becomes more accurate to better assess the total activation as while the precise figures will change, the principle of selective smelting will not. Smelting ILW parts of the reactor together with LLW parts to homogenise the activity will result in lowering the activity of the higher activity components with the aim of achieving the LLW limits.

A concern about selective smelting is the feasibility of being able to viably separate the sections in practice. Previous work has shown removing the FW ought to be feasible, for disposal purposes, using destructive methods. However, sectioning up the rest of the blanket may be very difficult in part due to all the other material present, such as the neutron multiplier and breeding material, and the design of the blankets. The design of the blanket modules could be optimised to allow for easier segregation of the module and easier removal of the FW as no current designs for blanket modules are optimised in this way [24]. This would make the segregation for selective smelting simpler while potentially lowering the amount of waste such as dust produced during the segregation process. Remote Handling capabilities may also limit the accuracy and availability of the selective smelting process along with the difficulty of physical identification of sections. These design issues may limit the impact selective smelting could have on ILW reduction.

While this analysis was only carried out for the HCPB design the same principle could be applied to other designs such as the Water Cooled Lithium Lead (WCLL).

As future DEMO designs become more mature and modelling capabilities improve this analysis can be refined further to include more of the reactor, representing real reactor components rather than generic slices. This would allow analysis of the material in this report to better predict the effectiveness of a selective smelting regime while also allowing highlighting key areas of improvement from a waste perspective.

C-14 Removal and other Smelting Concerns

The smelter required to carry out any smelting would likely have to be a Vacuum Induction Melter (VIM) smelter due to the need for very strict atmosphere control due to the liberation of H₃. At end of life the smelter itself would have to be classified as waste, a 30t capacity smelter has been suggested as more than adequate for DEMO needs [25]. This smelter is likely to be the largest active waste burden from the smelting process and will almost certainly be all LLW waste [27]. A very approximate estimate of mass would be around 1000t. This is very small compared to the mass of the reactor itself and also the rest of the waste handling facilities required to handle the reactor waste. It is also likely that a smelter of this size could service multiple future fusion reactors as well. It could also be used when demand from the fusion industry is low to treat some waste from the global fission industry providing multiple commercial avenues for DEMO moving forward [26].

Removal of C14 while a mature process, brief discussions with UK industry suggest it is time consuming taking up to several hours to achieve the desired removal factor, according to brief discussions with UK industry. The simplest decarburisation method is with an oxygen blow although due to the presence of liberated H₃ the O₂ blow may need to be delayed until after most of the H₃ has been removed lengthening the process further. This would result in longer exposure times for the furnace and in particular the furnace lining (normally a ceramic) which will likely become contaminated or partially enter the melt. These concerns could likely be mitigated if the facility is designed and built solely for nuclear waste management purposes where these issues could be analysed and controlled.

Radioactive slag will be produced during the process. Activity levels will vary depending on its chemical composition and interactions with the steel melt. Further work would need to be done on this to assess the amount of slag and level of contamination within it. This could be mitigated depending on the desired outcome of the smelting by casting the slag with the melt itself rather than treating them separately which would not cause problems if disposal was the end goal rather than re-casting and re-use.

CONCLUSIONS

Selective smelting would be capable, based on the data provided for this report, of reducing the EUROFER97 ILW in the BB/BBSS by 86% if C-14 removal is feasible in the UK and by 36% if under French regulations. For SS316 reductions of 24% and 74% are possible for regulations in the UK and France respectively. Combining these results and separating the waste streams for EUROFER97 and SS316 with the EUROFER97 going to the UK and the SS316 to France an 81% reduction in steel ILW

from the BB, BBSS and VV could be achieved. While there may be some concerns about the effectiveness of C-14 removal, selective smelting still represents a feasible and very promising process that could be part of the solution to minimising the waste legacy of DEMO.

Legislation currently varies across the world and is primarily concerned with the fission industry at the current time. This means that there is a potential possibility for different regulations regarding fusion waste in the future and it is vital that the fusion community are involved in future regulatory discussions about radioactive waste across the world both at the IAEA level and national levels during the creation of the next stream of disposal facilities.

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V-5. APPLICABILITY OF HEAVY WATER FISSION REACTOR WASTE MANAGEMENT EXPERIENCES TO FUSION ENERGY PRODUCTION

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Abstract

Fusion reactors, e.g. the ITER Tokamak reactor in France, various testing installations, and future fusion reactors, will employ tritium and deuterium as primary fuel and will use new reactor materials, such as beryllium blankets, tungsten divertors, Nb₃Sn alloy, and graphite composites. The selection of materials will have implications on waste management after tritiation and fusion neutron irradiation. The waste streams from the operation and decommissioning of fusion systems reactors are expected to contain significant amounts of tritium and neutron activation products that differ from fission reactors. The management of tritiated waste streams from fusion power generation may require new management processes and strategies. However, management of tritiated waste is a familiar scenario to the Canadian-designed CANDU Pressure-Tube Heavy Water Reactor (PT-HWR) and heavy-water research reactors. These reactors use heavy water as both coolant and neutron moderator, where the irradiation of heavy water generates tritium. The PT-HWR waste streams, including the reactor components, concrete, and wastewaters, have been characterized as containing higher tritium levels than that from the light water cooled reactors (LWR).

This paper presents a brief history of fusion research in Canada and compares typical waste stream characteristics between the Tokamak reactor and the PT-HWR such as tritiated wastewaters, neutron-activated reactor components, and concrete. The impact of tritiation and irradiated new reactor materials on waste classification is also discussed by comparing predictions of highly tritiated neutron-activated reactor components from the Tokamak reactor and the PT-HWR with less-tritiated components from selected small modular fission reactors.

INTRODUCTION

Canadian participation in fusion programs

Canada pioneered fusion research, starting in the late 1970s. In 1983, Canada constructed its first tokamak, STOR-1M, at University of Saskatchewan [1]. This was followed by a larger tokamak, STOR-M, in 1987, currently the only device in Canada devoted to magnetic fusion research. In parallel, the Canadian Centre for Fusion Magnetism (CCFM) operated an experimental tokamak in Varennes, a suburb of Montreal, starting in 1986 [2] [3]. This medium-sized machine, named Tokamak de Varennes, specialized in the study of plasma-wall interactions. For more than one decade, Tokamak de Varennes formed the kernel of CCFM activity. Due to the lack of funding, the operation of the Tokamak de Varennes was stopped, and the tokamak installation was transferred in 2001 to Canada's Science & Technology Museum. Canadian Nuclear Laboratories (CNL), formerly Atomic Energy of Canada Limited (AECL), was a founding member of Canadian Fusion Fuels Technology Project (CFFTP) in 1982 [4] [5]. The CFFTP represented part of Canada's overall effort in fusion development. The focus for CFFTP was technology for the tritium and deuterium nuclear fusion fuel cycle.

The initial cooperation for an international nuclear fusion project that was the foundation for ITER began in 1979. Canada had a long association with ITER from 1985 to 2000, and was a partner in the ITER project until 2004 [2]. After a 15 years hiatus, Canada signed an MOU with ITER in 2018 to identify the precise domains in which Canadian suppliers could export expertise and technologies on a commercial basis to the ITER Project [6]. In October 2020, a nuclear cooperation agreement (NCA) was signed between Canada and ITER for Canadian-supplied nuclear material (tritium), and tritium-related equipment and technology [7]. Canada has the largest reserve of pure tritium, which has been produced by and extracted from a large fleet of Canadian CANDU® Pressure Tube Heavy Water Reactors (PT-HWRs) since 1990. Canada is also internationally recognized for its contributions to fusion energy development, in particular hydrogen isotope separation and tritium handling expertise and technology, which was initially developed from the PT-HWRs. CNL, Canada's premier nuclear science and technology organization, has been collaborating with General Fusion, a private fusion reactor developer, since 2002. CNL is currently also working with private Small Modular fission Reactor (SMR) developers to host their first demonstration projects [8], which are considered bridging technologies before the deployment of fusion reactors.

Canadian heavy water reactors

Canada has pioneered the development of heavy water D₂O (or ²H₂O) moderated and cooled nuclear reactors since the 1960s (Figure V-5.1). There are 24 CANDU® PT-HWR units built in Canada and 12 units overseas [9]. Six prototype and research HWRs were built and tested by AECL. Currently, 19 CANDU® units are operational in Canada and 11 units overseas. CNL has extensive experience with tritium and tritiated materials from PT-HWR related research activities and 60 years of operation of the 135 MWth National Research Universal (NRU) reactor at CNL's Chalk River site (from 1957 Nov 3 to 2018 Mar 31). CNL is currently working on decommissioning prototype and research HWRs in Canada, which is theoretically relevant to the future fusion system decommissioning and waste management.

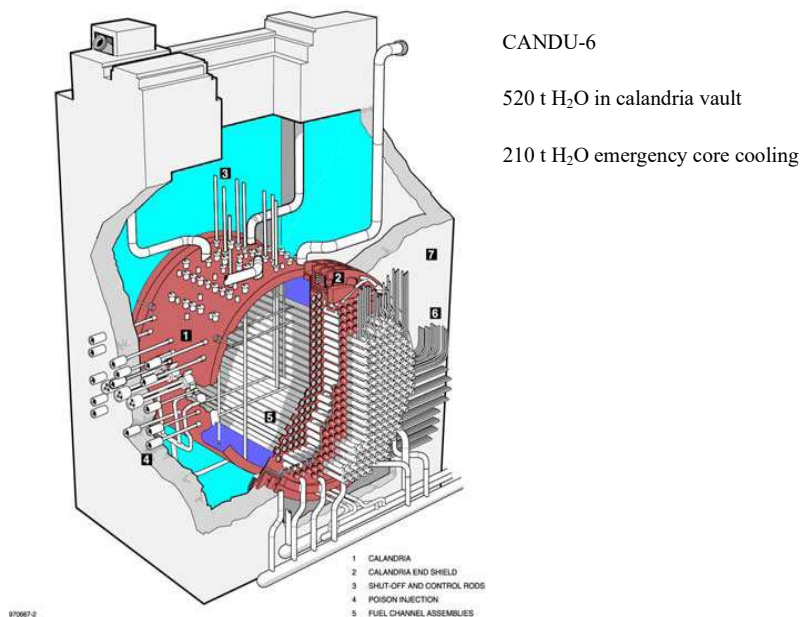


FIG. V-5.1. Broken-out section view of a CANDU® reactor [10].

HWR tritium removal technologies

Heavy water is an indispensable part of the PT-HWR design. Technologies for extraction of heavy water from light water were developed in tandem with the PT-HWR. These technologies were adapted for tritium reduction from the irradiated heavy water at the PT-HWR plants. The Darlington Tritium Removal Facility (DTRF) was constructed to reduce the average tritium dose rate per worker and tritium emission at CANDU reactors [11]. The DTRF started operation in 1990, and has since supported the

operation of multiple CANDU reactors in Canada. The facilities were designed to maintain a steady-state tritium level less than 370 GBq/kg in the moderator. The DTRF processes about 2,000 Mg D₂O/a and produces about 740 PBq/a elemental tritium with >99.9% purity [12]. The main components in the DTRF include:

- Main tritium extraction processes: The DTRF uses a vapour phase catalytic exchange (VPCE) process in the front end to transfer tritium from completely-vapourized D₂O to gaseous deuterium (D₂) in the form of DT gas [11] [13]. Elemental tritium is separated as T₂ from the D₂ gas using cryogenic distillation (CD).
- Tritium Immobilization System: The tritium immobilization system (TIS) is designed to draw off 45 kCi (4.5 grams) batches of tritium gas from the last column of the CD system and immobilize it on a titanium getter for long-term storage. The system is housed in two gloveboxes, each containing the necessary equipment to complete the full immobilization cycle. One of the gloveboxes has a tritium sales container airlock to allow filling of uranium beds and tritium gas sales containers. The gloveboxes are maintained under an argon blanket at slightly positive pressure. Globe Box Clean-Up System (GBCS): the GBCS is designed to purify the Argon atmosphere in the two tritium immobilization system gloveboxes and associated airlocks. Support Systems: Major systems are located in ten different rooms to be able to isolate the individual rooms in case of a tritium release. The ventilation system was designed such that airflow is maintained to flow from tritium-free areas into tritium-containing areas, and then to the DTRF building stack. An air clean up system (ACS) is used to remove hydrogen isotopes from room air in case of an accidental release. Process rooms in the DTRF are provided with hydrogen and tritium monitors. Tritium monitors are set up to automatically start up the ACS when the tritium levels in the rooms reach a threshold value. The Wolsong Tritium Removal Facility (WTRF) for four CANDU reactors in South Korea started operation in 2007 [14]. The WTRF employs a Liquid Phase Catalytic Exchange (LPCE) process developed at CNL, in lieu of a VPCE process at the DTRF, and a CD process. Alternative processes to the VPCE and LPCE have also been developed, such as Bi-thermal Ammonia-Hydrogen Exchange, Electrolysis, and Combined Electrolysis and Catalytic Exchange (CPCE) processes [15]. These technologies, however, have not been used for detritiation on industrial scale. CNL has been involved in design, operation, and continuous improvement of tritium removal processes in the DTRF and the WTRF. CNL also operates Tritium Facilities at CNL's Chalk River site. These facilities use similar technologies as a nuclear fusion fuel cycle.

Current Canadian fusion research

Although most fusion research projects ceased in Canada in 2003, fusion research has continued at Canadian universities and by private developers [2]. General Fusion stands out as a major private developer in Canada and in the international fusion community. The sources of funding for General Fusion come from both Canadian Government grants and private venture investments. The General Fusion design employs Magnetized Target Fusion (MTF) / Magnetically Insulated Inertial Confinement Fusion (MCIF) concepts, and a prototype installation has been built in Burnaby, British Columbia (Figure V-5.2) [2]. CNL also formed a partnership with General Fusion to advance fusion technology, specifically the process of extracting tritium from liquid metal to provide a continuous supply of tritium fuel by leveraging CNL's premier tritium infrastructure.

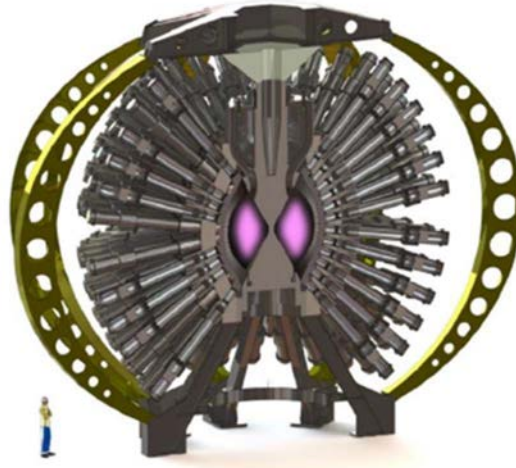
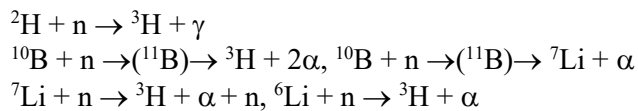


FIG. V-5.2. General Fusion magnetized target fusion concept.

The cooperation agreement signed between ITER and Canada in Oct. 2020 is awaiting approval. This agreement will affect growing activities on fusion technology development in Canada. Similar to many countries, Canada will need to understand the issues associated with fusion waste streams, management, and disposal, e.g. the waste management regulations, practices, and industry standards made for the current fission reactors may have to be re-examined for fusion waste streams. This paper describes and compares the HWR and ITER tokamak waste characteristics and management practices and makes suggestions for investigation of fusion waste streams from potential fusion reactors, including the General Fusion concept. The data of the ITER Tokamak is obtained from the public domain and may not be current; thus, it will be updated when the ITER data becomes available to Canada.

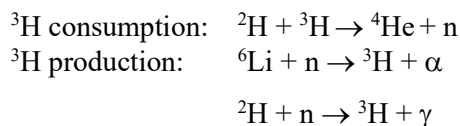
TRITIUM PRODUCTION FROM FISSION AND FUSION REACTORS

Tritium (^3H) is produced from heavy water moderator and coolant neutron irradiation through the following nuclear reactions:



Boron and lithium are neutron poisons in the coolant. In addition, tritium is produced from ternary fission of heavy metals (^{235}U , Pu, etc.). The irradiation of control rods containing neutron poisons such as B and Li also produce tritium. Neutron energy influences tritium production in the above reactions. More thermal (or slower) neutrons will result in higher tritium production in the coolant. Overall, the major tritium production is from deuterium and neutron reactions. The tritium production mechanisms are similar in Pressurized and Boiling Water Reactors (PWRs and BWRs) with light water as moderator and coolant. However, the natural heavy water abundance is only 0.0147 mol % in the light water, and neutrons are less thermalized in the PWR and BWR than in the HWR, resulting in much lower tritium production by neutron capture on deuterium.

The ITER Tokamak consumes ^3H in plasma, and at the same time produces ^3H by $^6\text{Li} + n$ reactions in breeding blankets, in order to be ^3H self-sufficient. To a lesser extent, ^3H is also produced by the neutron irradiation of in-vessel components cooling water.



The inventories of water that produce ^3H by neutron irradiation are shown in Table V-5.1. The tritium in these waters is also the sources of ^3H releases to air and liquid effluents. A larger water inventory indicates potential difficulty in controlling ^3H release, or higher ^3H release.

TABLE V-5.1. D_2O AND H_2O INVENTORIES IN PRIMARY HEAT TRANSPORT (PHT) SYSTEM THAT PRODUCE ^3H IN FISSION AND FUSION REACTORS

CANDU-6 (600 MWe)		PWR AP-600 [16] (600 MWe)	GE BWR INTERNATIONAL ATOMIC ENERGY AGENCY, Pressurized Water Reactor Simulator, Training Course Series No. 22, IAEA, Vienna (2003) [17] [18] (1,300 MWe)	ITER Tokamak [19] [20] (500 MWth)
Moderator	Coolant			
190 t D_2O	240 t D_2O	284 t H_2O	550 t H_2O *	950 t H_2O
* estimated by the reactor coolant clean-up system flowrate and processing time of the entire coolant volume.				

The presence of tritium in the fission and fusion reactors is compared in Table V-5.2. The ITER Tokamak cooling light water has a relatively lower ^3H inventory than the HWR moderator and coolant, but higher than the PWR coolant, indicating higher neutron flux in the Tokamak than in the PWR. A small amount of ^3H in the plasma permeates through the barriers and contributes to ^3H in the cooling water.

The total ^3H inventory in the 500 MWth ITER Tokamak is actually similar to that in the 600 MWe HWR, and much higher than that in the 600 MWe PWR. This indicates that a larger amount of ^3H could be present within a Tokamak reactor compared to an HWR and PWR on an equivalent energy basis.

TABLE V-5.2. ^3H IN FISSION AND FUSION REACTORS

Reactor	HWR (CANDU-6) [12] [21]		PWR (AP-600) [22] [23]	HWR/PWR [22]	500 MWth ITER Tokamak [24]-[27]	
Medium	Moderator r D_2O	Coolant D_2O	Moderator and Coolant H_2O	Fuel	In-vessel Coolant H_2O	Inside Vacuum Vessel (VV)
Steady state ^3H (Bq/kg)	3.10×10^{12}	7.5×10^{10}	9.25×10^7	Not available n.a.	1.78×10^9	Total ^3H during plasma: 3.6×10^{17} Bq or 1 kg ^3H consumption: 8.9×10^{15} Bq/d or 25 g/d ^3H production: 5.6×10^{13} Bq/d or 0.16 g/d ^3H per breeding blanket (1.3 m ²)
^3H (Bq)	5.89×10^{17}	1.8×10^{16}	2.63×10^{13}	n.a.	1.69×10^{15}	
^3H yield (Bq/MWe-a)	7.49×10^{13}	1.39×10^{13}	3.2×10^{10}	$7.4/9.2 \times 10^{11}$	n.a.	
^3H yield (Bq/500MWth-a)	1.12×10^{16}	2.29×10^{15}	5.28×10^{12}	$1.1/1.5 \times 10^{14}$	n.a.	

TRITIUM EMISSIONS

The ^3H release pathways for a fusion reactor such as the ITER Tokamak are basically similar to the fission reactors such as the HWR (see Table V-5.3). The gaseous release from the ITER Tokamak also includes ^{41}Ar from the activation of air ($<1.4 \times 10^{12}$ Bq/y) [25]. However, there will be no fission gases from a fusion reactor, which are of major environmental impact in addition to the waste streams. The elimination of fission gases by using fusion reactors is considered a significant advantage in protecting the environment and public health.

The tritium emissions from the HWR, PWR, BWR, and ITER Tokamak are summarized in Table V-5.4. Similar ^3H inventories are observed between the CANDU[®]-6 HWR and the ITER Tokamak in Table V-5.2. However, the estimated ^3H emission from the ITER Tokamak appears to be higher than the HWR. This higher ITER ^3H emission is probable because the total Tokamak emission also includes the emission from the tritium plant, which is not included in the source terms in Table V-5.2. Radioactive ^3H is primarily in its elemental form (or $^3\text{H}_2$ gas) in the ITER Tokamak. Gaseous $^3\text{H}_2$ is generally more difficult to recover than tritium oxide that is the primary ^3H form in the HWR and PWR and will cause higher ^3H emissions.

TABLE V-5.3. MAJOR ^3H EMISSION PATHWAYS

HWR	ITER Tokamak
— Vapor and gases in air. — Heavy water systems maintenance and operation leakage. — Boiler blowdown, ^3H by leakage through heat exchangers to the boiler side water. — Carryovers by spent fuels to spent fuel bays.	— Vapor and gases in air. — Heat transport systems (HTS) maintenance and operation leakage. — Tritium systems maintenance and operation leakage, etc.
Release of I_2 and noble gases (Ar, Kr, Xe, etc.) as fission products and air activation	Release of ^{41}Ar from activation of air ($<1.4 \times 10^{12}$ Bq/a)

TABLE V-5.4. ^3H EMISSIONS FROM FISSION AND FUSION REACTORS [22, [23, [25]

Reactor	HWR (Canada)	PWR	BWR	ITER Tokamak
Tritium release (Bq/MWe-a)	$(1 \text{ to } 3) \times 10^{11}$	7.7×10^9	1.8×10^9	Estimated to be up to 4×10^{12} Bq/MWth-a (99% as gaseous emission)

HIGHLY TRITIATED WATER WASTE MANAGEMENT

Highly tritiated water waste

Tritiated heavy and light water waste are produced from HWRs. These waste streams and their management include:

- D₂O leakage and vaporization due to PHT and Moderator operation and maintenance are captured
 - Mixed heavy water and light water waste are upgraded for reuse.
 - Radioactive ^3H is removed from heavy water at the Tritium Removal Facility (TRF), and the cleaned heavy water is returned for reuse.
 - Elemental ^3H as a by-product from the TRF is stored in containers, and is sold for commercial use or stored for future use.

- Fuel bays light water (H₂O) up to 3.7×10^8 Bq/kg ³H [28].
 - Stored for decay, immobilization, or awaiting detritiation with ³H recovery for commercial use.
- 1.25 kg/s of boiler blowdown (H₂O) per boiler at a CANDU-6 reactor plant, up to 3.7×10^4 Bq/kg ³H [29].
 - Discharged with a large flow of condenser cooling water.

The tritiated water waste estimation from the ITER Tokamak include

- Condensate from air and off-gas cryogenic detritiation systems and Tokamak Heat Transport System (HTS) water leakage:
 - In-vessel HTS – up to 1.78×10^9 Bq/kg ³H, vacuum vessel HTS – up to 3.57×10^7 Bq/kg ³H [25].
 - Routed to stored tanks for Water Detritiation System (WDS) for tritium removal and recovery or for further treatment if ³H > 7.4×10^4 Bq/kg [30].
 - Total – 30 m³/a from the HTS [25]. A higher value of 165 m³/a is reported in [31].

³H emission control

The ³H emission from a facility may be controlled by means of detritiation and administrative controls. These measures for an HWR are

- D₂O waste collection, upgrade, and reuse.
- Cryogenic detritiation systems to remove ³H from D₂O.
- Licensed airborne dose limits including ³H.
- Licensed liquid waste release limits including ³H.

The above control measures are also applied at the ITER Tokamak. For example, cryogenic detritiation systems will be used at various locations to remove elemental ³H, and a WDS will be built to treat ITER Tokamak tritiated water. The WDS is designed with a feed ³H concentration of $\sim 3.7 \times 10^7$ to $\sim 1.1 \times 10^{13}$ Bq/kg [30] [32, 33]. Effluents with ³H concentrations greater than the upper limit may be mixed with lower ³H concentration water. The WDS designed for ITER is based on the combined electrolysis and catalytic exchange (CECE) process. CNL has studied, developed, demonstrated, and applied the CECE process for tritium removal. The CECE process design parameters for the ITER applications have been assessed and improved for the effectiveness of tritium removal by using CNL's process simulation computer code, FMP-HWPsim [34].

OPERATION AND DECOMMISSIONING SOLID WASTE

Waste classification in Canada

Currently, HWRs and associated research activities generate the majority of waste in Canada. The classifications of radioactive waste streams follow the Canadian Standards Association (CSA) Standard N292.0:19 [35]. The standard is developed from IAEA GSG-1 [36] and LLW Repository Ltd. NWP-REP-134 [37]. Four classes of radioactive waste are recognized for radioactive waste management in Canada:

- (a). *“Low-level radioactive waste (LLW), a limit of 400 Bq/g on average (and up to 4,000 Bq/g for individual waste packages) for long-lived alpha emitting radionuclides can be considered in the classification process. For long-lived beta and/or gamma emitting radionuclides, such as ¹⁴C, ³⁶Cl, ⁶³Ni, ⁹³Zr, ⁹⁴Nb, ⁹⁹Tc, and ¹²⁹I, the allowable average activity concentrations can be*

considerably higher (up to tens of kBq/g) and can be specific to the site and disposal facility". This class also includes very-short-lived low-level radioactive waste (VS-LLW) and very-low-level radioactive waste (VLLW).

- (b). "Intermediate-level radioactive waste (ILW), a precise boundary between LLW and ILW cannot be provided, as limits on the acceptable level of activity concentration will differ between individual radionuclides or groups of radionuclides."*
- (c). "High-level radioactive waste (HLW) is used (i.e. irradiated) nuclear fuel that has been declared as radioactive waste and/or is waste that generates significant heat (typically more than 2 kW/m³) via radioactive decay. HLW typically has levels of activity concentration in the range of 10⁴ to 10⁶ TBq/m³."*
- (d). "Uranium mine and mill tailings are a specific type of radioactive waste generated during the mining and milling of uranium ore and the production of uranium concentrate."*

CSA Standard N292.0:19 also provides guidance on waste exemption, and clearance and storage decay. The exemption quantities and clearance limits are defined in the Canadian regulation [38].

The major waste streams in Canada include:

- High Level Waste (HLW): Used fuel;
- Intermediate Level Waste (ILW): Reactor components and ion exchange resins;
- Low Level Waste (LLW): Operational / technological waste from operation, maintenance, and decommissioning;
- Uranium mine and mill tailings: Waste from the operation of mining companies;
- Clearable waste: Treated as conventional waste from operation, maintenance, and decommissioning.

The waste classifications in Canada are defined to allow for classifying a broad range of waste streams. Further analysis will be needed to examine whether all fusion waste streams can be classified using the current CSA Standard N292.0:19 and governmental regulations.

Waste classification for ITER Tokamak operation in France

The ITER Tokamak is located in Cadarache, France. Thus, the French waste classifications are used to categorize the waste streams generated from the operation and decommissioning of ITER Tokamak. The waste classifications and ITER Tokamak waste streams are as follows [39]:

- HLW (HA): No fuel waste (or radioactive fuel waste) from the fusion reactions: $2\text{H} + 3\text{H} \rightarrow 4\text{He} + \text{n} + 17.6 \text{ MeV}$;
- ILW-Long Lived (ILW-LL or Type B): Reactor components, in-vessel dust, and ion exchange resins;
- LILW-Short Lived (LILW-SL or Type A): Operational / technological waste from operation, maintenance, and decommissioning;
- Very low-level waste (VLLW or TFA): Operational / technological waste from operation, maintenance, decommissioning;
- Purely tritiated waste: Waste from the tritium plant and fuelling system operation, maintenance, decommissioning;
- No clearable waste programs.

The current LLW disposal facilities in France all have established ³H activity limits and waste acceptance criteria. Expectations are that the waste streams from ITER Tokamak operations will be disposed of at the existing facilities. However, observing the current activity limits and waste acceptance criteria may be a challenge for the ITER Tokamak waste.

Neutron activation of reactor components

The majority of components in a nuclear reactor only become waste during the reactor decommissioning stage. Only a very small portion of irradiated reactor components with defects are replaced and discarded as waste during normal operation. The activities of activated reactor components waste are important parameters for waste management. Table V-5.5 summarizes the characteristics of activated reactor core components from the CANDU®-type Nuclear Power Demonstration (NPD) reactor, a Small Modular High Temperature Gas-cooled Reactor (SM-HTGR), and the ITER Tokamak.

The ITER neutron activation modelling does not consider ^3H retention / permeation in in-vessel components, as the ^3H permeation and retention from the plasma is assumed to be minor and the main ^3H arises from neutron-initiated nuclear reactions. The SM-HTGR is included in this comparison as it is one of competing technologies for the fusion reactor in Canada. The SM-HTGR modelling does not consider impurities in the beryllium and B_4C that are still in early design stages and ^3H permeation after it is generated. These calculations will be repeated when SM-HTGR material design details are available. The NPD reactor core results are obtained by using proprietary sampling and analysis technologies and are observed to be typically lower than that from neutron activation modelling (not presented here).

TABLE V-5.5. NEUTRON ACTIVATION OF REACTOR COMPONENTS CONTAINING HIGH ^3H

	NPD HWR Core (Sampling and Analysis)		SM-HTGR Core (SCALE-ORIGEN 2.2 + SERPENT2 Model)		ITER Tokamak Components (ITER A-lite [40]-[42])	Plasma MCNP Model	Facing Model
	Zr-alloy tube	Al tube	BeO reflector	B_4C absorber	Be plate	CuCrZr	W Divertor
Bq/g	@30a from shutdown (25a Op.)		@10a from shutdown (10a Op.)		@ 10a from shutdown (12a Plasma Op.)		
^3H	5.46×10^4	1.02×10^4	1.32×10^3	2.35×10^3	8.04×10^5	2.52×10^3	1.13×10^1
^{14}C	1.37×10^4	5.87×10^1	4.65×10^{-1}	2.19×10^{-3}	7.01×10^{-1}	1.26	1.99×10^{-1}
^{36}Cl	8.86×10^2	5.74			5.03×10^{-3}		
^{55}Fe	4.62×10^4	6.61×10^3			4.10×10^2	1.53×10^4	1.01×10^1
^{60}Co	7.21×10^4	1.61×10^4			3.94×10^1	1.38×10^5	2.92×10^2
^{59}Ni		7.18×10^2			1.49×10^{-2}	6.24	1.05×10^{-2}
^{63}Ni	1.79×10^6	8.86×10^4			1.82	6.98×10^4	1.45
^{93}Zr	7.50×10^3					1.74×10^{-2}	
^{94}Nb	4.78×10^1					4.12×10^{-1}	
^{152}Eu	9.00×10^2	4.49×10^2					

NPD sampling and analysis results are typically lower than that from modelling.

^3H permeation is not considered in ITER modelling.

The NPD HWR fuel channel Zr-alloy and Al liner have lower ^3H concentrations than the Tokamak beryllium plate after the ^3H in the plate is decay-corrected to 2.6×10^5 Bq/g at 30 years from the Tokamak shutdown. The Tokamak modelling does not consider the ^3H diffusional release from the beryllium plate after the shutdown. The actual ^3H concentrations in the ITER beryllium plate would be lower than the modelling, as observed with the NPD HWR. In comparison with the component of the HWR and ITER Tokamak, lower ^3H concentrations are seen in the SM-HTGR beryllium neutron reflector and B_4C

poisons. The ITER tungsten divertor shows the lowest ^3H concentrations generated from neutron activation among the core reactor components.

The reactor components in the NPD HWR are classified as LLW in Canada and are disposed of in-situ by entombing the entire reactor structure in concrete. Similar waste from fusion energy production may also fall in the LLW category, such as those having similar activities to the ITER Tokamak components. However, the CuCrZr alloy and tungsten are unique materials to the fusion reactors and are not found with the fission reactors. There may be activation products that are unique to the fusion reactors, depending on the impurities in these materials. Therefore, the current Canadian regulations and standards ought to be examined for potential gaps in managing the fusion reactor component waste.

The concrete structure in the NPD HWR shows higher ^3H activities than in the ITER Tokamak one (Table V-5.6). The activities of the other radionuclides are more or less similar between the two types of reactors. The NPD concrete structure is classified as VLLW, and disposed of together with the reactor components inside it as described above. The concrete bio-shield of the ITER Tokamak has activities close to the generic clearance levels and lower than the exemption quantities in the Canadian regulations. As France does not have a clearance program, the concrete will have to be disposed of as VLLW.

TABLE V-5.6. NEUTRON ACTIVATION OF REACTOR EXTERNAL STRUCTURE

	NPD HWR		ITER Tokamak [40]-[42]	
	Concrete	Mild Steel Liner	Concrete (bio-shield)	Vacuum Vessel SS316
Bq/g	Analysis@30a from shutdown (25a Op)		Modelling@10a from shutdown (12a plasma Op)	
^3H	3.22E+03	1.36E+01	6.23E-01	1.26E+00
^{14}C	5.49E+01	4.70E+00	1.82E+00	4.99E-01
^{36}Cl	3.97E-01	6.31E-01	1.24E-02	
^{41}Ca	1.74E+00		1.13E+01	
^{55}Fe	1.79E+02	1.00E+03	4.06E+02	1.02E+03
^{60}Co	7.27E+02	6.79E+02		1.04E+03
^{63}Ni	1.56E+02	2.54E+02		2.58E+02
^{152}Eu	2.65E-02	3.42E+00		

Tritiated waste management

The proposed tritiated waste management at ITER is compared to tritiated waste management processes at the HWR (CANDU) plant in Figure V-5.3 [43] [44]. The two tritiated waste management scenarios bear great similarities. Canada has significant experiences in the treatment, handling, transport and storage of tritiated waste resulting from the use of deuterium oxide as moderator and coolant that generate tritium in the CANDU reactors. Tritiated heavy water from leaks from the moderator and heat transport systems are recovered and cleaned before being transferred to the tritiated heavy water upgrader to remove light water (protium) and return heavy water to the reactor.

After ^3H reaches a certain level in the heavy water, tritiated heavy water is detritiated using cryogenic distillation in a Tritium Removal Facility and is returned for reuse in the reactor. Tritium recovered from heavy water detritiation is stored for commercial use. The solid waste from the detritiation facility operation have tritium contamination ranging from $3.7 \times 10^7 \text{ Bq/m}^3$ to 10^{12} Bq/m^3 or greater [43]. Waste contaminated by low levels of tritiated water do not pose unique management problems since these are handled in the same manner as waste from normal reactor operation [43]. While precious heavy water is processed for reuse, tritiated light water from the fuel bays is stored in large tanks.

Although it is beneficial by reducing tritiated waste water volume and recovering tritium for commercial use, light water detritiation has not been implemented in Canada, largely because of the considerable cost. The CECE process is well suited to the light water detritiation because of its high tritium-protium separation factor, which can split the feed into a tritium-free effluent stream and a tritiated heavy water product suitable for recycling through a heavy water upgrader [28].

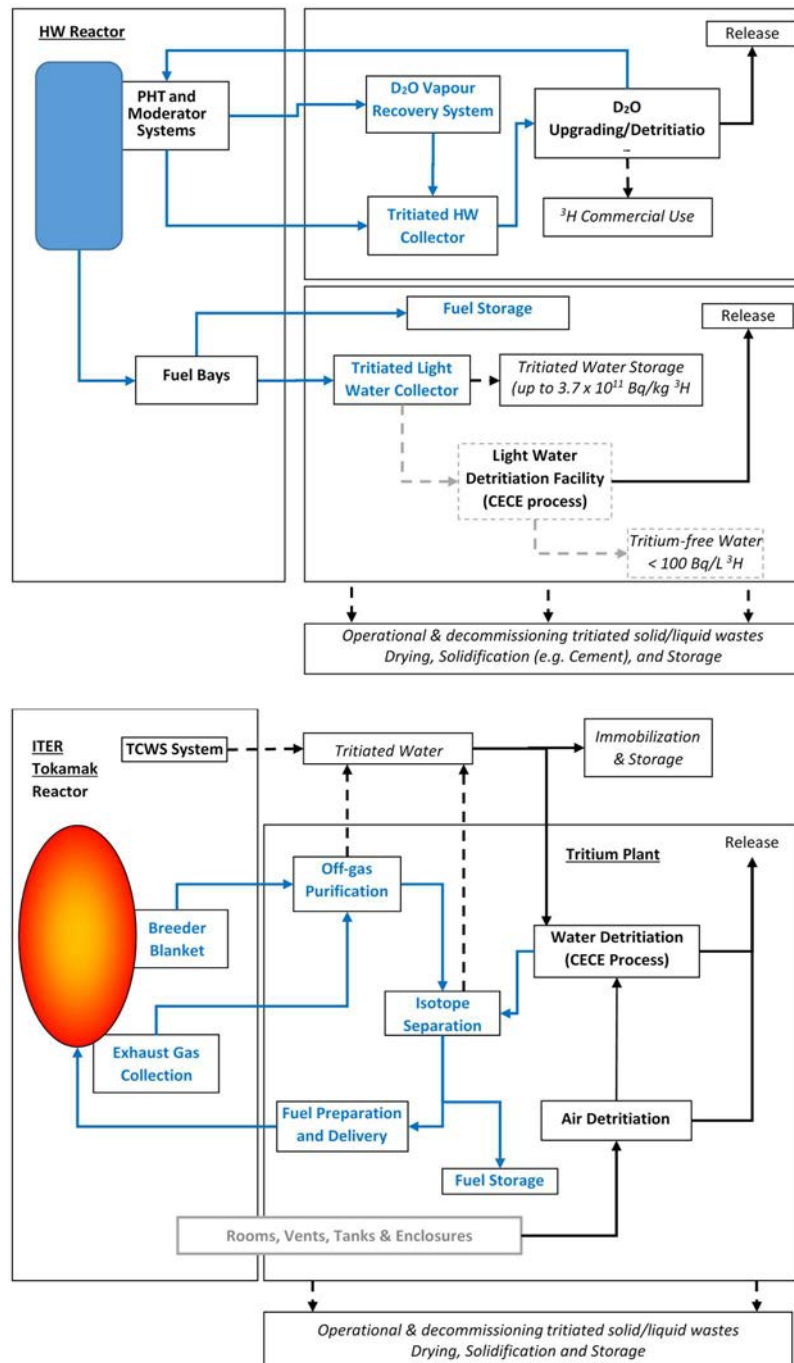


FIG. V-5.3. General management processes of tritiated waste.

SUMMARY

The ³H levels in the solid waste streams from the ITER Tokamak are either similar to or higher than those in the comparable HWR ones. In Canada, high ³H levels in the waste streams are not found to

significantly impact the waste classification. Activation product activities may be comparable in the same component materials from the HWR and Tokamak. However, the neutron activated fusion reactor components that are unique to the fusion may contain radionuclides that are not found within the fission reactors, and their impact on the waste classification system in Canada is still to be understood.

Tritiated heavy water has been detritiated for heavy water reuse at the power plants. The same approach is being considered and examined for highly tritiated light water in Canada. The ITER Tokamak has included a WDS plant for highly tritiated light water in its design. Both ITER and Canada could benefit from sharing the information from their light water detritiation projects.

The ^3H emissions from the ITER Tokamak are estimated to be higher than those from HWR and PWR. Vigorous air detritiation approaches are included in the ITER Tokamak design, and the actual ^3H emissions from ITER Tokamak, where conservatism is usually considered in modelling, are still to be seen after the Tokamak starts its active operation.

Purely tritiated waste is unique to the fusion applications and to the tritium removal facilities at the HWR plants. Radioactive ^3H in this waste stream may be recovered to facilitate its disposal. After all, the use of fusion for energy production is to be able to eliminate a number of waste streams produced from the fission reactors, such as used heavy metal fuel, and the entire suite of fission products. This nuclear waste avoidance has been a major drive in the fusion technology development.

General Fusion has an operational fusion device in Canada. The device has not started ^2H - ^3H (or D-T) fusion to produce plasma, and there are no immediate fusion waste issues. General Fusion's MTF technology is different from the ITER Tokamak; therefore, the MTF fusion reactor design and material selection will be different from the ITER Tokamak. These differences may limit the experiences and knowledge that can be learnt from the ITER project. Hence, it may be advantageous to examine potential waste streams from the General Fusion technology and their management strategies sooner rather than later.

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V-6. SUITABLE RECYCLING TECHNIQUES FOR DEMO ACTIVATED METALS

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Abstract

Recycling feasibility of activated materials (structural metals and breeder) generated by commercial nuclear fusion plants is one of the most debated issues concerning the demonstrability of the environmentally friendly management of fusion radioactive waste. ENEA and RINA Consulting CSM have been committed in EUROfusion tasks, related to the possibility of applying melting techniques for their recycling or, as an alternative, to disposal in shallow-land repositories for low-level radioactive waste (LLW). The melting process for removing elements such as carbon, chromium, manganese, tungsten and impurities (like Nb) has been analyzed from the theoretical (thermodynamic) point of view providing excellent results for C removal (Decontamination Factor DF~1000) for EUROFER and SS 316L(N) based on Vacuum Oxygen Decarburization (VOD). The study has also determined, based on steel making experience, the quantity of secondary waste produced (powders, filters, refractories, etc.) plus the CaCO_3 produced to recover most of ^{14}C removed from the melted bath, and the decarburized steels, the two main solid streams of the process. The results of the experimental tests on SS 316L(N) and EUROFER ingots prepared on purpose by Vacuum Induction Melting (VIM), confirmed the positive outcomes of theoretical analyses, with DFs around 100 that can be easily achieved. The process kinetics velocity found has shown that the process can be performed in reasonable times as per industrial practices. The DFs obtained would allow to respect the acceptance criteria of ^{14}C for some EU LLW disposal sites, if applied to the specific activity values of the most activated EUROFER components, calculated by activation codes.

FUSION POWER INDUSTRY SCENARIO

It is worthwhile to figure out the impact of fusion energy, in the case of its relevance in the energy market at the end of next century, from the management of radioactive waste point of view. To do that it is first required the definition of a fusion-power industry development scenario. This scenario has the start of a global fusion power industry in the second middle of the 21st century, suppose 2070, however, the precise date is not important. The important parameter for this scenario is the growth rate, which was assumed to be eight 1-GWe units per year. For comparison, the global fission industry began in the 1960s and grew at an average rate of at least 10 reactors per year (over 400 reactors in 40 years). Therefore, a global growth rate of 8 fusion power reactors per year is not an unreasonable assumption, and perhaps, it could be considered rather modest given that the world population is expected to increase of 25% by 2050 (9.7 billion according to UN Department of Economic and Social Affairs)³⁴ and the forecast of demand for electricity until 2050 will grow accordingly as confirmed by different independent organizations. On that perspective this paper refers to the IEA Energy Outlook 2019 [1]. Nevertheless, this growth rate is adequate for identifying the major waste management issues. At this growth rate, it takes 50 years to reach a level of 400 operating reactors and 125 years to reach a level of 1 000 operating reactors that is around the end of the next century. Other assumptions made for this scenario analysis are:

— Reactor life = 50 years;

³⁴ <https://www.un.org/development/desa/en/news/population/world-population-prospects-2019.html>

- Average reactor availability = 80%;
- Reactor decommissioning starts after 20-year dormancy;
- Decommissioning lasts 20 years.

Decommissioning waste begins to appear 70 years after the first reactor built – see Table V-6.1.

TABLE V-6.1. FUSION POWER INDUSTRY SCENARIO EVOLUTION FROM 2070 UP TO 2200

	2070	2120	2121	2141	2180	2200
Reactors in operation	8	408	416	576	888	1048
Reactors closed	0	0	8	168	480	640
Reactors starting decommissioning	0	0	0	8	320	480
Total reactors constructed	8	408	424	744	1368	1688

During the plant life, the radwaste from components replacement (blanket modules and divertor) will be generated but in this analysis, they are accounted at the end of decommissioning with the related radioactive waste. Extrapolating by a linear trend the growth of electricity up to 2040, as given in the IEA sustainable development scenario, we can assume that the electricity energy demand by 2200 will be about 124 PJ, (Fig. V-6.1).

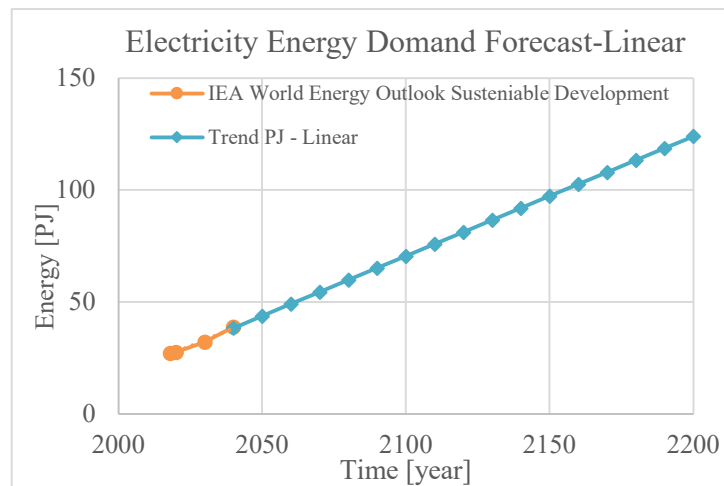


FIG. V-6.1. Electricity Energy Demand, forecast by linear extrapolation of IEA Energy Outlook 2019 [1].

Building eight 1-GWe fusion plants annually, a fleet of 1 000 D-T fusion power plants could provide from ~ 6% up to 10% of world electricity energy demand by ~2 200 as shown in Fig. V-6.2.

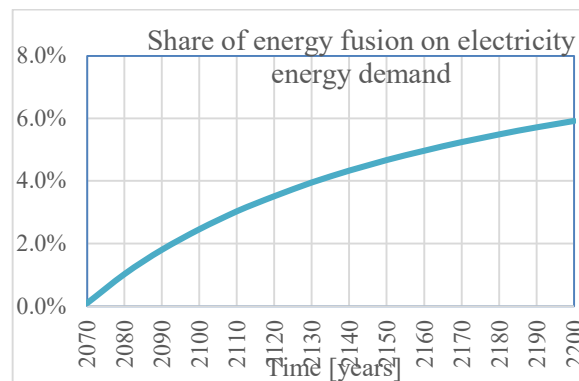


FIG. V-6.2. Fusion Share on Electricity Energy Demand; forecast by linear extrapolation of IEA Energy Outlook 2019 [1].

The result provided at the end of this century, roughly 3% can be compared with the results of an in-depth study on nuclear energy penetration scenarios from fusion into future low carbon global electricity [2]. Amongst the various scenarios considered there is one named “Harmony” which foresees a value of 4% for the fusion share in the global electricity market. Considering the many uncertainties behind this type of analysis, the hypothesis represented in Figure V-6.2 can be considered as acceptable to continue with the estimate of the masses, volumes of waste generated from fusion energy. The mass of materials in the DEMO Models WCLL and HCPB are taken from Table 8 of [3], assuming the plant life of 50 y and a plant availability of 80% that means 40 full power years (FPY) of operation. To evaluate the mass and volume of material generated as possible radioactive waste it was assumed to double the irradiation schedule of 20 FPY assumed for DEMO, as shown in the pictorial representation of Figure V-6.3 taken from [4].

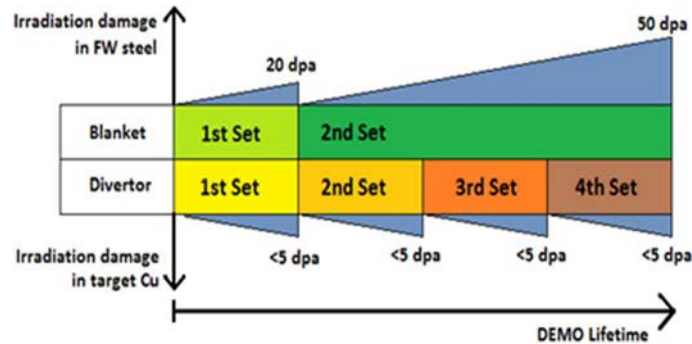


FIG. V-6.3. Irradiation Schedule for DEMO reactor [4].

The total replacement and decommissioning possible waste for DEMO according to the irradiation schedule lasting 20 FPY (reactor plant life 25 years) and 40 FPY (reactor plant life 50 years) (see next Table V-6.2.)

TABLE V-6.2. REPLACEMENT AND DECOMMISSIONING POTENTIAL RADWASTE FROM DEMO

DEMO Model	Replacement and decommissioning waste		Replacement and decommissioning metallic waste *	
	Mass (ton)	Volume (m ³)**	Mass (ton)	Volume (m ³) **
HCPB (20 FPY)	6.04E+07	1,53E+04	3.38E+07	4.19E+03
WCLL (20 FPY)	8.16E+07	1,69E+04	3.48E+07	4.32E+03
HCPB (40 FPY)	7.17E+07	1,72E+04	4.35E+07	5.38E+03
WCLL (40 FPY)	1.14E+08	2,03E+04	4.55E+07	5.63E+03

* Blanket/divertor/Shield/Manifolds/VV/ Magnet/Structure;

** volume calculated by dividing mass by actual density, assuming the best volume reduction

A large share of the radwaste mass is due to the bioshield 2.5E+07 ton or 1.04E+04 m³, which is at least 50% of the total waste at the best. DEMO will not be the size as a 1-GWe fusion power reactor, as it is designed to release 500 MW of net electricity [5]. Hence, the volumes of Table V-6.2. were scaled up of a factor ~1.4 to consider the larger size (reference PPCS Model A and B [6]). According to the data shown in Table V-6.1, 320 fusion reactors will complete the decommissioning by 2200 generating an amount of metallic radioactive waste as shown in Table V-6.3.

TABLE V-6.3. METALLIC WASTE GENERATED WITHIN 2200 BY FUSION INDUSTRY SCENARIO

Fusion reactor type	DEMO single reactor (m ³)	1 GWe Fusion Reactor (m ³)	320 1-GWe Fusion Reactors (m ³)
HCPB (40 FPY)	5.38E+03	7,56E+03	2,42E+06
WCLL (40 FPY)	5.63E+03	7,92E+03	2,54E+06

Considering a filling capacity of the shallow land burial (SLB) of 50% and a size of the single SLB of $1\text{E}+06\text{ m}^3$, next Figure V-6.4 shows the number of SLBs necessary to dispose all the metallic waste from fusion reactors up to 2200. Clearance of bioshield and complete recycling of the breeder material is assumed.

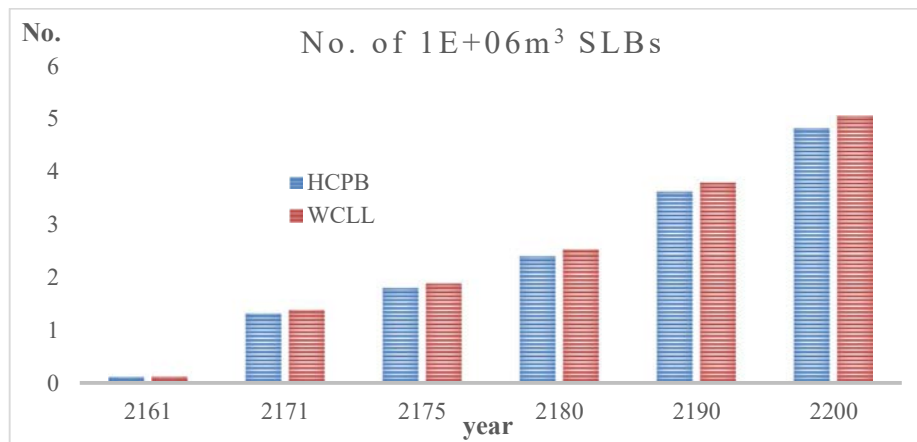


FIG. V-6.4. No. of SLBs required by fusion energy by 2200 if no recycling and clearance will be implemented.

It is also assumed that the SLB will be required just at the end of the decommissioning for all the metallic materials generated during the plant life, making the conservative assumption that no metallic radwaste were not recyclable, which disagrees with the R&D efforts put in place to promote recycling. It is obvious that each new deposit needs to be made available when the last one is completed, this without counting the territorial implications that in the proposed scenario provide for the possibility of transferring waste even beyond individual national borders. It is therefore likely that a larger number of deposits might be required, albeit of lesser capacity. If no recycling or clearance of metallic waste will be done, about 5 or more SLBs will be made available within 2200. This conclusion does not want to justify by itself the need for recycling, but simply to shape the dimensions of the disposal problems if this together with clearance were not pursued.

RECYCLING OF EU DEMO ACTIVATED METALS BY MELTING

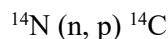
The preferential recycling treatment for metallic waste is based on melting, as it offers several key advantages:

- Detritiation of the material which will simplify further management steps and minimize the creation of secondary waste caused by off-gassing of tritium;
- Reduction of other key radionuclides, i.e. ^{14}C (calculated decontamination factor $\text{DF} \sim 1000$, experimental ~ 100), ^{53}Mn ;
- Transfer of volatile species to secondary waste (tritium DFs of ~ 40000 in a hydrogen atmosphere);
- Binding the remaining radioactivity in metal structure;
- Homogenisation of radionuclide activity distribution;
- Volume reduction of up to 10 times less;
- Reduction of surface to volume ratio;
- First step in a recycling process advantages.

The melting process is suitable for being performed remotely as the modern facilities, with high level of monitoring and automation, are available for performing sophisticated steelmaking operations under strictly controlled conditions. The current technologies allow high degree of flexibility and customization to satisfy the requirements of a large spectrum of production capacity and quality. A general trend in the modern steelmaking industry is toward a higher degree of robotization and remote control. This evolution is in line with the needs of processing potential radioactive waste.

THEORETICAL STUDY OF CARBON REMOVAL FROM EUROFER AND SS 316L(N) BY MELTING [7]

The study and experiments performed [7] were focused on the possibility to remove together with the stable isotope carbon also ^{14}C from typical structural steels foreseen in DEMO, i.e. EUROFER and SS316L(N). Isotopes behave chemically as the stable one. The ^{14}C isotope is not due to the transmutation of ^{12}C but is largely due to the following reaction:



other reaction is: $^{13}\text{C} (n, \gamma) ^{14}\text{C}$, but it is negligible considering the low isotopic abundance of ^{13}C (1.1%).

Furthermore, a minimum content of nitrogen is unavoidable for important mechanical and structural reasons. Hence, the interest for removing ^{14}C is dictated by fact that is one of the key nuclides for defining acceptance criteria for shallow-land disposal repositories for low-level radioactive waste. One may argue that decarburization is alternative to recycling of the steel because it aims to comply with waste acceptance criteria (WAC) for low-level waste in SLBs. On the other hand, the decarburization together with the removal of tritium can be considered a preparatory process pending the final decision about recycling or disposal which can be deferred for a few decades depending on the technical and economic framework for metals supply for fusion reactors. On the other hand, the reference scheme for the management of low-activation steels such as EUROFER or F82H foresees a decay period of few decades to exploit their ability to reduce the specific activity and dose rate of some orders of magnitude. The possible recycling as structural materials after 80-100 years would require a tailoring of the composition and the addition of the necessary carbon would not be a problem.

A first theoretical study proposed the decarburization of the steel accordingly with the process adopted in steelmaking to produce Ultra Low Carbon (ULC) Steel and High Chromium Steel (like stainless steel) [7]:

“The process is based on the reaction between Carbon and Oxygen, both dissolved in liquid steel, forming CO gas which is removed by operating under vacuum and/or with inert gas like Argon or Nitrogen. This process consists of blowing oxygen in the melt at high temperature ($T \geq 1600^\circ\text{C}$). Increasing the temperature increases the oxygen solubility and the reaction rate. In the steelworks, the decarburization process is performed by:

- Argon-Oxygen-Decarburization (AOD), where the partial pressure of CO is lowered by blowing Argon;
- Vacuum-Oxygen-Decarburization (VOD), with CO partial pressure lowered by dropping the total pressure.

The decarburization process parameters applied in most industrial plants are $T = 1700^\circ\text{C}$ and vacuum conditions ($P = 1\text{-}5\text{ mbar}$), that allow to obtain very low concentration of dissolved C (order of few ppm). Based on the industrial VOD process, the theoretical study was based on the following main operating conditions $T = 1700^\circ\text{C}$ and $P = 1\text{ mbar}$. Thermodynamic calculations were performed with commercial code ThermoCalc© - SLAG3 [8]. For EUROFER a Fe solution containing: C = 0.12%, Cr = 9%, Mn = 0.60%, Si = 0.05%, Al = 0.005 % was considered as main elements that can be involved in reactions with the oxygen blown into the melt. Another element considered was Tungsten ($W = 1.0 - 1.2\%$). All the results of the thermodynamic calculations were referred to 1 ton of steel. Details are given in [7]. As shown in Fig. V-6.5 a), 2.1 Nm^3 of stoichiometric O_2 per ton of steel is needed to obtain 1.15 ppm of C whilst, beyond this point, the oxygen blown reacts mainly with the other species (Cr, Mn, Si and Al) to form a mixture of oxides (commonly named “slag” in steelmaking operations) and pure Cr_2O_3 . After reaching 1.15 ppm of C, a further decrease of C is very difficult to reach: the volume of oxygen needed to reach C concentration of 0.5 ppm is $21\text{ Nm}^3/\text{t}$ of steel, with significant a) production of slag and Cr_2O_3 . The feasibility of Mn removal has been also evaluated.

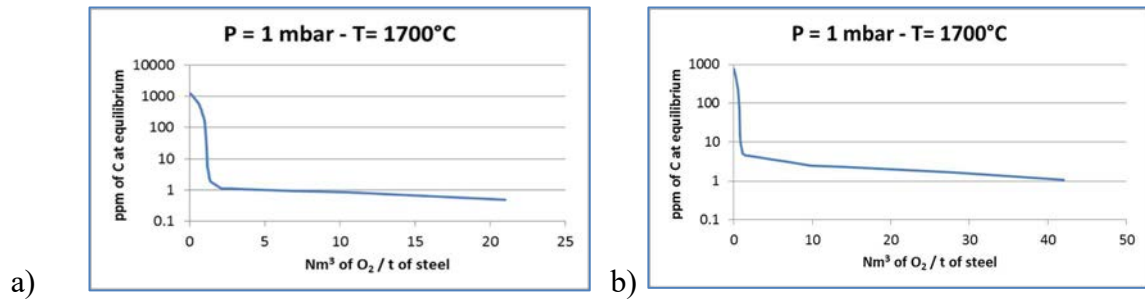


FIG. V-6.5. Calculated C equilibrium concentration in VOD vs O₂ blown in 1 ton of a) EUROFER, b) SS 316L(N).

It is partially oxidized, so that it is found in the molten steel and the slag (as MnO). With a large O₂ flow rate blown (21 Nm³/t steel), Mn concentration reduces of about a half (from 0.6% to 0.35%). The mass of Mn in the slag is limited ~ 3 kg/t steel at 21 Nm³/t steel. The removal of W is practically impossible as calculated W passing from steel into the slag is negligible. After blowing 21 Nm³ of O₂ per t of steel, the mass fraction of WO₂ in the slag is ~35 ppm, i.e. 0.003 kg of WO₂ removed per t of steel from an original quantity of 10-12 kg.

Similar analysis was done for the SS 316L(N) with an initial composition: Cr 16 – 18%; C < 0.08%; Mn < 2.0%; P < 0.045%; S < 0.03%; Mo 2.0 – 3.0 %; Ni 10 – 14.0%; Si < 1.0%; others 0.10 – 0.16%; Fe Balance. The decrease of C concentration is such that 1.1 Nm³ of stoichiometric O₂ per t of steel is sufficient to obtain 5.2 ppm of C. See Figure V-6.5b). The total CO gas produced does not vary significantly beyond the point where C in the steel is ~ 5.2 ppm, at which most of the O₂ blown react with the other elements of the steel to form slag and pure Cr₂O₃. Similarly, as the case of W for EUROFER, the Ni removal is not feasible, even at oxygen concentration of 1%. The possibility to form NiO as slag component has been investigated. The results showed that a tiny fraction of initial Ni concentration can be removed as NiO: 0.038 kg of Ni, from a starting value of 140 kg per ton. Mo removal was also analysed, but calculations showed that it is negligible.”

Mass balance with secondary waste assessment from 14C removal by VOD melting

According to [7]: “The CO produced during the decarburization process has to be converted to CO₂ and blocked in a matrix like CaCO₃ as shown in Fig. V-6.6 representing the mass balance including the assessment of secondary waste.

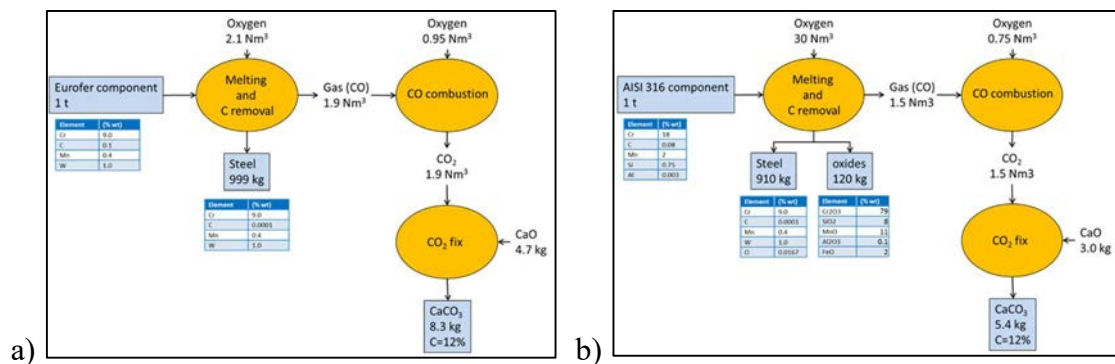


FIG. V-6.6. Mass balance (for 1 ton of steel) for the VOD process applied to C removal from a) EUROFER and b) SS 316L(N).

It was assumed 2.1 Nm³ of O₂ blown for 1 ton of EUROFER and 30 Nm³ for 1 ton of SS 316L(N) to reach a residual level of ~1 ppm of carbon. The formed phases after the VOD are decarburized steel and secondary waste, such as: gas, CaCO₃, dust, slag (oxides, Cr₂O₃) plus crucible refractory whose values are in Table V-6.4.”

TABLE V-6.4. SECONDARY WASTE GENERATED BY C REMOVAL VOD PROCESS [7]

Secondary waste from 1 ton	EUROFER		SS 316L(N)	
	k g	d m 3	k g	dm ³
Crucible refractories	9. 4	3. 4	9 .8	3.5
Metallic powders and filters	1 0. 5	5. 3	1 0 .5	5.3
CaCO ₃ after CO ₂ capture	8. 3	2. 8	5 .4	1.8
Other oxides as slag	0	0	1 2 0 .0	30.0
Total	2 8. 2	11 .4	1 4 5 .7	40.6
Share	3 %	8 %	1 5 %	28 %

Referring to the activation calculations of the EUROFER structures obtained with the FISPACT-II code, decarburization at levels of ~1 ppm would reduce the ¹⁴C level to a few tens of kBq/kg, i.e. 2-3 orders of magnitude <WACs typical of EU LLW radioactive waste deposits. For SS 316L(N), it would be sufficient to lower [C] to ~10 ppm to comply with those levels of ¹⁴C activity (few tens of kBq/kg) with a reduced secondary waste, as calculated for EUROFER (3% in mass). That is because SS 316L(N) is used mostly for the VV of DEMO which is less activated than EUROFER blanket or divertor structures.

TESTS ON C REMOVAL FROM EUROFER AND SS 316L(N) BY MELTING [9]

Decarburization tests were carried in a Balzers induction furnace in RINA Consulting – CSM workshops (see Fig. V-6.7) aiming:

- To evaluate O₂ max flow rate, allowing decarburisation without slag formation;
- To estimate process kinetics velocity, and consequently time for decarburisation.



FIG. V-6.7. Balzers Induction furnace.

Steel ingots of 3.5-6.3 kg of EUROFER and SS 316L(N) were prepared on purpose in VIM furnace with reference to the expected composition after irradiation (data provided by FISPACT-II). They were then melted in the Balzers oven injecting O₂ gas, as oxidizing agent, at the lowest possible flux (<5-10 l/h) under Ar cover atmosphere to minimize slag formation and flooding issues. Oxygen was inflated in the metal bath by a porous septum (Figure V-6.8) with alumina shaft specifically made (for better O₂ blowing and hence greater mixing).



FIG. V-6.8. Porous septum with alumina shaft.

Tests were repeated injecting O₂ gas, as oxidizing agent, at the lowest possible flux (<5-10 l/h) under argon cover atmosphere to minimize slag formation and flooding issues.

As general decarburization procedure for both selected alloys, the following steps were defined:

- Melting of the sample at about 1500°C under vacuum;
- Addition of argon until 800 mbar in the furnace;

- Addition (insertion) of the porous septum (or metamict tube) in the melted bath;
- Flushing of oxygen at minimum flow (about 5-10 l/h to reduce spurts production) for 15-20 min;
- Extraction of the porous septum and degassing phase to remove gaseous products formed during the decarburization phase;
- Second insertion of the porous septum and second decarburization phase;
- Casting;
- Sampling after ingot cooling for carbon analysis.

Technical sheets of the decarburization processes for both steels are given in the EUROfusion Tech. Report [10]. Different phases of the trial are shown in Figure V-6.9.



FIG. V-6.9. On the left: slag formed during melting phase; on the right decarburisation phase (different colour in the left picture is due to the presence of a shielded glass on the top of the Balzers furnace window).

The defined decarburization procedure was eventually adopted to perform additional decarburization trials, with different times and different oxygen flow rates. They provided the best results as shown in following Table V-6.5 and Table V-6.6 for EUROFER and SS 316L(N) respectively.

TABLE V-6.5. DECARBURIZATION TESTS WITH EUROFER (INITIAL C CONCENTRATION: 1200 PPM)

Test number	Ingot mass (kg)	O ₂ flow rate (l/h)	O ₂ specific flow rate (l/(h·kg))	Final [C] (ppm)	Time (min)	O ₂ total volume (l)	[C]end/[C]ini (ppm/ppm)	de-C efficiency (*)
1	3.6	10.0	2.8	200	20	3.3	0.167	0.833
2	3.6	10.0	2.8	130	25	4.2	0.108	0.892
3	3.6	13.7	3.8	39	20	4.6	0.033	0.967
4	3.5	15.0	4.3	10	25	6.3	0.008	0.992
5	3.5	15.0	4.3	6	25	6.3	0.005	0.995

(*) the decarburization efficiency is defined as: $(de-C) = 1 - [C]_{end}/[C]_{ini}$; Decontamination factor DF is given by $DF = [C]_{ini}/[C]_{end}$

TABLE V-6.6. DECARBURIZATION TESTS WITH SS 316L(N) (INITIAL C CONCENTRATION: 300 PPM)

Test number	Ingot mass (kg)	O ₂ flow rate (l/h)	O ₂ specific flow rate (l/(h·kg))	Final [C] (ppm)	Time (min)	O ₂ total volume (L)	[C]end/[C]ini (ppm/ppm)	de-C efficiency (*)
1	5.3	5	0.9	131	20	1.7	0.437	0.563
2	5.3	10	1.9	15	15	2.5	0.050	0.950
3	5.2	10	1.9	3	20	3.3	0.010	0.990

(*) the decarburization efficiency is defined as: $(de-C)_{eff} = 1 - [C]_{end}/[C]_{ini}$; Decontamination factor DF is given by $DF = [C]_{ini}/[C]_{end}$

Decarburization efficiency obtained was up to 99 % for both EUROFER and SS 316L(N). These results indicate that the practical O₂ volume to obtain decarburization with efficiency close to 100% is 1.5 times the theoretical volume for EUROFER and about 2.0 times for SS 316L(N) the theoretical volume calculated previously (see previous section). These are to be considered limits for decarburization without slag formation. Larger O₂ volume would causes oxidation of other chemical species in the steel. After confirmation of the theoretical analyses, the other aim of the tests was to evaluate the effect of O₂ flow rate on process efficiency for calculating the kinetics velocity, and consequently the decarburisation time. The decarburisation rate can be expressed by:

$$\frac{d[C]}{dt} = k \cdot ([C] - [C]_{final}) \quad (\text{eq. 1), where:}$$

$d[C]/dt$ is the variation in time of carbon concentration in steel (ppm/s);

$[C]$ is the actual concentration of carbon at time t (ppm);

$[C]_{final}$ is the theoretical minimum carbon concentration (ppm).

k is a kinetic parameter depending on the operating condition (1/s)

$[C]_{final}$ is the thermodynamic limit. As calculated in the paper [7], $[C]_{final}$, is ~1 ppm and ~5 ppm respectively.

In the adopted decarburisation regime, k depends on oxygen flow rate.

The integration of the eq. 1 gives the equation of the evolution of carbon concentration $[C]$, as function of time, initial and final carbon concentration $[C]_{initial}$, $[C]_{final}$.

$$[C] = [C]_{initial} \cdot e^{-kt} + [C]_{final} \cdot (1 - e^{-kt}) \quad (\text{eq. 2})$$

Being $[C]_{final} \ll [C]_{initial}$, the (eq. 2) can be approximated by the simplified formula.

$$\frac{[C]}{[C]_{initial}} = e^{-kt} \quad (\text{eq. 3})$$

The parameter k , in the eq. 3. can be estimated from the experimental results, which for EUROFER are shown in the following Figure V-6.10, the analogue graph for SS 316L(N) tests can be found in [10]. The values of the kinetic parameter k of (eq. 3) resulting from decarburisation tests with different oxygen flow rates were obtained by the correlating exponential graphs of Fig. V-6.10 below and shown in the following Table V-6.7 for both steels.

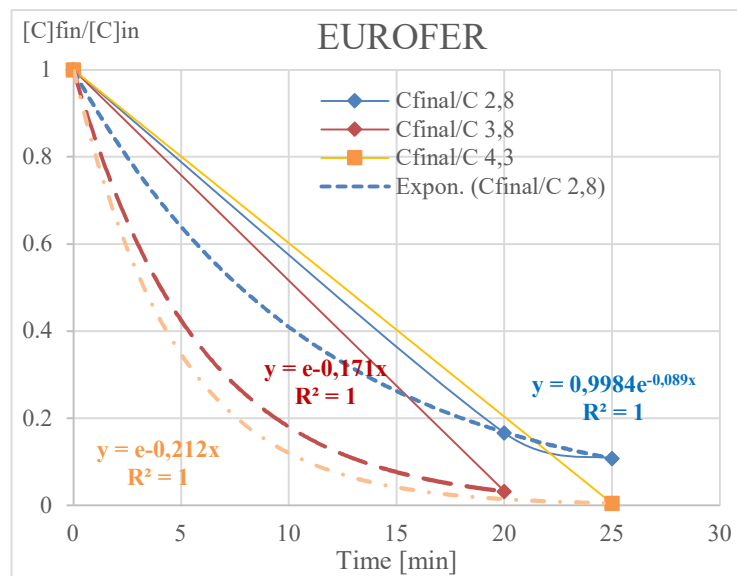


FIG. V-6.10. Experimental (exp) and calculated (mod) $[C]_{fin}/[C]_{in}$ values in the decarburization tests of EUROFER carried out with different O₂ flow rates.

TABLE V-6.7. KINETIC PARAMETER K (Eq. 3) FROM THE DECARBURIZATION TESTS

Test with EUROFER	O2 flow rate (l/h)	O2 specific flow rate (l/(h·kg))	k (1/min)
1, 2	10.0	2.8	0.09
3	13.7	3.8	0.17
4,5	15.0	4.3	0.20
Tests with SS 316L(N)	O2 flow rate (l/h)	O2 specific flow rate (l/(h·kg))	k (1/min)
1	5	0.9	0.04
2	10	1.9	0.22
3	10	1.9	0.23

The results give indications on the value of oxygen specific flow rate to perform deep decarburisation (de-C efficiency >99%) without slag, which are about 4.3 l/(kg·h) for the EUROFER and about 1.9 l/(kg·h) for AISI 316L(N)-IG. The corresponding kinetic parameters are: 0.20 L/min for EUROFER and 0.23 L/min for SS 316L(N). They provide practical indications on the specific flow rate of oxygen to be used to maximize the decarburisation rate. The necessary process time to obtain the target final value can be calculated from (eq. 3).

For example, to reduce the concentration of EUROFER to 0.001 of the initial concentration, considering the k value of 0.20 1/min, the necessary decarburisation time is:

$$t = -\frac{\ln(0.001)}{0.20} = 34.5 \text{ minutes}$$

For SS 316L(N) the same calculation gives a process time of 30 minutes, taking into account that C_{in} is equal to 300 ppm and make reference to best results of test 3 (see Table V-6.6), where $k = 0.23 \text{ min}^{-1}$.

CONCLUSIONS

Fusion Power Plants (FPPs) will generate a remarkable amount of metallic waste and breeders activated and contaminated by ^3H . It is important to reduce this quantity by recycling. Most challenging process on activated metals is the removal of ^3H and ^{14}C , even though this last one is not directly aimed at recycling but to prepare the component to be disposed of in a SLB if recycling would not be feasible after a reasonable interim storage. The experimental tests confirmed the feasibility of decarburization of both EUROFER and SS 316L(N) without slag formation, in agreement with the thermodynamic calculations. The DFs >100 achieved would allow to respect the acceptance criteria of ^{14}C for some EU LLW disposal sites, if applied to the specific activity values of the most activated EUROFER and SS316L(N) components.

ACKNOWLEDGEMENTS

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V-7. APPLICATION OF POWDER METALLURGY AND ADDITIVE MANUFACTURING TO REFABRICATION OF DEMO COMPONENTS/ STRUCTURES

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Abstract

The identification of components that can be manufactured starting from activated material recycled from DEMO fusion reactor, with the relative manufacturing techniques, is a decisive factor to support the environmental sustainability of the fusion radioactive waste management. The feasibility of the powder production from activated steels (SS 316L(N)-IG and EUROFER) for the re-fabrication by Additive Manufacturing (AM) Technology of components for fusion reactors has been studied in the EUROfusion SAE 3.6.2 T003 Task. The process chosen is the inert gas induction vacuum atomization (VIGA) to produce metal powders for AM processes, with the possibility to perform detritiation before the melting phase. The chemical-physical parameters of VIGA process have been defined, with the verification of its viability, in terms of times and dimensions compatible with the construction of an industrial facility. VIGA plant consists of a vacuum induction furnace to melt the charge in a refractory crucible and an atomization tower where an inert gas jet atomizes the metal into spherical oxidation-free micrometric particles with properties suitable for AM. Integrating AM into the refabrication of nuclear components, however, requires addressing several steps. The first one is the production of metal powders with suitable physicochemical characteristics. Then in their developing path, AM products need to achieve characteristics required by structural nuclear components for which nuclear qualification for licensing will be mandatory. 4kg of powder of SS 316L(N)-IG and EUROFER were prepared in the VIGA plant starting from ingots with the expected chemical composition after activation. Powders were characterized in terms of granulometry by SEM-EDS analysis, while chemical analysis confirmed original composition with no oxidation. Small mock-ups were produced from powder by Selective Laser Melting (SLM) and characterized positively on the point of view of hardness, surface roughness and micrograph analysis.

INTRODUCTION

Additive Manufacturing (AM) technologies for metallic and ceramic components have a great potential in the future production of components with various and even complicated shapes as observed by the continuous and strong increase of the related market. Through AM technologies it is possible to build a part using a layer-by-layer process directly from a digital model, without the use of moulds or dies that add time, waste material and expense to the production process. Metal powder metallurgy is a well-known technology that offers several advantages in comparison with other metal forming technologies such as forging and metal casting. The main advantages referred to metal powder metallurgy are more efficient material utilization, possibility to realize more complex shaped components and near-net-shape dimensional control, reducing or eliminating further mechanical operations. These advantages make this technology generally safer and with a higher level of automatization than traditional methods and, also, contribute to sustainability, making powder metallurgy a recognized greener technology. A new substantial push for the diffusion of metal powder technologies comes from the development of new technologies for exploiting the potentiality of metal powder metallurgy, in particular: Metal Injection Moulding and Additive Manufacturing.

POTENTIAL USE OF POWDER METALLURGY AND ADDITIVE MANUFACTURING FOR REFABRICATION OF STEEL COMPONENTS OF DEMO PLANTS

Because of the growing diffusion of AM technologies, for making metallic components, it is reasonably to believe that this technology can be exploited in the fabrication and re-fabrication of components for fusion power plants, like DEMO [1]. These plants will be characterized by the presence of many metallic components of large variety of size and shapes. Some of these components might be compatible in size with the current or emerging potentiality for AM technology. Smaller components, to be also frequently replaced, might take advantage from the availability of virtually wasteless manufacturing processes, such as AM. For example, some components of blanket modules which are expected to be replaced after 2 full power year of operation [2], might be candidate for exploiting fabrication or refabrication by activated metals using AM technology.

On the other side, melting technology is already considered a promising option for treatment and recycling of dismantled components of DEMO plants. Being a melting phase the first step of the production process of metal powder and AM the technique to use the powder for making the final components, the integration of these two processes in a complete manufacturing cycle purposely dedicated to the re-fabrication of DEMO components appears a logical solution.

All the metallic parts of the blanket modules of divertor cassettes might be re-fabricated starting from melting of dismantled activated components, atomisation for producing metal powder, eventually new component production with AM techniques.

This option has been already investigated and few examples of components for fusion power plant have been already realized with AM techniques. For example: sub-components of blanket modules, with hollow and double walled structures and components and with internal channel structures have been realized by means of Laser Beam Melting (LBM) and Selective Laser Sintering (SLS) [3]; thin- and double-walled flow channel inserts for dual-coolant lead lithium blankets have been successfully manufactured by means of Selective Laser Sintering [4].

These works and other confirm that AM technique is a promising technology for fabrication and re-fabrication of components of fusion power plants. Furthermore, process steps for powder production could be exploited to perform removal of key radionuclides. For example, recovery of tritium could be performed during the atomization process for producing powder or, alternatively, the large specific surface of the powder could be exploited to optimize the detritiation process.

ATOMIZATION PROCESS FOR ADDITIVE MANUFACTURING METAL POWDERS PRODUCTION

The atomization process for producing metal powders for AM is currently performed in an atomization tower, pouring the liquid metal at temperature around 1550 °C and high pressure (typically 30-40 bar). A gas jet impacts on the liquid stream, forming small metallic droplets, which rapidly are cooled and solidified. At the RINA CSM workshops, two ingots of SS 316L (N) -IG and EUROFER of well-defined chemical composition were produced in the VIM plant (Fig. V-7.1). Therefore, the ingots were introduced in the VIGA plant and powders have been produced (Fig. V-7.2).



FIG V-7.1. RINA - CSM VIM.



FIG. V-7.2. RINA - CSM VIGA plant.

The chemical composition of the EUROFER and AISI 316L(N)-IG ingots, analysed by ICP-MS, are shown in Table V-7.1 [9]; the powders SEM-EDS analysis is shown in Fig. V-7.3. Both ingots have a chemical composition close to the average of the HCPB WCLL models irradiated components after shutdown, as simulated by FISPACT calculations.

TABLE V-7.1. EUROFER AND AISI 316L(N)-IG INGOTS CHARACTERIZATION [9]

Composition (Wt%)	EUROFER	AISI 316L(N)-IG
C	0.12	0.03
N	0.032	0.097
O	0.0034	0.0042
Si	0.092	-
S	0.0045	<0.01
P	<0.006	0.025
Cr	9.10	17.80
Mn	0.49	1.87
Ni	0.016	12.57
Cu	0.0076	0.32
Mo	<0.01	2.63
Zr / Sn / Sb e B	<0.001	<0.001
Ta	0.12	<0.01
W	1.08	-
V	0.15	-
As	0.002	-
Al	<0.01	-
Ti	<0.01	0.1
Co	<0.01	0.05

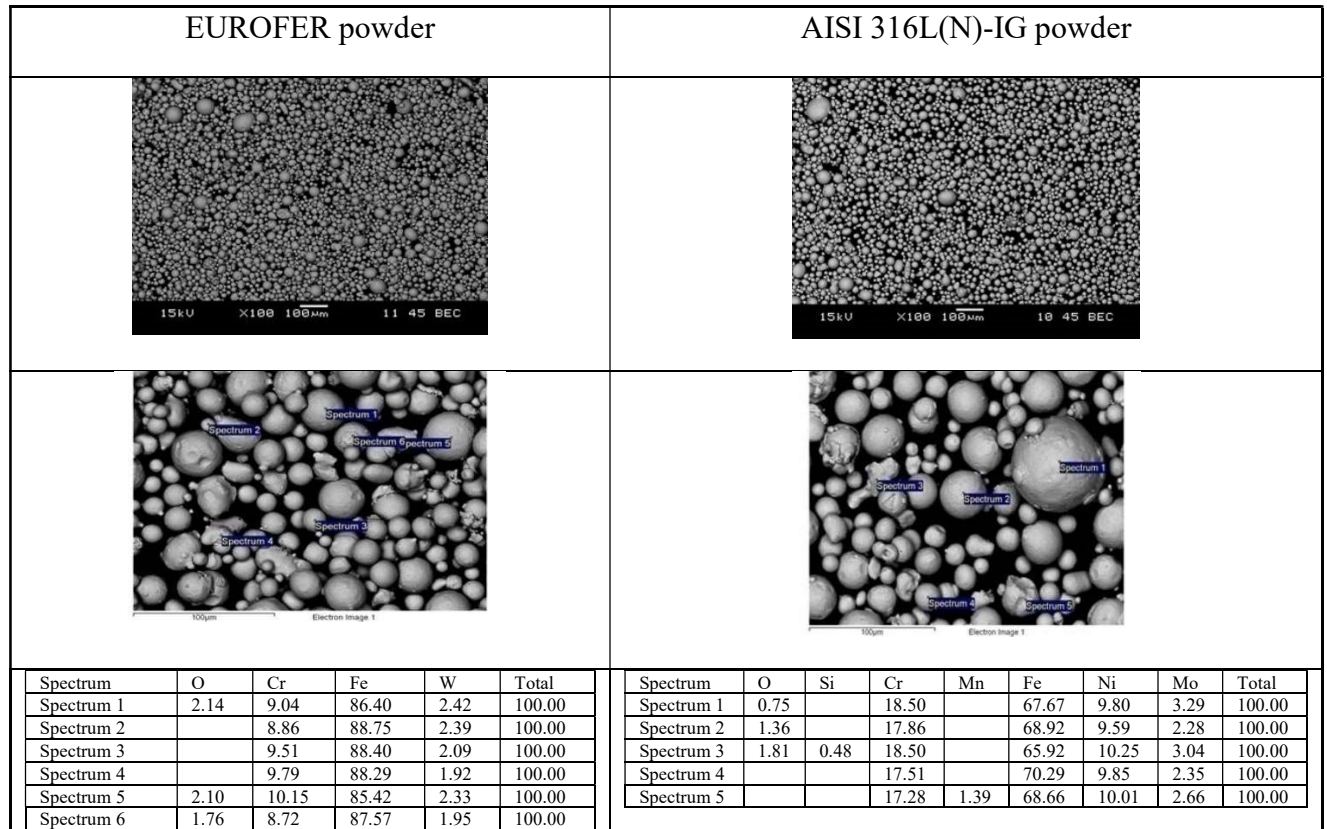


FIG. V-7.3. EUROFER and AISI 316L(N)-IG SEM-EDS analysis.

Powders have been characterised also in terms of laser granulometry (Fig. V-7.4).

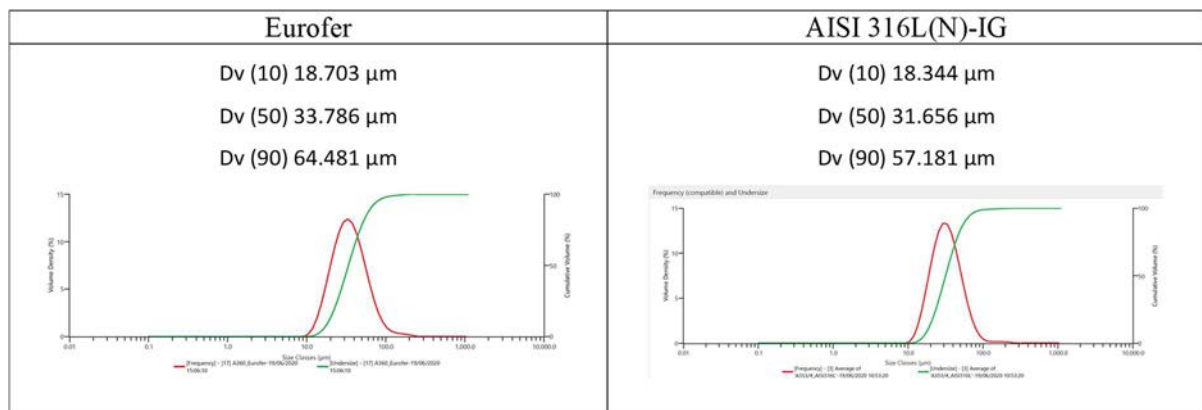


FIG V-7.4. Granulometry characterization of powders.

The powder granulometry of interest for Additive Manufacturing is in the range of 15-65 μm . About 8% of the EUROFER powder has been discarded, due to a granulometry higher than 65 μm and no EUROFER powders was lower than 15 microns. In the case of AISI 316L(N)-IG, about 3% of particles is below 15 microns and about 5% is above 65 microns. Therefore, both samples have about 92 % of powders suitable for Additive Manufacturing process.

Selective Laser Melting

There are several technologies for metal AM [14]:

“Schematically they can be divided in “direct” and “indirect” processes:

- In the “direct” processes the metal powder is completely melted and then solidified to form the final components. “Direct” metal AM technologies include: “Selective Laser Melting” (SLM), “Laser Metal Deposition” (LMD) and “Electron Beam Melting” (EBM);
- In the “indirect” processes a binder is used to join the particles of metal powder together and post processing is necessary to achieve the target density.

Direct technologies can produce denser parts because do not need binder and post-processing, hence reasonably can be considered the most promising technological routes for future metal AM technologies.

The first step in AM technology is the production of powders [14]; The metallic powder is produced from melted alloy, which is poured inside a so called “atomization” chamber, as explained above. Final step of refining, densification, thermo-mechanical treatment can be applied to give the target properties”.

Selective Laser Melting (SLM), used in this work for printing metal mock components, is a special additive manufacturing technology in which metal powders are arranged on a platform and selected portions of the platform are consolidated using thermal energy from a laser source. The SLM process belongs to the category of additive powder bed processes and is used to produce metal components. In Selective Laser Melting (SLM) powder is spread in thin layers, with layer thicknesses typically in the range 20 - 100 μm , over the work area and is melted by a localized laser beam following a predefined scan path.

The SLM process can be schematically divided into three steps:

- a) Powder bed formation: powder, fed from a hopper or storage tank, is spread on the work platform by a coater to form a layer of uniform thickness;
- b) Powder consolidation: a laser beam, guided by an optical system, scans the powder bed and selectively melts the powder;
- c) Lowering of the platform: after the completion of the first layer, the working platform is lowered by a distance equal to the thickness of the layer through the action of one or more pistons placed under it.

This sequence of operations is repeated until processing is complete. Powder not melted, in excess during layer laying, is collected in an overflow platform or accumulation tanks and can be sieved and loaded back into the machine for further processing. For this reason, one of the great advantages of this process is that it has virtually zero raw material rejection rates.

Selective laser melting of the produced powders for component refabrication

A deep study of literature demonstrates that EUROFER steels are weldable [10, 12], while AISI 316L is among the most workable materials through SLM. The current chemical composition of EUROFER steel has not been studied for SLM in the literature. Instead, the chemical composition of AISI 316L(N)-IG has a higher N content than the compositions conventionally used for SLM. In both cases the new variants of the chemical compositions have been deeply evaluated for their workability by SLM. Hence, both samples have powders suitable for AM and in line with commercial powder productions. To prepare mock-ups of AISI 316L(N) and EUROFER starting from powders produced, RINA has commissioned to AddMe Lab of Politecnico di Milano Department of Mechanical Engineering the production of specimens of both steels. As noted in [14]:

“An industrial SLM machine (200 W laser power) was used for assessing the processability of the produced powders and was fitted with a reduced build platform (RBV) system allowing to process small quantities of powder (<5 kg). The powders were processed under constant Ar flow with the initial O₂ less than 1000 ppm “.

During the SLM process, the laser is scanned to emit with a given peak power (P_{peak}) for a fixed exposure time (t_{on}) in μ s (microsecond) range and exposure points are spaced with a point distance (d_p). The laser jumps onto the consecutive spot on the scanned line with a hatch distance (d_h) and applies the exposure. At the end of each scanned line, the laser jumps to the adjacent one. The operation is continued until the layer is scanned completely. For a given layer, the system applies different hatch lines for the part volume and part external border, which use different parameter sets. As noted in [14]:

“The preliminary analysis assessed the flowability of the powder and consolidation of the material with the laser beam as a function of the main process parameters namely laser power, point and hatch distance, pulse duration, and the focal point “.

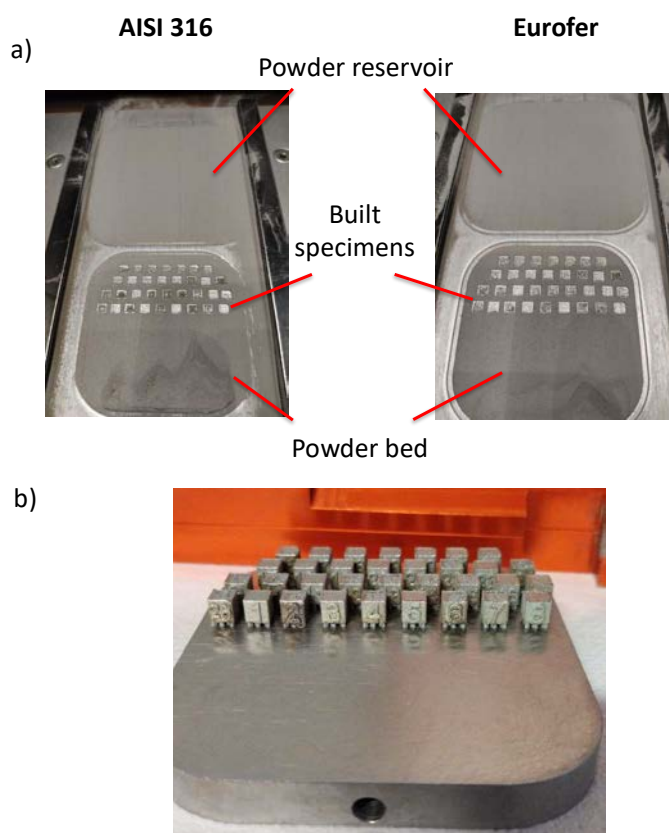


FIG. V-7.5. from reference [13]
a) The regular powder bed formed during SLM of AISI 316L(N)-IG and EUROFER.
b) Freestanding samples produced with the AISI316(N)-IG powder.

During the initial trials 33 specimens with different process parameters was consolidated with each material. Cubic samples with 5x5x5 mm³ dimensions were produced. In both cases a stable powder bed could be formed, and freestanding cubic samples was made without macroscopic defects [13, 14].

SPECIMEN CHARACTERIZATION TESTS

The qualification of materials made using AM techniques is necessary to verify that these materials have characteristics suitable for the construction of nuclear components. The optimization of the microstructure of these materials needs also be achieved taking into account post-processing (hot isostatic pressing, machining operations, etc.) technologies that can improve material qualities and make these

AM materials more similar to that produced by more traditional casting technology. It needs to be considered that there is still a long way to go before AM technologies could be applied even at large dimensional scales for nuclear grade component fabrication. The recycling of contaminated metal powders is undoubtedly another frontier to be crossed, but through research, which has been moving very quickly in recent years, into new powder formulations and new additive manufacturing techniques it will be probably possible to achieve these goals. To start the path toward this end, a series of mechanical tests were carried out to define the main characteristics of the materials produced by SLM.

Vickers-hardness test

Indentation tests were carried out, using the Vickers-hardness Tester FV-700, to determine the hardness of the SLM specimens of the three different metal alloys analysed.

To determine Vickers hardness (HV) according to ISO 6507, a pyramidal indenter (with a vertex angle of 136°) is driven into a specimen (workpiece) with a defined test load. The Vickers hardness HV is derived from the quotient of the applied test force F (expressed in Newton N) and the surface of the indentation left on the specimen. To calculate the surface area of the pyramidal impression, the average value of the two diagonals d_1 and d_2 (in mm) is used, because the base of Vickers impressions is often not an exact square. The recommended Vickers hardness range is given in the standard (ISO 6507). Depending on the applied test force and the sample material, it is between 1 and 3000 HV.

The hardness tester before use was calibrated on a 300HV specimen and then calibrated by loading at 10 kg_f, on one of the specimens, to test the hardness range. Following the calibrations and preliminary testing, measurements were taken and the values of the diagonals of the indented profile and the resulting hardness are reported Table V-7.2 below.

TABLE V-7.2. VICKERS-HARDNESS TEST RESULTS OF AISI316, AISI316L(N)-IG AND EUROFER SLM SPECIMENS [13]

Sample	HV	D1 (μm)	D2 (μm)	Average HV
AISI 316L	203	301.5	303.0	203
	200	304.0	305.5	
	205	301.5	300.5	
AISI 316L(N)-IG	200	307.3	301.3	196
	197	306.5	307.1	
	191	313.9	309.6	
EUROFER	257	270.0	267.5	249
	240	279.2	277.4	
	251	274.2	270.0	

Thus, these results are in line with the values reported in literature for steels prepared in SLM show that the measured material hardness is approximately HV200 for the AISI 316L and AISI 316L(N)-IG specimens, while it is 250 for the EUROFER specimens.

Roughness measurements

One of the most common defects of the SLM process products is given as the low surface quality of the printed components. In this study, therefore, the surface roughness of the samples was examined was evaluated (Figure V-7.6). To perform this analysis, it was necessary to achieve a surface scanning by stereomicroscope: the analysis consists in scanning a surface and then counting on a limited number of areas of interest, peaks and valleys. Through this count it is possible to determine the average roughness of the entire surface.



FIG. V-7.6. Left: view of SLM specimen section; Centre: embedded in the specimen holder; Right: Durometer Tester FV-700 used for Vickers-hardness.

The results were compared with the results obtained by performing the same analysis on a sample of AISI 316L also made with the SLM technique. In Figure V-7.7 and Figure V-7.8 are shown as example the Zone 1 profilometry in 3D stereo-microscope scanning for AISI 316L(N)-IG and Zone 3 for EUROFER specimens. The same analysis was done for all the specimens (AISI 316L, AISI 316L(N)-IG and EUROFER) produced by SLM.

In this analysis, 1260 points were collected in three different areas of the surface, each of which made it possible to derive an area the roughness R_a . This test is not standardized, nevertheless, it is useful for a preliminary evaluation of the surface quality of the specimens. Roughness (R_a) is measured as a standard index, that represent the standard deviation from the average value of the peaks and valleys measured.

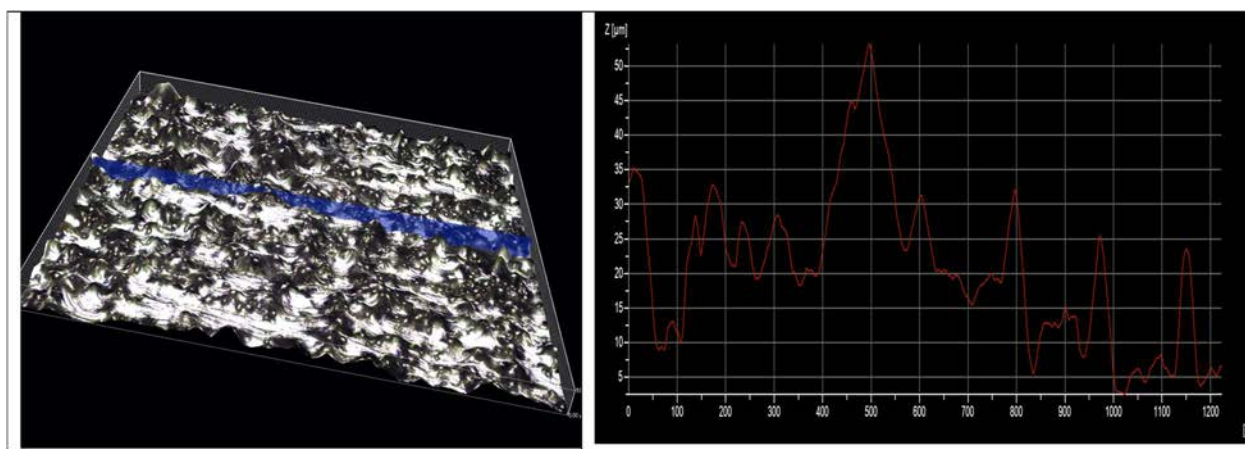


FIG. V-7.7. AISI 316L - Zone 1 profilometry 3D stereo-microscope scanning and 2D profile.

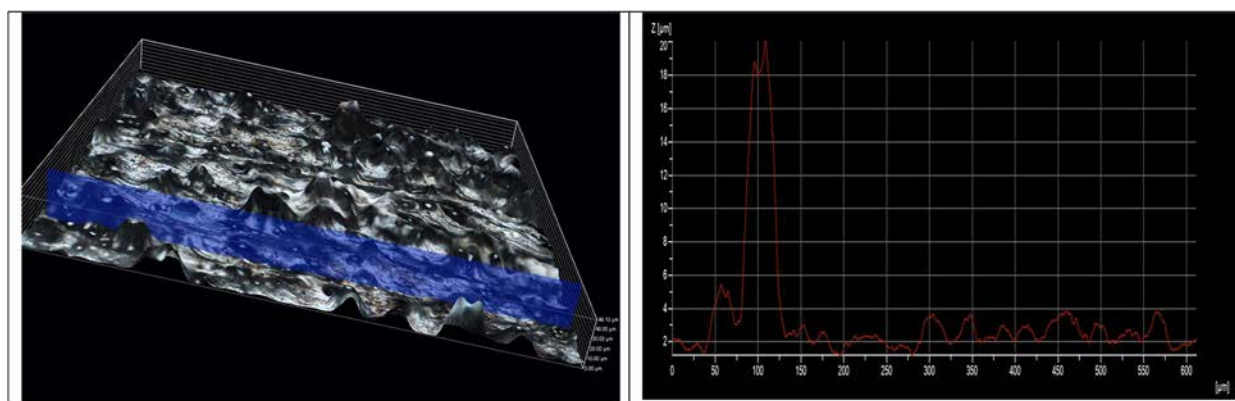


FIG. V-7.8. EUROFER - Zone 3 profilometry 3D stereo-microscope scanning and 2D profile.

By averaging the Ra's zone 2D profile of 3 different sectors of the specimens (right picture of previous Figure V-7.7 and V-7.8 viewable in blue colour), it was possible to derive the average Ra of the complete sample, shown in Table V-7.3 below. As a further analysis it was evaluated the number of positive peaks, measured with respect to the minimum value observed by the microscope: two cut-off values of 20 and 40 microns were defined. The table shows the number of peaks thus obtained and the percentage compared to the measurement points. This analysis confirms that the samples made of AISI 316 L(N)-IG and EUROFER are of excellent workmanship, as already indicated by the Ra index.

TABLE V-7.3. ROUGHNESS MEASUREMENTS RESULTS [13]

Material	Ra	Peaks > 20 μm		Peaks > 40 μm	
	μm	n.	%	n.	%
AISI 316 L	9.64	2103	0.556	118	0.031
AISI 316 L(N)-IG	4.22	811	0.215	0	0.000
EUROFER	5.73	282	0.075	73	0.019

Density measurements

Density measurements were performed by Archimedes' method according to ISO EN 623-2 on each type of steel. The method consists of measuring the weight of the sample first in air and then immersing it in distilled water using a precision hydrostatic balance, with thousandth of a gram sensitivity. The density of the sample is a function of the upward buoyant force that it receives once immersed in the liquid. Liquid density is known. First, the samples were placed in an oven at a temperature of 110°C for 1 hour to remove any possible trace of moisture in the porosities. The measurement was performed three times for each sample and the results were averaged.

TABLE V-7.4. AISI 316L(N)-IG AND EUROFER DENSITY MEASUREMENTS REFERRED TO STANDARD VALUES [13]

Material	T °C	$\rho_{\text{H}_2\text{O}}$ g/cm ³	w1 g	w2 g	ρ^* g/cm ³	ρ^* %
AISI 316L(N)-IG	20.0	1.00	1.063	0.929	7.91	99.1
EUROFER	20.0	1.00	1.064	0.926	7.72	98.3
AISI 316L	20.0	1.00	35.973	31.327	7.73	96.9

Note: w_1 as the weight of the sample in air, w_2 as the weight of the sample in water, and $\rho_{\text{H}_2\text{O}}$ as the density of distilled water, $\rho^* = \frac{w_1}{w_1 - w_2} * \rho_{\text{H}_2\text{O}}$

The relative densities of 316L and 316LN were calculated referring to the 316L steel bulk density equal to 7.98 kg/dm³ while the relative density of EUROFER was calculated referring to the EUROFER bulk density equal to 7.85 kg/dm³.

Micrographs analysis

Fig. V-7.9 shows high density specimens produced by both powder feedstocks using the same parameter set corresponding to 58 J/mm^3 energy density. This value is like the energy density used for producing AISI 316L parts with conventional chemistry. On both the materials the superficial relative density values were higher than 99.5%. This measure was done calculating the % of porosity on an internal sample surface with an image editing program.

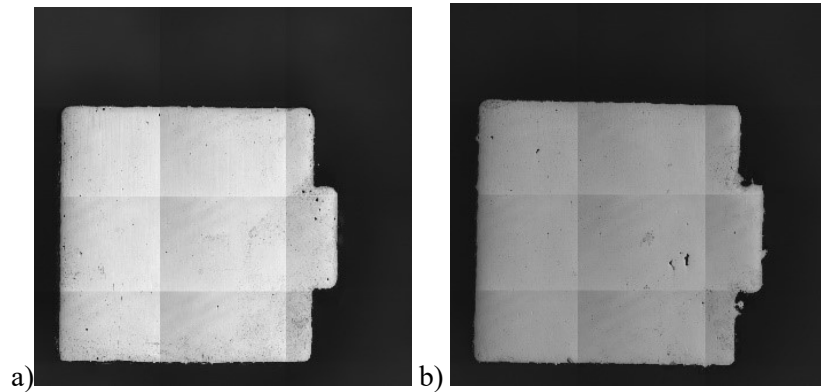


Fig. V-7.9. High density specimens obtained with a) AISI 316L(N)-IG, b) EUROFER.

The images confirm that freestanding components are without macro defects. Further investigations to characterize the material properties are undergoing.

CONCLUSIONS

First Selective Laser Melting (SLM) freestanding components were produced without macro defects using EUROFER metallic powder. The results show that the overall processability for EUROFER and AISI 316L(N)-IG materials are comparable to the alloy more conventionally processed by SLM. The good processability is owed to the correct chemical composition, correct powder size distribution and the choice of the process parameters [14]. The investigations to characterize the material properties show for both the materials relative density values higher than 99.5%. A macroscopic and qualitative analysis indicates good geometrical precision, bright surface quality.

While the benefit of AM is well understood, for these new advanced materials the long-term radiation stability under service conditions is largely unknown. Further characterization tests in this field of these AM materials will be considered with special attention to nuclear safety, qualification, and licensing issues for nuclear industry.

These first results encourage the continuation of research activities in the field of manufacturing of structural components and not for fusion reactors by AM using powders generated from recycled activated materials [14].

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